

TITLE 14B

BUILDING CODE*

* **Editor's note** – Title 14B was added by Coun. J. 4-10-19, p. 100029, Art. III, § 1. This title incorporates provisions of the 2018 International Building Code[®], which is published by International Code Council, Inc. (ICC). These provisions are incorporated by reference and not reprinted in the ordinance. A free, read-only document which compiles the model code provisions and amendments adopted by the City of Chicago, hosted by ICC, can be viewed at: <https://codes.iccsafe.org/content/CHIBC2019P1>. For frequent users of the code, this compiled version of the Chicago Building Code is available for purchase from ICC in print and electronic formats. Provisions governing effective dates for provisions of this title are found in Section 14A-1-105.

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CHAPTER 14B-1

SCOPE AND PURPOSE

14B-1-001 Adoption of the International Building Code by reference.

14B-1-002 Citations.

14B-1-003 Global modifications.

14B-1-100 Chapter 1.

14B-1-001 Adoption of the International Building Code by reference.

The *International Building Code*, 2018 edition, first printing, and all erratum thereto identified by the publisher (hereinafter referred to as "IBC") is adopted by reference and shall be considered part of the requirements of this title except as modified by the specific provisions of this title.

If a conflict exists between a provision modified by this title and a provision adopted without modification, the modified provision shall control.

14B-1-002 Citations.

Provisions of IBC which are incorporated into this title by reference may be cited as follows:

14B-[IBC chapter number]-[IBC section number]

Provided that the appendices shall be deemed to be part of a new Chapter 36.

14B-1-003 Global modifications.

The following modifications shall apply to each provision of IBC incorporated into this title:

1. Replace each occurrence of "*International Codes*" with "*Chicago Construction Codes*."
2. Replace each occurrence of "*International Building Code*" with "*Chicago Building Code*."
3. Replace each occurrence of "ASME A17.1/CSA B44" with "Chapter 3 of the *Chicago Conveyance Device Code*."
4. Replace each occurrence of "NFPA 70" with "the *Chicago Electrical Code*."
5. Replace each occurrence of "*International Fire Code*" with "*Chicago Fire Prevention Code*."
6. Replace each occurrence of "*International Fuel Gas Code*" with "*Chicago Fuel Gas Code*."
7. Replace each occurrence of "*International Mechanical Code*" with "*Chicago Mechanical Code*."
8. Replace each occurrence of "*International Energy Conservation Code*" with "*Chicago Energy Conservation Code*."
9. Replace each occurrence of "*International Plumbing Code*" with "*Chicago Plumbing Code*."
10. Replace each occurrence of "*International Existing Building Code*" with "*Chicago Building Rehabilitation Code*."
11. Replace each occurrence of "*International Property Maintenance Code*" with "*Chicago Minimum Requirements for Existing Buildings*."
12. Delete each bracketed designation of ICC code development committee responsibility preceding a section number.

14B-1-100 Chapter 1.

The provisions of Chapter 1 of IBC are not adopted. The following is adopted as Chapter 1:

"CHAPTER 1. SCOPE AND PURPOSE

101. GENERAL

101.1 Title.

This Title 14B of the Municipal Code of Chicago shall be known as the *Chicago Building Code*, hereinafter referred to as "this code."

101.2 Scope.

The provisions of this code shall apply to the initial construction of *buildings* and *structures* and to other building- and property-related activities, including demolition, maintenance and *rehabilitation*, as specifically referenced in the other *Chicago Construction Codes*.

101.3 Intent.

The purpose of this code is to establish the minimum requirements to provide a reasonable level of safety, public health and general welfare through structural strength, *means of egress* facilities, stability, sanitation, adequate light and ventilation, energy conservation, and safety to life and property from fire, explosion and other hazards, and to provide a reasonable level of safety to fire fighters and emergency responders during emergency operations."

CHAPTER 14B-2

DEFINITIONS AND MEASUREMENTS

14B-2-201 General.

14B-2-202 Definitions.

14B-2-203 Measurements.

14B-2-201 General.

The provisions of Section 201 of IBC are not adopted. The following language is adopted as Section 201:

"201. GENERAL

201.1 Definitions.

The definitions in Section 202 shall apply to italicized words throughout this code, except where specifically limited to a particular chapter or section. Unless the context requires otherwise, the definitions in Section 202 shall also apply to non-italicized words throughout this code.

201.2 Interchangeability.

Words used in the present tense include the future; words stated in the masculine gender include the feminine and neuter; the singular number includes the plural and the plural includes the singular.

201.3 Words defined in other codes.

Where non-italicized words are not defined in this code and are defined in the *Chicago Construction Codes Administrative Provisions*, *Chicago Conveyance Device Code*, *Chicago Electrical Code*, *Chicago Fire Prevention Code*, *Chicago Fuel Gas Code*, *Chicago Mechanical Code*, *Chicago Energy Conservation Code*, *Chicago Plumbing Code*, *Chicago Building Rehabilitation Code*, or *Chicago Minimum Requirements for Existing Buildings*, such words shall have the meanings ascribed to them in those codes.

201.4 Words not defined.

Where italicized words are not defined in Section 202 or non-italicized words are not defined in Section 202 or any of the codes referenced in Section 201.3, such words shall have the meaning given in the latest edition of Merriam Webster's Collegiate Dictionary as the context implies."

14B-2-202 Definitions.

The provisions of Section 202 of IBC are adopted by reference with the following modifications:

1. Revise the definition of "accessible" to read:

"**ACCESSIBLE.** A *site, building, facility* or portion thereof that complies with Section 907.5.2.3, Chapters 10 and 11 and Appendix E."

2. Revise the definition of "accessible means of egress" to read:

"**ACCESSIBLE MEANS OF EGRESS.** A continuous and unobstructed way of egress travel, from any *accessible* point in a *building* or *facility* to a *public way*, that complies with Section 1009."

3. Revise the definition of "accessible route" to read:

"**ACCESSIBLE ROUTE.** A continuous, unobstructed path that complies with Chapter 11 connecting all *accessible elements* and *spaces* of a *building* or *facility*. Interior *accessible routes* may include *corridors*, floors, *ramps*, elevators, lifts, skywalks, tunnels and clear floor space at fixtures. Exterior accessible routes may include parking access aisles, *curb ramps*, crosswalks at *vehicular ways*, *pedestrian walkways*, *ramps*, and lifts."

4. Revise the definition of "accreditation body" to read:

"ACCREDITATION BODY. A third-party organization that is independent of the grading and inspection agencies, and the lumber mills, and that initially accredits and subsequently monitors, on a continuing basis, the competency and performance of a grading or inspection agency related to carrying out specific tasks."

5. Insert the following definitions:

"ADAPTABILITY or ADAPTABLE (for Chapter 11). The ability of certain *building spaces* and *elements*, such as cabinetry, countertops, sinks and grab bars, to be added or altered so as to accommodate the needs of individuals with different types or degrees of *disability*.

ADAPTABLE DWELLING UNIT (for Chapter 11). A *dwelling unit* constructed and equipped so it can be converted with minimal *structural change* for use by persons with different types or degrees of *disability*."

6. Revise the definition of "addition" to read:

"ADDITION. As defined in Chapter 2 of the *Chicago Building Rehabilitation Code*."

7. Revise the definition of "adhered masonry veneer" to read:

"ADHERED MASONRY VENEER. *Veneer* secured and supported through the adhesion of a bonding material applied to a *backing*."

8. Delete the definition of "adobe construction" (and related definitions).

9. Revise the definition of "aerosol container" to read:

"AEROSOL CONTAINER. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

10. Revise the definition of "aerosol product" to read:

"AEROSOL PRODUCT. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

11. Insert the following definition:

"AFFORDABLE UNIT (for Chapter 11). A *dwelling unit* or *sleeping unit* required to be affordable, as that term is defined in Chapter 2-44 of the *Municipal Code*, pursuant to Chapter 2-44 of the *Municipal Code*, an agreement between the *owner* and the *City*, or an agreement between the *owner* and another governmental entity."

12. Revise the definition of "alteration" to read:

"ALTERATION. As defined in Chapter 2 of the *Chicago Building Rehabilitation Code*."

13. Delete the definition of "anchor building."

14. Revise the definition of "anchored masonry veneer" to read:

"ANCHORED MASONRY VENEER. *Veneer* secured with mechanical fasteners to a *backing*."

15. Revise the definition of "approved" to read:

"APPROVED. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

16. Revise the definition of "approved agency" to read:

"APPROVED AGENCY. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

17. Delete the definition of "approved source."

18. Revise the definition of "area, building" to read:

"AREA, BUILDING. See "Building area."

19. Insert the following definition:

"ASSISTIVE LISTENING SYSTEM (ALS). An amplification system using transmitters, receivers and coupling devices to bypass the acoustical space between a sound source and a listener by means of induction loop, radio frequency, infrared or direct-wired equipment."

20. Revise the definition of "atrium" to read:

"ATRIUM. A vertical space that is closed at the top connecting two or more *stories* in a Group I-2 or I-3 occupancy or three or more *stories* in any other occupancy. "

21. Revise the definition of "attic" to read:

"ATTIC. The unfinished space between the ceiling framing of the highest *story* that contains *occupiable space* and the underside of the *roof structure*. An *attic* with a clear height of 81 inches (2057 mm) or more between the top of the ceiling framing and underside of the *roof structure* shall be considered an additional *story*."

22. Revise the definition of "baled cotton" to read:

"**BALED COTTON.** As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

23. Revise the definition of "baled cotton, densely packed" to read:

"**BALED COTTON, DENSELY PACKED.** As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

24. Delete the definition of "base flood."

25. Delete the definition of "base flood elevation."

26. Delete the definition of "basement (for flood loads)."

27. Revise the definition of "basement" to read:

"**BASEMENT.** A *story* that is not a *story above grade plane*."

28. Insert the following definition:

"**BED-AND-BREAKFAST ESTABLISHMENT.** An entire owner-occupied *building*, with not more than four *stories above grade plane*, or an owner-occupied *dwelling unit* containing 11 or fewer *guest rooms* that are regularly made available for rent or for hire on a *transient* basis."

29. Delete the definition of "boarding house."

30. Revise the definition of "boiling point" to read:

"**BOILING POINT.** As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

31. Revise the definition of "building" to read:

"**BUILDING.** Any *structure* utilized or intended for supporting or sheltering any occupancy. Except for purposes of Chapter 11, a single *structure* may be divided into two or more *buildings* by *fire walls*."

32. Revise the definition of "building area" to read:

"**BUILDING AREA.** The value determined in accordance with Section 203.4."

33. Revise the definition of "building height" to read:

"**BUILDING HEIGHT.** The value determined in accordance with Section 203.3."

34. Delete the definition of "building line."

35. Revise the definition of "building official" to read:

"**BUILDING OFFICIAL.** As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

36. Insert the following definition:

"**BUILT ENVIRONMENT (for Chapter 11).** Those parts of the physical environment which are designed, constructed or altered by people, including all *public facilities* and *multi-story housing*."

37. Revise the definition of "capacitor energy storage system" to read:

"**CAPACITOR ENERGY STORAGE SYSTEM.** As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

38. Insert the following definition:

"**CARPORT.** A *structure* or portion of a *structure* that is open on at least two sides and in which not more than five motor vehicles used by the occupants of the *building* or *buildings* on the premises are stored or kept."

39. Revise the definition of "ceiling limit" to read:

"**CEILING LIMIT.** As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

40. Revise the definition of "certificate of compliance" to read:

"**CERTIFICATE OF COMPLIANCE.** A certificate stating that materials and products meet specified standards or that work was done in compliance with *permitted construction documents*."

41. Revise the definition of "change of occupancy" to read:

"**CHANGE OF OCCUPANCY.** As defined in Chapter 2 of the *Chicago Building Rehabilitation Code*."

42. Insert the following definitions:

"**CHICAGO BUILDING CODE.** As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

CHICAGO BUILDING REHABILITATION CODE. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

CHICAGO CONSTRUCTION CODES. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

CHICAGO CONSTRUCTION CODES ADMINISTRATIVE PROVISIONS. Title 14A of the *Municipal Code*.

CHICAGO CONVEYANCE DEVICE CODE. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*.

CHICAGO ELECTRICAL CODE. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*.

CHICAGO ENERGY CONSERVATION CODE. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*.

CHICAGO FIRE PREVENTION CODE. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*.

CHICAGO FUEL GAS CODE. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*.

CHICAGO MECHANICAL CODE. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*.

CHICAGO MINIMUM REQUIREMENTS FOR EXISTING BUILDINGS. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*.

CHICAGO PLUMBING CODE. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*.

CHICAGO ZONING ORDINANCE. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

43. Delete the definition of "children's play structure."

44. Insert the following definitions:

"CIRCULATION PATH (for Chapter 11). An exterior or interior way of passage provided for pedestrian travel including but not limited to: *pedestrian walkways*, hallways, courtyards, elevators, platform lifts, *ramps*, *stairways* and landings.

CITY DATUM. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*.

CITY FIRE ALARM BOX. A device wired directly to the *City's* emergency communications system and capable of transmitting an *alarm signal*."

45. Delete the definition of "climate zone."

46. Insert the following definition:

"CLOSED-CIRCUIT TELEPHONE (for Chapter 11). A telephone with a dedicated line such as a house phone, courtesy phone or phone that must be used to gain entry to a *facility*."

47. Revise the definition of "closed system" to read:

"CLOSED SYSTEM. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

48. Delete the definition of "coastal A zone."

49. Delete the definition of "coastal high-hazard area."

50. Revise the definition of "combustible dust" to read:

"COMBUSTIBLE DUST. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

51. Revise the definition of "combustible fibers" to read:

"COMBUSTIBLE FIBERS. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

52. Revise the definition of "combustible liquid" to read:

"COMBUSTIBLE LIQUID. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

53. Revise the definition of "common path of egress travel" to read:

"COMMON PATH OF EGRESS TRAVEL. That portion of *exit access* travel distance measured from the most remote point of each room, area or space to that point where the occupants have separate and distinct access to two *exits* or *exit access* doorways or to an exterior *exit* door at the level of *exit discharge*."

54. Insert the following definition:

"COMMON USE AREAS or COMMON AREAS (for Chapter 11). Areas, including interior and exterior rooms, *spaces* or *elements*, that are held out for use by all tenants and *owners* in *public facilities* and *multi-story housing*, including but not limited to: residents of an apartment building or condominium complex, occupants of an office building or the guests of such residents or occupants. *Common use areas or common areas* include but are not limited to lobbies, elevators, hallways, laundry rooms, swimming pools, storage rooms, recreation areas, parking garages, building offices, conference rooms, patios, restrooms, telephones, drinking fountains, restaurants, cafeterias, delicatessens and stores."

55. Revise the definition of "compressed gas" to read:

"COMPRESSED GAS. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

56. Revise the definition of "construction documents" to read:

"CONSTRUCTION DOCUMENTS. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*. Also see "permitted construction documents."

57. Insert the following definition:

"CONVEYANCE DEVICE. As defined in Chapter 2 of the *Chicago Conveyance Device Code*."

58. Revise the definition of "corrosive" to read:

"CORROSIVE. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

59. Revise the definition of "court" to read:

"COURT. An open space on a *lot*, other than a *yard* or portion of a *yard*, that is unobstructed from its lowest level to the sky and bounded, in whole or part, by *abutting property lines*, exterior *building walls* or other enclosing devices.

Inner Court. A *court* other than an *outer court*.

Outer Court. A *court* with at least 30 feet (9144 mm) or 25 percent of its perimeter, whichever is less, abutting and open to a *yard* at least 3 feet (914 mm) in width or a *public way* at least 15 feet (4572 mm) in width. "

60. Delete the definition of "covered mall building" (and related definitions).

61. Revise the definition of "cryogenic fluid" to read:

"CRYOGENIC FLUID. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

62. Insert the following definition:

"CURB RAMP (for Chapter 11). A *ramp* that cuts through or is built up to the curb. *Curb ramps* can be perpendicular or parallel, or a combination of perpendicular and parallel *ramps*."

63. Revise the definition of "dangerous" to read:

"DANGEROUS. As defined in Chapter 2 of the *Chicago Building Rehabilitation Code*."

64. Revise the definition of "day box" to read:

"DAY BOX. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

65. Insert the following definition:

"DECK. An open, unroofed and uncovered floor *structure*, other than an *occupiable rooftop*, designed or used for more than incidental occupancy, which may or may not include an exterior *means of egress*."

66. Revise the definition of "decorative glass" to read:

"DECORATIVE GLAZING. A carved, leaded or *Dalle glass* or glazing material whose purpose is decorative or artistic, not functional; whose coloring, texture or other design qualities or components cannot be removed without destroying the glazing material and whose surface, or assembly into which it is incorporated, is divided into segments."

67. Revise the definition of "deferred submittal" to read:

"DEFERRED SUBMITTAL. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

68. Revise the definition of "deflagration" to read:

"DEFLAGRATION. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

69. Delete the definition of "design flood."

70. Delete the definition of "design flood elevation."

71. Revise the definition of "detached building" to read:

"DETACHED BUILDING. A separate single-story *building*, without a *basement* or crawl space, used for the storage or use of *hazardous materials* and located an acceptable distance from all other *structures*."

72. Revise the definition of "detonation" to read:

"DETONATION. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

73. Insert the following definition:

"DISABILITY (for Chapter 11). A physical or mental impairment that substantially limits one or more major life activities; or a record or history of such an impairment; or regarded as having such an impairment."

74. Revise the definition of "dormitory" to read:

"DORMITORY. A space in a *building* where group sleeping accommodations are provided in one room, or in a series of closely associated rooms, for persons not members of the same family group or *household*."

75. Delete the definition of "dry floodproofing."

76. Delete the definition of "dwelling."

77. Insert the following definition:

"EAVE HEIGHT. The vertical distance from the ground surface adjacent to the *building* to the roof eave line at a particular wall. If the vertical distance varies along the wall, eave height shall be the average height."

78. Insert the following definition:

"ELEMENT (for Chapter 11). An architectural, mechanical (including plumbing) or electrical component of a *building, facility, space, site, or public right-of-way.*"

79. Delete the definition of "emergency escape and rescue opening."

80. Delete the definition of "emergency voice/alarm communications."

81. Insert the following definition:

"EMERGENCY WARNING SYSTEM (for Chapter 11). A fire alarm or smoke or heat detector system used to activate audible and visual emergency alarms."

82. Revise the definition of "employee work area" to read:

"EMPLOYEE WORK AREA. All or any portion of a *space* used only by employees and only for work. *Corridors*, toilet rooms, bathing rooms, locker rooms, kitchenettes and break rooms are not employee work areas."

83. Insert the following definition:

"ENTRANCE (for Chapter 11). Any access point to a *building* or portion of a *building* or *facility* or *multi-story housing* used for entering. An *entrance* includes the approach *pedestrian walkway*, the vertical access leading to the *entrance* platform, the *entrance* platform itself, vestibule, if provided, the entry door or gate and the hardware of the entry door or gate."

84. Insert the following definition:

"ENVIRONMENTAL BARRIER (for Chapter 11). An *element* or *space* of the *built environment* which limits accessibility to or use of the *built environment* by individuals with *disabilities.*"

85. Revise the definition of "exhausted enclosure" to read:

"EXHAUSTED ENCLOSURE. As defined in Chapter 2 of the *Chicago Fire Prevention Code.*"

86. Insert the following definition:

"EXHIBITION AREA. A *building*, room or space used for *temporary* exhibition of goods, wares, merchandise or equipment other than a Group M occupancy."

87. Revise the definition of "existing building" to read:

"EXISTING BUILDING. As defined in Chapter 2 of the *Chicago Building Rehabilitation Code.*"

88. Revise the definition of "existing structure" to read:

"EXISTING STRUCTURE. As defined in Chapter 2 of the *Chicago Building Rehabilitation Code.*"

89. Revise the definition of "exit access" to read:

"EXIT ACCESS. That portion of a *means of egress* system that leads from any *occupiable space* to an *exit.*"

90. Revise the definition of "explosion" to read:

"EXPLOSION. As defined in Chapter 2 of the *Chicago Fire Prevention Code.*"

91. Revise the definition of "explosive" to read:

"EXPLOSIVE. As defined in Chapter 2 of the *Chicago Fire Prevention Code.*"

92. Insert the following definition:

"EXTERIOR BALCONY. An exterior floor *structure*, other than an *occupiable rooftop*, that is attached to the *exterior wall* of a *building*, with all *means of egress* requiring travel through the *building*, and that is open to the atmosphere on at least one side."

93. Revise the definition of "facility" to read:

"FACILITY. All or any portion of *buildings, structures, site* improvements, *elements* and pedestrian or vehicular routes located on a *site.*"

94. Insert the following definition:

"FENCE. A *structure*, independent of other *structures*, forming a barrier at ground level between *lots*, between a *lot* and a *public way* or between portions of the same *lot.*"

95. Insert the following definition:

"FIRE ALARM BOX, CITY. See "City fire alarm box."

96. Insert the following definition:

"FIRE CODE OFFICIAL. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

97. Revise the definition of "fire separation distance" to read:

"FIRE SEPARATION DISTANCE. The horizontal distance measured from the building face or element to one of the following:

1. The closest *abutting property line*.
2. The far boundary of a *public way* adjoining the *lot*.
3. An imaginary line between two *buildings* on the same *lot*.

The distance shall be measured at right angles from the face of a wall or edge of a building element."

98. Revise the definition of "fire wall" to read:

"FIRE WALL. A fire-resistance-rated wall having protected openings, which restricts the spread of fire and extends continuously from the foundation to or through the roof. "

98.1 Revise the definition of "fireworks" to read:

"FIREWORKS. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

99. Revise the definition of "flammable gas" to read:

"FLAMMABLE GAS. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

100. Revise the definition of "flammable liquefied gas" to read:

"FLAMMABLE LIQUEFIED GAS. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

101. Revise the definition of "flammable liquid" to read:

"FLAMMABLE LIQUID. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

102. Revise the definition of "flammable material" to read:

"FLAMMABLE MATERIAL. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

103. Revise the definition of "flammable solid" to read:

"FLAMMABLE SOLID. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

104. Revise the definition of "flammable vapors or fumes" to read:

"FLAMMABLE VAPORS OR FUMES. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

105. Revise the definition of "flash point" to read:

"FLASH POINT. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

106. Delete the definition of "flood damage-resistant materials."

107. Delete the definition of "flood, design."

108. Delete the definition of "flood elevation, design."

109. Delete the definition of "flood hazard area."

110. Delete the definition of "flood hazard areas, special."

111. Delete the definition of "flood insurance rate map (FIRM)."

112. Delete the definition of "flood insurance study."

113. Delete the definition of "floodway."

114. Revise the definition of "floor area, gross" to read:

"FLOOR AREA, GROSS. The value determined in accordance with section 203.5.1."

115. Revise the definition of "floor area, net" to read:

"FLOOR AREA, NET. The value determined in accordance with section 203.5.2."

116. Delete the definition of "food court."

117. Revise the definition of "foster care facilities" to read:

"FOSTER CARE FACILITIES. Facilities that provide care on a *24-hour basis* to more than five children that are less than 3 years of age."

118. Revise the definition of "gas cabinet" to read:

"GAS CABINET. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

119. Revise the definition of "gaseous hydrogen system" to read:

"GASEOUS HYDROGEN SYSTEM. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

120. Insert the following definition:

"GOVERNMENTAL UNIT. State agencies, as defined in the State Auditing Act, 30 ILCS 5/, circuit courts, units of local government and their officers, boards of election commissioners, public colleges and universities and school districts."

121. Revise the definition of "grade plane" to read:

"GRADE PLANE. A reference plane established in accordance with Section 203.2."

122. Insert the following definition:

"GROSS FLOOR AREA. See "Floor area, gross."

123. Delete the definition of "gross leasable area."

124. Revise the definition of "group home" to read:

"GROUP HOME. A facility for social rehabilitation, substance abuse or mental health problems that contains a group housing arrangement that provides *custodial care* but does not provide *medical care*."

125. Revise the definition of "guest room" to read:

"GUEST ROOM. A room or suite of rooms used or intended to be used for rent or for hire by one or more guests for living or sleeping purposes and located within a *dwelling unit* or a *sleeping unit*."

126. Revise the definition of "habitable space" to read:

"HABITABLE SPACE. A space in a *building* for recreation, working, sleeping, eating or cooking. Bathrooms, toilet rooms, closets, halls, incidental storage or utility spaces and similar areas are not considered habitable spaces."

127. Revise the definition of "hazardous materials" to read:

"HAZARDOUS MATERIALS. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

128. Revise the definition of "hazardous production material (HPM)" to read:

"HAZARDOUS PRODUCTION MATERIAL (HPM). As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

129. Revise the definition of "health hazard" to read:

"HEALTH HAZARD. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

130. Revise the definition of "height, building" to read:

"HEIGHT, BUILDING. See "Building height."

131. Revise the definition of "highly toxic" to read:

"HIGHLY TOXIC. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

132. Revise the definition of "high-rise building" to read:

"HIGH-RISE BUILDING. A *building* greater than 80 feet (24.4 m) in *building height*."

133. Revise the definition of "historic buildings" to read:

"HISTORIC BUILDINGS. As defined in Chapter 2 of the *Chicago Building Rehabilitation Code*."

134. Insert the following definition:

"HOTEL. A *building* kept, used, maintained as, advertised or held out to the public to be an inn, hotel, motel, family hotel, apartment hotel, boarding house, *lodging house*, *dormitory* or other place where sleeping or rooming accommodations are furnished for hire or rent on a *transient basis*, either with or without meals, excluding *bed-and-breakfast establishments*."

135. Insert the following definition:

"HOUSEHOLD. As defined in Chapter 17-17 of the *Chicago Zoning Ordinance*."

136. Delete the definition of "hurricane-prone regions."

137. Revise the definition of "hydrogen fuel rooms" to read:

"HYDROGEN FUEL GAS ROOM. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

138. Insert the following definition:

"ILLINOIS ACCESSIBILITY CODE. Rules adopted by the State of Illinois pursuant to the Environmental Barriers Act, 410 ILCS 25."

139. Revise the definition of "immediately dangerous to life and health (IDLH)" to read:

"IMMEDIATELY DANGEROUS TO LIFE AND HEALTH (IDLH). As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

140. Revise the definition of "incompatible materials" to read:

"INCOMPATIBLE MATERIALS. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

141. Revise the definition of "inert gas" to read:

"INERT GAS. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

142. Delete the definition of "jurisdiction."

143. Delete the definition of "limit of moderate wave action."

144. Revise the definition of "liquid" to read:

"LIQUID. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

145. Revise the definition of "live/work unit" to read:

"LIVE/WORK UNIT. A *dwelling unit* in which a significant portion of the space includes a nonresidential use that is operated by a resident."

146. Revise the definition of "lodging house" to read:

"LODGING HOUSE. A *building* with not more than four *stories above grade plane* or a *dwelling unit* where one or more occupant is primarily permanent in nature and rent is paid for *guest rooms*."

147. Insert the following definition:

"LOFT. A floor level located above the main floor level within a *dwelling unit* or *sleeping unit*, open to the main floor on at least one side and used as a living or sleeping space."

148. Revise the definition of "lot" to read:

"LOT. As defined in Chapter 17-17 of the *Chicago Zoning Ordinance*.

Lot of record. As defined in Chapter 17-17 of the *Chicago Zoning Ordinance*.

Zoning lot. As defined in Chapter 17-17 of the *Chicago Zoning Ordinance*."

149. Revise the definition of "lot line" to read:

"LOT LINE. See "Property line."

150. Insert the following definition:

"LOT OF RECORD. See "Lot."

151. Revise the definition of "lower flammable limit (LFL)" to read:

"LOWER FLAMMABLE LIMIT (LFL). As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

152. Delete the definition of "lowest floor."

153. Insert the following definition:

"LOW-SLOPED ROOF. A roof with a slope of less than two units vertical in 12 units horizontal (17 percent slope)."

154. Insert the following definition:

"MAILBOXES (for Chapter 11). Receptacles for the receipt of documents, packages or other deliverable matter. *Mailboxes* include, but are not limited to, post office boxes and receptacles provided by commercial mail- receiving agencies, apartment *facilities* or schools."

155. Delete the definition of "mall building, covered" and "mall building, open."

156. Insert the following definition:

"MARKED CROSSING (for Chapter 11). A crosswalk or other identified path intended for pedestrian use in crossing a *vehicular way*."

157. Revise the definition of "means of egress" to read:

"MEANS OF EGRESS. A continuous and unobstructed path of vertical and horizontal egress travel from any *occupiable space* in a *building* or *structure* to a *public way*. A means of egress consists of three separate and distinct parts: the *exit access*, the *exit* and the *exit discharge*."

158. Insert the following definition:

"MEANS OF EGRESS (for Chapter 11). A continuous and unobstructed way of egress travel from any point in a *building* or *facility* that provides an accessible route to an *area of refuge*, a *horizontal exit*, or a *public way*."

159. Insert the following definition:

"MEAN ROOF HEIGHT. The average of the *eave height* and the vertical distance from *grade plane* to the highest point for the highest roof surface, except that for roof slopes of less than 10 percent, the *mean roof height* is permitted to be taken as the *eave height*."

160. Insert the following definition:

"MULTI-STORY HOUSING (for Chapter 11). Any *building* of four or more *stories* containing ten or more *dwelling units* or *sleeping units* constructed to be held out for sale or lease by any *person* to the *public*. Multi-story housing includes, but is not limited to, the following *building* types: apartment buildings, condominium buildings, convents, housing for the elderly and monasteries."

161. Insert the following definition:

"NET FLOOR AREA. See "Floor area, net."

162. Insert the following definition:

"OCCUPIABLE ROOFTOP. An exterior floor *structure* or walking surface, intended for human occupancy, other than occasional use by maintenance and service personnel, that is installed above a *roof deck*. "

163. Revise the definition of "occupiable space" to read:

"OCCUPIABLE SPACE. A room or enclosed space intended for human occupancy in which individuals reside; congregate for amusement, education or similar purposes; or are engaged at labor, and which is required to be equipped with *means of egress* and light and *ventilation* facilities by this code. All *habitable spaces* are *occupiable spaces*. A space which is only required by this code to be provided with means of access by maintenance and service personnel is not an *occupiable space*."

164. Revise the definition of "open system" to read:

"OPEN SYSTEM. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

165. Revise the definition of "operating building" to read:

"OPERATING BUILDING. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

166. Revise the definition of "organic peroxide" to read:

"ORGANIC PEROXIDE. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

167. Revise the definition of "owner" to read:

"OWNER. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

168. Revise the definition of "oxidizer" to read:

"OXIDIZER. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

169. Revise the definition of "oxidizing gas" to read:

"OXIDIZING GAS. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

170. Insert the following definition:

"PARKING FACILITY (except for Chapter 11). A multi-level *structure* in which each level is used primarily for the purpose of storing private motor vehicles, and which does not necessarily have enclosing walls."

171. Revise the definition of "penthouse" to read:

"PENTHOUSE. An enclosed, *rooftop structure* used for sheltering mechanical and electrical equipment, tanks, elevators and related machinery, and vertical *shaft* openings."

172. Revise the definition of "permit" to read:

"PERMIT. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

172.1. Insert the following definition

"PERMITTED CONSTRUCTION DOCUMENTS. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

173. Revise the definition of "person" to read:

"PERSON. As defined in Section 1-4-090 of the *Municipal Code*."

174. Revise the definition of "physical hazard" to read:

"PHYSICAL HAZARD. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

175. Revise the definition of "physiological warning threshold level" to read:

"PHYSIOLOGICAL WARNING THRESHOLD LEVEL. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

176. Insert the following definition:

"PLANNED DEVELOPMENT. A *lot* subject to zoning regulations in accordance with Chapter 17-8 of the *Chicago Zoning Ordinance*."

177. Revise the definition of "platform" to read:

"PLATFORM. A raised area used for worship; the presentation of music, plays or other entertainment; the head table for special guests; the raised area for lecturers and speakers; boxing and wrestling rings; theater-in-the-round; and similar purposes wherein, other than horizontal sliding curtains, there are no overhead hanging curtains, drops, scenery or theatrical effects other than lighting and sound."

178. Delete the definition of "polypropylene siding."

179. Insert the following definition:

"PORCH. An unheated roofed or covered *structure*, containing a *stairway* used for ingress and egress and additional floor space, that is separated from heated areas of the *building* by a fire-resistance rated *exterior wall* and unprotected openings.

Open porch. A *porch* that is open to the atmosphere on at least one side.

Enclosed porch. Any *porch* other than an *open porch*."

180. Delete the definition of "prestressed masonry."

180.1. Revise the definition of "primary structural frame" by adding Item 5 to read:

"5. Bearing walls in *light-frame construction*."

181. Revise the definition of "private garage" to read:

"PRIVATE GARAGE. A *building* or portion of a *building* in which not more than five motor vehicles used by the occupants of the *building* or *buildings* on the premises are stored or kept, without provisions for repairing or servicing such vehicles for profit, inclusive of any attached enclosed walkway."

182. Insert the following definition:

"PROPERTY LINE. As defined in Chapter 17-17 of the *Chicago Zoning Ordinance*.

Abutting property line. As defined in Chapter 17-17 of the *Chicago Zoning Ordinance*.

Front property line. As defined in Chapter 17-17 of the *Chicago Zoning Ordinance*.

Rear property line. As defined in Chapter 17-17 of the *Chicago Zoning Ordinance*.

Side property line. As defined in Chapter 17-17 of the *Chicago Zoning Ordinance*."

183. Insert the following definition:

"PUBLIC (for Chapter 11). Any group of people who are users of the *building* or employees of the *building*. The term "public" does not include those people who are employed by the *owner* of a *building* for the sole purpose of construction or *alteration* of a *building* during the time in which the *building* is being constructed or altered."

184. Insert the following definitions:

"PUBLIC FACILITY (for Chapter 11). A public facility includes any of the following:

1. Any *building*, *structure* or *site* improvement which is: owned by or on behalf of a *governmental unit*; leased, rented or used in whole or in part, by a *governmental unit*; or financed, in whole or in part, by a grant or a loan made or guaranteed by a *governmental unit*.

2. Any *building* or *structure* or *site* improvement used or held out for use or intended for use by the *public* or by employees for one or more of, but not limited to, the following: the purpose of gathering, recreation, transient lodging, education, employment, institutional care, or the purchase, rental, sale or acquisition of any goods, personal property or services; places of public display or collection; social service establishments; and stations used for specified public transportation.

3. A *public right-of-way*.

PUBLICLY-OWNED BUILDING (for Chapter 11). Any *building* or *facility* owned by the State of Illinois or any *governmental unit*.

PUBLIC RIGHT-OF-WAY (for Chapter 11). Public land or property, usually in interconnected corridors, that is acquired for or dedicated to transportation purposes."

185. Revise the definition of "public way" to read:

"PUBLIC WAY. As defined in Chapter 17-17 of the *Chicago Zoning Ordinance*."

186. Revise the definition of "public-use areas" to read:

"PUBLIC-USE AREAS. Interior or exterior rooms or spaces that are made available to the general public. *Employee work areas* are not public-use areas."

187. Revise the definition of "pyrophoric" to read:

"PYROPHORIC. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

188. Revise the definition of "pyrotechnic composition" to read:

"PYROTECHNIC COMPOSITION. As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

189. Delete the definition of "record drawings."

190. Revise the definition of "registered design professional" to read:

"REGISTERED DESIGN PROFESSIONAL. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

191. Revise the definition of "registered design professional in responsible charge" to read:

"REGISTERED DESIGN PROFESSIONAL IN RESPONSIBLE CHARGE. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

192. Insert the following definition:

"REGISTERED GEOTECHNICAL ENGINEER. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

193. Delete the definition of "religious worship, place of."

194. Revise the definition of "repair" to read:

"REPAIR. As defined in Chapter 2 of the *Chicago Building Rehabilitation Code*."

195. Revise the definition of "reroofing" to read:

"REROOFING. As defined in Chapter 2 of the *Chicago Building Rehabilitation Code*."

196. Delete the definition of "residential aircraft hangar."

197. Revise the definition of "restricted entrance" to read:

"RESTRICTED ENTRANCE. An *entrance* that is made available for *common use* on a controlled basis, but not public use, and that is not a *service entrance*. Such *entrances* shall include, but are not limited to, employee- only *entrances*."

198. Revise the definition of "roof recover" to read:

"ROOF RECOVER. As defined in Chapter 2 of the *Chicago Building Rehabilitation Code*."

199. Revise the definition of "roof repair" to read:

"ROOF REPAIR. As defined in Chapter 2 of the *Chicago Building Rehabilitation Code*."

200. Revise the definition of "roof replacement" to read:

"ROOF REPLACEMENT. As defined in Chapter 2 of the *Chicago Building Rehabilitation Code*."

201. Insert the following definition:

"ROOFTOP, OCCUPIABLE. See "Occupiable rooftop."

202. Insert the following definition:

"SCAFFOLDING. As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

203. Delete the definition of "self-luminous."

204. Revise the definition of "service entrance" to read:

"SERVICE ENTRANCE. An *entrance* intended primarily for delivery of goods or services."

205. Revise the definition of "site" to read:

"**SITE.** A *lot* or a designated portion of a *public way*."

206. Revise the definition of "sleeping unit" to read:

"**SLEEPING UNIT.** A single unit that provides rooms or spaces for one or more persons, includes permanent provisions for sleeping and can include provisions for living, eating and either sanitation or kitchen facilities but not both. Such rooms and spaces that are part of a *dwelling unit* occupied by a single *household* are not *sleeping units*."

207. Delete the definition of "soft contained play equipment structure."

208. Revise the definition of "solid" to read:

"**SOLID.** As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

209. Insert the following definition:

"**SPACE (for Chapter 11).** A definable area, such as a room, toilet room, hall, assembly area, entrance, storage room, alcove, courtyard or lobby."

210. Delete the definition of "special flood hazard area."

211. Revise the definition of "special inspection" to read:

"**SPECIAL INSPECTION.** As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

212. Revise the definition of "special inspector" to read:

"**SPECIAL INSPECTOR.** As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

213. Revise the definition of "specified compressive strength of masonry" to read:

"**SPECIFIED COMPRESSIVE STRENGTH OF MASONRY, f'_m .** Minimum compressive strength, expressed as force per unit of net cross-sectional area, required of the *masonry* used in construction by the *permitted construction documents*, and upon which the project design is based. Whenever the quantity f'_m is under the radical sign, the square root of numerical value only is intended and the result has units of pounds per square inch (psi) (MPa)."

214. Revise the definition of "spray room" to read:

"**SPRAY ROOM.** As defined in Chapter 2 of the *Chicago Fire Prevention Code*."

215. Revise the definition of "stage" to read:

"**STAGE.** A space utilized for entertainment or presentations, which includes overhead hanging curtains, drops, scenery or theatrical effects other than lighting and sound."

216. Revise the definition of "start of construction" to read:

"**START OF CONSTRUCTION.** As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions*."

217. Delete the definition of "storm shelter" (and related definitions).

218. Revise the definition of "story" to read:

"**STORY.** That portion of a *building* included between the upper surface of a floor and the upper surface of the floor or roof next above (see "Attic," "Basement," "Building height," "Grade plane," "Loft" and "Mezzanine").

For purposes of Chapters 16 and 23, a *story* is measured as the vertical distance from top to top of two successive tiers of beams or finished floor surfaces and, for the topmost *story*, from the top of the floor finish to the top of the ceiling joists or, where there is not a ceiling, to the top of the roof rafters."

219. Revise the definition of "story above grade plane" to read:

"**STORY ABOVE GRADE PLANE.** Any *story* having its finished floor surface entirely above *grade plane*, or in which the finished surface of the floor next above is:

1. More than 6 feet (1829 mm) above *grade plane*; or
2. More than 12 feet (3658 mm) above the adjacent finished ground level at any point."

220. Revise the definition of "structural observation" to read:

"**STRUCTURAL OBSERVATION.** The visual observation of the structural system by a *registered design professional* for general conformance to the *permitted construction documents*."

221. Revise the definition of "substantial damage" to read:

"**SUBSTANTIAL DAMAGE.** As defined in Chapter 2 of the *Chicago Building Rehabilitation Code*."

222. Revise the definition of "substantial improvement" to read:

"**SUBSTANTIAL IMPROVEMENT.** As defined in Chapter 2 of the *Chicago Building Rehabilitation Code.*"

223. Insert the following definition:

"**TELECOMMUNICATIONS EQUIPMENT AREA.** An area or enclosed room within a *building* where electronic equipment used for the transmission of audio, video and data, power equipment (e.g., dc converters, inverters and batteries), technical support equipment (e.g., computers) and conductors dedicated solely to the operation of the equipment are located, including support rooms served by the same *ventilation* system."

223.1. Insert the following definition:

"**TELEPHONE EXCHANGE.** A building or portion of a building that primarily houses communications switching equipment and that is owned or operated by a person whose primary business is the providing of communication services other than radio or television."

224. Insert the following definition:

"**TEMPORARY.** As defined in Chapter 2 of the *Chicago Construction Codes Administrative Provisions.*"

225. Insert the following definition:

"**TEMPORARY (for Chapter 11).** A *building* or *element* of a *building* which is not permanent and is designed to be used only for a short period of time for some special purpose. Temporary *buildings* or *facilities* include, but are not limited to, reviewing stands, temporary classrooms, bleacher areas, stages, platforms and daises, fixed furniture systems, wall systems, exhibit areas, temporary banking *facilities* and temporary health screening *facilities.*"

226. Insert the following definition:

"**TEMPORARY OVERNIGHT SHELTER.** A *building*, or portion thereof, in which sleeping accommodations are provided for no more than 12 hours per day, for three or more individuals who are not related to the *owner*, operator, manager or other occupants by blood or by marriage, without provisions for *medical care.*"

227. Revise the definition of "tires, bulk storage of" to read:

"**TIRES, BULK STORAGE OF.** As defined in Chapter 2 of the *Chicago Fire Prevention Code.*"

228. Revise the definition of "townhouse" to read:

"**TOWNHOUSE.** A *dwelling unit* constructed in a group of four or more attached *dwelling units* in which each *dwelling unit* extends from the foundation or a *horizontal assembly* with a *fire-resistance rating* of not less than 4 hours to the roof and with a *public way, yard* or *outer court* on at least two sides."

229. Revise the definition of "toxic" to read:

"**TOXIC.** As defined in Chapter 2 of the *Chicago Fire Prevention Code.*"

230. Delete the definition of "tsunami design geodatabase."

231. Delete the definition of "tsunami design zone."

232. Insert the following definition:

"**TYPE C UNIT.** A *dwelling unit* designed and constructed for limited *accessibility* in accordance with this code and the provisions for *Type C units* (visitable units) in ICC A117.1."

233. Revise the definition of "unstable (reactive) material" to read:

"**UNSTABLE (REACTIVE) MATERIAL.** As defined in Chapter 2 of the *Chicago Fire Prevention Code.*"

234. Revise the definition of "use (material)" to read:

"**USE (MATERIAL).** As defined in Chapter 2 of the *Chicago Fire Prevention Code.*"

235. Insert the following definition:

"**VEHICULAR WAY (for Chapter 11).** A route provided for vehicular traffic, such as a street, driveway or parking facility."

236. Revise the definition of "water-reactive material" to read:

"**WATER-REACTIVE MATERIAL.** As defined in Chapter 2 of the *Chicago Fire Prevention Code.*"

237. Delete the definition of "windborne debris region."

238. Insert the following definition:

"**WORK-AREA EQUIPMENT.** Any machine, instrument, engine, motor, pump, conveyor or other apparatus used to perform work. This term shall only apply to equipment that is permanently installed or built-in in *employee work areas*. Work-area equipment does not include passenger elevators and other accessible means of vertical transportation."

239. Revise the definition of "yard" to read:

"YARD. A four-sided open space that is unobstructed from the ground to the sky, with at least one side abutting and open to a *public way* at least 15 feet (4572 mm) in width, at least one side being at least 75 percent of a *property line* for the *lot* on which the *building* is situated and, except as specifically allowed by this code, on the same *lot* on which the *building* is situated."

240. Insert the following definition:

"ZONING LOT. See "Lot."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 1; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 1; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 1; Amend Coun. J. 1-26-22, p. 43756, Art. II, § 1; Amend Coun. J. 7-20-22, p. 50878, § 1)

14B-2-203 Measurements.

The following language is adopted as a new Section 203:

"203. MEASUREMENTS

203.1 Scope.

Grade plane, *building height*, *building area* and floor area shall be established in accordance with Sections 203.2 through 203.5, respectively.

203.2 Grade plane.

With respect to each *building*, *grade plane* shall be a reference plane representing the average finished ground level adjoining the *building* at *exterior walls*. Where the finished ground level slopes away from the *exterior walls*, the reference plane shall be established by the lowest points within the area between the *building* and the *property line* or, where the *property line* is more than 6 feet (1829 mm) horizontally from the *building*, between the *building* and a point 6 feet (1829 mm) horizontally from the *building*.

Exceptions:

1. In no case shall *grade plane* be lower than the top of the established curb at the midpoint of the *front property line* of the *lot* on which the *building* is located.
2. For all *structures* on a *zoning lot* not exceeding 10,000 square feet (929 m²) in area, "grade," determined in accordance with Section 17-17-0265 of the *Chicago Zoning Ordinance*, shall be allowed to be used as *grade plane*.

203.3 Building height.

Building height shall be the vertical distance from *grade plane* to the mean elevation of the highest roof plane. For purposes of this code, *building height* is not the same as *mean roof height*.

Exceptions:

1. *Building height* shall be measured to the highest walking surface of an *occupiable rooftop* located above the highest *story*.
2. *Building height* shall be measured to the top of a parapet wall that exceeds 42 inches (1067 mm) above the highest point of a *low-sloped roof* or *occupiable rooftop*.
3. Unoccupied *rooftop structures* shall not be considered in determining *building height* in accordance with Section 1510.1.1.
4. Rooftop access *penthouses* and other *rooftop structures* associated with *occupiable rooftops* shall be considered in determining *building height* in accordance with Section 1513.
5. Dormers that do not have a *low-sloped roof*, are no higher than the highest point of the roof plane in which they are located and do not exceed one-third of the horizontal area of the roof plane in which they are located shall not be considered in determining *building height*.
6. For *buildings* without *low-sloped roofs* or *occupiable rooftops*, *building height* may be measured to the mean elevation of all roof surfaces above the finished floor of the highest *story*.
7. Where there is no *occupiable rooftop* above and the highest roof plane is provided with *continuous insulation* installed entirely above and not exceeding 12 inches (305 mm) in height above the structural *roof deck* and required parapet walls, *building height* may be measured to the highest point of the structural *roof deck*.

203.4 Building area.

Building area shall be the sum of the horizontal area included within surrounding *exterior walls*, or *exterior walls* and *fire walls*, exclusive of *courts* for each *story* above *grade plane* and the horizontal area of *mezzanines*, *equipment platforms* and *lofts*. Areas of the *building* not provided with surrounding walls, such as patios and *exterior balconies*, shall be included in the *building area* if such areas are included within the horizontal projection of the roof, floor or walking surface above. The floor area of *basements* shall be included in *building area* where required by Section 506.1.3.

Exception: The horizontal area of the portion of any *story*, *mezzanine* or *loft* with a ceiling height of less than 5 feet (1524 mm) shall be excluded from the calculation of *building area*.

203.5 Floor area.

Each reference to "floor area" in this code shall mean *gross floor area* unless the provision specifically references *net floor area*.

203.5.1 Gross floor area.

Gross floor area shall be measured in the manner provided for determining *building area* in Section 203.4 with a further deduction for the horizontal area of *shafts* without openings and the thickness of interior walls enclosing such *shafts*. There shall be no deduction for *corridors*, *stairways*, *ramps*, closets, mechanical rooms, elevator shafts, toilet rooms, the thickness of interior walls, columns, or other features.

203.5.1.1 Separated mixed occupancies.

The *gross floor area* of a mixed occupancy separated in accordance with Section 508.4 shall be the horizontal area included within surrounding *exterior walls*, *fire walls* and *fire barriers*. Areas of the occupancy not provided with surrounding walls, such as patios and *exterior balconies*, shall be included in the *gross floor area* if such areas are included within the horizontal projection of the roof, floor or walking surface above.

203.5.1.2 Unseparated mixed occupancies.

The *gross floor area* of an occupancy that is not separated in accordance with Section 508.4 shall be measured to the centerline of interior walls and partitions that separate the occupancy from other occupancies in the same *story*.

203.5.2 Net floor area.

Net floor area shall be the actual occupiable horizontal area, not including columns, permanently attached equipment and furnishings that are 28 inches (711 mm) or greater above the floor, the thickness of interior walls and unoccupied accessory areas such as *aisles*, *corridors*, *stairways*, *ramps*, individual wellness/lactation rooms, toilet rooms, mechanical rooms and closets."

(Amend Coun. J. 7-24-19, p. 3646, § 25; Amend Coun. J. 2-19-20, p. 14473, Art. II, § 2; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 2; Amend Coun. J. 1-26-22, p. 43756, Art. II, § 2)

CHAPTER 14B-3

OCCUPANCY CLASSIFICATION AND USE

14B-3-300 Defined terms.

14B-3-301 General.

14B-3-302 Occupancy classification and use designation.

14B-3-303 Assembly Group A.

14B-3-304 Business Group B.

14B-3-305 Educational Group E.

14B-3-306 Factory Group F.

14B-3-307 High-hazard Group H.

14B-3-308 Institutional Group I.

14B-3-309 Mercantile Group M.

14B-3-310 Residential Group R.

14B-3-311 Storage Group S.

14B-3-312 Utility and Miscellaneous Group U.

14B-3-300 Defined terms.

Where provisions of Chapter 3 of IBC are adopted by reference, the following terms shall be reset in italic type: "building" and "structure."

14B-3-301 General.

The provisions of Section 301 of IBC are adopted by reference with the following modification:

1. Revise Section 301.1 to read:

"301.1 General.

The provisions of this chapter shall control the classification of all *buildings* as to occupancy and use."

14B-3-302 Occupancy classification and use designation.

The provisions of Section 302 of IBC are adopted by reference with the following modifications:

1. Revise Section 302.1 to read:

"302.1 Occupancy classification.

Occupancy classification is the formal designation of the primary purpose of the *building, structure* or portion thereof. *Buildings* shall be classified into one or more of the occupancy groups listed in this section based on the nature of the hazards and risks to occupants generally associated with the intended purpose of the *building*. *Structures* that are not *buildings* shall be classified as Group U. An area, room or space that is intended to be occupied at different times for different purposes shall comply with all applicable requirements associated with each potential purpose. *Buildings* containing multiple occupancy groups shall comply with Section 508. Where a *building* is proposed for a purpose that is not specifically listed in this section, such *building* shall be classified in the occupancy it most nearly resembles based on the fire safety and relative hazard as *approved* by the *building official*.

1. Assembly (see Section 303): Groups A-1, A-2, A-3, A-4 and A-5.
2. Business (see Section 304): Group B.
3. Educational (see Section 305): Groups E-1 and E-2.
4. Factory and Industrial (see Section 306): Groups F-1 and F-2.
5. High Hazard (see Section 307): Groups H-1, H-2, H-3, H-4 and H-5.
6. Institutional (see Section 308): Groups I-1, I-2, I-3 and I-4.
7. Mercantile (see Section 309): Group M.
8. Residential (see Section 310): Groups R-1, R-2, R-3, R-4 and R-5.
9. Storage (see Section 311): Groups S-1 and S-2.
10. Utility and Miscellaneous (see Section 312): Group U."

14B-3-303 Assembly Group A.

The provisions of Section 303 of IBC are adopted by reference with the following modifications:

1. Delete Section 303.1.4.
2. Revise Section 303.3 to insert, alphabetically: "Dance halls (including food or drink consumption)."
3. Revise Section 303.4 to replace "Indoor tennis courts (without spectator seating)" with "Indoor sports and fitness facilities (without spectator seating)."
4. Revise Section 303.4 to insert, alphabetically: "Trading floors."

14B-3-304 Business Group B.

The provisions of Section 304 of IBC are adopted by reference with the following modifications:

1. Revise Section 304.1 to replace "Electronic data processing" with "Electronic data processing (with routine human occupancy)."
2. Add a new Section 304.2 to read:

"304.2 Telecommunications equipment areas.

A *telecommunication equipment area* that is associated with an occupancy in the same *building* is not considered a separate occupancy."

14B-3-305 Educational Group E.

The provisions of Section 305 of IBC are not adopted. The following language is adopted as Section 305:

"305. EDUCATIONAL GROUP E

305.1 Educational Group E.

Educational Group E occupancy includes, among others, the use of a *building* or *structure*, or a portion thereof, for educational purposes through the 12th grade or to provide day care services to children.

305.2 Educational Group E-1.

Educational Group E-1 occupancy includes the use of a *building* or *structure*, or a portion thereof, to educate six or more children, three years of age or older, at any one time, through the 12th grade.

Exception: Classrooms and auditoriums, which are accessory to an assembly occupancy and have *occupant loads* of less than 100 per room or space, shall be classified as Group A-3 occupancies.

305.3 Educational Group E-2.

Educational Group E-2 occupancy includes the use of a building or structure, or a portion thereof, for providing supervision, or *personal care services* for fewer than 24 hours per day to more than three children.

Exceptions:

1. Rooms and spaces accessory to an assembly occupancy providing such day care during events shall be classified as part of the primary occupancy.
2. A *facility* having three or fewer children receiving such day care shall be classified as part of the primary occupancy.
3. A *facility* within a *dwelling unit* occupied as a residence by the operator and having eight or fewer children receiving such day care shall be classified as a Group R-2, R-3 or R-5 occupancy.
4. A *facility* where any children under two years of age occupy a level other than the *level of exit discharge* shall be classified as a Group I-4 occupancy.
5. A *facility* with more than 30 children under two years of age shall be classified as a Group I-4 occupancy."

14B-3-306 Factory Group F.

The provisions of Section 306 of IBC are adopted by reference with the following modifications:

1. Revise Section 306.2 to replace "Laundries" with "Laundries (other than pick-up and delivery stations and self- service)."
2. Revise Section 306.2 to insert, alphabetically: "Recycling plants."
3. Revise Section 306.2 to insert, alphabetically: "Tire storage [under 10,000 cubic feet (283 m³)]."
4. Revise the first sentence of Section 306.3 to read:

"Factory industrial uses that involve the fabrication or manufacturing of noncombustible materials that during finishing, packing or processing do not involve a significant fire hazard may only be classified as F-2 occupancies with the written approval of the *fire code official* and may include the following:"

5. Revise Section 306.3 to insert, alphabetically: "Chilled water plants."
6. Revise Section 306.3 to delete "Metal products (fabrication and assembly)."
7. Revise Section 306.3 to insert, alphabetically: "Water pumping stations."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 3)

14B-3-307 High-hazard Group H.

The provisions of Section 307 of IBC are adopted by reference with the following modifications:

1. Revise note e to Table 307.1(1) to delete "approved" and "Section 5003.9.10 of."
2. Revise note i to Table 307.1(1) to delete "Section 603.3.2 of."
3. Revise note k to Table 307.1(1) to replace "approved" with "*approved*."
4. Revise note m to Table 307.1(1) to delete "Section 5003.1.2 of."
5. Revise note p to Table 307.1(1) to delete "Sections 5705.5 and 5705.5.1 of."
6. Revise note d to Table 307.1(2) to delete "approved."
7. Revise note e to Table 307.1(2) to delete "approved."
8. Revise note g to Table 307.1(2) to delete "approved."
9. Revise note i to Table 307.1(2) to delete "Section 5003.1.2 of."
10. Revise item 1 in Section 307.1.1 to delete "Section 416 and."
11. Revise item 17 in Section 307.1.1 to delete "Section 428 and Chapter 38 of."
12. Revise Section 307.7 by replacing "Section 415.11" with "the *Chicago Fire Prevention Code*."

14B-3-308 Institutional Group I.

The provisions of Section 308 of IBC are adopted by reference with the following modifications:

1. In provisions adopted by reference, reset the following term in italic type: "24-hour basis" and "custodial care."
- 1.1. Revise Section 308.2 by replacing "Alcohol and drug centers" with "Alcohol and drug abuse treatment facilities."
2. Revise Section 308.2.3 to read:

"308.2.3 Six to 16 individuals receiving custodial care.

A facility housing not fewer than six and not more than 16 individuals receiving *custodial care* shall be classified as Group R-4."

3. Revise Section 308.2.4 to read:

"308.2.4 Five or fewer individuals receiving custodial care.

A facility with five or fewer individuals receiving *custodial care* shall be classified as Group R-3 provided an *automatic sprinkler system* is installed in accordance with Section 903.3.1.3."

4. Revise Section 308.3.1.1 to read:

"308.3.1.1 Condition 1.

This occupancy condition shall include facilities that provide nursing or *medical care* but do not provide emergency care, surgery, obstetrics or in-patient stabilization units for psychiatric or detoxification purposes, including but not limited to *nursing homes* and *foster care facilities*."

5. Revise Section 308.3.1.2 to read:

"308.3.1.2 Condition 2.

This occupancy condition shall include facilities that provide nursing or *medical care* and could provide emergency care, surgery, obstetrics or in-patient stabilization units for psychiatric or detoxification purposes, including but not limited to *hospitals*."

6. Revise Section 308.3.2 to read:

"308.3.2 Five or fewer individuals receiving medical care.

A facility with five or fewer individuals receiving nursing or *medical care* shall be classified as Group R-3 provided an *automatic sprinkler system* is installed in accordance with Section 903.3.1.3."

7. Revise Section 308.4.1 to read:

"308.4.1 Condition 1.

This occupancy condition shall include *buildings* in which free movement is allowed from sleeping areas, and other spaces where access or occupancy is permitted, to the exterior via *means of egress* without restraint. A Condition 1 facility is permitted to be constructed in accordance with requirements for a Group R occupancy where *approved* by the *fire code official*."

8. Revise Section 308.5.1 to read:

"308.5.1 Classification as Group E-2.

A child day care facility that provides care for less than 30 children under 2 years of age, where the rooms in which the children are cared for are located on a *level of exit discharge* serving such rooms, shall be classified as Group E-2."

9. Delete Section 308.5.2.

10. Revise Section 308.5.4 to read:

"308.5.4 Five or fewer individuals receiving care in a dwelling unit.

A facility such as the above within a *dwelling unit* and having five or fewer individuals receiving *custodial care* shall be classified as a Group R-2, R-3 or R-5 occupancy."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 4)

14B-3-309 Mercantile Group M.

The provisions of Section 309 of IBC are adopted by reference without modification.

14B-3-310 Residential Group R.

The provisions of Section 310 of IBC are not adopted. The following language is adopted as Section 310:

"310. RESIDENTIAL GROUP R

310.1 Residential Group R.

Residential Group R occupancy includes, among others, the use of a *building* or *structure*, or a portion thereof, for sleeping purposes when not classified as an Institutional Group I.

310.2 Residential Group R-1.

Residential Group R-1 occupancies contain *dwelling units* or *sleeping units* where the occupants are primarily *transient* in nature, including:

Hotels with accommodations for more than 10 *transient* occupants

Congregate living facilities with accommodations for more than 10 *transient* occupants

Temporary overnight shelters

310.3 Residential Group R-2.

Residential Group R-2 occupancies contain one or more *sleeping units* or more than three *dwelling units* (including *live/work units*) where the occupants are primarily permanent in nature, including:

Apartment houses (including condominium and cooperative apartments)

Congregate living facilities (nontransient) with more than 16 occupants

Townhouses

310.4 Residential Group R-3.

Residential Group R-3 occupancies where the occupants are primarily permanent in nature and not classified as Group R-1, R-2, R-4, R-5 or I, including:

Bed-and-breakfast establishments

Care facilities that provide accommodations for 5 or fewer individuals receiving care

Congregate living facilities (nontransient) with 16 or fewer occupants

Congregate living facilities with accommodation for 10 or fewer *transient* occupants

Hotels with accommodations for 10 or fewer *transient* occupants

310.4.1 Care facilities within a dwelling.

Care facilities for five or fewer individuals receiving care that are within a single-family dwelling are permitted to comply with the requirements for Group R-5, provided an *automatic sprinkler system* is installed in accordance with Section 903.3.1.3.

310.5 Residential Group R-4.

Residential Group R-4 occupancy shall include *buildings*, *structures* or portions thereof for more than five but not more than 16 individuals, excluding staff, who reside on a *24-hour basis* in a supervised residential environment and receive *custodial care*. *Buildings* of Group R-4 shall be classified as one of the occupancy conditions specified in Section 310.5.1 or 310.5.2. This group shall include, but not be limited to, the following:

Alcohol and drug abuse treatment facilities

Assisted living facilities

Congregate care facilities

Group homes

Halfway houses

Residential board and care facilities

Social rehabilitation facilities

Group R-4 occupancies shall meet the requirements for construction as defined for Group R-3, except as otherwise provided for in this code.

310.5.1 Condition 1.

This occupancy condition shall include *buildings* in which all individuals receiving *custodial care* are capable of responding to an emergency situation to complete *building* evacuation without any assistance.

310.5.2 Condition 2.

This occupancy condition shall include *buildings* in which there are any individuals receiving *custodial care* who require

verbal or physical assistance while responding to an emergency situation to complete *building* evacuation.

310.6 Residential Group R-5.

Residential Group R-5 occupancy shall include *buildings* with no more than four *stories above grade plane* containing one, two or three *dwelling units* (including *live/work units*), with or without an attached *private garage*, and no other occupancy, where each *dwelling unit* is primarily occupied on a non-*transient* basis by a single *household*. Group R-5 shall also include accessory *buildings* with no more than two *stories above grade plane* located on the same *lot* as Group R-5 *dwelling units*.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 5)

14B-3-311 Storage Group S.

The provisions of Section 311 of IBC are adopted by reference with the following modifications:

1. Revise Section 311.2 to insert, alphabetically: "Computers and office machines."
2. Revise Section 311.2 to insert, alphabetically: "Electronic data processing (with only incidental human occupancy)."
3. Revise the first paragraph of Section 311.3 to read:

"Storage Group S-2 occupancies shall include open or enclosed parking garages, other than *repair garages*, *private garages* and *carports*.

Storage Group S-2 occupancies may also include, with the written approval of the *fire code official*, buildings used for the storage of noncombustible materials with no more than one layer of combustible packaging or protection. Such products are permitted to have a negligible amount of plastic *trim*, such as knobs, handles or film wrapping. *Approved* Group S-2 storage uses may include storage of the following:"

4. Revise Section 311.3 to delete "Fresh fruits and vegetables in nonplastic trays or containers," "Ivory" and "Parking garages, open or enclosed."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 6)

14B-3-312 Utility and Miscellaneous Group U.

The provisions of Section 312 of IBC are adopted by reference with the following modifications:

1. Revise the list in Section 312.1 to read:

"*Agricultural buildings*

Equipment *structures* with a *gross floor area* of less than 1,500 square feet (139 m²)

Lumber yards (open exterior type only)

Parking facilities

Private garages and carports

Stables"

2. Revise Section 312.1.1 to read:

"312.1.1 Greenhouses.

Greenhouses not classified as another occupancy shall be classified as Use Group U."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 7)

CHAPTER 14B-4

SPECIAL DETAILED REQUIREMENTS BASED ON USE AND OCCUPANCY

14B-4-401 Scope.

14B-4-402 IBC Section 402.

14B-4-403 High-rise buildings.

14B-4-404 Atriums.

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14B-4-406 Motor-vehicle-related occupancies.

- 14B-4-407 Group I-2.
- 14B-4-408 Group I-3.
- 14B-4-409 Motion picture projection rooms.
- 14B-4-410 Stages, platforms and technical production areas.
- 14B-4-411 Special amusement buildings.
- 14B-4-412 Aircraft-related occupancies.
- 14B-4-413 Combustible storage.
- 14B-4-414 Hazardous materials.
- 14B-4-415 IBC Section 415.
- 14B-4-416 IBC Section 416.
- 14B-4-417 IBC Section 417.
- 14B-4-418 IBC Section 418.
- 14B-4-419 Live/work units.
- 14B-4-420 Groups I-1, R-1, R-2, R-3, R-4 and R-5.
- 14B-4-421 IBC Section 421.
- 14B-4-422 Ambulatory care facilities.
- 14B-4-423 IBC Section 423.
- 14B-4-424 IBC Section 424.
- 14B-4-425 IBC Section 425.
- 14B-4-426 IBC Section 426.
- 14B-4-427 IBC Section 427.
- 14B-4-428 IBC Section 428.

14B-4-401 Scope.

The provisions of Section 401 of IBC are adopted by reference without modification.

14B-4-402 IBC Section 402.

The provisions of Section 402 of IBC are not adopted.

14B-4-403 High-rise buildings.

The provisions of Section 403 of IBC are adopted by reference with the following modifications:

1. Revise item 2 in the exception to Section 403.1 to read:

"2. *Parking facilities* in accordance with Section 406.9."
2. Delete items 4 and 5 in the exception to Section 403.1.
3. Revise item 1 (including the exception) in Section 403.2.1.1 to read:

"1. For *buildings* not greater than 400 feet (122 m) in *building height*, the *fire-resistance rating* of the building elements in Type IA construction, other than the *primary structural frame*, shall be permitted to be reduced to the minimum *fire-resistance ratings* for the building elements in Type IB."
4. Revise item 2 in Section 403.2.1.1 to read:

"2. In other than Group F, H, and S occupancies, the *fire-resistance rating* of the building elements in Type IB construction, other than the *primary structural frame*, shall be permitted to be reduced to the *fire-resistance ratings* in Type IIA."
5. Revise Section 403.2.1.2 by replacing "420 feet (128 m)" with "400 feet (122 m)."
6. Revise Section 403.2.3 by replacing "420 feet (128 m)" with "400 feet (122 m)."
7. Revise Table 403.2.4 by replacing "420 feet" with "400 feet."

8. Revise the exception to Section 403.3 to read:

"Exception: An *automatic sprinkler system* shall not be required in *parking facilities* accordance with Section 406.9."

9. Delete Section 403.3.1 and its subsections.

10. Revise the title and first sentence of Section 403.3.2 to read:

"403.3.2 Secondary water supply to required fire pumps.

In *buildings* that are more than 300 feet (91.4 m) in *building height*, required fire pumps shall be supplied by connections to not fewer than two water mains located in different streets."

11. Revise the title and first sentence of Section 403.3.3 to read:

"403.3.3 Secondary on-site water supply.

An automatic secondary on-site water supply having a capacity not less than the hydraulically calculated sprinkler demand, including the hose stream requirement, shall be provided for *high-rise buildings* assigned to *Seismic Design Category C* or *D* as determined by Section 1613."

11.1. Add a new Section 403.3.5 to read:

"403.3.5 Standpipe system.

Where *building height* exceeds 500 feet (152.4 m), the connection to the source of supply shall be located in the lowest *story* of the building, and the piping system shall be completely looped, without dead ends."

12. Revise Section 403.4.4 to read:

"403.4.4 Voice communication systems.

Voice communication systems shall be provided in accordance with Section 907.2.12.2."

13. Delete Section 403.4.5.

14. Revise Section 403.4.6 to read:

"403.4.6 Fire command.

A *fire command center* complying with Section 911 shall be provided in a location acceptable to the *fire code official*."

15. Revise exception 1 to item 1 in Section 403.4.7 by deleting "or suite."

16. Revise Section 403.4.8.1, excluding the exception, to read:

"403.4.8.1 Equipment room.

If the *standby power system* or *emergency power system* includes a generator set inside a *building*, the generator set shall be located in a separate room enclosed with 2-hour *fire barriers* constructed in accordance with Section 707 or *horizontal assemblies* constructed in accordance with Section 711, or both. System supervision with manual start and transfer features shall be provided at the *fire command center*."

17. Revise Section 403.4.8.2 to read:

"403.4.8.2 Fuel line piping protection.

Fuel lines supplying a generator set inside a *building* shall be of double wall pipe with leak detection provided in the interstitial space and separated from areas of the *building* other than the room the generator is located in by an assembly that has a *fire-resistance rating* of not less than 2 hours."

18. Revise Section 403.4.8.3 to read:

"403.4.8.3 Standby power loads.

The following are classified as standby power loads:

1. [Reserved.]
2. Elevators.
3. [Reserved.]

19. Revise Section 403.4.8.4 to read:

"403.4.8.4 Emergency power loads.

The following are classified as emergency power loads:

1. Exit signs and *means of egress* illumination required by Chapter 10.
2. Elevator car lighting.

3. Voice communication systems.
4. Automatic fire detection systems.
5. *Fire alarm systems.*
6. Electrically powered fire pumps.
7. Power and lighting for the *fire command center* required by Section 403.4.6.
8. Sump pumps required by Section 403.4.9.3."

20. Add a new Section 403.4.8.5 to read:

"403.4.8.5 Diesel generator.

In *high-rise buildings* that are more than 400 feet (122 m) in *building height*, a diesel-driven generator with an on-premises fuel supply sufficient for not less than 4 hours of full-demand operation shall be provided to supply standby and emergency power loads.

The fuel source for generators in *high-rise buildings* that are 400 feet (122 m) or less in *building height* shall be either diesel fuel or natural gas."

21. Add a new Section 403.4.9 to read:

"403.4.9 Electrical equipment rooms.

Electrical equipment rooms in *high-rise buildings* shall comply with Sections 403.4.9.1 through 403.9.3.

403.4.9.1 Main electrical service equipment.

All main electrical service equipment shall be installed in vaults or outdoor locations.

403.4.9.2 Construction.

Electrical equipment rooms shall be enclosed with *fire barriers* constructed in accordance with Section 707 or *horizontal assemblies* constructed in accordance with Section 711, or both. The fire-resistance rating of the enclosure shall be in accordance with Section 110.26 of the *Chicago Electrical Code*.

403.4.9.3 Sump pump.

Where electrical service equipment is installed in the lowest *story*, the electrical equipment room shall be provided with a sump pump."

22. Add a new exception to Section 403.5.1 to read:

"Exception: In Group R-2 occupancies with an automatic smoke detection system that activates the occupant notification system in accordance with Section 907.5 installed throughout all interior *corridors* serving *dwelling units*, required *interior exit stairways* shall be separated by a distance not less than 15 feet (4572 mm)."

23. Delete Section 403.5.2.

24. Revise the second sentence of Section 403.5.3 to read:

"*Stairway* doors that are locked from the *stairway* side shall be capable of being unlocked simultaneously, without unlatching, upon a signal from the *fire command center* and automatically, either by smoke detectors or sprinkler waterflow devices."

25. Revise Section 403.5.3.1 to read:

"403.5.3.1 Stairway communication system.

A two-way communication system shall be provided in accordance with Section 907.2.12.2.1."

26. Delete Section 403.5.4.

27. Delete Section 403.5.5.

28. Delete Section 403.5.6.

29. Revise Section 403.6 and its subsections to read:

"403.6 Elevators.

Elevator installation and operation in *high-rise buildings* shall comply with Chapter 30 and Section 403.6.1.

403.6.1 Fire service access elevator.

A minimum of one fire service access elevator shall be provided in accordance with Section 3007. Each fire service access elevator shall have a capacity of not less than 3,500 pounds (1588 kg) and shall comply with Section 3002.4."

14B-4-404 Atriums.

The provisions of Section 404 of IBC are adopted by reference with the following modifications:

1. Revise the exception to Section 404.2 by deleting "*approved*."
2. Revise Section 404.3 to read:

"404.3 Automatic sprinkler protection.

An automatic sprinkler system shall be installed throughout the entire building.

Exception: Where the ceiling of the *atrium* is more than 55 feet (16 764 mm) above the floor, sprinkler protection at the ceiling of the *atrium* is not required."

3. Revise exception 1 to Section 404.6 by replacing "smoke partition" with "smoke-resistive barrier."
4. Revise Section 404.7 to read:

"404.7 Standby power.

Equipment required to provide smoke control shall be provided with standby power."

14B-4-405 Underground buildings.

The provisions of Section 405 of IBC are adopted by reference with the following modifications:

1. Revise exception 1 to Section 405.1 by replacing "One- and two-family *dwelling*s" with "Group R-5 occupancies."
2. Delete Section 405.5 and its subsections.
3. Revise Section 405.7 and its subsections to read:

"405.7 Means of egress.

Means of egress shall be in accordance with Section 405.7.1.

405.7.1 Number of exits.

Each floor level shall be provided with not fewer than two *exits*. Where compartmentation is required by Section 405.4, each compartment shall have not fewer than one vertical *exit* and not fewer than one *exit access* doorway into an adjoining compartment."

4. Revise Section 405.8 and its subsections to read:

"405.8 Standby and emergency power.

A standby power system complying with Section 2702 shall be provided for the standby power loads specified in Section 405.8.1. *An emergency power system* complying with Section 2702 shall be provided for the emergency power loads specified in Section 405.8.2.

405.8.1 Standby power loads.

The following are classified as standby power loads:

1. Smoke control system.
2. [Reserved.]
3. Elevators, as required in Section 3003.

405.8.2 Emergency power loads.

The following are classified as emergency power loads:

1. [Reserved.]
2. *Fire alarm systems*.
3. Automatic fire detection systems.
4. Elevator car lighting.
5. Means of egress and exit sign illumination as required by Chapter 10.
6. Fire pumps."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 8)

14B-4-406 Motor-vehicle-related occupancies.

The provisions of Section 406 of IBC are adopted by reference with the following modifications:

1. In provisions adopted by reference, reset the following terms in italic type: "accessible"; "approved"; "building official"; "carport"; "open parking garage"; "private garage" and "repair garage."

1.1. Revise Section 406.1 by adding a final sentence to read:

"*Parking facilities* shall also comply with Section 406.9."

1.2. Add a new Section 406.1.1 to read:

"406.1.1 Covered loading spaces and driveways.

Loading spaces and driveways which are used for the storage of unattended motor vehicles and extend more than 5 feet (1524 mm) into a building shall be classified as parking garages or carports.

Exception: Interior parking and loading spaces that occupy no more than 20 percent of the overall floor area in a Group F or S occupancy and which are separated from other areas by a change in level, curb, guard or other fixed barrier."

2. Revise Section 406.2.4 by replacing "sloped" with "sloped at least one-quarter inch per foot (6.35 mm per 305 mm)."

3. Revise exception 1 to Section 406.2.4 to read:

"1. Asphalt parking surfaces shall be permitted at ground level for *open parking garages* and *carports*."

4. Revise Section 406.2.8 to read:

"406.2.8 Separation from other occupancies.

Motor-vehicle-related occupancies shall be separated from other occupancies in the same building in accordance with Sections 406.2.8.1 through 406.2.8.4.

406.2.8.1 Private garages and carports.

Private garages and carports shall be separated from other occupancies in the same building in accordance with Section 406.3.2.

406.2.8.2 Public garages.

Public garages shall be separated from other occupancies in the same building with *fire barriers* and *horizontal assemblies* of noncombustible construction having a *fire-resistance rating* of not less than 4 hours.

Exception: Where the *building* (including the parking garage) is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1, the required *fire-resistance rating* shall be not less than 3 hours.

406.2.8.3 Repair garages.

Repair garages shall be separated from other occupancies in the same building with *fire barriers* and *horizontal assemblies* of noncombustible construction having a *fire-resistance rating* of not less than 4 hours.

406.2.8.4 Motor fuel-dispensing facilities.

Motor fuel-dispensing facilities shall not be located in a building containing other uses or occupancies."

5. Revise Section 406.2.9.1, including its exception, to read:

"406.2.9.1 Elevation of ignition sources.

Equipment and appliances having an ignition source and located in *private garages*, *repair garages*, motor fuel- dispensing facilities and parking garages shall be elevated such that the source of ignition is not less than 18 inches (457 mm) above the floor surface on which the equipment or appliance rests. For the purpose of this section, rooms or spaces that are not part of the living space of a *dwelling unit* or *sleeping unit* and that communicate directly with a *private garage* through openings shall be considered to be part of the *private garage*."

6. Revise Section 406.3 and its subsections to read:

"406.3 Private garages and carports.

Private garages and *carports* shall comply with Sections 406.2 and 406.3.

406.3.1 Classification.

Private garages and *carports* shall be classified as Group U occupancies.

Exception: *Private garages* and *carports* on the same *lot* as a Group R-5 occupancy shall be classified as a Group R-5 occupancy.

406.3.2 Area.

Each *private garage* or *carport* of Type I, II, IIIA or IV construction shall be not greater than 1,000 square feet (93 m²) in *gross floor area*. Each *private garage* or *carport* of Type IIIB or VA construction shall be not greater than 800 square feet

(74.3 m²) in *gross floor area*. Each *private garage* or *carport* of Type VB construction shall be not greater than 600 square feet (55.7 m²) in *gross floor area*.

406.3.3 Height.

A building of Type IIB, IIIB, or VB construction containing a *private garage* shall not exceed one *story* or 12 feet (3658 mm) in *building height*.

406.3.4 Separation.

Private garages and *carports* shall be separated from other *buildings* and occupancies in accordance with Sections 406.3.4.1 and 406.3.4.2. *Carports* may be separated from *buildings* in accordance with Section 406.3.4.3.

406.3.4.1 Detached private garages.

Any *exterior wall* of a detached *private garage* which is located less than 6 feet (1829 mm) from another building on the same *lot* shall have a *fire-resistance rating* of not less than 1 hour. Openings shall be protected in accordance with Section 716.

Exception: There may be one doorway not exceeding 21 square feet (1.95 m²) in area, protected with a self-closing and self-latching door not less fire-resistive than a solid wood door 1 3/4 inches (44.5 mm) thick.

406.3.4.2 Attached private garages.

Private garages and *carports* attached to another occupancy shall be separated from all other areas of the *building* by *fire barriers* and *horizontal assemblies* having a *fire-resistance rating* of not less than 1 hour. Door openings between a *private garage* and other occupancies shall be equipped with either solid wood doors or solid or honeycomb core steel doors not less than 1 3/8 inches (34.9 mm) in thickness, or doors in compliance with Section 716.2.2.1 with a fire protection rating of not less than 20 minutes. Doors shall be *self-closing* and self-latching.

Only one attached *private garage* or *carport* shall be allowed per *building*.

406.3.4.3 Carports.

A separation is not required between an enclosed occupancy and a *carport*, provided that the *carport* is entirely open on two or more sides and there are not enclosed areas above.

406.3.5 Automatic sprinkler system.

Private garages located in a *basement* and with a capacity for four or more motor vehicles shall be equipped with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2.

Exception: Where the *private garage* is open to the atmosphere for a height of not less than 18 inches (457 mm) around not less than 50 percent of its perimeter."

7. Delete the last sentence of Section 406.4.

8. Revise Section 406.4.3 to read:

"406.4.3 Ramps.

Vehicle ramps shall not be considered as vertical components of the *means of egress* connecting more than one *story* unless designated pedestrian paths are marked. Vehicle ramps that are utilized for vertical egress as well as for parking shall not exceed a slope of 1:15 (6.67 percent)."

8.1. Add new Sections 406.4.4 through 406.4.6 to read:

"406.4.4 Construction.

Public parking garages shall be of Type I, II, or IV construction. Public parking garages of Type IIB or IV construction shall only be permitted in single-*story* buildings. Floors supporting motor vehicles shall have a *fire-resistance rating* of at least 1 hour.

406.4.5 Height and area.

Public parking garages shall be limited to the allowable heights and areas specified in Sections 504 and 506 for Group S-2 occupancies. Roof parking shall be allowed.

406.4.6 Vertical tiers.

The number of vertical tiers of parking shall be limited to the values specified in Table 406.4.6. In the case of a parking garage having a continuous spiral floor, each 9 feet 6 inches (2896 mm) of height, or portion thereof, shall be considered to be a tier.

TABLE 406.4.6

PUBLIC PARKING GARAGE HEIGHT (IN TIERS)

TYPE OF CONSTRUCTION	HEIGHT (in tiers)	
	Ramp access	Mechanical access

TYPE OF CONSTRUCTION	HEIGHT (in tiers)	
	Ramp access	Mechanical access
IA	Unlimited	Unlimited
IB	3 tiers	3 tiers ^a
IIA	3 tiers	2 tiers ^a
IIB	1 tier	1 tier ^a
IV	1 tier	1 tier ^a

For SI: 1 square foot = 0.0929 m².

- a. One additional tier of mechanical access parking is allowed in buildings equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1."

9. Delete Section 406.5.1.

10. Revise Section 406.5.3 to read:

"406.5.3 Mixed occupancies and uses.

Mixed uses shall be allowed in the same building as an *open parking garage* subject to the provisions of Sections 406.2.8 and 508.1.

Exception: The grade-level tier is permitted to contain an office, waiting and toilet rooms having a total combined area of not more than 1,000 square feet (93 m²). Such area need not be separated from the *open parking garage*."

11. Delete Sections 406.5.4 and 406.5.4.1 and Table 406.5.4

12. [Reserved.]

13. Delete Section 406.5.5.

14. Revise Section 406.5.6 by replacing "adjacent *lot line*" with "*abutting property line*."

Revise Section 406.5.7 to read:

"406.5.7 Means of egress.

Where persons other than parking attendants are permitted, *open parking garages* shall meet the *means of egress* requirements of Chapter 10. Where persons other than parking attendants are not permitted, there shall be not fewer than two *exit stairways*. Each *exit stairway* shall be not less than 30 inches (762 mm) in width. Personnel lifts shall be permitted to be installed for use of employees only, provided that they are completely enclosed by noncombustible materials."

15. Revise Section 406.5.8 to read:

"406.5.8 Fire protection systems.

An *open parking garage* shall be equipped with a standpipe system as required by Section 905.3. An *open parking garage* shall be equipped with an *automatic sprinkler system* as required by Section 903.2.10."

16. Delete item 2 in Section 406.5.11.

16.1. Delete Section 406.6.1.

17. Revise the exception to Section 406.6.2 to read:

"Exception: Mechanical ventilation shall not be required for *private garages*."

18. Revise Section 406.7.2 to read:

"406.7.2 Canopies.

Canopies under which fuels are dispensed shall have a clear, unobstructed height of not less than 13 feet 6 inches (4115 mm) to the lowest projecting element in the vehicle drive-through area. *Canopies* and their supports over pumps shall be of noncombustible materials or construction providing 1-hour fire resistance."

19. Revise Section 406.8.2.2 by replacing "an approved" with "a central."

20. Add a new Section 406.9 to read:

"406.9 Parking facilities.

Parking facilities shall comply with the provisions of Sections 406.2, 406.4 and 406.5 or 406.6, as modified by Section 406.9.

406.9.1 Occupancy.

Buildings containing *parking facilities* shall not contain any other occupancy.

Exceptions:

1. The grade-level tier is allowed to contain an office, waiting and toilet rooms having a total combined area of not more than 1,000 square feet (93 m²). Such area need not be separated from the *parking facility*.
2. No more than 50 percent of the floor area of the first *story above grade plane* is allowed to contain Group A, B or M occupancies that exit directly to a *public way*, are provided with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2 and are separated from the remainder of the *structure* by *fire barriers* and horizontal assemblies having a *fire-resistance rating* of not less than 3 hours.

906.9.2 Type of construction.

Parking facilities shall be of Type IA construction.

906.9.3 Fire protection systems.

An *automatic sprinkler system* is not required in *parking facilities* except where four or more levels are enclosed on all sides. Standpipes are not required in *parking facilities*, except a manual dry standpipe system shall be required in *parking facilities* that exceed 55 feet (16.7 m) in *building height*. A *fire alarm system* is not required in *parking facilities* that do not exceed 55 feet (16.7 m) in *building height*.

906.9.4 Emergency lighting.

A System III emergency lighting system shall be provided in *parking facilities* in accordance with Article 700 of the *Chicago Electrical Code*."

(Amend Coun. J. 7-24-19, p. 3646, § 27; Amend Coun. J. 2-19-20, p. 14473, Art. II, § 9; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 4; Amend Coun. J. 1-26-22, p. 43756, Art. II, § 3)

14B-4-407 Group I-2.

The provisions of Section 407 of IBC are adopted by reference with the following modifications:

1. Revise Section 407.3 to read:

"407.3 Corridor wall construction.

Corridor walls shall be constructed as *smoke barriers* in accordance with Section 709."

2. Revise Section 407.4 by deleting "Sections 403 and 404 of."
3. Revise Section 407.4.4.2 to read:

"407.4.4.2 Separation.

Care suites shall be separated from other portions of the *building*, including other care suites, by a *smoke barrier* complying with Section 709."

14B-4-408 Group I-3.

The provisions of Section 408 of IBC are adopted by reference with the following modification:

1. Delete Section 408.9.

14B-4-409 Motion picture projection rooms.

The provisions of Section 409 of IBC are adopted by reference with the following modification:

1. Revise Section 409.3.3 by replacing "*approved*" with "*intended*."

14B-4-410 Stages, platforms and technical production areas.

The provisions of Section 410 of IBC are adopted by reference with the following modifications:

1. Revise item 3 in the exception to Section 410.2.1 by deleting "*approved*."
2. Revise Section 410.2.2, including its exception, to read:

"410.2.2 Technical production areas: galleries, gridirons and catwalks.

Beams designed only for the attachment of portable or fixed theater equipment, gridirons, galleries and catwalks shall be constructed of noncombustible materials and a *fire-resistance rating* shall not be required. These areas shall not be considered to be floors, *stories*, *mezzanines* or levels in applying this code."

3. Revise Section 410.2.5 to read:

"410.2.5 Proscenium curtain.

Where a proscenium wall is required to have a *fire-resistance rating*, the *stage* opening shall be provided with a fire curtain

complying with NFPA 80 or horizontal sliding doors complying with Section 716 having a fire protection rating of not less than 1 hour."

4. Revise Section 410.2.7.1 by deleting "*approved*."
5. Delete Section 410.2.7.2.

14B-4-411 Special amusement buildings.

The provisions of Section 411 of IBC are adopted by reference with the following modifications:

1. Delete Section 411.5.
2. Revise Section 411.6 and its subsections to read:

"411.6 Exit marking.

Exit signs shall be installed at the required *exit* or *exit access doorways* of amusement buildings in accordance with this section and Section 1013. Directional exit markings, acceptable to the *fire code official*, shall be provided. Where mirrors, mazes or other designs are utilized that disguise the path of egress travel such that they are not apparent, *listed* low-level exit signs that comply with Section 1013, and directional path markings *listed* in accordance with UL 1994, shall be provided and located not more than 8 inches (203 mm) above the walking surface and on or near the path of egress travel. Such markings shall become visible in an emergency. The directional exit marking shall be activated by the automatic fire detection system and the *automatic sprinkler system* in accordance with Section 907.2.11."

14B-4-412 Aircraft-related occupancies.

The provisions of Section 412 of IBC are adopted by reference with the following modifications:

1. Delete Section 412.2.2.1.
2. Delete exception 2 to Section 412.2.3.2.
3. Revise exception 3.1 to Section 412.2.3.2 to read:
"3.1. Voice communication systems control units."
4. Revise Section 412.3.2 to read:

"412.3.2 Construction.

Aircraft hangars shall be of Type I or II construction. Where hangars have *basements*, floors over *basements* shall be of Type IA construction and shall be made tight against seepage of water, oil or vapors. There shall not be openings or communication between *basements* and the hangar. Access to *basements* shall be from the exterior only."

5. Revise Table 412.3.6 by changing the values for Types III, IV and V construction to "NP" and adding a legend below the table stating "NP = Not permitted."
6. Delete Section 412.4 and its subsections.
7. Revise Section 412.5 by replacing "Section 416" with "the *Chicago Fire Prevention Code*."
(Amend Coun. J. 10-7-20, p. 21791, Art. II, § 5)

14B-4-413 Combustible storage.

The provisions of Section 413 of IBC are adopted by reference with the following modifications:

1. Revise the exception to Section 413.2 to read:

"Exception: Neither fire-resistance-rated construction nor opening protectives are required in any of the following locations:

1. [Reserved.]
2. Group U occupancies.
3. Within individual *dwelling units* in Group R occupancies."

14B-4-414 Hazardous materials.

The provisions of Section 414 of IBC are adopted by reference with the following modifications:

1. Delete Section 414.1 and its subsections.
2. Delete the exception to Section 414.2.
3. Revise Table 414.2.2 by replacing each "1" in the last column with "2".

3.1. Revise Table 414.2.2 by replacing the rows for "below grade plane" as follows:

Below grade plane	1	12.5	2	2
	2	5	1	2
	Lower than 2	Not allowed	Not allowed	Not allowed

3.2. Delete the exception to Section 414.2.4.

4. Notwithstanding Section 14B-1-003(5), retain "International Fire Code" in footnote c to Table 414.2.5(1).

4.1. Revise Table 414.2.5(2) by deleting "Tables 5704.3.6.3(4) through 5704.3.6.3(8) and 5704.3.7.5.1 of".

5. Delete Sections 414.3 through 414.6 and their subsections.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 10; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 4)

14B-4-415 IBC Section 415.

The provisions of Section 415 of IBC are not adopted.

14B-4-416 IBC Section 416.

The provisions of Section 416 of IBC are not adopted.

14B-4-417 IBC Section 417.

The provisions of Section 417 of IBC are not adopted.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 11)

14B-4-418 IBC Section 418.

The provisions of Section 418 of IBC are not adopted.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 12)

14B-4-419 Live/work units.

The provisions of Section 419 of IBC are adopted by reference with the following modifications:

1. Revise the exception to Section 419.1 to read:

"Exception: *Dwelling units* or *sleeping units* that include workspace that is less than 20 percent of the area of the *dwelling unit* or 200 square feet (18.6 m²), whichever is greater, are permitted to be classified as *dwelling units* with accessory occupancies in accordance with Section 508.2."

2. Revise items 1 and 2 in Section 419.1.1 to read:

"1. The *live/work unit* shall be not less than 800 square feet (74.3 m²) and not greater than 3,000 square feet (279 m²) in *gross area*.

2. The nonresidential area shall be not more than 50 percent of the area of each *live/work unit*."

3. Revise Section 419.2, excluding the exception, to read:

"419.2 Occupancies.

Live/work units shall be classified as a Group R-2 or R-5 occupancy. Separation requirements found in Sections 420 and 508 shall not apply within the *live/work unit* where the *live/work unit* is in compliance with Section 419. Nonresidential uses that would otherwise be classified as either a Group H or S occupancy shall not be permitted in a *live/work unit*. *Live/work units* shall not be used for the commercial production, sale or distribution of food or drink for consumption on- or off-premises."

4. Delete Section 419.5.

5. Revise Section 419.9 to read:

"419.9 Plumbing facilities.

The nonresidential area of the *live/work unit* shall be provided with minimum plumbing facilities as specified by the *Chicago Plumbing Code*, based on the function of the nonresidential area. Where the nonresidential area of the *live/work unit* is required to

be *accessible* by Section 1107.6.2.1, the plumbing facilities and fixtures specified by the *Chicago Plumbing Code* shall be *accessible*. Plumbing facilities shall be allowed to serve both residential and nonresidential areas within a *live/work unit*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 13)

14B-4-420 Groups I-1, R-1, R-2, R-3, R-4 and R-5.

The provisions of Section 420 of IBC are adopted by reference with the following modifications:

1. Revise Section 420.1 to read:

"420.1 General.

Occupancies in Groups I-1, R-1, R-2, R-3, R-4 and R-5 shall comply with the provisions of Sections 420.1 through 420.10 and other applicable provisions of this code."

2. Revise the exceptions to Section 420.2 to read:

"Exception: Within Group R *congregate living facilities*, where *sleeping units* are arranged in suites containing no more than five *sleeping units* each, walls between the *sleeping units* and shared living spaces are not required to be constructed as *fire partitions*."

3. Delete the exception to Section 420.3.
4. Revise Section 420.4 by replacing "in accordance with" with "where required by."
5. Revise Section 420.5 by replacing "R-3 and R-4" with "R-3, R-4 and R-5."
6. Revise item 1 in Section 420.6.1 by replacing "15 net square feet (1.4 m²)" with "30 net square feet (2.8 m²)."
7. Revise item 5 in Section 420.8 by deleting "Section 505 of."
8. Delete Section 420.8.1.
9. Revise Section 420.9 to read:

"420.9 Group R cooking facilities.

In Group R occupancies, cooking appliances used for domestic cooking operations and located within a *dwelling unit* or *sleeping unit* shall be in accordance with the *Chicago Mechanical Code*.

420.9.1 Shared cooking facilities.

In Group R occupancies, rooms or indoor spaces that contain cooking facilities with domestic cooking appliances shared by the occupants of more than one *dwelling unit* or *sleeping unit* shall be in accordance with all of the following criteria:

1. The types of domestic cooking appliances permitted shall be limited to ovens, cooktops, ranges, coffee makers, warmers and microwaves.
 2. The space containing the domestic cooking facilities shall be arranged so as not to obstruct access to a required *exit*.
 3. Domestic cooking hoods installed and constructed in accordance with section 505 of the *Chicago Mechanical Code* shall be provided over cooktops or ranges.
 4. [Reserved.]
 5. A timer shall be provided that automatically deactivates the cooking appliances within a period of not more than 120 minutes.
 6. A portable fire extinguisher shall be provided. Installation shall be in accordance with Section 906 and the extinguisher shall be located within a 30-foot (9144 mm) distance of travel from each domestic cooking appliance."
10. Delete Section 420.10 and its subsections.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 14; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 6))

14B-4-421 IBC Section 421.

The provisions of Section 421 of IBC are not adopted.

14B-4-422 Ambulatory care facilities.

The provisions of Section 422 of IBC are adopted by reference with the following modifications:

1. In provisions adopted by reference, reset the following term in italic type: "ambulatory care facilities" and "fire alarm system."
2. Revise Section 422.3.1 by deleting "Sections 403 and 404 of."

14B-4-423 IBC Section 423.

The provisions of Section 423 of IBC are not adopted.

14B-4-424 IBC Section 424.

The provisions of Section 424 of IBC are not adopted.

14B-4-425 IBC Section 425.

The provisions of Section 425 of IBC are not adopted.

14B-4-426 IBC Section 426.

The provisions of Section 426 of IBC are not adopted.

14B-4-427 IBC Section 427.

The provisions of Section 427 of IBC are not adopted.

14B-4-428 IBC Section 428.

The provisions of Section 428 of IBC are not adopted.

CHAPTER 14B-5

GENERAL BUILDING HEIGHTS AND AREAS

14B-5-501 General.

14B-5-502 Building address.

14B-5-503 General building height and area limitations.

14B-5-504 Building height and number of stories.

14B-5-505 Mezzanines and equipment platforms.

14B-5-506 Building area.

14B-5-507 IBC Section 507.

14B-5-508 Mixed use and occupancy.

14B-5-509 Incidental uses.

14B-5-510 IBC Section 510.

14B-5-501 General.

The provisions of Section 501 of IBC are adopted by reference with the following modification:

1. Revise Section 501.1 by adding a second paragraph to read:

"Nothing in this chapter shall be interpreted to prevent the application of other limitations provided by the *Chicago Zoning Ordinance*."

14B-5-502 Building address.

The provisions of Section 502 of IBC are adopted by reference with the following modification:

1. Revise Section 502.1 to read:

"502.1 Address identification.

Address identification shall be provided in accordance with Chapter 10-4 of the *Municipal Code*."

14B-5-503 General building height and area limitations.

The provisions of Section 503 of IBC are adopted by reference with the following modifications:

1. Revise Section 503.1 to read:

"503.1 General.

Unless otherwise specifically modified in Chapter 4 and this chapter, *building height*, number of *stories above grade plane* and *building area* shall not exceed the limits specified in Sections 504 and 506 based on the type of construction as determined by Section 602 and the occupancies as determined by Section 302 except as modified hereafter. *Building height*, number of *stories above grade plane* and *building area* provisions shall be applied independently. For the purposes of determining *building area* limitations, *building height* limitations and type of construction, each portion of a *structure* separated by one or more *fire walls* complying with Section 706 shall be considered to be a separate *building*."

2. Revise Section 503.1.1 to read:

"503.1.1 Special industrial occupancies.

Single-story *buildings* and *structures* of Type I or II construction used exclusively for the production and distribution of electricity, gas or steam or for low-hazard industrial processes that require large areas and unusual *building heights* to accommodate craneways or special machinery and equipment shall be exempt from the *building height* and *building area* limitations specified in Sections 504 and 506."

3. Revise Section 503.1.2 to read:

"503.1.2 Buildings on same lot.

Two or more *buildings* on the same *lot* shall be regulated as separate *buildings*.

Exception: Two or more *buildings* on the same *lot of record* may be considered as portions of one *building* where the *building height* and number of *stories above grade plane* of each *building* and the aggregate *building area* of the *buildings* are within the limitations specified in Sections 504 and 506. The provisions of this code applicable to the aggregate *building* shall be applicable to each *building*."

4. Delete Section 503.1.3.

5. Revise Section 503.1.4 and its subsections to read:

"503.1.4 Occupiable rooftops.

A roof level or portion thereof shall be permitted to be used as an *occupiable rooftop* provided the occupancy of the roof is an occupancy that is permitted by Table 504.4 for the *story* immediately below the roof. The area of the *occupiable rooftop* shall not be included in the *building area* as regulated by Section 506.

Exception: The occupancy located on an *occupiable rooftop* shall not be limited to the occupancies allowed on the *story* immediately below the *occupiable rooftop* where the building is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2 and occupant notification in accordance with Section 907.5 is provided in the area of the *occupiable rooftop*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 15)

14B-5-504 Building height and number of stories.

The provisions of Section 504 of IBC are adopted by reference with the following modifications:

1. Revise Section 504.1, excluding the exception, to read:

"504.1 General.

The allowable *building height*, in feet, and the allowable number of *stories above grade plane* of a building shall be determined based on the type of construction, occupancy classification and whether there is an *automatic sprinkler system* installed throughout the building."

2. Delete Section 504.1.1.

3. Delete Section 504.1.2.

4. Revise Section 504.3 to read:

"504.3 Height in feet.

The maximum *building height*, in feet, of a building shall not exceed the limits specified in Table 504.3.

Exception: Towers, spires, minarets, steeples and other *rooftop structures* with a footprint not exceeding 25 percent of the ground area of the *building* on which they are erected shall be constructed of materials consistent with the required type of construction of the *building* except where other construction is permitted by Section 1510.2.4. Such *rooftop structures* shall not be used for habitation or storage. The *rooftop structures* shall not extend more than 60 feet (18.3 m) above the allowable *building height* where of noncombustible materials and shall not extend more than 20 feet (6096 mm) above the allowable *building height* where of combustible materials (see Chapter 15 for additional requirements)."

5. Replace Table 504.3 as shown:

TABLE 504.3

ALLOWABLE BUILDING HEIGHT IN FEET ABOVE GRADE PLANE ^a

OCCUPANCY CLASSIFICATION	TYPE OF CONSTRUCTION									
	SEE FOOTNOTES	TYPE I		TYPE II		TYPE III		TYPE IV	TYPE V	
		A	B	A	B	A	B	HT	A	B
OCCUPANCY CLASSIFICATION	TYPE OF CONSTRUCTION									
	SEE FOOTNOTES	TYPE I		TYPE II		TYPE III		TYPE IV	TYPE V	
		A	B	A	B	A	B	HT	A	B
A, B, E, F, H-4 ^c , M, S, U	NS ^b	80	80	65	30	55	30	65	30	15
	S	UL	150	85	45	70	45	85 ^g	45	30
H-1, H-2, H-3, H-5	NS ^{c, d}	80	80	65	30	55	NP	65	NP	NP
	S	UL								
I	NS ^b	80	80	65	30	55	30	65	30	NP
	S	UL	150	85						
R	NS ^b	80	80	65	30	55	30	65	30 ^e	20
	S13D	40	40	40	40	40	40	40	40	35
	S13R	55	55	55	45	55	45	55	45 ^f	35
	S	UL	150	85	45	70	45	85 ^g	45 ^f	35

For SI: 1 foot = 304.8 mm.

UL = Unlimited.

NS = *Buildings* not equipped throughout with an *automatic sprinkler system*.

S = *Buildings* equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1.

S13R = *Buildings* equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.2.

S13D = *Buildings* equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.3.

a. See Chapters 4 and 5 for specific exceptions to the allowable *building height* in this table.

b. See Section 903.2 for the minimum thresholds for protection by an *automatic sprinkler system* for specific occupancies.

c. New Group H occupancies are required to be protected by an *automatic sprinkler system* in accordance with Section 903.2.5.

d. The NS value is only for use in evaluation of *existing buildings* in accordance with the *Chicago Building Rehabilitation Code*.

e. Group R occupancies of Type VA construction shall be allowed to be 40 feet in *building height*, provided that the finished floor level of the highest *occupiable space* shall not exceed 30 feet above *grade plane*.

f. Group R occupancies of Type VA construction equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2 shall be allowed to be 55 feet in *building height*, provided that the finished floor level of the highest *occupiable space* shall not exceed 40 feet above *grade plane*.

g. 120 feet for evaluation of *buildings* constructed or permitted for construction before January 1, 1951, that are equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1.

6. Revise Section 504.4 to read:

"504.4 Number of stories above grade plane.

The maximum number of *stories above grade plane* of a *building* shall not exceed the limits specified in Table 504.4."

7. Replace Table 504.4 as shown:

TABLE 504.4

ALLOWABLE NUMBER OF STORIES ABOVE GRADE PLANE ^{a, b}

OCCUPANCY CLASSIFICATION	SEE NOTES	TYPE OF CONSTRUCTION				
		TYPE I	TYPE II	TYPE III	TYPE IV	TYPE V

		A	B	A	B	A	B	HT	A	B
OCCUPANCY CLASSIFICATION	SEE NOTES	TYPE OF CONSTRUCTION								
		TYPE I		TYPE II		TYPE III		TYPE IV	TYPE V	
		A	B	A	B	A	B	HT	A	B
A-1	NS	UL	5	3	1	2	NP	2	1	NP
	S-13	UL	6	4	2	3	1	3	1	NP
A-2	NS	UL	10	3	1	2	NP	2	1	NP
	S-13	UL	11	4	2	3	1	3	1	NP
A-3	NS	UL	10	3	1	2	NP	2	1	NP
	S-13	UL	11	4	2	3	1	3	1	NP
A-4	NS	UL	10	3	1	2	NP	2	1	NP
	S-13	UL	11	4	2	3	1	3	1	NP
A-5	NS	UL	UL	UL	UL	1	1	UL	1	1
	S-13	UL	UL	UL	UL	1	1	UL	1	1
B	NS	UL	11	6	1	4	2	5	1	NP
	S-13	UL	12	7	2	5	3	6	2	1
E-1	NS	UL	5	3	1	1	1	1	1	NP
	S-13	UL	6	4	1	2	1	2	1	NP
E-2	NS	UL	5	3	1	1	1	1	1	NP
	S-13	UL	6	4	1	2	1	2	1	NP
F-1	NS	UL	5	4	1	3	2	4	1	1
	S-13	UL	6	5	2	4	3	5	1	1
F-2	NS	UL	5	4	1	3	2	4	1	1
	S-13	UL	6	5	2	4	3	5	1	1
H-1	NS ^{c, d}	1	1	1	1	1	1	1	1	NP
	S-13	1	1	1	1	1	1	1	1	NP
H-2	NS ^{c, d}	UL	3	2	1	2	1	1	NP	NP
	S-13	UL	3	2	1	2	1	1	NP	NP
H-3	NS ^{c, d}	UL	5	3	1	3	2	3	NP	NP
	S-13	UL	5	3	1	3	2	3	NP	NP
H-4	NS ^{c, d}	UL	5	4	1	3	2	4	1	NP
	S-13	UL	6	5	1	4	3	5	1	NP
H-5	NS ^{c, d}	4	4	3	1	3	2	3	NP	NP
	S-13	4	4	3	1	3	2	3	NP	NP
I-1, Condition 1	NS ^{d, e}	UL	9	3	1	2	1	2	1	NP
	S-13	UL	10	4	2	3	1	3	1	NP
I-1, Condition 2	NS ^{d, e}	UL	4	3	1	1	NP	1	1	NP
	S-13	UL	5	4	1	1	NP	1	1	NP
I-2, Condition 1	NS ^f	UL	4	3	1	1	NP	1	1	NP
	S-13	UL	5	4	1	1	NP	1	1	NP
I-2, Condition 2	NS ^f	UL	4	3	1	1	NP	1	1	NP
	S-13	UL	5	4	1	1	NP	1	1	NP
I-3	NS ^{d, e}	UL	9	2	1	1	NP	1	1	NP
	S-13	UL	10	3	1	1	NP	1	1	NP
I-4	NS	UL	9	3	1	1	NP	1	1	NP
	S-13 ^{d, g}	UL	10	4	1	2	NP	3	1	NP
M	NS	UL	11	6	1	4	2	5	1	NP

	S-13	UL	12	7	2	5	3	6	1	1
R-1 ^h	NS ^d	UL	11	4	1	4	2	4	2	1
	S-13R	4	4	4	2	4	2	4	2	1
	S-13	UL	12	6	2	5	3	5	3	2
R-2	NS	UL	11	4	1	4	2	4	2	1
	S-13R	4	4	4	2	4	3	4	4	2
	S-13	UL	12	6	2	5	3	5	4	2
R-3 ^h	NS	UL	11	4	1	4	2	4	2	1
	S-13D	3	3	3	2	3	2	3	3	2
	S-13R	4	4	4	2	4	3	4	4	2
	S-13	UL	12	6	2	5	3	5	4	2
R-4 ^h	NS ^d	UL	11	4	1	4	2	4	NP	NP
	S-13D	3	3	3	2	3	2	3	3	2
	S-13R	4	4	4	2	4	3	4	3	2
	S-13	UL	12	6	2	5	3	5	3	2
R-5	NS	UL	11	4	1	4	3	4	3	2
	S-13D	3	3	3	3	3	3	3	3	2
	S-13R	4	4	4	3	4	3	4	4	2
	S-13	UL	12	6	3	5	3	5	4	2
S-1	NS	UL	5	4	1	3	2	5	1	1
	S-13	UL	6	5	2	4	3	6	1	1
S-2	NS	UL	5	4	1	3	2	5	1	1
	S-13	UL	6	5	2	4	3	6	1	1
U	NS	UL	5	4	1	3	2	4	1	1
	S-13	UL	6	5	1	4	3	5	1	1

UL = Unlimited.

NP = Not permitted.

NS = *Buildings* not equipped throughout with an *automatic sprinkler system*.

S = *Buildings* equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1.

S13R = *Buildings* equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.2.

S13D = *Buildings* equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.3.

- See Chapters 4 and 5 for specific exceptions to the allowable number of *stories above grade plane* in this table.
- See Section 903.2 for the minimum thresholds for protection by an *automatic sprinkler system* for specific occupancies.
- New Group H occupancies are required to be protected by an *automatic sprinkler system* in accordance with Section 903.2.5.
- The NS value is only for use in evaluation of existing buildings in accordance with the *Chicago Building Rehabilitation Code*.
- New Group I-1 and I-3 occupancies are required to be protected by an automatic sprinkler system in accordance with Section 903.2.6. For new Group I-1 occupancies, Condition 1, see Exception 1 of Section 903.2.6.
- New and existing Group I-2 occupancies are required to be protected by an automatic sprinkler system in accordance with Section 903.2.6 and the *Chicago Minimum Requirements for Existing Buildings*.
- For new Group I-4 occupancies, see Exceptions 2 and 3 of Section 903.2.6.
- New Group R-1, R-3 and R-4 occupancies are required to be protected by an *automatic sprinkler system* in accordance with Section 903.2.8.

(Amend Coun. J. 7-24-19, p. 3646, §§ 28, 29; Amend Coun. J. 2-19-20, p. 14473, Art. II, § 16; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 5; Amend Coun. J. 1-26-22, p. 43756, Art. II, § 4)

14B-5-505 Mezzanines and equipment platforms.

The provisions of Section 505 of IBC are adopted by reference with the following modifications:

- Revise Section 505.2 to read:

"505.2 Mezzanines.

A *mezzanine* or *mezzanines* in compliance with Section 505.2 shall be considered a portion of the *story* below. Such *mezzanines* shall not contribute to the number of *stories above grade plane* as regulated by Section 503.1. The area of the *mezzanine* shall be

included in determining the *building area* and the *fire area*. The clear height above and below the *mezzanine* floor construction shall be not less than 7 feet (2134 mm)."

2. Revise Section 505.2.1 to read:

"505.2.1 Area limitation.

The aggregate area of a *mezzanine* or *mezzanines* within a room shall be not greater than twenty percent of the floor area of that room or space in which they are located. The enclosed portion of a room shall not be included in a determination of the floor area of the room in which the *mezzanine* is located. In determining the allowable *mezzanine* area, the area of the *mezzanine* shall not be included in the floor area of the room.

Exceptions:

1. The aggregate area of *mezzanines* in *buildings* of Type I or II construction for special industrial occupancies in accordance with Section 503.1.1 shall be not greater than two-thirds of the floor area of the room.
2. [Reserved.]
3. The aggregate area of a *mezzanine* within a *dwelling unit* that is located in a *building* equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2 shall not be greater than one-half of the floor area of the room, provided that both:
 - 3.1. Except for enclosed closets and bathrooms, the *mezzanine* shall be open to the room in which such mezzanine is located.
 - 3.2. The opening to the room shall be unobstructed except for walls not more than 48 inches (1219 mm) in height, columns and posts."
3. Revise Section 505.2.1.1 by replacing "two-thirds" with "one-half."
4. Revise the exceptions to Section 505.2.3 to read:
 - "1. In Group B, F, M and S occupancies in buildings equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1, *mezzanines* or portions thereof are not required to be open to the room in which the *mezzanines* are located, provided that the occupant load of the aggregate area of the enclosed space is not greater than 10.
 2. [Reserved.]
 3. *Mezzanines* or portions thereof are not required to be open to the room in which the *mezzanines* are located provided that the aggregate floor area of the enclosed space is not greater than 10 percent of the allowable *mezzanine* area.
 4. In industrial facilities, *mezzanines* used for control equipment are permitted to be glazed on all sides.
 5. [Reserved.]
 6. *Mezzanines* in accordance with Exception 3 to Section 505.2.1."
5. Revise Section 505.3 by replacing "*stories*" with "*stories above grade plane*."
6. Revise Section 505.3.1 by replacing "two-thirds" with "one-half."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 17)

14B-5-506 Building area.

The provisions of Section 506 of IBC are adopted by reference with the following modifications:

1. Revise Section 506.1 and its subsections to read:

"506.1 General.

The allowable *building area* of a *building* shall be determined based on the type of construction, occupancy classification, whether there is an *automatic sprinkler system* installed throughout the *building* and the amount of building frontage on a *public way* or *yard*.

506.1.1 [Reserved.]

506.1.2 [Reserved.]

506.1.3 Basements.

Basements shall be included in *building area* where the total area of such *basements* exceeds the allowable *building area* for a one-story above grade plane building of the same occupancy and construction type.

506.1.3.1 Maximum area.

Every *basement* exceeding 40,000 square feet (3716 m²) in area shall be subdivided by *fire barriers* into areas not exceeding 40,000 square feet (3716 m²), provided where the building is equipped throughout with an *automatic*

sprinkler system, the maximum undivided area may be increased to 80,000 square feet (7432 m²)."

2. Replace Table 506.2 as shown:

TABLE 506.2

ALLOWABLE AREA FACTOR (A_f = NS, S1, S13R, S13D or SM, as applicable) IN SQUARE FEET ^{a, b}

OCCUPANCY CLASSIFICATION	SEE NOTES	CONSTRUCTION TYPE								
		TYPE I		TYPE II		TYPE III		TYPE IV	TYPE V	
		A	B	A	B	A	B	HT	A	B
OCCUPANCY CLASSIFICATION	SEE NOTES	CONSTRUCTION TYPE								
		TYPE I		TYPE II		TYPE III		TYPE IV	TYPE V	
		A	B	A	B	A	B	HT	A	B
A-1	NS	UL	UL	16,000	8,500	6,000	NP	6,000	3,000	NP
	S1	UL	UL	64,000	34,000	24,000	12,000	24,000	12,000	NP
	SM	UL	UL	48,000	25,500	18,000	9,000	18,000	9,000	NP
A-2	NS	UL	UL	16,000	8,500	6,000	5,000	6,000	3,000	NP
	S1	UL	UL	64,000	34,000	24,000	20,000	24,000	12,000	NP
	SM	UL	UL	48,000	25,500	18,000	15,000	18,000	9,000	NP
A-3	NS	UL	UL	16,000	8,500	6,000	5,000	6,000	3,000	NP
	S1	UL	UL	64,000	34,000	24,000	20,000	24,000	12,000	NP
	SM	UL	UL	48,000	25,500	18,000	15,000	18,000	9,000	NP
A-4	NS	UL	UL	16,000	8,500	6,000	5,000	6,000	3,000	NP
	S1	UL	UL	64,000	34,000	24,000	20,000	24,000	12,000	NP
	SM	UL	UL	48,000	25,500	18,000	15,000	18,000	9,000	NP
A-5	NS	UL	UL	UL	UL	UL	UL	UL	UL	UL
	S1	UL								
	SM	UL								
B	NS	UL	UL	20,000	6,000	10,000	8,000	10,000	6,000	4,000
	S1	UL	UL	UL	UL	UL	32,000	40,000	24,000	16,000
	SM	UL	UL	60,000	18,000	30,000	24,000	30,000	18,000	NP
E-1	NS	UL	UL	16,000	5,000	8,000	3,000	10,000	3,000	NP
	S1	UL	UL	64,000	20,000	32,000	12,000	40,000	12,000	NP
	SM	UL	UL	48,000	15,000	24,000	9,000	30,000	9,000	NP
E-2	NS	UL	UL	16,000	5,000	8,000	3,000	10,000	3,000	NP
	S1	UL	UL	64,000	20,000	32,000	12,000	40,000	12,000	NP
	SM	UL	UL	48,000	15,000	24,000	9,000	30,000	9,000	NP
F-1	NS	UL	UL	16,000	4,000	8,000	4,000	8,000	5,000	3,000
	S1	UL	UL	UL	UL	32,000	16,000	32,000	20,000	12,000
	SM	UL	UL	48,000	12,000	24,000	12,000	24,000	15,000	NP
F-2	NS	UL	UL	20,000	10,000	10,000	6,000	10,000	6,000	4,000
	S1	UL	UL	UL	UL	40,000	24,000	40,000	24,000	16,000
	SM	UL	UL	60,000	30,000	30,000	18,000	30,000	18,000	NP
H-1	NS ^{c, d}	21,000	16,500	11,000	6,000	9,500	5,000	10,500	5,000	NP
	S1	21,000	16,500	11,000	6,000	9,500	5,000	10,500	5,000	NP
	SM	NP	NP	NP	NP	NP	NP	NP	NP	NP
H-2	NS ^{c, d}	21,000	16,500	11,000	6,000	9,500	5,000	10,500	NP	NP
	S1	21,000	16,500	11,000	6,000	9,500	5,000	10,500	NP	NP
	SM	42,000	16,500	11,000	NP	9,500	NP	NP	NP	NP

H-3	NS ^{c, d}	UL	60,000	20,000	6,000	10,000	5,000	10,000	NP	NP
	S1	UL	60,000	20,000	6,000	10,000	5,000	10,000	NP	NP
	SM	UL	60,000	20,000	NP	10,000	5,000	10,000	NP	NP
H-4	NS ^{c, d}	UL	UL	20,000	6,000	10,000	8,000	10,000	6,000	NP
	S1	UL	UL	80,000	24,000	40,000	32,000	40,000	24,000	NP
	SM	UL	UL	60,000	NP	30,000	24,000	30,000	NP	NP
H-5	NS ^{c, d}	UL	UL	16,000	4,000	8,000	4,000	8,000	NP	NP
	S1	UL	UL	64,000	16,000	32,000	16,000	32,000	NP	NP
	SM	UL	UL	48,000	NP	24,000	12,000	24,000	NP	NP
I-1	NS ^{d, e}	UL	55,000	16,000	4,000	6,000	4,000	6,000	3,000	NP
	S1	UL	220,000	64,000	16,000	24,000	16,000	24,000	12,000	NP
	SM	UL	165,000	48,000	12,000	18,000	12,000	18,000	9,000	NP
I-2	NS ^{d, f}	UL	UL	12,000	6,000	6,000	NP	6,000	3,000	NP
	S1	UL	UL	48,000	24,000	24,000	NP	24,000	12,000	NP
	SM	UL	UL	36,000	NP	NP	NP	NP	9,000	NP
I-3	NS ^{d, e}	UL	UL	12,000	6,000	6,000	NP	6,000	3,000	NP
	S1	UL	UL	48,000	24,000	24,000	NP	24,000	12,000	NP
	SM	UL	UL	36,000	NP	NP	NP	NP	9,000	NP
I-4	NS ^{d, g}	UL	55,000	16,000	4,000	6,000	4,000	6,000	3,000	NP
	S1	UL	220,000	64,000	16,000	24,000	16,000	24,000	12,000	NP
	SM	UL	165,000	48,000	12,000	18,000	12,000	18,000	9,000	NP
M	NS	UL	UL	16,000	6,000	8,000	6,000	8,000	5,000	3,000
	S1	UL	UL	UL	UL	UL	24,000	32,000	20,000	12,000
	SM	UL	UL	48,000	18,000	24,000	18,000	24,000	15,000	NP
R-1 ^h	NS ^d	UL	UL	24,000	10,000	10,000	8,000	10,000	5,000	3,000
	S13R	UL	UL	48,000	20,000	20,000	16,000	20,000	10,000	6,000
	S1	UL	UL	96,000	40,000	40,000	32,000	40,000	20,000	12,000
	SM	UL	UL	72,000	30,000	30,000	24,000	30,000	15,000	9,000
R-2	NS	UL	UL	24,000	10,000	10,000	8,000	10,000	5,000	3,000
	S13R	UL	UL	48,000	20,000	20,000	16,000	20,000	10,000	6,000
	S1	UL	UL	96,000	40,000	40,000	32,000	40,000	20,000	12,000
	SM	UL	UL	72,000	30,000	30,000	24,000	30,000	15,000	9,000
R-3 ^h	NS	UL	UL	24,000	10,000	10,000	8,000	10,000	5,000	3,000
	S13D	UL	UL	48,000	20,000	20,000	16,000	20,000	10,000	6,000
	S13R	UL	UL	48,000	20,000	20,000	16,000	20,000	10,000	6,000
	S1	UL	UL	96,000	40,000	40,000	32,000	40,000	20,000	12,000
	SM	UL	UL	72,000	30,000	30,000	24,000	30,000	15,000	9,000
R-4 ^h	NS ^d	UL	UL	24,000	10,000	10,000	8,000	10,000	5,000	3,000
	S13D	UL	UL	48,000	20,000	20,000	16,000	20,000	10,000	6,000
	S13R	UL	UL	48,000	20,000	20,000	16,000	20,000	10,000	6,000
	S1	UL	UL	96,000	40,000	40,000	32,000	40,000	20,000	12,000
	SM	UL	UL	72,000	30,000	30,000	24,000	30,000	15,000	9,000
R-5	NS	UL	UL	24,000	10,000	10,000	8,000	10,000	5,000	3,000
	S13D	UL	UL	48,000	20,000	20,000	16,000	20,000	10,000	6,000
	S13R	UL	UL	48,000	20,000	20,000	16,000	20,000	10,000	6,000
	S1	UL	UL	96,000	40,000	40,000	32,000	40,000	20,000	12,000
	SM	UL	UL	72,000	30,000	30,000	24,000	30,000	15,000	9,000
S-1	NS	UL	35,500	19,000	8,500	14,000	8,000	18,000	5,000	3,000
	S1	UL	UL	UL	UL	56,000	32,000	72,000	20,000	12,000
	SM	UL	106,500	57,000	25,500	42,000	24,000	54,000	15,000	NP

S-2	NS	UL	79,000	28,500	13,000	21,000	10,000	27,000	6,000	4,000 ⁱ
	S1	UL	UL	UL	UL	84,000	40,000	108,000	24,000	16,000 ⁱ
	SM	UL	237,000	85,500	39,000	63,000	30,000	81,000	18,000	NP
U	NS	UL	30,000	16,000	6,000	8,000	6,000	8,000	5,000	1,000
	S1	UL	120,000	64,000	24,000	32,000	24,000	32,000	20,000	4,000
	SM	UL	90,000	48,000	18,000	24,000	18,000	24,000	15,000	NP

For SI: 1 square foot = 0.0929 m².

UL = Unlimited.

NP = Not permitted.

NS = *Buildings* not equipped throughout with an *automatic sprinkler system*.

S1 = *Buildings* a maximum of one *story above grade plane* equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1.

SM = *Buildings* two or more *stories above grade plane* equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1.

S13R = *Buildings* equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.2.

S13D = *Buildings* equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.3.

- a. See Chapters 4 and 5 for specific exceptions to the allowable area factor in this table.
- b. See Section 903.2 for the minimum thresholds for protection by an *automatic sprinkler system* for specific occupancies.
- c. New Group H occupancies are required to be protected by an *automatic sprinkler system* in accordance with Section 903.2.5.
- d. The NS value is only for use in evaluation of *existing buildings* in accordance with the *Chicago Building Rehabilitation Code*.
- e. New Group I-1 and I-3 occupancies are required to be protected by an *automatic sprinkler system* in accordance with Section 903.2.6. For new Group I-1 occupancies, Condition 1, see Exception 1 of Section 903.2.6.
- f. New and existing Group I-2 occupancies are required to be protected by an *automatic sprinkler system* in accordance with Section 903.2.6 and Section 1103.5 of the *Chicago Fire Prevention Code*.
- g. New Group I-4 occupancies see Exceptions 2 and 3 of Section 903.2.6.
- h. New Group R-1, R-3 and R-4 occupancies are required to be protected by an *automatic sprinkler system* in accordance with Section 903.2.8.
- i. Type VB construction is not allowed for motor-vehicle-related occupancies other than *private garages*.

3. Revise Equation 5-2 by deleting the second sentence from the description of S_a.

4. Revise Section 506.2.4 to read:

"506.2.4 Mixed-occupancy, multistory buildings.

Each *story* of a mixed-occupancy *building* with more than one *story above grade plane* shall individually comply with the applicable requirements of Section 508.1. For *buildings* with more than three *stories above grade plane*, the *building area* shall be such that the aggregate sum of the ratios of the actual floor area of each *story* divided by the allowable area of such *stories*, determined in accordance with Equation 5-3 based on the applicable provisions of Section 508.1, shall not exceed three."

5. Delete the exception to Section 506.2.4.

6. Revise Section 506.3 to replace "public way" with "*public way*."

7. Revise Section 506.3.1 to read:

"506.3.1 Minimum percentage of perimeter.

To qualify for an area factor increase based on frontage, a *building* shall have not less than 25 percent of its perimeter abut a *public way, court or yard*. Such *court* or *yard* shall be either on the same *lot* or dedicated for public use and shall have direct access, not less than 20 feet (6096 mm) wide, from a *public way*."

8. Revise Section 506.3.2 to read:

"506.3.2 Minimum frontage distance.

To qualify for an area factor increase based on frontage, the *public way, court or yard* adjacent to the *building* perimeter shall have a minimum distance (*W*) of 20 feet (6096 mm) measured at right angles from the *building* face to any of the following:

1. The closest *abutting property line*.
2. The far side of a *public way*.
3. The exterior face of an adjacent *building* on the same *lot*.
4. The opposite side of a *court*.

Where the value of *W* is greater than 30 feet (9144 mm), a value of 30 feet (9144 mm) shall be used in calculating the area factor

increase based on frontage, regardless of the actual width of the *public way, court or yard*. Where the value of W varies along the perimeter of the *building*, the calculation performed in accordance with Equation 5-5 shall be based on the weighted average calculated in accordance with Equation 5-4."

9. Revise the definitions of values for Equation 5-4 by replacing "public way or open space" with "*public way, court or yard*" and inserting "but not greater than 30 feet (9144 mm)" at the end of the definition of W_n .

10. Delete the exception to Section 506.3.2.

11. Revise Equation 5-5 to read:

$$I_f = (F/P)(W/30)(S) \quad \text{(Equation 5-5)}$$

where:

I_f = Area factor increase due to frontage.

F = *Building* perimeter that fronts on a *public way or yard* having a minimum width of 20 feet (6096 mm) (feet).

P = Perimeter of entire *building* (feet).

W = Width of *public way or yard* (feet) in accordance with Section 506.3.2.

S = Where *building* is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1, $S = 2$; where *building* is not equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2, $S = 1$.

(Amend Coun. J. 7-24-19, p. 3646, § 30; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 7; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 6)

14B-5-507 IBC Section 507.

The provisions of Section 507 of IBC are not adopted.

14B-5-508 Mixed use and occupancy.

The provisions of Section 508 of IBC are adopted by reference with the following modifications:

1. Delete exception 1 to Section 508.1.

1.1. Revise exception 2 to Section 508.1 by replacing "Table 415.6.2" with "the *Chicago Fire Prevention Code*."

2. Revise Section 508.2.2 to read:

"508.2.2 Allowable building height.

The allowable *building height* and number of *stories above grade plane* of the *building* containing accessory occupancies shall be in accordance with Section 504 for the main occupancy of the *building*."

3. Revise Section 508.2.3 to read:

"508.2.3 Allowable building area.

The allowable area of the *building* shall be based on the applicable provisions of Section 506 for the main occupancy of the *building*. Aggregate accessory occupancies shall not occupy more than 25 percent of the floor area of the *story* nor more than 10 percent of the total floor area of the main occupancy in which they are located and shall not exceed the tabular area values for nonsprinklered *buildings* in Table 506.2 for each such accessory occupancy.

Exception: The *building official* and *fire code official* may jointly approve accessory occupancies exceeding the 10 percent and 25 percent limits when in their opinion such accessory occupancies do not unduly increase the hazard."

4. Revise the exceptions to Section 508.2.4 to read:

"Exceptions:

1. Group H occupancies shall not be classified as accessory occupancies and shall be separated from all other occupancies in accordance with Section 508.4.

2. Group I-1, R-1, R-2, R-3, R-4 and R-5 *dwelling units* and *sleeping units* shall not be classified as accessory occupancies and shall be separated from other *dwelling or sleeping units* and from other occupancies contiguous to them in accordance with the requirements of Section 420.

3. Motor-vehicle-related occupancies shall not be classified as accessory occupancies and shall be separated from other occupancies in accordance with Section 406."

5. Revise Section 508.3.2 to read:

"508.3.2 Allowable building area, height and number of stories above grade plane.

The allowable *building area*, *building height* and number of *stories above grade plane* of the building or portion thereof shall be based on the most restrictive allowances for the occupancy groups under consideration for the type of construction of the building in accordance with Section 503.1."

6. Revise the exceptions to Section 508.3.3 to read:

"Exceptions:

1. Group H occupancies shall be separated from all other occupancies in accordance with Section 508.4.
2. Group I-1, R-1, R-2, R-3, R-4 and R-5 *dwelling units* and *sleeping units* shall be separated from other *dwelling* or *sleeping units* and from other occupancies contiguous to them in accordance with the requirements of Section 420.
3. Group A occupancies with an aggregate occupant load of 300 or more persons shall be separated from other main occupancies by construction providing a *fire-resistance rating* of not less than 2 hours in *buildings* that are not protected throughout by an *automatic sprinkler system* in accordance with Section 903.3.1.1.
4. Group A occupancies with an aggregate occupant load of less than 300 persons shall be separated from other main occupancies by construction providing a *fire-resistance rating* of not less than 1 hour in *buildings* that are not protected throughout by an *automatic sprinkler system* in accordance with Section 903.3.1.1.
5. Motor-vehicle-related occupancies shall be separated from other occupancies in accordance with Section 406."

7. Replace Table 508.4 as shown:

TABLE 508.4

REQUIRED SEPARATION OF OCCUPANCIES (HOURS) ^f

OCCUPANCY	A, E		I-1 ^a , I-3, I-4		I-2		R ^a		F-2, S-2 ^b , U		B ^e , M		F-1, S-1		H-1		H-2		H-3, H-4		H-5	
	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS
OCCUPANCY	A, E		I-1 ^a , I-3, I-4		I-2		R ^a		F-2, S-2 ^b , U		B ^e , M		F-1, S-1		H-1		H-2		H-3, H-4		H-5	
	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS	S	NS
A, E	N ^h	N ^{g,h}	2	3	3	NP	1 ^h	2	1 ^h	2	1 ^h	2	1 ^h	2	NP	NP	4	NP	4	NP	4	NP
I-1 ^a , I-3, I-4	—	—	N	N	2	NP	2	NP	3	4	3	4	3	4	NP	NP	4	NP	4	NP	4	NP
I-2	—	—	—	—	N	N	2	NP	3	NP	2	NP	2	NP	NP	NP	4	NP	4	NP	4	NP
R ^a	—	—	—	—	—	—	N	N	1	2	1	2	1	2	NP	NP	4	NP	4	NP	4	NP
F-2, S-2 ^b , U	—	—	—	—	—	—	—	—	N	N	1	2	1	2	NP	NP	3	4	4	4	4	NP
B ^e , M	—	—	—	—	—	—	—	—	—	—	N	N	1	2	NP	NP	3	4	4	4	3	NP
F-1, S-1	—	—	—	—	—	—	—	—	—	—	—	—	N	N	NP	NP	3	4	4	4	3	NP
H-1	—	—	—	—	—	—	—	—	—	—	—	—	—	—	N	NP	NP	NP	NP	NP	NP	NP
H-2	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	N	NP	2	NP	1	NP
H-3, H-4	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	2 ^d	NP	1	NP
H-5	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	N	NP

S = *Buildings* equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1.

NS = *Buildings* not equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1.

N = No separation requirement.

NP = Not Permitted.

a. See Section 420.

b. The required separation from parking garages and other motor-vehicle-related occupancies shall be in accordance with Section 406.

c. [Reserved.]

d. Separation is not required between occupancies of the same classification.

e. See Section 422.2 for *ambulatory care facilities*.

f. Occupancy separations that serve to define *fire area* limits established in Chapter 9 for requiring *fire protection systems* shall also comply with Section 707.3.10 and

Table 707.3.10 in accordance with Section 901.7.

g. Group A occupancies with an aggregate occupant load of less than 300 persons shall be separated from other occupancies by construction providing *fire-resistance rating* of not less than one 1 hour in *buildings* that are not protected throughout by an *automatic sprinkler system* in accordance with Section 903.3.1.1.

h. Group A occupancies with an aggregate occupant load of 300 or more persons shall be separated from other occupancies by construction providing *fire-resistance rating* of not less than 2 hours in *buildings* that are not protected throughout by an *automatic sprinkler system* in accordance with Section 903.3.1. 1.

8. Revise Section 508.4.3, including the exception, to read:

"508.4.3 Allowable building height and number of stories above grade plane.

Each separated occupancy shall comply with the *building height* limitations and *story above grade plane* limitations based on the type of construction of the *building* in accordance with Section 503.1."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 18; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 8; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 7; Amend Coun. J. 1-26-22, p. 43756, Art. II, § 5; Amend Coun. J. 1-24-24, p. 8634, Art. IV, § 1)

14B-5-509 Incidental uses.

The provisions of Section 509 of IBC are adopted by reference with the following modifications:

1. Replace Table 509 as shown:

TABLE 509

INCIDENTAL USES

ROOM OR AREA	SEPARATION AND/OR PROTECTION
ROOM OR AREA	SEPARATION AND/OR PROTECTION
Furnace room where any piece of equipment is over 400,000 Btu per hour input	2 hours
Rooms with boilers where the largest piece of equipment is over 15 psi and 10 horsepower	2 hours
Furnace rooms and boiler rooms in <i>buildings</i> having an <i>occupant load</i> exceeding 200 persons	2 hours
Furnace rooms and boiler rooms in Group I occupancies	2 hours
Furnace rooms and boiler rooms in other than Group R-5 occupancies	1 hour
Refrigerant machinery room	1 hour
Hydrogen fuel gas rooms, not classified as Group H	1 hour in Group B, F, M, S and U occupancies; 2 hours in Group A, E, I and R occupancies.
Incinerator rooms	2 hours and provide <i>automatic sprinkler system</i>
Paint shops, not classified as Group H, located in occupancies other than Group F	2 hours; or 1 hour and provide <i>automatic sprinkler system</i>
In Group E occupancies, laboratories and vocational shops not classified as Group H	1 hour
In Group I-2 occupancies, laboratories not classified as Group H	1 hour and provide <i>automatic sprinkler system</i>
In ambulatory care facilities, laboratories not classified as Group H	1 hour
Laundry rooms over 100 square feet	1 hour
Group I-3 cells and Group I-2 patient rooms equipped with padded surfaces	1 hour
In Group I-2, physical plant maintenance shops	1 hour
Waste, recycling and linen collection rooms with containers that have an aggregate volume of 10 cubic feet or greater in <i>ambulatory care facilities</i> or Group I-2 occupancies	1 hour
Waste, recycling and linen collection rooms over 100 square feet	1 hour
Storage rooms over 100 square feet in <i>ambulatory care facilities</i> and Group A, E, I, R-1, R-2, R-3 or R-4 occupancies	2 hours or 1 hour and provide <i>automatic sprinkler system</i>
Storage rooms used for combustible material in buildings with an <i>exhibition area</i>	2 hours and provide <i>automatic sprinkler system</i>

Stationary storage battery systems having an energy capacity greater than the threshold quantity specified in the <i>Chicago Fire Prevention Code</i>	1 hour in Group B, F, M, S and U occupancies; 2 hours in Group A, E, I and R occupancies.
<i>Telecommunication equipment areas</i> over 50 square feet and not over 150 square feet	2 hours or 1 hour and provide <i>automatic sprinkler system</i>
<i>Telecommunication equipment areas</i> over 150 square feet	2 hours ^a
Electrical installations and transformers	See Sections 110.26 through 110.34 and Sections 450.8 through 450.48 of the <i>Chicago Electrical Code</i> for protection and separation requirements.

For SI: 1 square foot = 0.0929 m², 1 pound per square inch (psi) = 6.9 kPa, 1 British thermal unit (Btu) per hour = 0.293 watts, 1 horsepower = 746 watts, 1 gallon = 3.785 L, 1 cubic foot = 0.0283 m³.

- a. *Telecommunication equipment areas* over 150 square feet require an automatic fire extinguishing system in accordance with section 903.2.11.7.
2. Revise the exception to Section 509.1 to read:
- "**Exception:** Incidental uses within and serving a single *dwelling unit* or *sleeping unit* are not required to comply with this section."
3. Revise Section 509.3 to read:
- "**509.3 Area limitations.**
- Incidental uses shall not occupy more than 10 percent of the *gross floor area* of the primary occupancy in which they are located."
4. Delete the second sentence in Section 509.4.1.
5. Delete Section 509.4.2 and its subsections.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 19; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 8; Amend Coun. J. 12-14-22, p. 58278, Art. III, § 1)

14B-5-510 IBC Section 510.

The provisions of Section 510 of IBC are not adopted.

CHAPTER 14B-6

TYPES OF CONSTRUCTION

- 14B-6-601 General.**
- 14B-6-602 Construction classification.**
- 14B-6-603 Combustible material in Types I and II construction.**
- 14B-6-604 Combustible material in Types III and IV construction.**
- 14B-6-605 Basement construction.**

14B-6-601 General.

The provisions of Section 601 of IBC are adopted by reference with the following modification:

1. Replace Table 601 as shown:

TABLE 601

FIRE-RESISTANCE RATING REQUIREMENTS FOR BUILDING ELEMENTS (HOURS)

BUILDING ELEMENT	TYPE I		TYPE II		TYPE III		TYPE IV	TYPE V	
	A	B	A	B	A	B	HT	A	B
BUILDING ELEMENT	TYPE I		TYPE II		TYPE III		TYPE IV	TYPE V	
	A	B	A	B	A	B	HT	A	B

Primary structural frame ^f	3 ^{a, b}	2 ^{a, b}	1 ^b	0	1 ^b	0	HT	1 ^b	0
Bearing walls									
Exterior ^{e, f}	3	2	1	0	2	2	2	1	0
Interior	3 ^a	2 ^a	1	0	1	0	1/HT	1	0
Nonbearing walls and partitions - Exterior	See Table 602								
Nonbearing walls and partitions - Interior ^d	0	0	0	0	0	0	Note i	0	0
Floor construction and associated secondary members	2	2	1	0	1 ^g	0	HT	1 ^g	0
Roof construction and associated secondary members	1.5 ^b	1 ^b	1 ^b	0 ^c	1 ^{b, c, h}	0	HT	1 ^{b, c, h}	0

For SI: 1 foot = 304.8 mm.

- Roof supports: *Fire-resistance ratings of primary structural frame* and bearing walls are permitted to be reduced by 1 hour where supporting a roof only.
- Except in Group F-1, H, M and S-1 occupancies, fire protection of structural members in roof construction shall not be required, including protection of *primary structural frame* members, roof framing and decking where every part of the roof construction is 20 feet or more above any floor immediately below.
- In all occupancies, heavy timber complying with Section 2304.11 shall be allowed where a 1-hour or less *fire-resistance rating* is required.
- Not less than the *fire-resistance rating* required by other sections of this code.
- Not less than the *fire-resistance rating* based on *fire separation distance* (see Table 602).
- Not less than the *fire-resistance rating* as referenced in Section 704.10.
- In single-family dwellings, the floor construction over *basements* and unexcavated spaces below the first *story above grade plane* is not required to have a *fire-resistance rating*.
- In buildings of exclusively Group R-2, R-3, R-4 or R-5 occupancy with no more than four *stories above grade plane*, the required *fire-resistance rating* of roof construction and associated secondary members shall be reduced to 30 minutes.
- See Section 2304.11.2.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 20)

14B-6-602 Construction classification.

The provisions of Section 602 of IBC are adopted by reference with the following modifications:

- Replace Table 602 as shown:

TABLE 602

FIRE-RESISTANCE RATING REQUIREMENTS FOR EXTERIOR WALLS BASED ON FIRE SEPARATION DISTANCE^{a, d, g}

FIRE SEPARATION DISTANCE = X (feet)	TYPE OF CONSTRUCTION	OCCUPANCY GROUP H^e	OCCUPANCY GROUP F, S^f	OCCUPANCY GROUP A, B, E, M, I, R, U^h
$X < 3^b$	VA, VB	NP	NP	2 ^{h, j}
	Others	3	2	2
$3 \leq X < 5$	All	3	2	1
$5 \leq X < 10$	IA	3	2	1
	Others	2	1	1
$10 \leq X < 30$	IA, IB	2	1 ^k	1 ^{c, k}
	IIB, VB	1	0	0
	Others	1	1 ^k	1 ^{c, k}
$X \geq 30$	All	0	0	0

For SI: 1 foot = 304.8 mm.

- a. Load-bearing *exterior walls* shall also comply with the *fire-resistance rating* requirements of Table 601.
- b. See Section 706.1.1 for party walls.
- c. *Open parking garages* complying with Section 406 shall not be required to have a *fire-resistance rating*.
- d. The *fire-resistance rating* of an *exterior wall* is determined based upon the *fire separation distance* of the *exterior wall* and the *story* in which the wall is located.
- e. For special requirements for Group H occupancies, see the *Chicago Fire Prevention Code*.
- f. For special requirements for Group S aircraft hangars, see Section 412.3.1.
- g. Where Table 705.8 permits nonbearing *exterior walls* with unlimited area of unprotected openings, the required *fire-resistance rating* for the exterior walls is 0 hours.
- h. A single-story detached *private garage*, *carport* or storage building not exceeding 600 square feet (56 m²) in *building area*, not exceeding 12 feet (3658 mm) in *building height* and with a roof slope of 2:12 or greater shall be permitted in Type VA or Type VB construction when the *fire separation distance* is 2 feet (610 mm) or greater and the *exterior wall* shall not be required to have a *fire-resistance rating*.
- i. [Reserved.]
- j. Any *exterior wall* with a *fire separation distance* of less than 3 feet shall have a *fire-resistance rating* of not less than 2 hours and shall be built of noncombustible materials, materials allowed by Section 603 or *fire-retardant treated wood* framing complying with Section 2303.2 with a noncombustible *exterior wall covering*.
- k. The required *fire-resistance rating* of *exterior walls* with a *fire separation distance* of 10 feet or greater shall only be required to be rated for exposure to fire from the inside in accordance with Section 705.5.

2. Delete the second sentence of Section 602.3.

3. Revise Section 602.4 and its subsections to read:

"602.4 Type IV.

Type IV construction is that type of construction in which the *exterior walls* are of noncombustible materials and the interior building elements are of solid wood, laminated wood, heavy timber (HT) or *structural composite lumber* (SCL) without concealed spaces. The minimum dimensions for permitted materials including solid timber, *structural glued-laminated timber*, *structural composite lumber* (SCL), and *cross-laminated timber* and details of Type IV construction shall comply with the provisions of this section and Section 2304.11. Interior walls and partitions and noncombustible floor and roof assemblies with not less than a 1-hour *fire-resistance rating* or heavy timber complying with Section 2304.11.2.2 shall be permitted.

602.4.1 [Reserved.]

602.4.2 [Reserved.]

602.4.3 Exterior structural members.

Where a *fire separation distance* of 20 feet (6096 mm) or more is provided, wood columns and arches conforming to heavy timber sizes complying with Section 2304.11 shall be permitted to be used externally."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 21; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 9)

14B-6-603 Combustible material in Types I and II construction.

The provisions of Section 603 of IBC are adopted by reference with the following modifications:

1. Revise item 1 and its subparts in Section 603.1 to read:

"1. *Fire-retardant-treated wood* shall be permitted in:

1.1. Nonbearing partitions where the required *fire-resistance rating* is 1 hour or less.

1.2. [Reserved.]

1.3. Roof construction, including girders, trusses, framing and decking of one-story buildings that do not exceed the height and area limitations for Type IIB construction."

2. Add an exception to item 6 in Section 603.1 to read:

"Exception: Combustible millwork shall not be permitted in *exterior wall* openings that are within 5 feet (1524 mm) horizontally and 3 feet (914 mm) vertically of an opening in an adjacent *story*."

3. Revise item 13 in Section 603.1 to read:

"13. Combustible *exterior wall coverings*, *exterior balconies* and similar projections and bay or oriel windows in accordance with Chapter 14 and Section 705.2."

4. Revise item 14 in Section 603. 1 to read:

"14. Blocking such as for handrails, millwork, cabinets and window and door frames in *exterior walls* or in interior walls and partitions where the required *fire-resistance rating* is 1 hour or less."

5. Revise item 19 in Section 603.1 to read:

"19. Heavy timber as permitted by Note c to Table 601 and Sections 602.4.3 and 705.2."

6. Revise item 25 in Section 603.1 to read:

"25. Materials exposed within plenums complying with the *Chicago Mechanical Code* and *Chicago Electrical Code*."

7. Add a new item 27 to Section 603.1 to read:

"27. Exterior fiber-reinforced polymer as permitted by Section 2613.5."

8. Revise Section 603.1.2 to read:

"603.1.2 Piping.

The use of combustible piping materials shall be permitted where installed in accordance with the limitations of Section 903, the *Chicago Mechanical Code* and the *Chicago Plumbing Code*."

9. Revise Section 603.1.3 by replacing "this code" with "the *Chicago Electrical Code*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 22)

14B-6-604 Combustible material in Types III and IV construction.

The following language is adopted as a new Section 604:

"604. COMBUSTIBLE MATERIAL IN TYPES III AND IV CONSTRUCTION

604.1 Allowable materials.

Combustible materials shall be permitted in the *exterior walls* of buildings of Type III or IV construction where combustible materials in the same application are permitted in the *exterior walls* of buildings of Type I or II construction by Section 603.1."

14B-6-605 Basement construction.

The following language is adopted as a new Section 605:

"605. BASEMENT CONSTRUCTION

605.1 General.

In all *buildings*, *basement* construction shall comply with the requirements of Sections 605.2 through 605.4.

605.2 Basement columns and bearing walls.

In *buildings* of Type III or V construction with three or more *stories above grade plane*, all *basement* columns and *basement* bearing walls shall be of noncombustible material or of heavy timber construction.

605.3 Multiple basement stories.

In *buildings* with more than one *basement story*, the floor construction separating the highest *basement* from the first *story above grade plane* and all construction below that level shall be of Type IA construction.

605.4 Floor construction.

Except in *buildings* of Group R-5 occupancy, floor construction over *basements* shall have a *fire-resistance rating* of not less than 1 hour."

CHAPTER 14B-7

FIRE AND SMOKE PROTECTION FEATURES

14B-7-701 General.

14B-7-702 Multiple use fire assemblies.

14B-7-703 Fire-resistance ratings and fire tests.

14B-7-704 Fire-resistance rating of structural members.

14B-7-705 Exterior walls.

14B-7-706 Fire walls.

14B-7-707 Fire barriers.

14B-7-708 Fire partitions.

14B-7-709 Smoke barriers.

- 14B-7-710 IBC Section 710.**
- 14B-7-711 Floor and roof assemblies.**
- 14B-7-712 Vertical openings.**
- 14B-7-713 Shaft enclosures.**
- 14B-7-714 Penetrations.**
- 14B-7-715 Fire-resistant joint systems.**
- 14B-7-716 Opening protectives.**
- 14B-7-717 Ducts and air transfer openings.**
- 14B-7-718 Concealed spaces.**
- 14B-7-719 Fire-resistance requirements for plaster.**
- 14B-7-720 Thermal- and sound-insulating materials.**
- 14B-7-721 Prescriptive fire resistance.**
- 14B-7-722 Calculated fire resistance.**

14B-7-701 General.

The provisions of Section 701 of IBC are adopted by reference with the following modification:

1. Revise Section 701.1 to read:

"701.1 Scope.

The provisions of this chapter shall govern the materials, systems and assemblies used for structural *fire resistance* and fire-resistance-rated construction, separation of adjacent spaces to safeguard against the spread of fire and smoke within a *building* and the spread of fire to or from *buildings*."

14B-7-702 Multiple use fire assemblies.

The provisions of Section 702 of IBC are adopted by reference without modification.

14B-7-703 Fire-resistance ratings and fire tests.

The provisions of Section 703 of IBC are adopted by reference with the following modifications:

1. Delete Section 703.2.5.
2. Revise the numbered portion of Section 703.3 to read:
 - "1. *Fire-resistance* designs documented in a test report prepared by an *approved agency* or test results compiled in the UL Online Certifications Directory, Intertek Directory of Building Products, GA 600 or another *approved* source.
 2. Prescriptive designs of fire-resistance-rated building elements, components or assemblies as prescribed in Section 721.
 3. *Approved* calculations in accordance with Section 722.
 4. *Approved* engineering analysis based on a comparison of building element, component or assembly designs having *fire-resistance ratings* as determined by the test procedures set forth in ASTM E119 or UL 263."
3. Delete the last sentence of Section 703.4.
4. Revise the first sentence of Section 703.7 to read:

"Where there is an accessible concealed floor, floor-ceiling or *attic* space and within mechanical and electrical rooms, *fire walls*, *fire barriers*, *fire partitions*, *smoke barriers* and smoke partitions or any other wall required to have protected openings or penetrations shall be effectively and permanently identified with signs or stenciling in the concealed space or mechanical or electrical room."

14B-7-704 Fire-resistance rating of structural members.

The provisions of Section 704 of IBC are adopted by reference with the following modifications:

1. Delete the exception to Section 704.1.
2. Revise Section 704.3 to read:

"704.3 Protection of the primary structural frame other than columns.

Members of the *primary structural frame*, other than columns, that are required to have protection to achieve a *fire-resistance rating* and support more than two floors or one floor and roof, or support a load-bearing wall or a nonload-bearing wall more than two *stories* high, shall be provided individual encasement protection by protecting them on all sides for the full length, including connections to other structural members, with materials having the required *fire-resistance rating*."

3. Revise Section 704.11 to read:

"704.11 Bottom flange protection.

Fire protection is not required at the bottom flange of lintels, shelf angles and plates, spanning not more than 6 feet 4 inches (1931 mm) whether part of the *primary structural frame* or not, and from the bottom flange of lintels, shelf angles and plates not part of the *primary structural frame*, regardless of span."

4. Revise Section 704.12 by deleting "approved."

5. Revise Section 704.13.1 to read:

"704.13.1 Fire-resistance rating.

The application of SFRM shall be consistent with the *fire-resistance rating* and the listing, including, but not limited to, minimum size of structural element, minimum thickness and dry density of the applied SFRM, method of application, substrate surface conditions and the use of bonding adhesives, sealants, reinforcing or other materials."

14B-7-705 Exterior walls.

The provisions of Section 705 of IBC are adopted by reference with the following modifications:

1. Revise Section 705.2, its subsections, and related tables to read:

"705.2 Projections.

Cornices, eave overhangs, *exterior balconies*, *decks* and similar projections extending beyond the *exterior wall* shall conform to the requirements of this section and Section 1405. Egress balconies shall comply with Section 1021. *Porches*, *exterior exit stairways* and *exterior exit ramps* shall comply with Section 1027. Projections shall not extend any closer to the line used to determine the *fire separation distance* than shown in Table 705.2.

Exception: *Buildings* on the same *lot* and considered as portions of one *building* in accordance with Section 705.3 are not required to comply with this section for projections between the *buildings*.

TABLE 705.2

MINIMUM DISTANCE OF PROJECTION

FIRE SEPARATION DISTANCE - FSD (feet)	MINIMUM DISTANCE FROM LINE USED TO DETERMINE FSD
0 to less than 2	Projections not permitted ^a
2 to less than 3	24 inches ^a
3 to less than 5	2/3 FSD
5 or greater	40 inches

For SI: 1 foot = 304.8 mm; 1 inch = 25.4 mm.

a. Detached *private garages* and buildings of Group R-3 or R-5 occupancy located within 28 inches (711 mm) of a *property line* are permitted to have roof eave projections not exceeding 4 inches (102 mm).

705.2.1 Types I and II construction.

Projections from walls of Type I or II construction shall be of materials allowed by Table 705.2.1. Projections not listed in the table shall conform to the type of construction and *fire-resistance rating* required for the *building* to which they are attached.

705.2.2 Type III, IV or V construction.

Projections from walls of Type III, IV or V construction shall be of noncombustible materials or combustible materials as allowed by Table 705.2.1 or Table 705.2.2. Projections not listed in the table shall conform to the type of construction and *fire-resistance rating* required for the *building* to which they are attached.

705.2.3 Combustible projections.

Decks, *exterior balconies*, *exterior exit stairways*, *porches*, and similar projections of combustible construction shall not exceed 50 percent of the *building's* perimeter on each floor."

2. Insert Tables 705.2.1 and 705.2.2 as shown:

TABLE 705.2.1

PROJECTIONS FROM WALLS OF ANY TYPE OF CONSTRUCTION ^a

Type of Projection	Material Type	Fire Separation Distance (feet) ^b			
		0 to less than 3	3 to less than 5	5 to less than 10	10 or greater
Type of Projection	Material Type	Fire Separation Distance (feet) ^b			
		0 to less than 3	3 to less than 5	5 to less than 10	10 or greater
Cornices, eave overhangs, sunshades, bay windows, oriel windows and similar decorative projections on <i>buildings</i> not exceeding 40 feet in <i>building height</i>	U	No	Yes	Yes	Yes
	P	Yes	Yes	Yes	Yes
Cornices, eave overhangs, sunshades, bay windows, oriel windows and similar decorative projections on <i>buildings</i> greater than 40 feet in <i>building height</i>	U	No	No	No	No
	UNC	No	Yes	Yes	Yes
	P	No	No	No	Yes
	PNC	Yes	Yes	Yes	Yes
Gutters and downspouts on <i>buildings</i> not exceeding 40 feet in <i>building height</i>	U	No	Yes	Yes	Yes
	UNC	Yes	Yes	Yes	Yes
Gutters and downspouts on <i>buildings</i> greater than 40 feet in <i>building height</i>	U	No	No	No	No
	UNC	Yes	Yes	Yes	Yes
Exterior balconies, each not exceeding 100 square feet in area, on <i>buildings</i> not exceeding 55 feet in <i>building height</i>	U	No	Yes	Yes	Yes
	UNC	Yes	Yes	Yes	Yes
Exterior balconies, each not exceeding 100 square feet in area, on <i>buildings</i> greater than 55 feet in <i>building height</i>	U	No	No	No	No
	UNC	Yes	Yes	Yes	Yes
Decks laid directly on the ground without airspace below	U	Yes	Yes	Yes	Yes
Decks not exceeding 400 square feet in area	U	No	Yes	Yes	Yes
Decks, any area, not more than 6 feet above adjoining ground	U	No	No	Yes	Yes
Decks, any area, any height	UNC	Yes	Yes	Yes	Yes
Weather-protected entries not exceeding 50 square feet in area	U	Yes	Yes	Yes	Yes
Open porches, serving no more than 1 story above grade plane and not more than 200 square feet in area	U	No	Yes	Yes	Yes
Other open porches and exterior exit stairways, serving more than 1 but no more than 4 stories above grade plane	U	No	No	No	No
	UNC	Yes	Yes	Yes	Yes
Exterior exit stairways, other than fire escapes, serving any story more than 4 stories above grade plane	PNC	Yes	Yes	Yes	Yes
Fire escapes	UNC	Yes	Yes	Yes	Yes
Pickets, handrails and similar vertical elements not more than 42 inches in height above the adjacent walking surface	U	No	Yes	Yes	Yes
	UNC	Yes	Yes	Yes	Yes

For SI: 1 foot = 304.8 mm; 1 inch = 25.4 mm; 1 square foot = 0.0929 m².

U = any material not less fire resistive than untreated wood or plastic composites that comply with ASTM D7032 and Section 2612, unprotected

P = any material, protected with construction providing *fire-resistance rating* as required for *exterior wall*

UNC = noncombustible, unprotected

PNC = noncombustible, protected with construction providing *fire-resistance rating* as required for *exterior wall*

a. Where material type U is allowed, material types P, UNC and PNC shall also be allowed; where material type P is allowed, material types UNC and PNC shall also be allowed; where material type UNC is allowed, material type PNC shall also be allowed.

b. Egress balconies shall also comply with Section 1021. *Exterior exit stairways* and *exterior exit ramps* shall also comply with Section 1027.

TABLE 705.2.2

PROJECTIONS FROM WALLS OF TYPE III, IV OR V CONSTRUCTION ^a

Type of Projection	Material Type	Fire Separation Distance (feet) ^b			
		0 to less than 3	3 to less than 5	5 to less than 10	10 or greater
Type of Projection	Material Type	Fire Separation Distance (feet) ^b			
		0 to less than 3	3 to less than 5	5 to less than 10	10 or greater
<i>Exterior balconies</i> , each not exceeding 100 square feet in area, on <i>buildings</i> not exceeding 55 feet in <i>building height</i>	U	No	Yes	Yes	Yes
	UNC, HT	Yes	Yes	Yes	Yes
<i>Exterior balconies</i> individually exceeding 100 square feet in area	P, HT, UNC	Yes	Yes	Yes	Yes
<i>Decks</i> , any area, any height	HT	Yes	Yes	Yes	Yes
<i>Open porches</i> on buildings of Group R occupancy serving no more than 3 <i>stories above grade plane</i> , not projecting more than 10 feet from the building and not providing more than 150 feet of floor surface, exclusive of stairs, per <i>dwelling unit</i>	U	No	Yes	Yes	Yes
	P, UNC	Yes	Yes	Yes	Yes
<i>Repair</i> or replacement of <i>enclosed porches</i> existing on December 31, 2018, serving no more than 3 <i>stories above grade plane</i>	P1	Yes	Yes	Yes	Yes
	U	No	No	Yes	Yes
<i>Repair</i> or replacement of <i>porches</i> existing on January 20, 1950, serving no more than 4 <i>stories above grade plane</i>	U	Yes	Yes	Yes	Yes

For SI: 1 foot = 304.8 mm; 1 inch = 25.4 mm; 1 square foot = 0.0929 m².

U = any material not less fire resistive than untreated wood or plastic composites that comply with ASTM D7032 and Section 2612, unprotected

P = any material, protected with construction providing *fire-resistance rating* as required for *exterior wall*

P1 = any material, protected with construction providing *fire-resistance rating* of at least one hour

HT = heavy timber construction, complying with Section 2304.11

UNC = noncombustible, unprotected

PNC = noncombustible, protected with construction providing *fire-resistance rating* as required for *exterior wall*

a. Where material type U is allowed, material types P, HT, UNC and PNC shall also be allowed; where material type P is allowed, material types HT, UNC and PNC shall also be allowed; where material type UNC is allowed, material type PNC shall also be allowed.

b. Egress balconies shall also comply with Section 1021. *Exterior exit stairways* and *exterior exit ramps* shall also comply with Section 1027.

3. Revise exception 1 to Section 705.3 to read:

"1. Two or more buildings on the same *lot of record* in accordance with the exception to Section 503.1.2."

4. Revise Section 705.5 to read:

"705.5 Fire-resistance ratings.

Exterior walls shall be fire-resistance rated in accordance with Tables 601 and 602 and this section. For other than Group H occupancies, the required *fire-resistance rating* of *exterior walls* with a *fire separation distance* of 10 feet (3048 mm) or greater shall only be required to be rated for exposure to fire from the inside. The required *fire-resistance rating* of *exterior walls* with a *fire separation distance* of less than 10 feet (3048 mm) shall be rated for exposure to fire from both sides."

5. Delete Section 705.7.

6. Delete Equation 7-1.

7. Delete Figure 705.7.

8. Delete Table 705.8.

9. Insert Tables 705.8(1) and 705.8(2) as shown:

TABLE 705.8(1)

MAXIMUM AREA OF EXTERIOR WALL OPENINGS BASED ON FIRE SEPARATION DISTANCE AND DEGREE OF OPENING PROTECTION FOR BUILDINGS NOT COVERED BY TABLE 705.8(2)

FIRE SEPARATION DISTANCE (feet)	DEGREE OF OPENING PROTECTION	ALLOWABLE AREA ^a
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FIRE SEPARATION DISTANCE (feet)	DEGREE OF OPENING PROTECTION	ALLOWABLE AREA ^a
0 to less than 3 ^{b, c, k}	Unprotected, Nonsprinklered (UP, NS)	Not Permitted ^k
	Unprotected, Sprinklered (UP, S) ⁱ	Not Permitted ^k
	Protected (P)	Not Permitted ^k
3 to less than 5 ^e	Unprotected, Nonsprinklered (UP, NS)	Not Permitted
	Unprotected, Sprinklered (UP, S) ⁱ	15%
	Protected (P)	15%
5 to less than 10 ^e	Unprotected, Nonsprinklered (UP, NS)	10%
	Unprotected, Sprinklered (UP, S) ⁱ	25%
	Protected (P)	25%
10 to less than 15 ^{e, g}	Unprotected, Nonsprinklered (UP, NS)	15%
	Unprotected, Sprinklered (UP, S) ⁱ	45%
	Protected (P)	45%
15 to less than 20 ^{f, g, j}	Unprotected, Nonsprinklered (UP, NS)	25%
	Unprotected, Sprinklered (UP, S) ⁱ	75%
	Protected (P)	75%
20 to less than 25 ^g	Unprotected, Nonsprinklered (UP, NS)	45%
	Unprotected, Sprinklered (UP, S) ⁱ	No Limit
	Protected (P)	No Limit
25 to less than 30 ^g	Unprotected, Nonsprinklered (UP, NS)	70%
	Unprotected, Sprinklered (UP, S) ⁱ	No Limit
	Protected (P)	No Limit
30 or greater	Unprotected, Nonsprinklered (UP, NS)	No Limit
	Unprotected, Sprinklered (UP, S) ⁱ	No Limit
	Protected (P)	No Limit

For SI: 1 foot = 304.8 mm.

UP, NS = Unprotected openings in *buildings* not equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1.

UP, S = Unprotected openings in *buildings* equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2.

P = Openings protected with an opening protective assembly in accordance with Section 705.8.2.

a. Values indicated are the percentage of the area of the *exterior wall*, per *story*.

b. For the requirements for *fire walls* of buildings with differing heights, see Section 706.6.1.

c. For openings in a *fire wall* for buildings on the same lot, see Section 706.8.

d. [Reserved.]

e. Unprotected openings shall not be permitted for openings with a *fire separation distance* of less than 15 feet for Group H-2 and H-3 occupancies.

f. [Reserved.]

g. The area of openings in an open parking structure with a fire separation distance of 10 feet or greater shall not be limited.

h. [Reserved.]

i. Not applicable to Group H-1, H-2 and H-3 occupancies.

j. [Reserved.]

k. For openings between S-2 parking garage and Group R-2 building, see Section 705.3, Exception 2.

TABLE 705.8(2)

MAXIMUM AREA OF EXTERIOR WALL OPENINGS BASED ON FIRE SEPARATION DISTANCE AND DEGREE OF OPENING PROTECTION FOR PRIVATE GARAGES AND GROUP R-2, R-3, R-4 AND R-5 OCCUPANCIES UP TO 4 STORIES ABOVE GRADE PLANE ^a

FIRE SEPARATION DISTANCE (feet)	DEGREE OF OPENING PROTECTION	ALLOWABLE AREA ^b
FIRE SEPARATION DISTANCE (feet)	DEGREE OF OPENING PROTECTION	ALLOWABLE AREA ^b
	Unprotected, Nonsprinklered (UP, NS)	Not Permitted

0 to less than 2	Unprotected, Sprinklered (UP, S)	Not Permitted
	Protected (P)	Not Permitted
2 to less than 3	Unprotected, Nonsprinklered (UP, NS)	5%
	Unprotected, Sprinklered (UP, S)	10%
	Protected (P)	10%
3 to less than 5	Unprotected, Nonsprinklered (UP, NS)	25%
	Unprotected, Sprinklered (UP, S)	25%
	Protected (P)	25%
5 or greater	Unprotected, Nonsprinklered (UP, NS)	No Limit
	Unprotected, Sprinklered (UP, S)	No Limit
	Protected (P)	No Limit

For SI: 1 foot = 304.8 mm.

UP, NS = Unprotected openings in *buildings* not equipped throughout with an *automatic sprinkler system*.

UP, S = Unprotected openings in *buildings* equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2.

P = Openings protected with an opening protective assembly in accordance with Section 705.8.2.

- a. Table 705.8(2) is applicable to detached *private garages* and *carports*, *private garages* and *carports* attached to *buildings* exclusively containing a Group R-2, R-3, R-4 or R-5 occupancy with no more than four *stories above grade plane*, and *buildings* exclusively containing a Group R-2, R-3, R-4 or R-5 occupancy with no more than four *stories above grade plane*. All other *buildings* shall be subject to the requirements of Table 705.8(1).
- b. Values indicated are the percentage of the area of the *exterior wall*, per *story*. For *basements*, both above- and below-grade portions of the *exterior wall* shall be included in determining the allowable area.

10. Revise Section 705.8.1 by replacing "Table 705.8" with "Table 705.8(1) or 705.8(2), as applicable" and inserting exceptions 3 and 4 to read:

"3. Where *fire separation distance* is based on an imaginary line between *buildings* on the same *lot*, the maximum area of protected exterior wall openings shall be the greater of the value in Table 705.8(1) or 10 percent for *stories* that are at least 15 feet (4572 mm) vertically above the roof of adjacent *buildings* on the same *lot*.

4. Where *fire separation distance* is based on an imaginary line between *buildings* on the same *lot*, the maximum area of protected and unprotected exterior wall openings shall not be limited for *stories* that are at least 50 feet (15.24 m) vertically above the roof of adjacent *buildings* on the same *lot*."

11. Revise Section 705.8.3 by replacing "*approved materials*" with "*materials allowed by this code*."

12. Revise the description of A_p in Equation 7-2 to read: "Actual area of protected openings."

13. Add new exception 4 to Section 705.8.5 to read:

"4. This section shall not apply to Group R-5 buildings."

14. Revise Section 705.8.6 to read:

"Where *fire separation distance* is based on an imaginary line between *buildings* on the same *lot* and is less than 15 feet (4572 mm), every opening that is less than 15 feet (4572 mm) vertically above the roof of adjacent *buildings* on the same *lot* shall be provided with opening protectives having a *fire protection rating* of not less than 3/4 hour.

Exceptions:

1. Opening protectives are not required where the roof assembly of the adjacent *building* has a *fire-resistance rating* of not less than 2 hours for a minimum distance of 15 feet (4572 mm) horizontally from the *exterior wall* facing the imaginary line and the entire length and span of the supporting elements for the fire-resistance-rated roof assembly has a *fire-resistance rating* of not less than 2 hours.
2. *Buildings* on the same *lot* that are considered as portions of one *building* in accordance with Section 705.3. exception 1, are not required to comply with Section 705.8.6.
3. Group R buildings that are four *stories* or less *above grade plane*."

15. Delete exception 5 to Section 705.11 and its subparts.

16. [Reserved.]

17. Add new Section 705.12 to read:

"705.12 Easements.

The *building official* may adopt rules for considering permanent and perpetual easements or restrictive declarations on adjoining *lots* as a means of adjusting requirements in this section based on *fire separation distance*."

14B-7-706 Fire walls.

The provisions of Section 706 of IBC are adopted by reference with the following modifications:

1. Revise Section 706.1.1 to read:

"706.1.1 Party walls.

Any wall located on a *property line* between adjacent *buildings*, which is used or adapted for joint service between the two *buildings*, shall be constructed as a *fire wall* in accordance with Section 706. Party walls shall be constructed without openings and shall create separate *buildings*.

Exceptions:

1. *Fire walls* are not required on *property lines* dividing a *building* on a single *zoning lot* for ownership purposes where the aggregate height and area of the portions of the *building* located on both sides of the *property line* do not exceed the maximum height and area requirements of this code and a perpetual recorded easement or recorded contractual agreement, such as a condominium declaration, authorizes the owners of portions of the *building* located on either side of the *property line* to access the other side for purposes of maintaining the *building* and its systems.
2. Party walls dividing a *structure* on a single *zoning lot* into separate *buildings* are permitted to have protected openings in accordance with Section 716 where a perpetual recorded easement or recorded contractual agreement, such as a condominium declaration, authorizes the owners of portions of the *building* located on either side of the *property line* to access the other side for purposes of egress and maintaining the *building* and its systems."

2. Delete Section 706.2.

3. Revise Section 706.3, including the exception, to read:

"706.3 Materials.

Fire walls shall be of noncombustible materials."

4. Revise Section 706.4 to read:

"706.4 Fire-resistance rating.

Fire walls shall have a *fire-resistance rating* of not less than 4 hours."

5. Delete Table 706.4.

6. Revise the exceptions to Section 706.5 to read:

"Exceptions:

1. [Reserved.]
2. *Fire walls* shall be permitted to terminate at the interior surface of noncombustible exterior sheathing, exterior siding or other noncombustible exterior finishes provided that the sheathing, siding or other exterior noncombustible finish extends a horizontal distance of not less than 4 feet (1220 mm) on both sides of the *fire wall*. Unprotected openings are not permitted within the horizontal distance of 4 feet (1220 mm).
3. *Fire walls* shall be permitted to terminate at the interior surface of noncombustible exterior sheathing where the *building* on each side of the *fire wall* is protected by an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 or 903.3.1.2."

7. Revise item 1 in Section 706.5.1 to read:

"1. The *exterior walls* on both sides of the *fire wall* shall be of noncombustible construction and have a 1-hour *fire-resistance rating* with 3/4-hour protection where opening protection is required by Section 705.8. The *fire-resistance rating* of the *exterior wall* shall extend not less than 4 feet (1220 mm) on each side of the intersection of the *fire wall* to *exterior wall*. *Exterior wall* intersections at *fire walls* that form an angle equal to or greater than 180 degrees (3.14 rad) do not need *exterior wall* protection."

8. Revise item 2 in Section 706.5.1 by replacing "*lot line*" with "*property line*."

9. Revise Section 706.5.2, including the exceptions, to read:

"706.5.2 Horizontal projecting elements.

Fire walls shall extend to the outer edge of horizontal projecting elements such as *porches*, *exterior balconies*, roof overhangs, *canopies*, *marquees* and similar projections that are within 4 feet (1220 mm) of the *fire wall*."

10. Delete exception 2 to Section 706.6.

11. Revise Exception 3 to Section 706.6 by replacing "Class B roof covering" with "Class A *roof covering*."

12. Delete exceptions 4 and 5 to Section 706.6.

13. Add a new exception 7 to Section 706.6 to read:

"7. Offset construction in accordance with Section 706.6.3."

13.1. Revise Section 706.6.1 to read:

"Where a *fire wall* separates *buildings* having different roof levels, the *fire wall* shall terminate at a point not less than 30 inches (762 mm) vertically above the lower roof level provided that within 15 feet (4572 mm) above the lower roof level the *exterior wall* shall have a *fire-resistance rating* of not less than 1 hour and opening protectives shall have a *fire protection rating* of not less than 3/4 hour. The *exterior wall* above the *fire wall* shall also comply with Sections 602 and 705 based on *fire separation distance*.

Exception: Where the *fire wall* terminates at the underside of the *roof covering* or *roof deck* of the lower roof, provided that:

1. The lower roof assembly within 15 feet (4572 mm) horizontally of the *fire wall* has not less than a 2-hour *fire-resistance rating* and the entire length and span of supporting elements for the rated roof assembly has a *fire-resistance rating* of not less than 2 hours.

2. Openings in the lower roof shall not be located within 15 feet (4572 mm) horizontally of the *fire wall*."

14. Add a new Section 706.6.3 to read:

"706.6.3 Offset construction.

Where a *fire wall* is offset at an intermediate floor level, the offset floor and its supporting construction shall have a *fire-resistance rating* of not less than 4 hours. *Interior exit stairways* and *interior exit ramps* shall not penetrate the horizontal offset of a *fire wall*."

15. Revise Section 706.7 by deleting "*approved for fireblocking*."

16. Revise Section 706.11 by replacing "*lot line*" with "*property line*."

(Amend Coun. J. 6-25-21, p. 32180, Art. II, § 10; Amend Coun. J. 1-26-22, p. 43756, Art. II, § 7)

14B-7-707 Fire barriers.

The provisions of Section 707 of IBC are adopted by reference with the following modifications:

1. Revise Section 707.3.2 by replacing "Section 1023.1" with "Section 1023.2."
2. Revise Section 707.3.5 by replacing "Section 1026.1" with "Section 1026.2."
3. Revise Section 707.3.10 by deleting "*fire walls*."
4. Revise Table 707.3.10 by deleting "FIRE WALLS" from the title.
5. Revise Table 707.3.10 by adding "U" to the third row and deleting the fourth row.
6. Delete both exceptions to Section 707.5.1.
7. Revise Section 707.9 to read:

"707.9 Voids at intersections.

The voids created at the intersection of a *fire barrier* and a nonfire-resistance-rated *roof assembly* or a nonfire-resistance-rated *exterior wall* assembly shall be filled. The material or system used to fill the void shall be securely installed in or on the intersection for its entire length so as not to dislodge, loosen or otherwise impair its ability to accommodate expected building movements and to retard the passage of fire and hot gases."

(Amend Coun. J. 10-7-20, p. 21791, Art. II, § 11)

14B-7-708 Fire partitions.

The provisions of Section 708 of IBC are adopted by reference with the following modifications:

1. Delete items 2 and 4 in Section 708.1.
2. Delete the exceptions to Section 708.3.
3. Delete exceptions 2 and 4 to Section 708.4.
4. Delete the exception to Section 708.4.1.
5. Add a new item 3 to Section 708.4.2 to read:

"3. In Group R-2 occupancies not more than four *stories above grade plane* and not exceeding 60 feet (18 288 mm) in *building height*, the *attic space* shall be subdivided by *draftstops* into areas not exceeding 3,000 square feet (279 m²) or above every two *dwelling units*, whichever is smaller."

6. Delete the exceptions to Section 708.4.2.

14B-7-709 Smoke barriers.

The provisions of Section 709 of IBC are adopted by reference with the following modifications:

1. Revise the last sentence of Section 709.4 to read:

"Smoke-barrier walls used to enclose areas of refuge in accordance with Section 1009.6.4 or to enclose elevator lobbies in accordance with Section 405.4.3, 3007.6.2, or 3008.6.2 shall comply with Section 709.4.2."

2. Revise the first sentence of Section 709.4.2 to read:

"*Smoke-barrier* walls used to enclose areas of refuge in accordance with Section 1009.6.4, shall form an effective membrane enclosure that terminates at a *fire barrier* wall having a level of *fire protection rating* not less than 1 hour, another *smoke barrier* wall or an outside wall."

14B-7-710 IBC Section 710.

The provisions of Section 710 of IBC are not adopted.

14B-7-711 Floor and roof assemblies.

The provisions of Section 711 of IBC are adopted by reference with the following modifications:

1. Delete the exception to Section 711.2.3.
2. Delete the exception to Section 711.2.4.3.

14B-7-712 Vertical openings.

The provisions of Section 712 of IBC are adopted by reference with the following modifications:

1. Revise Section 712.1.2 to read:

"712.1.2 Individual dwelling unit.

Unconcealed and unprotected vertical openings totally within an individual residential *dwelling unit* and connecting four *stories* or less shall be permitted."

2. Revise Section 712.1.3.2 by replacing "approved" with "*approved*."
3. Revise Section 712.1.5.2 by replacing "an *approved* material or system" with "a material or system engineered."
- 3.1. Revise Section 712.1.7 to read:

"712.1.7 Atriums.

Atriums complying with Section 404 that connect two or more *stories* in a Group I-2 or Group I-3 occupancy or three or more *stories* in any other occupancy shall be allowed.

Exceptions:

1. *Atriums* shall not be allowed within Group H occupancies.
2. Interior balconies within Group A occupancies, *mezzanines* and *equipment platforms* that comply with Section 505, and *lofts* that comply with Section 1207.5 shall not be considered *stories* for purposes of this section.
4. Revise Section 712.1.8 by replacing "Approved vertical" with "Vertical."
5. Revise Section 712.1.9 to read:

"712.1.9 Two-story openings.

In other than Groups H, I-2 and I-3, a vertical opening that is not used as one of the applications listed in Sections 712.1.1 through 712.1.8 or 712.1.10 through 712.1.15 shall be permitted if the opening complies with either of the following items:

1. Does not connect more than two *stories* and:
 - 1.1. Does not penetrate a horizontal assembly that separates fire areas or *smoke barriers* that separate *smoke compartments*.
 - 1.2. Is not concealed within the construction of a wall or a floor/ceiling assembly.
 - 1.3. Is not open to a corridor in Group I and R occupancies.
 - 1.4. The *building* is equipped throughout with an *automatic sprinkler system* in accordance with Section

903.3.1.1 or 903.3.1.2.

1.5. Is separated from floor openings and air transfer openings serving other floors by construction conforming to required shaft enclosures.

1.6. Does not connect to a *story* containing a *mezzanine*.

2. Does not connect more than two *stories* and:

2.1. Is located in a building of Type IA, IB, or IIA construction.

2.2. One of the *stories* is a *level of exit discharge*.

2.3. Where the *building* contains a Group R occupancy, the *building* is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 24; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 11)

14B-7-713 Shaft enclosures.

The provisions of Section 713 of IBC are adopted by reference with the following modifications:

1. Add a new exception to Section 713.4 to read:

"Exception: Where shafts contain only noncombustible pipes, noncombustible conduit or iron, steel or aluminum ducts requiring floor penetrations not exceeding 9 square feet (0.8 m²) per *story*, the shaft enclosures connecting any number of *stories* shall have a *fire-resistance rating* of not less than 1 hour."

2. Revise section 713.11, including the exception, by deleting "*approved*."

3. Revise the second sentence of Section 713.13.1 to read:

"A single shaft enclosure shall be permitted to contain recycling and waste chutes."

4. Revise the first sentence of Section 713.13.4 to read:

"Waste or linen chutes shall discharge into an enclosed room separated by *fire barriers* with a *fire-resistance rating* not less than 2 hours and constructed in accordance with Section 707 or *horizontal assemblies* constructed in accordance with Section 711, or both."

5. Revise Section 713.13.6 by deleting "*approved*."

14B-7-714 Penetrations.

The provisions of Section 714 of IBC are adopted by reference with the following modifications:

1. Revise Section 714.4.1.1 by deleting "*approved*."

2. Revise exception 1.2 to Section 714.4.2 by deleting "cellulose loose fill."

3. Revise exception 2 to Section 714.4.2 by replacing "any material" with "any material allowed by the *Chicago Electrical Code*."

4. Revise exception 3 to Section 714.4.2 by replacing "any size or type" with "any size or type allowed by the *Chicago Electrical Code*."

5. Revise exception 4 to Section 714.4.2 by deleting "*approved*."

6. Delete the exceptions to Section 714.5.1.

7. Revise Section 714.5.1.1 by deleting "*approved*."

Revise Section 714.5.1.2 by replacing "an *approved*" with "a."

8. Revise exception 3 to Section 714.5.2 by replacing "any size or type" with "any size or type allowed by the *Chicago Electrical Code*."

9. Revise exception 4 to Section 714.5.2 by replacing "any material" with "any material allowed by the *Chicago Electrical Code*."

10. Revise Section 714.5.4 by replacing "an *approved*" with "a."

11. Revise Section 714.6.1 by replacing "an *approved*" with "a."

12. Revise Section 714.6.2 by replacing "an *approved*" with "a."

14B-7-715 Fire-resistant joint systems.

The provisions of Section 715 of IBC are adopted by reference with the following modifications:

1. Revise Section 715.1 by replacing "an *approved*" with "a."

2. Delete item 4 in the exception to Section 715.1.
3. Revise Section 715.4 by replacing "an *approved*" with "a."
4. Revise the first sentence of the exception to Section 715.4 to read:

"Voids created at the intersection of the exterior curtain wall assemblies and such floor assemblies where the vision glass extends to the finished floor level shall be allowed to be sealed with a material to prevent the interior spread of fire."
5. Revise Section 715.4.1 by replacing "an *approved*" with "a."
6. Revise Section 715.4.2 by replacing "An *approved*" with "A."

14B-7-716 Opening protectives.

The provisions of Section 716 of IBC are adopted by reference with the following modifications:

1. Revise Table 716.1(2) by changing the rows for the first two types of assemblies as shown:

Revisions to TABLE 716.1(2)

TYPE OF ASSEMBLY	REQUIRED WALL ASSEMBLY RATING (hours)	MINIMUM FIRE DOOR AND FIRE SHUTTER ASSEMBLY RATING (hours)	DOOR VISION PANEL SIZE ^b	FIRE-RATED GLAZING MARKING DOOR VISION PANEL ^{c, e}	MINIMUM SIDELIGHT/TRANSOM ASSEMBLY RATING (hours)		FIRE-RATED GLAZING MARKING SIDELIGHT/TRANSOM PANEL	
					Fire protection	Fire resistance	Fire protection	Fire resistance
TYPE OF ASSEMBLY	REQUIRED WALL ASSEMBLY RATING (hours)	MINIMUM FIRE DOOR AND FIRE SHUTTER ASSEMBLY RATING (hours)	DOOR VISION PANEL SIZE ^b	FIRE-RATED GLAZING MARKING DOOR VISION PANEL ^{c, e}	MINIMUM SIDELIGHT/TRANSOM ASSEMBLY RATING (hours)	Fire protection	Fire resistance	FIRE-RATED GLAZING MARKING SIDELIGHT/TRANSOM PANEL
					Fire protection	Fire resistance	Fire protection	Fire resistance
Fire walls and fire barriers having a required fire-resistance rating greater than 1 hour	4	See Note f.	Not Permitted	Not Permitted	Not Permitted	NP	Not Permitted	NP
	4 ^g	3 ^a	100 sq. in.	D-H-W-240	Not Permitted	4	Not Permitted	W-240
	3	3 ^a	100 sq. in.	D-H-W180	Not Permitted	3	Not Permitted	W-180
	2	1-1/2	100 sq. in. ^b	< 100 sq. in. = D-H-90 >100 sq. in.= D-H-W-90	Not Permitted	2	Not Permitted	W-120
	1-1/2	1-1/2	100 sq. in. ^b	< 100 sq. in. = D-H-90 >100 sq. in.= D-H-W-90	Not Permitted	1-1/2	Not Permitted	W-90
Enclosures for shafts, interior exit stairways and interior exit ramps	2	1-1/2	100 sq. in. ^{b, c}	< 100 sq. in. = D-H-90 > 100 sq. in.= D-H-T-W-90	Not Permitted	2	Not Permitted	W-120

2. Revise Table 716.1(2) by deleting the rows for "Horizontal exits in fire walls" and the rows for assemblies with a required wall assembly rating of 0.5 hours.
3. Add new notes f and g to Table 716.1(2) to read:

"f. Two doors or shutters, each with a *fire-resistance rating* of 3 hours, installed on opposite sides of the same opening, with or without a vestibule. One door or shutter shall be automatic and the other shall be either automatic or self-closing.

g. Where *buildings* on both sides of a *fire wall* are equipped with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2."
4. Revise the numbered portion of Section 716.1.1 to read:
 - "1. Designs documented in a test report prepared by an *approved agency* or test results compiled in the UL Online Certifications Directory, Intertek Directory of Building Products, or another *approved* source.
 2. *Approved* calculations performed in an *approved* manner.
 3. *Approved* engineering analysis based on a comparison of opening protective designs having *fire protection ratings* as

determined by the test procedures set forth in NFPA 252, NFPA 257, UL 9, UL 10B or UL 10C."

5. Revise Section 716.2.1 by replacing "Approved *fire door*" with "*Fire door*."
6. Revise Section 716.5.2 by replacing "Approved fire-protection-rated" with "Fire-protection-rated."
7. Delete item 8 in Section 716.2.6.6.
8. Revise Section 716.2.9.1 by deleting "approved and."
9. Revise Section 716.3.5.1 to read:

"716.3.5.1 Fire window frames.

Fire window frames shall only be used for intended applications."

(Amend Coun. J. 6-25-21, p. 32180, Art. II, § 12)

14B-7-717 Ducts and air transfer openings.

The provisions of Section 717 of IBC are adopted by reference with the following modifications:

1. Revise the title of Section 717.1.1 to "Transfer ducts."
2. Delete Section 717.2.1.
3. Revise the first sentence of Section 717.4 to read:

"Fire and smoke *dampers* shall be provided with a means of access that is large enough to allow inspection and maintenance of the *damper* and its operating parts."

4. Revise Section 717.5.1 to read:

"717.5.1 Fire walls.

Ducts and air transfer openings permitted in *fire walls* in accordance with Section 706.11 shall be protected with *listed* 3-hour-rated *fire dampers* installed on both sides of the *fire wall* in accordance with their listing."

5. Revise exception 2 to Section 717.5.2 by replacing "an *approved*" with "a."
6. Revise Section 717.5.3 by replacing "fire and smoke *dampers*" with "*fire dampers*."
7. Delete exceptions 2, 3 and 4 to Section 717.5.2.
8. Revise exception 5 to Section 717.5.2 to read:

"5. *Fire dampers* are not required in kitchen and clothes dryer exhaust systems."

9. Revise Section 717.5.4 by replacing "Ducts and air transfer openings" with "Air transfer openings."
10. Delete the exceptions to Section 717.5.4.
11. Delete Section 717.5.4.1.
12. Revise Section 717.5.5 by replacing "at each point" with "adjacent to each point."
13. Add a new exception to Section 717.5.6 to read:

"Exception: *Fire dampers* are not required in residential kitchen and clothes dryer exhaust systems."

14. Delete Section 717.5.7.
15. Revise Section 717.6.1 by deleting "*approved*."
16. Revise the exception to Section 717.6.1 by replacing "three floors" with "four floors."
17. Revise Section 717.6.2 by deleting "*approved*."
18. Revise Section 717.6.3 to read:

"717.6.3 Nonfire-resistance-rated floor assemblies.

Duct systems constructed of materials in accordance with the *Chicago Mechanical Code* that penetrate nonfire- resistance-rated floor assemblies shall be protected by any of the following methods:

1. A shaft enclosure in accordance with Section 713.
2. The *annular space* around the penetrating duct is protected with a noncombustible material that resists the free passage of flame and the products of combustion.

Exception: *Fire dampers* are not required in ducts within individual residential *dwelling units*."

19. Add a new exception to Section 717.7.

"Exception: Within individual Group R *dwelling units* or *sleeping units*."

14B-7-718 Concealed spaces.

The provisions of Section 718 of IBC are adopted by reference with the following modifications:

1. Delete item 8 in Section 718.2.1.
2. Revise Section 718.2.1.3 to read:

"718.2.1.3 Loose-fill insulation material.

Loose-fill insulation material, insulating foam sealants and caulk materials shall not be used as a fireblock."

3. Revise Section 718.2.6 by replacing "approved wood of natural decay resistance" with "*naturally durable wood*."
4. Revise Section 718.2.7 by replacing "an *approved* material" with "a noncombustible material."
5. Delete the exception to Section 718.4.

14B-7-719 Fire-resistance requirements for plaster.

The provisions of Section 719 of IBC are adopted by reference with the following modifications:

1. Revise Section 719.3 by deleting "*approved*."
2. Revise Section 719.4 by deleting "*approved*."

14B-7-720 Thermal- and sound-insulating materials.

The provisions of Section 720 of IBC are adopted by reference with the following modification:

1. Revise Section 720.5 by replacing "*approved* roof coverings" with "a Class A or Class B *roof covering*."

14B-7-721 Prescriptive fire resistance.

The provisions of Section 721 of IBC are adopted by reference with the following modification:

1. Delete the word "approved" wherever it appears.

14B-7-722 Calculated fire resistance.

The provisions of Section 722 of IBC are adopted by reference with the following modifications:

1. Revise Section 722.1 by inserting the following before the first sentence:
"The procedures described in Section 722 may only be used where *approved* by the *building official*."
2. Revise Section 722.5.2.1 by replacing "*approved*" with "tested."
3. Revise Section 722.5.2.1.2 by replacing "*approved*" with "tested."
4. Revise Section 722.5.2.2 by replacing "*approved*" with "tested."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 25)

CHAPTER 14B-8

INTERIOR FINISHES

14B-8-801 Scope.

14B-8-802 General.

14B-8-803 Wall and ceiling finishes.

14B-8-804 Interior floor finish.

14B-8-805 Combustible materials in Types I and II construction.

14B-8-806 Decorative materials and trim.

14B-8-807 Insulation.

14B-8-808 Acoustical ceiling systems.

14B-8-801 Scope.

The provisions of Section 801 of IBC are adopted by reference without modification.

14B-8-802 General.

The provisions of Section 802 of IBC are adopted by reference with the following modification:

1. Delete Section 802.4.

14B-8-803 Wall and ceiling finishes.

The provisions of Section 803 of IBC are adopted by reference with the following modifications:

1. Revise Section 803.3 to read:

"803.3 Heavy timber exemption.

In *buildings* protected throughout by an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2, exposed portions of structural elements complying with the requirements for heavy timber construction in Section 602.4 or Section 2304.11 shall not be subject to *interior finish* requirements except in *interior exit stairways*, *interior exit ramps*, and *exit passageways*."

2. Replace Table 803.13 as shown:

TABLE 803.13

INTERIOR WALL AND CEILING FINISH REQUIREMENTS BY OCCUPANCY ^k

GROUP	SPRINKLERED ^l			NONSPRINKLERED		
	Interior exit stairways and ramps, exit passageways and interior portions of the exit discharge ^a	Corridors and enclosure for exit access stairways and ramps	Rooms and enclosed spaces ^c	Interior exit stairways and ramps, exit passageways and interior portions of the exit discharge ^a	Corridors and enclosure for exit access stairways and ramps	Rooms and enclosed spaces ^c
GROUP	SPRINKLERED ^l			NONSPRINKLERED		
	Interior exit stairways and ramps, exit passageways and interior portions of the exit discharge ^a	Corridors and enclosure for exit access stairways and ramps	Rooms and enclosed spaces ^c	Interior exit stairways and ramps, exit passageways and interior portions of the exit discharge ^a	Corridors and enclosure for exit access stairways and ramps	Rooms and enclosed spaces ^c
A-1, A-2	A	B	C	A	A	B ^c
A-3, A-4, A-5	A	B	C	A	A	C
B, E, M, R-1	A	C ^m	C	A	B	C
R-4	B	C	C	A	B	B
F	A	C	C	A	C	C
H	A	B	C ^g	A	A	B
I-1	A	C	C	A	B	B
I-2	A	B	B ^{h, i}	A	A	A
I-3	A	A ^j	C	A	A	B
I-4	A	B	B ^{h, i}	A	A	B
R-2	A	C	C	A	B	C
R-3, R-5	C	C	C	C	C	C
S	A	C	C	A	B	C
U	No restrictions			No restrictions		

For SI: 1 inch = 25.4 mm, 1 square foot = 0.0929 m².

- a. Class C interior finish materials shall be permitted for wainscoting or paneling of not more than 1,000 square feet of applied surface area in the grade-level lobby where applied directly to a noncombustible base or over furring strips applied to a noncombustible base and fireblocked as required by Section 803.15.1.
- b. [Reserved.]
- c. Requirements for rooms and enclosed spaces shall be based on spaces enclosed by partitions. Where *fire-resistance rating* is required for structural elements, the enclosing partitions shall extend from the floor to the ceiling. Partitions that do not comply with this shall be considered to be enclosing spaces and the rooms or spaces on both sides shall be considered to be one room or space. In determining the applicable requirements for rooms and enclosed spaces, the specific occupancy thereof shall be the governing factor regardless of the group classification of the *building or structure*.
- d. [Reserved.]
- e. Class C interior finish materials shall be permitted in places of assembly with an *occupant load* of 300 persons or less.
- f. [Reserved.]
- g. Class B material is required where the *building* exceeds two *stories above grade plane*.
- h. Class C interior finish materials shall be permitted in administrative spaces.
- i. Class C interior finish materials shall be permitted in rooms with a capacity of four persons or less.
- j. Class B materials shall be permitted as wainscoting extending not more than 48 inches above the finished floor in *corridors, exit access stairways and exit access ramps*.
- k. Finish materials as provided for in other sections of this code.
- l. Applies when protected by an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 or 903.3.1.2.
- m. *Corridors in ambulatory care facilities* shall be provided with Class A or B materials.

3. Revise Section 803.15.1 by replacing "1 3/4 inches (44 mm)" with "1 3/4 inches (44 mm) in depth."

4. Revise Section 803.15.1.1 by deleting "inorganic or."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 26)

14B-8-804 Interior floor finish.

The provisions of Section 804 of IBC are adopted by reference with the following modification:

1. Revise Section 804.4.2 to read:

"804.4.2 Minimum critical radiant flux.

In all occupancies, *interior floor finish* and floor covering materials within enclosures for *interior exit stairways, interior exit ramps, exit passageways*, lobbies which are part of the *exit discharge, corridors* and rooms or spaces not separated from *corridors* by partitions extending from the floor to the underside of the ceiling shall withstand a minimum critical radiant flux. The minimum critical radiant flux shall be not less than Class I throughout Groups I-1, I-2, I-3 and R-4, not less than Class I within enclosures for *interior exit stairways, interior exit ramps, exit passageways* and lobbies which are part of the *exit discharge* in Groups A, B, E, F, H, I-4, M, R-1, R-2 and S, and not less than Class II in *corridors* and spaces not separated from *corridors* by partitions extending from the floor to the underside of the ceiling in Groups A, B, E, F, H, I-4, M, R-1, R-2 and S.

Exception: In areas other than within enclosures for *interior exit stairways, interior exit ramps or exit passageways* and lobbies which are part of the *exit discharge*, where a *building* is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2, Class II materials are permitted in any area where Class I materials are required and materials complying with DOC FF-1 "pill test" (CPSC 16 CFR Part 1630) or with ASTM D2859 are permitted in any area where Class II materials are required."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 27; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 12)

14B-8-805 Combustible materials in Types I and II construction.

The provisions of Section 805 of IBC are adopted by reference without modification.

14B-8-806 Decorative materials and trim.

The provisions of Section 806 of IBC are adopted by reference with the following modifications:

1. Revise item 3 in Section 806.1 to read:

"3. Furnishings or other objects shall not be placed to obstruct the *means of egress*."

2. Revise item 4 in Section 806.1 to read:

"4. The permissible amount of decorative live vegetation and noncombustible decorative materials shall not be limited."

3. Revise the exceptions to Section 806.2 by deleting "*approved*."

4. Delete Section 806.3.

14B-8-807 Insulation.

The provisions of Section 807 of IBC are adopted by reference without modification.

14B-8-808 Acoustical ceiling systems.

The provisions of Section 808 of IBC are adopted by reference without modification.

CHAPTER 14B-9

FIRE PROTECTION AND LIFE SAFETY SYSTEMS

14B-9-901 General.

14B-9-902 Fire pump and riser room size.

14B-9-903 Automatic sprinkler systems.

14B-9-904 Alternative automatic fire-extinguishing systems.

14B-9-905 Standpipe systems.

14B-9-906 Portable fire extinguishers.

14B-9-907 Fire alarm and detection systems.

14B-9-908 IBC Section 908.

14B-9-909 Smoke control systems.

14B-9-910 Smoke and heat removal.

14B-9-911 Fire command center.

14B-9-912 Fire department connections.

14B-9-913 Fire pumps.

14B-9-914 Emergency responder safety features.

14B-9-915 Carbon monoxide detection.

14B-9-916 Gas detection systems.

14B-9-917 IBC Section 917.

14B-9-918 IBC Section 918.

14B-9-919 City fire alarm box.

14B-9-901 General.

The provisions of Section 901 of IBC are adopted by reference with the following modifications:

1. Add a new Section 901.1.1 to read:

"901.1.1 Modification of reference standards.

The *fire code official* shall be authorized to promulgate rules, not inconsistent with this code, which modify the following referenced standards for the design of *fire protection systems* for consistency with existing installations and the operations of the Chicago Fire Department: NFPA 10, NFPA 13, NFPA 13D, NFPA 13R, NFPA 14, NFPA 20, NFPA 25, NFPA 72 and NFPA 92. Where the *fire code official* has promulgated such rules and posted them on the city's public web site, references to such standards in the *Chicago Construction Codes* shall mean such standards as modified by the *fire code official's* rules.

- 1.1. Revise Section 901.2 by replacing "reduction to the provisions" with "reduction under the provisions."

2. Revise Section 901.3 to read:

"901.3 Modifications.

It is unlawful for any *person* to remove, modify or disable for more than 24 hours any *automatic sprinkler system*, automatic fire-extinguishing system, standpipe system or fire pump installed or maintained under the provisions of this code or the *Chicago Fire Prevention Code* without a *permit* from the *fire code official*. It is unlawful for any *person* to remove, modify or disable for more than 24 hours a fire alarm and detection system, smoke control system or carbon monoxide detection system without a *permit* from

the building official. It is unlawful for any person to modify a fire command center without approval by the fire code official and a permit from the building official."

3. Revise Section 901.4 by replacing "local fire department" with "Chicago Fire Department."

4. Revise Section 901.5 to read:

"901.5 Acceptance tests.

Fire protection systems shall be tested in accordance with the requirements of this code and the *Chicago Fire Prevention Code*. Where required, the tests shall be conducted in the presence of the *fire code official*. Tests required by this code, the *Chicago Fire Prevention Code* and the standards listed in this code shall be conducted at the expense of the *owner*. It shall be unlawful to occupy portions of a structure until the required *fire protection systems* within that portion of the *structure* have been tested and accepted."

5. Revise Section 901.6 and its subsections to read:

"901.6 Supervisory service.

Where required, *fire protection systems* shall be monitored by a *supervising station* in accordance with NFPA 72 or connected to a *city fire alarm box* in accordance with Section 919. A provider of supervising station services shall notify the *fire code official* no less than 48 hours before terminating services to a *building*.

901.6.1 Automatic sprinkler systems.

Automatic sprinkler systems shall be monitored by a *supervising station*.

Exceptions:

1. A *supervising station* is not required for *automatic sprinkler systems* protecting Group R-5 occupancies.
2. Limited area systems in accordance with Section 903.3.8.

901.6.3 Fire alarm systems.

Fire alarm systems required by the provisions of Section 907.2 of this code shall be monitored by a *supervising station* in accordance with Section 907.6.6 of this code.

Exceptions:

1. Single- and multiple-station smoke alarms required by Section 907.2.10.
2. Smoke detectors in Group I-3 occupancies.
3. Supervisory service is not required for *automatic sprinkler systems* in Group R-5 occupancies.

901.6.4 Group H.

Supervision and monitoring of emergency alarm, detection and automatic fire-extinguishing systems in Group H occupancies shall be in accordance with the *Chicago Fire Prevention Code*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 28; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 13)

14B-9-902 Fire pump and riser room size.

The provisions of Section 902 of IBC are adopted by reference with the following modification:

1. Revise Section 902.1.2 by deleting "with an approved sign."

14B-9-903 Automatic sprinkler systems.

The provisions of Section 903 of IBC are adopted by reference with the following modifications:

1. Add a new Section 903.1.2 to read:

"903.1.2 Where prohibited.

Automatic sprinkler systems shall not be installed in hazardous chemical rooms or in hazardous chemical storage *buildings*, where the chemical has the ability to react with water to produce substances or violent chemical reactions which increase the hazard. When the presence of water will not add to the hazard, an *automatic sprinkler system* shall be required in such hazardous chemical room or building."

2. Revise Section 903.2 by deleting "Approved."
3. Delete the exception to Section 903.2.
4. Delete items 3 and 4 in Section 903.2.1.1.
5. Revise the numbered items in Section 903.2.1.2 to read:

- "1. The *fire area* exceeds 12,000 square feet (1115 m²).
2. The *fire area* has an *occupant load* of 300 or more.
3. The *fire area* has an *occupant load* of 100 or more and is located on a floor more than four feet (1219 mm) below the lowest *level of exit discharge* serving such occupancy."

6. Delete item 3 in Section 903.2.1.3.

7. Add a new item 4 in Section 903.2.1.3 to read:

"4. The Group A-3 occupancy contains one or more *exhibition areas* exceeding 5,000 square feet (464 m²) in aggregate."

8. Delete item 3 in Section 903.2.1.4.

9. Revise Section 903.2.1.6, including the exception, to read:

"903.2.1.6 Assembly occupancies on roofs.

Where an *occupiable rooftop* has an assembly occupancy with an *occupant load* of 300 or more, all floors between the *occupiable rooftop* and the *level of exit discharge* shall be equipped with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2."

10. Revise Section 903.2.1.7 by replacing "shall be provided" with "shall be provided throughout the *building*."

11. Revise Section 903.2.2 to read:

"903.2.2 Group B.

An *automatic sprinkler system* shall be provided for Group B occupancies as provided in this section.

903.2.2.1 Ambulatory care facilities.

An *automatic sprinkler system* shall be installed throughout the entire floor containing an *ambulatory care facility* where either of the following conditions exist at any time:

1. Four or more care recipients are incapable of self-preservation.
2. One or more care recipients that are incapable of self-preservation are located at other than the level of exit discharge serving such a facility.

In *buildings* where ambulatory care is provided on levels other than the *level of exit discharge*, an *automatic sprinkler system* shall be installed throughout the entire floor as well as all floors between the level of ambulatory care and the nearest *level of exit discharge*, the *level of exit discharge*, and all floors below the *level of exit discharge*.

903.2.2.2 Telephone exchanges.

An automatic fire-extinguishing system shall be installed throughout *buildings* containing *telephone exchanges*."

12. Revise Section 903.2.3 to read:

"903.2.3 Group E.

An *automatic sprinkler system* shall be provided for Group E occupancies as follows:

1. Throughout all Group E *fire areas* and all connected Group A *fire areas* in buildings where the aggregate area of all Group E and connected Group A *fire areas* is 7,200 square feet (669 m²) or greater.
2. Throughout *buildings* containing Group E occupancies owned by, operated by or constructed with funding from a *governmental unit*.
3. Throughout *buildings* containing Group E-2 occupancies.

Exception: Where every room or space in a Group E-2 occupancy normally subject to client occupancy, other than toilet rooms, has not less than one door leading directly to an *exit* or directly to the outside of the *building*."

13. Revise the title and first sentence of Section 903.2.4 to read:

"903.2.4 Group F.

An *automatic sprinkler system* shall be provided throughout all *buildings* containing a Group F occupancy where one of the following conditions exists:"

13.1 Revise item 4 in Section 903.2.4 to read:

"4. A Group F-1 *fire area* used for the manufacture of upholstered furniture or mattresses exceeds 2,500 square feet (232 m²)."

14. Revise Section 903.2.4.1 by replacing "*fire areas*" with "*buildings*."

15. Add a new Section 903.2.4.2 to read:

"903.2.4.2 Electric generation plants.

An automatic fire-extinguishing system shall be provided throughout any *structure* that is owned or operated by any person whose primary business is the generation or transmission of electrical power and that houses one or more electrical generators, electrical transformers, or switching equipment for a service area beyond the *building* in which the equipment is located."

16. Revise exception 2 to Section 903.2.6 to read:

"2. An *automatic sprinkler system* is not required where Group I-4 day care facilities are at the *level of exit discharge* and where every room where care is provided has not fewer than one exterior *exit* door."

17. Delete exception 3 to Section 903.2.6.

17.1. Revise item 4 in Section 903.2.7 to read:

"4. A Group M occupancy where the area used for the display and sale of upholstered furniture or mattresses exceeds 5,000 square feet (464 m²)."

18. Add a new item 5 to Section 903.2.7 to read:

"5. A Group M *fire area* is located in a *basement*."

19. Add new exceptions to Section 903.2.8 to read:

"Exceptions:

1. Buildings of Group R-5 occupancy.

2. Buildings of Group R-2 occupancy, other than *congregate living facilities*, with not more than four *stories above grade plane* and containing not more than ten *dwelling units* where each *fire area* contains not more than two *dwelling units* and all required vertical *exits* are interior *exit stairways*."

20. Revise Section 903.2.8.1 to read:

"903.2.8.1 Group R-3 and R-5.

An *automatic sprinkler system* installed in accordance with Section 903.3.1.3 shall be permitted in Group R-3 and R-5 occupancies."

20.1. Revise items 4 and 5 in Section 903.2.9 to read:

"4. A Group S-1 *fire area* used for the storage of *commercial motor vehicles* where the *fire area* exceeds 5,000 square feet (464 m²).

5. A Group S-1 occupancy where the area used for the storage of upholstered furniture or mattresses exceeds 2,500 square feet (232 m²)."

20.2. In Section 903.2.10.1, reset the following term in italic type: "commercial motor vehicle."

21. Revise Section 903.2.10 to read:

"903.2.10 Group S-2 parking garages.

An *automatic sprinkler system* shall be provided throughout buildings containing parking garages in accordance with Section 406.6 where any of the following conditions exists:

1. Where the floor area of the parking garage exceeds 12,000 square feet (1115 m²).

2. Where the parking garage has a storage capacity of more than 20 vehicles and is located beneath other occupancies.

3. Where the parking garage is in a *basement*."

22. Revise the title of Section 903.2.10.1 to "Commercial-vehicle parking garages."

22.1. In Section 903.2.10.1, reset the following term in italic type: "commercial motor vehicle."

22.2. Revise Section 903.2.11 by replacing "903.2.11.6" with "903.2.11.8."

23. Revise Section 903.2.11.1.2 by deleting "*approved*."

24. Revise Section 903.2.11.1.3 by deleting "*approved*."

25. Revise Section 903.2.11.3 to read:

"903.2.11.3 Buildings more than 70 feet in height.

An *automatic sprinkler system* shall be installed throughout buildings that are more than 70 feet (21.3 m) in *building height*.

Exception: *Parking facilities*, in accordance with Section 906.9."

26. Renumber existing Section and Table 903.2.11.6 to 903.2.11.8 and delete the following rows in the table: covered and open

mall buildings, Group H-5 HPM exhaust ducts, flammable finishes, drying rooms, children's play structures, buildings containing laboratory suites, unlimited area buildings, smoke-protected assembly seating, and the last row.

27. Add a new Section 903.2.11.6 to read:

"903.2.11.6 Shops and storerooms.

An *automatic sprinkler system* shall be installed in all rooms and spaces used as paint shops, wastepaper baling or storage, and other storage uses having a similar degree of combustibility.

Exceptions:

1. Shops and storerooms located in a Group R-5 occupancy.
2. Shops and storerooms not exceeding 1,000 square feet (93 m²) in a building not exceeding one story above *grade plane*."

28. Add a new Section 903.2.11.7 to read:

"903.2.11.7 Telecommunications equipment areas.

Every *fire area* within an *existing structure* containing a *telecommunication equipment area* that exceeds 150 square feet (13.93 m²) must be equipped with an automatic fire-extinguishing system."

29. Revise Section 903.2.12 by deleting "Chapter 33 of."

30. Revise Section 903.3.1.1.1 to read:

"903.3.1.1.1 Exempt locations.

Automatic sprinklers shall not be required in the following rooms or areas where such rooms or areas are protected with an automatic fire detection system in accordance with Section 907.2 that will respond to visible or invisible particles of combustion. Sprinklers shall not be omitted from a room merely because it is damp, of fire-resistance-rated construction or contains electrical equipment.

1. A room where the application of water, or flame and water, constitutes a serious life or fire hazard.
2. A room or space where sprinklers are considered undesirable because of the nature of the contents, where *approved by the fire code official*.
3. Generator, transformer and switchboard rooms separated from the remainder of the building by walls and floor/ceiling or roof/ceiling assemblies having a *fire-resistance rating* of not less than 2 hours.
4. [Reserved.]
5. Elevator machine rooms, elevator shafts and machinery spaces. (Automatic sprinklers shall be provided in elevator pits.)"

31. Delete Section 903.3.1.1.2.

32. Revise Section 903.3.1.2 to read:

"903.3.1.2 NFPA 13R sprinkler systems.

Automatic sprinkler systems installed in Group R-5; Group R-3; and Group R-4, Condition 1 shall be permitted to be installed throughout in accordance with NFPA 13D, provided the building shall not exceed 45 feet (13.7 m) in *building height*."

33. Delete Section 903.3.1.2.1.

34. Delete item 3 (and subitems) in Section 903.3.1.2.3.

35. Revise Section 903.3.1.3 to read:

"903.3.1.3 NFPA 13D sprinkler systems.

Automatic sprinkler systems installed in Group R-5; Group R-3; Group R-4, Condition 1; and *townhouses* shall be permitted to be installed throughout in accordance with NFPA 13D, provided the building shall not exceed 45 feet (13.7 m) in *building height*."

36. Revise the last sentence of Section 903.3.5 to read:

"For connections to the public waterworks systems, the water supply test used for design of fire protection systems shall be acceptable to the *fire code official* and not less than one year old."

37. Revise Section 903.3.8.4 by deleting "*approved*."

37.1 Insert a new Section 903.3.9 to read:

"903.3.9 Concealed sprinkler heads.

Where concealed sprinkler heads are used, cover plates shall have a brass, chrome or other unpainted metallic finish."

38. Revise exception 1 to Section 903.4 by replacing "one- and two-family dwellings" with "Group R-5 occupancies."

39. Revise Section 903.4 by adding a sentence, following the exceptions, to read:

"All valves in supply pipes to *automatic sprinkler systems* shall be secured open."

40. Revise Section 903.4.1, excluding the exceptions, to read:

"903.4.1 Monitoring.

Alarm, supervisory and trouble signals shall be distinctly different and shall be automatically transmitted to a *supervising station*."

41. Revise Section 903.4.2 to read:

"903.4.2 Alarms.

An electric bell, located on the exterior of the building above a fire department connection, shall be connected to each *automatic sprinkler system*. The bell shall be activated by water flow equivalent to the flow of a single sprinkler of the smallest orifice size installed in the system. Where a *fire alarm system* is installed, actuation of the *automatic sprinkler system* shall actuate the building *fire alarm system*.

A sign shall be placed at a conspicuous location near the bell and shall state: "SPRINKLER ALARM-WHEN BELL RINGS CALL FIRE OR POLICE DEPARTMENT."

42. Revise Section 903.4.3 by deleting "*Approved*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 29; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 14; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 13)

14B-9-904 Alternative automatic fire-extinguishing systems.

The provisions of Section 904 of IBC are adopted by reference with the following modifications:

1. Revise Section 904.2 to read:

"904.2 Where permitted.

Automatic fire-extinguishing systems installed as an alternative to the required *automatic sprinkler systems* of Section 903 shall be *approved* by the *fire code official*. As a condition of approval, the *fire code official* may require signs indicating the type of alternative automatic fire-extinguishing system be installed in prominent locations at each building entrance."

2. Revise Section 904.2.2 to read:

"904.2.2 Commercial hood and duct systems.

Each required commercial kitchen exhaust hood and duct system required by the *Chicago Fire Prevention Code* or *Chicago Mechanical Code* to have a Type I hood shall be protected with an automatic fire-extinguishing system installed in accordance with this code."

3. Revise Section 904.11.1.4 by replacing "*an approved*" with "*a*."

4. Revise the exception to Section 904.12 by deleting "Section 304.1 of."

5. Revise Section 904.13 by deleting item 3.

6. Revise Section 904.13.1.1 by deleting "*approved*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 30)

14B-9-905 Standpipe systems.

The provisions of Section 905 of IBC are adopted by reference with the following modifications:

1. Add new Sections 905.2.1, 905.2.2, 905.2.3 and 905.2.4 to read:

"905.2.1 Water supply.

Where wet standpipes are installed in *buildings* that are 80 feet (24.4 m) or less in *building height* and the *building* is protected throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2, the water supply for the building may be designed to provide 100 pounds per square inch (689 kPa) pressure with 500 gallons per minute (31.6 L/s) flowing from the two hydraulically most remote outlets in the standpipe system. The standpipe system shall be designed for full standpipe flow in accordance with NFPA 14.

905.2.2 Multiple systems required.

In *buildings* where the floor level of the highest *story* is more than 275 feet (83.8 m) above *grade plane*, two or more separate standpipe systems shall be provided. The lowest standpipe system shall serve the portion up to and including the *story* which is 275 feet (83.8 m) or less above *grade plane*. Above 275 feet (83.8 m) additional standpipe systems shall be provided to serve zones of not more than 20 *stories* each.

905.2.3 Express risers.

In *buildings* where the floor level of the highest *story* is more than 275 feet (83.8 m) above *grade plane*, individual standpipe systems, as required by Section 905.2.1, shall be supplied by two or more separate risers from the zone fire pump located in the lowest *story* of the building, except that the number of express risers shall not exceed the number of risers serving that zone; or by individual fire pumps located on the lowest *story* of each zone taking suction under head from a tank automatically filled by two or more risers from the adjacent lower zone. Such a tank shall have a capacity sufficient to supply the largest pump in the zone at its rated capacity for at least 30 minutes but not less than 10,000 gallons (37,854 L).

905.2.4 Pressure reducing devices.

Where the pressure at nozzles is more than 65 pounds per square inch (448 kN/m²), the 1 1/2-inch hose valve shall be provided with a device or disk to limit the force of the hose stream to 90 pounds (0.4 kN) so that the nozzle can be safely handled by one person."

2. Revise the exception to Section 905.3 to read:

"Exception: Standpipe systems are not required in Group R-2, R-3, R-4 and R-5 occupancies not more than four *stories above grade plane*."

3. Revise Section 905.3.1 to read:

"905.3.1 Height.

Class III standpipe systems shall be installed throughout buildings which exceed 55 feet in *building height* or where four or more *stories* are above *grade plane* or two or more *stories* are *basements*.

Exceptions:

1. Class I standpipes are allowed in buildings equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2.
2. [Reserved.]
3. [Reserved.]
4. Class I manual dry standpipes are allowed in *parking garages* and *parking facilities* that are subject to freezing temperatures, provided that the hose connections are located as required for Class II standpipes in accordance with Section 905.5."

4. Revise exception 2 to Section 905.3.2 by deleting "automatic dry and semiautomatic dry standpipes or."

5. Delete Section 905.3.3.

6. Revise Section 905.3.4.1 by deleting "*approved*."

7. Revise Section 905.3.6 to read:

"905.3.6 Helistops and heliports.

Buildings with a rooftop *helistop* or *heliport* shall be equipped with a Class I or III standpipe system extended to the roof level on which the *helistop* or *heliport* is located."

8. Revise Section 905.3.7 by deleting "Chapter 36 of."

9. Revise Section 905.3.8 to read:

"905.3.8 Rooftop gardens, landscaped roofs, and occupiable rooftops.

Buildings or *structures* that have *vegetative roofs*, rooftop gardens, landscaped roofs or *occupiable rooftops* and that are equipped with a standpipe system shall have the standpipe system extended to the roof level on which the *vegetative roof*, rooftop garden, landscaped roof or *occupiable rooftop* is located."

10. Revise Section 905.4 to read:

"905.4 Location of Class I standpipe hose connections.

Class I standpipe hose connections shall be provided in all of the following locations:

1. In every required *interior exit stairway*, a hose connection shall be provided for each *story* above and below *grade plane*. Hose connections shall be located at the floor landing.
2. [Reserved.]
3. In every *exit passageway*, at the entrance from the *exit passageway* to other areas of a building.
4. [Reserved.]
5. Where the roof has a slope less than four units vertical in 12 units horizontal (33.3-percent slope), a hose connection shall be located to serve the roof or at the highest landing of an *interior exit stairway* with access to the roof provided in accordance with Section 1011.12.
6. Where the most remote portion of a nonsprinklered floor or *story* is more than 150 feet (45 720 mm) from a hose

connection or the most remote portion of a sprinklered floor or *story* is more than 200 feet (60 960 mm) from a hose connection, the *fire code official* is authorized to require that additional hose connections be provided in *approved* locations."

11. Revise the exception to Section 905.4.1 to delete "*approved*."
12. Revise Section 905.5.3 by replacing "*approved*" with "*permitted*."
13. Revise Section 905.7.1 by deleting "*in an approved manner*."
14. Revise Section 905.9 by adding the following after the last sentence:
"All valves in supply pipes to standpipe systems shall be secured open."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 31; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 15))

14B-9-906 Portable fire extinguishers.

The provisions of Section 906 of IBC are adopted by reference with the following modifications:

1. Add new exceptions 3 and 4 to item 1 in Section 906.1 to read:

"3. In Group R-2 occupancies not more than three *stories above grade plane* and having a floor area not exceeding 3,000 square feet (279 m²).

4. In one-story buildings of B, F, M, or S occupancy having a building area not exceeding 3,000 square feet (279 m²)."
2. Revise item 4 in Section 906.1 to read:

"4. On each floor of structures under construction, except Group R-3, R-4 and R-5 occupancies, in accordance with Chapter 33."
3. Delete item 5 in Section 906.1.
4. Delete Table 906.1.
5. Revise Section 906.4 to read:

"906.4 Cooking equipment fires.

Fire extinguishers provided for the protection of cooking equipment shall be of a type compatible with the automatic fire-extinguishing system agent. Cooking equipment involving solid fuels or vegetable or animal oils and fats shall be protected by a Class K-rated portable extinguisher in accordance with the *Chicago Fire Prevention Code*."

6. Revise Section 906.6 to read:

"906.6 Unobstructed and unobscured.

Portable fire extinguishers shall not be obstructed or obscured from view. Where a fire extinguisher is likely to be obscured by piles of stock, lumber or otherwise, a sign shall be installed and maintained which will indicate the location of the fire extinguisher in a manner legible at a distance of at least 50 feet (15.2 m)."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 32)

14B-9-907 Fire alarm and detection systems.

The provisions of Section 907 of IBC are adopted by reference with the following modifications:

1. Revise Section 907.1.1 to read:

"907.1.1 Construction documents.

Construction documents for *fire alarm systems* shall be of sufficient clarity to indicate the location, nature and extent of the work proposed and show in detail that it will conform to the provisions of this code, the *Chicago Electrical Code* and the *Chicago Fire Prevention Code*."

2. Revise Section 907.1.2 to read:

"907.1.2 Fire alarm shop drawings.

Shop drawings for *fire alarm systems* shall be prepared in accordance with NFPA 72 and submitted for a separate *permit* prior to system installation."

3. Revise Section 907.1.3 by deleting "*and approved*."

4. Add a new Section 907.1.3.1 to read:

"907.1.3.1 Notification appliances.

Occupant notification appliances shall be red. Where *approved* by the *fire code official*, occupant notification appliances may be

concealed until activated."

5. Revise Section 907.2 to read:

"907.2 Where required-new buildings and structures.

A fire alarm system installed in accordance with the provisions of this code and NFPA 72 shall be provided in new buildings and structures in accordance with Sections 907.2.1 through 907.2.23 and provide occupant notification in accordance with Section 907.5, unless other requirements are provided by another section of this code.

Not fewer than one *manual fire alarm box* at each *exit* on the *level of exit discharge* shall be provided to initiate a fire alarm signal for fire alarm systems employing automatic fire detectors or waterflow detection devices. Where other sections of this code allow elimination of *manual fire alarm boxes* due to sprinklers, *manual fire alarm boxes* shall be installed at each *exit* on the *level of exit discharge*.

Exception: The *manual fire alarm box* is not required for fire alarm systems dedicated to elevator recall control and supervisory service."

6. Revise Section 907.2.1 and its subsections to read:

"907.2.1 Group A.

A manual fire alarm system that activates the occupant notification system in accordance with Section 907.5 shall be installed in Group A occupancies where the *occupant load* due to the assembly occupancy is 300 or more. Group A occupancies not separated from one another in accordance with Section 707.3.10 shall be considered as a single occupancy for the purposes of applying this section. Portions of Group E occupancies occupied for assembly purposes shall be provided with a fire alarm system as required for the Group E occupancy.

Exception: *Manual fire alarm boxes* are not required where the *building* is equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 and the occupant notification appliances will activate in accordance with Section 907.5 upon sprinkler water flow."

7. Revise Section 907.2.2 by replacing "500" with "300."

8. Revise the exception to Section 907.2.2 to read:

"Exception: *Manual fire alarm boxes* are not required where the *building* is equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 and the occupant notification appliances will activate in accordance with Section 907.5 upon sprinkler water flow."

9. Delete the exception to 907.2.2.1.

10. Revise Section 907.2.3, including the exceptions, to read:

"907.2.3 Group E.

A manual fire alarm system shall be installed in Group E occupancies. Where *automatic sprinkler systems* or smoke detectors are installed, such systems or detectors shall be connected to the *building* fire alarm system."

11. Revise the exception to Section 907.2.4 to read:

"Exception: *Manual fire alarm boxes* are not required where the *building* is equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 and the occupant notification appliances will activate in accordance with Section 907.5 upon sprinkler water flow."

12. Revise Section 907.2.5 by deleting "Chapters 60, 62 and 62, respectively, of."

13. Revise the exceptions to Section 907.2.6 to read:

"Exception: Occupant notification systems are not required to be activated where private operating mode signaling installed in accordance with NFPA 72 is permitted by the *fire code official* and staff evacuation responsibilities are included in the fire safety and evacuation plan required by the *Chicago Fire Prevention Code*."

14. Revise Section 907.2.6.1.1 by replacing "smoke alarms" with "smoke alarms, heat alarms and heat detectors" in both the heading and body.

15. Revise Section 907.2.6.3.1 to read:

"907.2.6.3.1 System initiation.

Actuation of an automatic fire-extinguishing system, *automatic sprinkler system*, a *manual fire alarm box* or a fire detector shall initiate a private operating mode fire alarm signal that automatically notifies staff throughout the *facility*."

16. Revise Section 907.2.6.3.2 to read:

"907.2.6.3.2 Manual fire alarm boxes.

Manual fire alarm boxes are not required to be located in accordance with Section 907.4.2 where *manual fire alarm boxes* are provided at staff-attended locations having direct supervision over areas where *manual fire alarm boxes* have been omitted."

17. Revise the exceptions to Section 907.2.7 to read:

"Exception: *Manual fire alarm boxes* are not required where the *building* is equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 and the occupant notification appliances will activate in accordance with Section 907.5 upon sprinkler water flow."

18. Delete Section 907.2.7.1.

19. Delete exception 2 (and subitems) to Section 907.2.8.1.

19.1. Revise Section 907.2.8.3 by replacing "smoke alarms" with "smoke alarms, heat alarms and heat detectors" in both the heading and body.

20. Revise Section 907.2.9.1 to read:

"907.2.9.1 Manual fire alarm system.

A manual fire alarm system that activates the occupant notification system in accordance with Section 907.5 shall be installed in Group R-2 occupancies where the *building* contains more than 16 *dwelling units* or *sleeping units*."

21. Revise exception 2 to Section 907.2.9.1 to read:

"2. *Manual fire alarm boxes* are not required where the *building* is equipped throughout with an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 or 903.3.1.2 and the occupant notification appliances will activate in accordance with Section 907.5 upon a sprinkler water flow."

22. Revise exception 3 to Section 907.2.9.1 by deleting "approved."

22.1. Revise Section 907.2.9.2 by replacing "smoke alarms" with "smoke alarms, heat alarms and heat detectors" in both the heading and body.

23. Delete Section 907.2.9.3.

24. Add a new item 4 to Section 907.2.10.1 to read:

"4. At the uppermost ceiling of each *interior exit stairway*."

25. Revise Section 907.2.10 by inserting the following at the end: "Listed heat detectors complying with UL 521 or heat alarms complying with UL 539 shall be installed in accordance with Sections 907.2.10. and 907.2.10.8 and NFPA 72."

25.1. Revise Section 907.2.10.2 to read:

"907.2.10.2 Groups R-2, R-3, R-4, R-5 and I-1.

Single- or multiple-station smoke alarms shall be installed and maintained in Groups R-2, R-3, R-4, R-5 and I-1 regardless of *occupant load* at all of the following locations:

1. On the ceiling or wall outside of and within 15 feet (4572 mm) of each room used for sleeping purposes.
2. In each room used for sleeping purposes.
3. In each *story* within a *dwelling unit*, including *basements* but not including crawl spaces and uninhabitable *attics*. In *dwelling units* or *dwelling units* with split levels and without an intervening door between the adjacent levels, a smoke alarm installed on the upper level shall suffice for the adjacent lower level provided that the lower level is less than one full *story* below the upper level.
4. At the uppermost ceiling of each *interior exit stairway*."

25.2. Revise Section 907.2.10.6 to read:

"907.2.10.6 Power source.

In new construction, required smoke alarms, heat alarms and heat detectors shall receive their primary power from the building wiring where such wiring is served from a commercial source and shall be equipped with a battery backup. Smoke alarms with integral strobes that are not equipped with battery backup shall be connected to an emergency electrical system in accordance with Section 2702. Smoke alarms, heat alarms and heat detectors shall emit a signal when the batteries are low. Wiring shall be permanent and without a disconnecting switch other than as required for overcurrent protection.

Exception: Smoke alarms, heat alarms and heat detectors are not required to be equipped with battery backup where they are connected to an emergency electrical system that complies with Section 2702."

25.3. Insert a new Section 907.2.10.8 to read:

"907.2.10.8 Heat alarms and heat detectors.

A heat detector or heat alarm shall be installed in a *private garage* that is attached to or directly below a *dwelling unit* or *sleeping unit* and shall comply with the following:

1. The heat alarm or heat detector shall be rated for the ambient environment.
2. The heat alarm or heat detector shall either be interconnected with the smoke alarms within the attached *dwelling*

unit or *sleeping unit* or the heat detector shall be connected to an audible occupant notification device located within the *dwelling unit* or *sleeping unit* and within 20 feet of the door nearest to the *private garage* and additional audible occupant notification devices, as necessary, so that occupant notification is clearly audible in all sleeping rooms over background noise levels with all intervening doors closed. Audible occupant notification is not required within the *private garage*.

3. Heat alarms and heat detectors shall be installed in accordance with the manufacturer's instructions."

26. Revise items 3 and 4 in Section 907.2.11.2 to read:

"3. Activate directional *exit* markings that will become apparent in an emergency.

4. Alarm signals shall produce a sound that is distinctive from other sounds used during normal operation."

27. Delete Section 907.2.11.3.

28. Revise Section 907.2.12 to read:

"907.2.12 High-rise buildings.

High-rise buildings shall be provided with an automatic smoke detection system in accordance with Section 907.2.12.1 and voice communication systems in accordance with Section 907.2.12.2.

Exceptions:

1. Airport traffic control towers in accordance with Sections 412 and 907.2.21.
2. *Parking facilities* in accordance with Section 406.9.
3. *Buildings* with an occupancy in Group A-5 in accordance with Section 303.1.
4. Low-hazard special industrial occupancies in accordance with Section 503.1.1.
5. [Reserved.]
6. In Group I-1 and I-2 occupancies, the alarm shall sound at a *constantly attended location* and private mode signaling shall be broadcast throughout the facility."

29. Revise Section 907.2.12.1.1 to read:

"907.2.12.1.1 Area smoke detection.

Area smoke detectors shall be provided in accordance with this section. Smoke detectors shall be connected to an automatic fire alarm system. The activation of any detector required by this section shall activate the occupant notification appliances in accordance with Section 907.5. In addition to smoke detectors required by Sections 907.2.1 through 907.2.9, smoke detectors shall be located as follows:

1. In each mechanical equipment, electrical, transformer, telephone equipment or similar room that is not provided with sprinkler protection.
2. In each elevator machine room, machinery space, control room and control space and in elevator lobbies.
3. Where enclosed elevator lobbies are not provided, in *corridors* adjoining elevators."

30. Delete Section 907.2.12.1.2.

31. Revise Section 907.2.12.2 to read:

"907.2.12.2 Voice communication systems.

Voice communication systems shall be provided in accordance with Sections 907.2.12.2.1 through 907.2.12.2.3.

907.2.12.2.1 Fire department two-way communication system.

A two-way fire department communication system shall be provided to operate between a *fire command center* complying with Section 911 and emergency two-way stations located at not less than every fifth level and every landing for a level numbered with a multiple of 5 and the uppermost landing within every *interior exit stairway*.

907.2.12.2.2 Selective one-way communication system.

A selective (multiple-channel) one-way communication system shall be provided to operate between a *fire command center* complying with Section 911 and the following areas:

1. Elevators cars.
2. Elevator lobbies.
3. Within *interior exit stairways* at every fifth level and every landing for a level numbered with a multiple of 5.
4. Office areas exceeding 5,000 square feet (465 m²).

5. *Corridors*, at intervals not exceeding 75 feet (22.9 m).
6. Outside of doors leading to *exit stairways*.
7. Inside the main entrance of each *dwelling unit* and *sleeping unit*.
8. Parking garages exceeding 5,000 square feet (465 m²).

907.2.12.2.3 Combined system.

The two-way fire department communication system may be combined with the one-way system where *approved* by the *fire code official*."

32. Delete Section 907.2.12.3.
33. Delete Section 907.2.13.
34. Revise Section 907.2.14 by deleting "Section 3206.5 of."
35. Revise Section 907.2.15 by replacing "an *approved*" with "a."
36. Revise Section 907.2.18 to read:

"907.2.18 Deep underground buildings.

Where the lowest *occupiable space* is more than 60 feet (18 288 mm) below the finished floor of the lowest *level of exit discharge*, the *structure* shall be equipped throughout with a manual fire alarm system."

37. Delete Section 907.2.19.
38. Delete Section 907.2.20.
39. Revise the exceptions to Section 907.3.1 to read:

"Exception: In occupancies not required to be equipped with a fire alarm system, actuation of a smoke detector shall activate a visible and an audible signal in a prominent location. Smoke detector trouble conditions shall activate a visible or audible signal in a prominent location and shall be identified as air duct detector trouble."

40. Revise Section 907.4.2.1 by replacing "200 feet (60 960 mm)" with "100 feet (30.5 m)."
41. Revise Section 907.4.2.4 by replacing "an *approved*" with "a."
42. Revise Section 907.4.3 by deleting "*approved*."
43. Revise Section 907.4.3.1 by replacing "*approved*" with "accepted."
44. Delete the exception to Section 907.5.
45. Revise Section 907.5.1 to read:

"907.5.1 Presignal feature.

Fire alarm systems in the following buildings shall be provided with a presignal feature:

1. Group I occupancies where the floor area of any *story* exceeds 8,000 square feet (743 m²).
2. Group I occupancies in buildings with more than three *stories*.
3. *High-rise buildings*.

A presignal feature shall not be installed in other building types unless *approved* by the *fire code official*. Where a presignal feature is provided, a signal shall be annunciated at a *constantly attended location* acceptable to the *fire code official* so that occupant notification can be activated in the event of fire or other emergency."

46. Delete Section 907.5.2.2 and its subsections.
47. Revise Section 907.5.2.3.1, including its exception, to read:

"907.5.2.3.1 Public use areas, common use areas and employee work areas.

Visible alarm notification appliances shall be provided in *public use areas* and *common use areas*. Where *employee work areas* have audible alarm coverage, visible alarms shall be provided at time of initial installation."

48. Revise Table 907.5.2.3.2 by replacing "6 to 25" with "2 to 25."
49. Revise Section 907.5.2.3.3 by adding the following at the end:

"Visible alarm notification appliances, arranged so the flashing light beam is visible in all *occupiable spaces* within the unit, shall be installed by the *owner* within *dwelling units* and *sleeping units* on an as-needed basis at the request of an occupant with a *disability*."

50. Revise Section 907.6.3.1 to read:

"907.6.3.1 Annunciation.

The initiating device status shall be annunciated at an on-site location acceptable to the *fire code official*."

51. Revise Section 907.6.4 by replacing "300 feet (91 440 mm)" with "200 feet (61 m)."
52. Revise Section 907.6.4.1 by replacing "an *approved* location" with "a location acceptable to the *fire code official*."
53. Revise Section 907.6.6 by deleting "*approved*."
54. Revise exception 3 to Section 907.6.6 by replacing "one- and two-family dwellings" with "Group R-5 occupancies."
55. Revise Section 907.6.6.1 by deleting "unless *approved* by the *fire chief*."
56. Revise Section 907.6.6.2 to read:

"907.6.6.2 Termination of monitoring service.

Termination of fire alarm monitoring services shall be in accordance with Section 901.6."

57. Revise Section 907.7.2 to read:

"907.7.2 Record of completion.

A record of completion in accordance with NFPA 72 verifying that the system has been installed and tested in accordance with the *permit* shall be provided to the *fire code official*."

58. Revise Section 907.7.3 to read:

"907.7.3 Instructions.

Operating, testing and maintenance instructions and record drawings ("as-builts") and equipment specifications shall be provided on site or at an *approved* off-site location."

59. Revise Section 907.8 by deleting "Section 907.8 of."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 33; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 16; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 14; Amend Coun. J. 12-14-22, p. 58278, Art. III, § 2)

14B-9-908 IBC Section 908.

The provisions of Section 908 of IBC are not adopted.

(Amend Coun. J. 10-7-20, p. 21791, Art. II, § 17)

14B-9-909 Smoke control systems.

The provisions of Section 909 of IBC are not adopted. The following language is adopted as Section 909:

"909. SMOKE CONTROL SYSTEMS

909.1 Smoke control for atriums.

Where required by Section 404.5, *atriums* shall be provided with a smoke control system meeting the following requirements:

1. If the volume of the *atrium* does not exceed 600,000 cubic feet (16 990 m³), the exhaust system shall provide not less than 6 air changes per hour.
2. If the volume of the *atrium* exceeds 600,000 cubic feet (16 990 m³), the exhaust system shall provide not less than 4 air changes per hour.
3. Supply openings shall be sized for at least one-half the exhaust capacity.
4. Exhaust shall be at the top of the *atrium* and intake at the bottom.
5. Smoke detectors shall be provided at the ceiling of the *atrium* and at the perimeter of the *atrium* on each floor within 15 feet (4572 mm).
6. The smoke control system shall have completely automatic control. The automatic-control sequences shall be initiated by activation of either the *automatic sprinkler system* or smoke detectors in the *atrium* area.

909.2 Acceptance.

Buildings, or portions thereof, required by this code to comply with this section shall not be issued a certificate of occupancy until such time that the *fire code official* determines that the provisions of this section have been fully complied with and that the fire department has received satisfactory instruction on the operation, both automatic and manual, of the system and a written maintenance program complying with the requirements of the *Chicago Fire Prevention Code* has been submitted to and accepted by the *fire code official*.

Exception: Where phased construction is permitted, a temporary certificate of occupancy shall be allowed provided that

those portions of the *building* to be occupied meet the requirements of this section and that the remainder does not pose a significant hazard to the safety of the proposed occupants or adjacent *buildings*.

909.3 Optional smoke control systems.

Optional smoke control systems shall comply with Appendix S."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 34; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 18)

14B-9-910 Smoke and heat removal.

The provisions of Section 910 of IBC are adopted by reference with the following modifications:

1. Revise Section 910.2, excluding the exceptions, to read:

"910.2 Where required.

Smoke and heat vents or a mechanical smoke removal system shall be installed as required by Sections 910.2.1 through 910.2.3."

2. Revise exception 1 to Section 910.2 by deleting "*approved*."

3. Revise Section 910.2.1 to read:

"910.2.1 Group F-1 or S-1.

A mechanical smoke removal system installed in accordance with Section 910.4 shall be installed in *buildings* and portions thereof used as a Group F-1 or S-1 occupancy having more than 50,000 square feet (4645 m²) of undivided area.

Exception: Group S-1 aircraft repair hangars."

4. Revise Section 910.2.2 to read:

"910.2.2 High-piled combustible storage.

A mechanical smoke removal system installed in accordance with Section 910.4 shall be installed in *buildings* and portions thereof containing high-piled combustible storage."

5. Insert a new Section 910.2.3 to read:

"910.2.3 Exhibition areas.

Automatic smoke and heat vents in accordance with Section 910.3 shall be installed on the topmost occupied *story of exhibition areas* with a floor area in excess of 10,000 square feet (929 m²). Where the upper surface of a *story* containing an *exhibition area* is not a *roof assembly*, a mechanical smoke removal system in accordance with Section 910.4 shall be installed."

6. Revise Section 910.3.2 by adding the following sentence at the end:

"Vents shall be spaced no greater than 100 feet (30.5 m) apart, on center."

7. Revise Section 910.4.1 by deleting "*approved*."

14B-9-911 Fire command center.

The provisions of Section 911 of IBC are adopted by reference with the following modifications:

1. Revise Section 911.1.1 to read:

"911.1.1 Location and access.

The location and accessibility of the *fire command center* shall be reviewed and accepted by the *fire code official*. The *fire command center* shall comply with the following criteria:

1. All panels shall be located in the line of sight immediately inside the primary entrance to the *building* which is facing the street of the building address. The *building* address number shall be conspicuously displayed above or next to the primary entrance doors.
2. Panels shall not be located in a separate room.
3. A six foot (1829 mm) clear working space shall be provided in front of all panels.
4. Panels shall be mounted vertically on a wall.
5. Panels shall be mounted so that the bottom edge is no less than 18 inches (457 mm) above the floor and the top edge is no more than 74 inches (1880 mm) above the floor.
6. Darkened, tinted or opaque glass shall not be used to cover or obscure the panels.

Exception: Slightly-tinted glass, as *approved* by the *fire code official*.

7. Panel doors shall be completely removable from the hinges.

8. Where a stairway door locking system is used, the required release switch shall be clearly marked.
9. Where panel doors are locked, an elevator-type key box, containing the panel door key, shall be provided in the immediate vicinity of the panels.
10. All panels shall be clearly labeled at the top to indicate the purpose or function, such as "detection," "voice communication," "elevators," or "generator."

2. Delete Section 911.1.2.
3. Delete Section 911.1.3.
4. Revise Section 911.1.4 to read:

"911.1.4 Layout review.

A layout of the *fire command center* and all features required by this section to be contained therein shall be submitted to the *fire code official* for review and acceptance prior to installation."

5. Delete Section 911.1.5.
6. Revise Section 911.1.6 to read:

"911.1.6 Required features.

The *fire command center* shall comply with NFPA 72 and shall contain all of the following features:

1. [Reserved.]
2. The fire department communications system.
3. Fire detection and alarm system annunciator.
4. Annunciator unit visually indicating the location of the elevators and whether they are operational.
5. Status indicators and controls for air distribution systems.
6. The fire fighter's control panel required by Section 909.16 for smoke control systems installed in the building.
7. Controls for unlocking *interior exit stairway* doors simultaneously.
8. Sprinkler valve and waterflow detector display panels.
9. Emergency and standby power status indicators.
10. A telephone for fire department use.
11. Fire pump status indicators.
12. Schematic building plans indicating the typical floor plan and detailing the building core, *means of egress*, fire protection systems, fire fighter air replenishment system, fire-fighting equipment and fire department access and the location of *fire walls, fire barriers, fire partitions, smoke barriers* and smoke partitions.
13. A Building Information Card that contains, information specified in rules adopted by the *fire code official*.
14. [Reserved.]
15. Generator supervision devices, manual start and transfer features.
16. Public address system, where specifically required by other sections of this code.
17. Elevator fire recall switch in accordance with the *Chicago Conveyance Device Code*.
18. Elevator emergency or standby power selector switch(es), where emergency or standby power is provided."

14B-9-912 Fire department connections.

The provisions of Section 912 of IBC are adopted by reference with the following modifications:

1. Revise Section 912.2 to read:

"912.2 Location.

With respect to hydrants, driveways, buildings and landscaping, fire department connections shall be so located that the fire apparatus and hose connected to supply the system will not obstruct access to buildings for other fire apparatus. At least one fire department connection shall be provided on each street exposure where the building is located closer than 100 feet (30.5 m) to the curb. If any street exposure is more than 250 feet (76.2 m) long, there shall be two fire department connections at least 200 feet (61.0 m) apart. Where a building has no street frontage, the fire department connection shall be located at the main entrance. The location of fire department connections shall be reviewed and accepted by the *fire code official*.

Exceptions:

1. Where a building has more than two street exposures, fire department connections are only required for two of the street exposures.
2. Where two adjacent street exposures have a combined length not greater than 250 feet (76.2 m), one fire department connection shall be allowed."

2. Revise Section 912.2.1 by deleting "approved."
3. Revise Section 912.2.2 by replacing "an *approved*" with "a" and deleting the last sentence.
4. Revise Section 912.3 by deleting "be approved and shall."
5. Revise Section 912.4 by deleting the last sentence.
6. Revise the exception to Section 912.4 by deleting "*approved* by the *fire code official*."
7. Revise Section 912.4.1 to read:

"912.4.1 Breakaway fire department connection caps.

The fire code official is authorized to require breakaway caps on fire department connections for water-based *fire protection systems*."

8. Revise Section 912.4.2 to read:

"912.4.2 Clear space around connections.

A working space with a radius of 5 feet (1524 mm) shall be provided and maintained in front of wall-mounted fire department connections and around the circumference of free-standing fire department connections, except as otherwise required or *approved* by the *fire code official*."

9. Revise Section 912.4.3 by deleting "Section 312 of."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 35; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 19)

14B-9-913 Fire pumps.

The provisions of Section 913 of IBC are adopted by reference with the following modifications:

1. Revise exception 1 to Section 913.2.1 to read:

"1. In buildings less than 55 feet (16.8 m) in *building height*, separation by 1-hour *fire barriers* constructed in accordance with Section 707 or 1-hour *horizontal assemblies* constructed in accordance with Section 711, or both, shall be permitted in buildings equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2."
2. Delete item 4 in Section 913.4.

14B-9-914 Emergency responder safety features.

The provisions of Section 914 of IBC are adopted by reference with the following modifications:

1. Add a new exception to Section 914.1.1 to read:

"Exception: Markings shall not be required on shaftway openings that are readily discernible as openings onto a shaftway by the construction or arrangement."
2. Revise Section 914.2 to read:

"914.2 Equipment room identification.

Fire protection equipment shall be clearly identified. Rooms containing controls for air-conditioning systems, sprinkler risers and valves or other fire detection, suppression or control elements shall be identified for the use of the fire department. Signs required to identify fire protection equipment and equipment location shall be constructed of durable materials, permanently installed and readily visible."

14B-9-915 Carbon monoxide detection.

The provisions of Section 915 of IBC are adopted by reference with the following modifications:

1. Revise Section 915.1 by replacing "Chapter 11 of the *International Fire Code*" with "the *Chicago Minimum Requirements for Existing Buildings*."
2. Revise Section 915.1.1 by replacing "through 915.1.6" with "through 915.1.5."
3. Revise the exception to Section 915.1.3 by replacing "an approved location" with "a location that is monitored at all times when the building is occupied."
4. Revise exception 2.1 to Section 915.1.4 by replacing "an approved" with "a."

5. Revise exception 4 to Section 915.1.5 by replacing "an approved" with "a."
6. Delete Section 915.1.6.
7. Revise Section 915.2.3 to read:

"915.2.3 Group E and I-4 occupancies.

Carbon monoxide detectors shall be installed in classrooms in Group E and I-4 occupancies. Carbon monoxide alarm signals shall be automatically transmitted to an on-site location that is staffed by school personnel.

Exception: Carbon monoxide alarm signals shall not be required to be automatically transmitted to an on-site location that is staffed by school personnel in Group E or I-4 occupancies with an occupant load of 30 or less."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 36)

14B-9-916 Gas detection systems.

The provisions of Section 916 of IBC are adopted by reference with the following modifications:

1. Revise Section 916.2 to read:

"916.2 Permits.

Permits shall be required as set forth in the *Chicago Fire Prevention Code*."

2. Revise Section 916.4 to read:

"916.4 Power connections.

Gas detection systems shall be permanently connected to the building electrical power supply."

3. Revise Section 916.5 by replacing "an *approved*" with "a central."
4. Revise Section 916.6 by deleting "approved."
5. Revise items 2 and 3 of Section 916.7 to read:

"2. For toxic gases that are not HPM, sample analysis shall be performed at intervals not exceeding 5 minutes in accordance with the *Chicago Fire Prevention Code*.

3. Where a less frequent or delayed sampling interval is *approved* by the *fire code official*."

6. Revise Section 916.10 to read:

"916.10 Fire alarm system connections.

Gas sensors and gas detection systems shall not be connected to fire alarm systems unless *approved* by the *fire code official* and connected in accordance with the fire alarm equipment manufacturer's instructions."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 37)

14B-9-917 IBC Section 917.

The provisions of Section 917 of IBC are not adopted.

14B-9-918 IBC Section 918.

The provisions of Section 918 of IBC are not adopted.

14B-9-919 City fire alarm box.

The following language is adopted as a new Section 919:

"919. CITY FIRE ALARM BOX

919.1 Placement.

A *city fire alarm box* shall be placed within 100 feet (30.5 m) from the principal entrance of the following occupancies:

1. Group A-1 occupancies.
2. Group I occupancies.
3. Group A-2, A-3 and A-4 occupancies with an *occupant load* of more than 1,000.
4. Day care centers operating between the hours of 9:00 p.m. and 6:00 a.m.

Exception: Where all rooms occupied by the day care facilities and programs have a direct exterior exit.

5. Group E-2 occupancies having a capacity of 100 or more children.
6. Group E-1 occupancies having an *occupant load* of more than 100.
7. Housing intended primarily for students through the undergraduate level, including apartments, dormitories and fraternity and sorority houses.
8. Public utility structures.

919.2 Connection.

The *fire alarm systems* required for the occupancies listed in Section 919.1 shall be directly connected to the *city fire alarm box*.

Exception: Where the *fire alarm system* is directly connected to a UL-listed central station fire alarm service and use of private supervisory service is not prohibited by the *fire code official*.

919.2.1 Other occupancies.

Fire alarm systems for occupancies other than those listed in Section 919.1 shall not be connected to a *city fire alarm box*.

Exception: Occupancies listed in Section 919.1 with an *occupant load* less than provided may be connected to a *city fire alarm box*.

919.3 Multiple connections.

The *fire alarm systems* of no more than one *building* shall be connected to any one *city fire alarm box*.

Exception: *Fire alarm systems* in a *building* that was permitted to be connected to a common *city fire alarm box* prior to July 1, 1971."

CHAPTER 14B-10

MEANS OF EGRESS

14B-10-1001 Administration.

14B-10-1002 Maintenance and plans.

14B-10-1003 General means of egress.

14B-10-1004 Occupant load.

14B-10-1005 Means of egress sizing.

14B-10-1006 Number of exits and exit access doorways.

14B-10-1007 Exit and exit access doorway configuration.

14B-10-1008 Means of egress illumination.

14B-10-1009 Accessible means of egress.

14B-10-1010 Doors, gates and turnstiles.

14B-10-1011 Stairways.

14B-10-1012 Ramps.

14B-10-1013 Exit signs.

14B-10-1014 Handrails.

14B-10-1015 Guards.

14B-10-1016 Exit access.

14B-10-1017 Exit access travel distance.

14B-10-1018 Aisles.

14B-10-1019 Exit access stairways and ramps.

14B-10-1020 Corridors.

14B-10-1021 Egress balconies.

14B-10-1022 Exits.

14B-10-1023 Interior exit stairways and ramps.

14B-10-1024 Exit passageways.

14B-10-1025 IBC Section 1025.

14B-10-1026 Horizontal exits.

14B-10-1027 Exterior exit stairways and ramps.

14B-10-1028 Exit discharge.

14B-10-1029 Assembly.

14B-10-1030 IBC Section 1030.

14B-10-1031 Frontage and fire department access requirements for Group E-1.

14B-10-1001 Administration.

The provisions of Section 1001 of IBC are adopted by reference with the following modifications:

1. Revise Section 1001.1 by replacing "an *approved*" with "a."
2. Delete Section 1001.2.

14B-10-1002 Maintenance and plans.

The provisions of Section 1002 of IBC are adopted by reference with the following modifications:

1. Revise Section 1002.1 to read:

"1002.1 Maintenance.

Means of egress shall be maintained in accordance with the *Chicago Minimum Requirements for Existing Buildings* and *Chicago Fire Prevention Code*."

2. Revise Section 1002.2 to read:

"1002.2 Fire safety and evacuation plans.

Fire safety and evacuation plans shall be provided for all occupancies and buildings where required by the *Chicago Fire Prevention Code*. Such fire safety and evacuation plans shall comply with the applicable provisions of the *Chicago Fire Prevention Code*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 38)

14B-10-1003 General means of egress.

The provisions of Section 1003 of IBC are adopted by reference with the following modifications:

1. Revise Section 1003.3.1 to read:

"1003.3.1 Headroom.

Protruding objects are permitted to extend below the minimum ceiling height required by Section 1003.2 where a minimum headroom of 80 inches (2032 mm) is provided over any *circulation paths*, including walks, *corridors*, *aisles* and passageways. Not more than 50 percent of the ceiling area of a *means of egress* shall be reduced in height by protruding objects.

A barrier shall be provided where the vertical clearance above a *circulation path* is less than 80 inches (2032 mm) high above the finished floor. The leading edge of such a barrier shall be located 27 inches (686 mm) maximum above the finished floor.

Exceptions:

1. Door closers and stops that project no more than 4 inches (102 mm) shall not reduce headroom to less than 78 inches (1981 mm).
 2. Spaces within *dwelling units* and *sleeping units*, other than *Accessible units* and *Type A units*, and *means of egress* from a single unit shall comply with Section 1207."
2. Revise Section 1003.4 to read:

"1003.4 Slip-resistant surface.

Circulation paths of the *means of egress* shall have a slip-resistant surface that is securely attached.

Exception: Where a mat is used on a *temporary* or seasonal basis, the mat shall be securely attached or have a backing designed to be non-slip."

3. Revise the last sentence of Section 1003.5 before the exceptions to read:

"Where a *ramp* is used for a difference in elevation that is 6 inches (152 mm) or less, it shall be equipped with either *handrails* or floor finish materials that contrast with adjacent floor finish materials."

4. Revise exceptions 1 and 2 to Section 1003.5 to read:

"1. A single step with a maximum riser height of 8 inches (203 mm) is permitted at exterior doors not required to be *accessible* by Chapter 11 in Groups F, H, R-2, R-3, R-5, S and U.

2. A *stair* with a single riser or with two risers and a tread is permitted at locations not required to be *accessible* by Chapter 11 where the risers and treads comply with Section 1011.5, the minimum depth of the tread is 13 inches (330 mm) and a *handrail* is provided in accordance with Section 1014."

5. Revise Section 1003.7 to read:

"1003.7 Elevators, escalators and moving walks.

Elevators, escalators and moving walks shall not be used as a component of a required *means of egress*.

Exception: Elevators used as an *accessible means of egress* in accordance with Section 1009.4."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 39)

14B-10-1004 Occupant load.

The provisions of Section 1004 of IBC are adopted by reference with the following modifications:

1. Revise the last sentence of Section 1004.5 to read:

"Where an intended function is not listed in Table 1004.5, selection of a listed function that most nearly resembles the intended function shall be *approved* by the *building official*."

1.1. Revise Table 1004.5 by changing the value for "Baggage claim" under "Airport terminal" from "20 gross" to "50 gross".

1.2. Revise Table 1004.5 by changing the value for "Gaming floors (keno, slots, etc.)" under "Assembly" from "11 gross" to "15 gross".

2. Revise Table 1004.5 by adding a new entry under "Assembly without fixed seats" to read:

"Outdoor areas (greater than 400 square feet) | 20 net"

3. Revise Table 1004.5 by changing the value for "Business areas" from "150 gross" to "100 net."

4. Revise Table 1004.5 by revising the entries under "Day care" to read:

"Day care |
Children under 2 | 55 net
All others | 35 net"

5. Revise Table 1004.5 by revising the entries under "Exercise rooms" to read:

"Exercise rooms |
With equipment | 50 gross
Open floor | 20 net"

6. Revise Table 1004.5 by replacing "Kitchens, commercial" with "Kitchens and service areas, commercial."

7. Revise Table 1004.5 by deleting the row for "Mall buildings-covered and open."

8. Revise Table 1004.5 by revising the entries under "Mercantile" to read:

"Mercantile |
First story above grade plane | 30 net
Basement sales floor | 30 net
Other floors | 60 net
Storage, stock, shipping areas | 300 gross"

9. Revise Table 1004.5 by adding a row for "Power plants" to read:

"Power plants | 400 gross"

10. Revise Table 1004.5 by revising the entries under "Residential" to read:

"Residential		
General		125 gross
Shared outdoor areas		20 net"

11. Revise note a to Table 1004.5 to read:

"a. *Net floor area* or *gross floor area* in square feet per occupant."

12. Revise Section 1004.5.1 to read:

"1004.5.1 Increased occupant load.

Where *approved* by the *building official*, the *occupant load* permitted in any *building*, or portion thereof, may be increased from that number established for the occupancies in Table 1004.5, provided that all other requirements of the code are met based on such modified number and the *occupant load* does not exceed one occupant per 7 square feet (0.65 m²) of occupiable floor space. An *aisle*, seating or fixed equipment diagram substantiating any increase in *occupant load* shall be submitted."

13. Revise Section 1004.6 by deleting the last sentence.

14. Revise Section 1004.7 to read:

"1004.7 Outdoor areas.

Where occupants are provided access to *yards*, *patios*, *decks*, *exterior balconies*, *occupiable rooftops*, *courts* and similar outdoor areas, the outdoor areas shall be provided with *means of egress* as required by this chapter. The *occupant load* of such outdoor areas shall be calculated in accordance with the anticipated use. Where the path of egress travel from the outdoor areas passes through the *building*, *means of egress* requirements for the *building* shall be based on the sum of the *occupant loads* of the *building* plus the outdoor areas.

Exceptions:

1. Outdoor areas accessed exclusively for maintenance need only have one *means of egress*.
2. Outdoor areas associated with Group R-5 occupancies and located not more than 15 feet (4572 mm) above *grade plane* need only have one *means of egress* and shall not be assigned an *occupant load*.
3. Outdoor areas only accessed from an individual *dwelling unit* in Group R-2, R-3 or R-5 occupancies need only have one *means of egress* and shall not be assigned an *occupant load*."

15. Revise Section 1004.8 to read:

"1004.8 Concentrated business use areas.

The *occupant load factor* for concentrated business use shall be applied to telephone call centers, trading floors, data processing centers and similar business use areas with a higher density of occupants than would normally be expected in a typical business occupancy environment. The *occupant load* for concentrated business use areas shall be the actual *occupant load*, but not less than one occupant per 50 square feet (4.65 m²) of *net floor area*."

16. Revise Section 1004.9 to read:

"1004.9 Posting of maximum capacity signs.

Maximum capacity signs shall be posted where required by Section 802 of the *Chicago Construction Codes Administrative Provisions*."

17. Insert a new Section 1004.10 to read:

"1004.10 Limited areas used for assembly-type functions.

Where a Group B, F or M occupancy contains limited areas used for functions classified as either "assembly" or "assembly without fixed seats" in accordance with Table 1004.5, the *occupant load* for those areas shall be determined in accordance with Section 1004.10.1 and the cumulative *occupant load* for the occupancy shall be determined in accordance with Section 1004.10.2.

1004.10.1 Occupant load for egress from the room or area.

Means of egress directly from a room or area used in full for assembly-type functions shall be designed for the *occupant load* determined in accordance with Section 1004.3 and Sections 1004.5 through 1004.8.

1004.10.2 Cumulative occupant load.

The cumulative *occupant load* for the Group B, F or M occupancy shall be determined in accordance with Section 1004.10.2.1 through 1004.2.3.

1004.10.2.1 Rooms up to 150 square feet.

For individual rooms or spaces with a *gross floor area* of 150 square feet (13.9 m²) or less, the *occupant load* shall be determined by applying the *occupant load factor* in Table 1004.5 that is applicable to the predominant function of adjacent spaces within the same suite and same *story* which are not used for assembly-type functions.

1004.10.2.2 Rooms up to 750 square feet.

For individual rooms or spaces with a *gross floor area* greater than 150 square feet (13.9 m²) but not greater than 750 square feet (69.7 m²), where the aggregate *gross floor area* of such rooms or spaces within the suite and *story* does not exceed 10 percent of the *gross floor area* of the suite or *story*, the *occupant load* shall be determined by applying the *occupant load* factor in Table 1004.5 that is applicable to the predominant function of adjacent spaces within the same suite and same *story* which are not used for assembly-type functions.

1004.10.2.3 Non-public meeting and training rooms.

For rooms and spaces used with tables and chairs for meetings, conferences and training primarily attended by individuals employed in the same building and not open to the general public, the *occupant load* shall be determined by applying the *occupant load* factor for "classroom areas" in Table 1004.5."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 40; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 20; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 15; Amend Coun. J. 1-24-24, p. 8634, Art. IV, § 2)

14B-10-1005 Means of egress sizing.

The provisions of Section 1005 of IBC are adopted by reference with the following modifications:

1. Revise Section 1005.2 by adding the following sentence at the end:

"The width of *means of egress* components shall be measured in the clear at the narrowest point of the *means of egress* component under consideration, excepting projections permitted by this chapter."

2. Revise exception 1 to Section 1005.3.1 by deleting "and an *emergency voice/alarm communication system* in accordance with Section 907.5.2.2."

- 2.1. Delete exception 2 to Section 1005.3.1.

3. Revise exception 1 to Section 1005.3.2 by deleting "and an *emergency voice/alarm communication system* in accordance with Section 907.5.2.2."

- 3.1. Delete exception 2 to Section 1005.3.2.

4. Revise Section 1005.6 by replacing "two adjacent stories" with "*stories* immediately above and below."

5. Revise Section 1005.7.1 to read:

"1005.7.1 Doors.

Doors in any position shall not reduce the required width by more than one-quarter.

Exceptions:

1. [Reserved.]
2. The restrictions on door swing shall not apply to doors within individual *dwelling units* and *sleeping units* of Group R-2, R-3 and R-5 occupancies.
3. The restrictions on door swing shall not apply to doors to rooms or spaces which are not *occupiable spaces*.
4. In Group E-1 occupancies, door swing shall not reduce the required width of *corridors* by any amount."

6. Revise Section 1005.7.2 to read:

"1005.7.2 Other projections.

Handrail projections shall be in accordance with the provisions of Section 1014.8. Other projections such as trim and similar features shall be allowed to project into the required width not more than 1 1/2 inches (38 mm) on each side.

Exception: Projections are allowed in corridors within Group I-2 Condition 1 in accordance with Section 407.4.3."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 41; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 21; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 16)

14B-10-1006 Number of exits and exit access doorways.

The provisions of Section 1006 of IBC are adopted by reference with the following modifications:

1. Revise Section 1006.1 by replacing "occupied roofs" with "*occupiable rooftops*."
2. Revise Section 1006.2 by deleting "or *basement*."
3. Revise Section 1006.2.1 by replacing "values listed in Table 1006.2.1" with "values listed in Table 1006.2.1 or the floor area of the space exceeds 4,000 square feet (372 m²)."

4. Replace Table 1006.2.1 as shown:

TABLE 1006.2.1

SPACES WITH ONE EXIT OR EXIT ACCESS DOORWAY

OCCUPANCY	MAXIMUM OCCUPANT LOAD OF SPACE	MAXIMUM COMMON PATH OF EGRESS TRAVEL DISTANCE (feet)	
		Without Sprinkler System (feet)	With Sprinkler System (feet)
OCCUPANCY	MAXIMUM OCCUPANT LOAD OF SPACE	MAXIMUM COMMON PATH OF EGRESS TRAVEL DISTANCE (feet)	
		Without Sprinkler System (feet)	With Sprinkler System (feet)
A ^c , E	49	75	75 ^a
B, M	49	75	115 ^a
F	49	75	115 ^a
H-1, H-2, H-3	3	NP	25 ^b
H-4, H-5	10	NP	75 ^b
I-1, I-2 ^d , I-4	10	NP	75 ^a
I-3	10	NP	100 ^a
R-1	10 ^h	60	75 ⁱ 115 ^a
R-2	10 ^h	60	75 ⁱ 115 ^a
R-3 ^e , R-5 ^e	10 ^h	60	75 ^{g, i} 115 ^a
R-4 ^e	10 ^h	60	75 ^{g, i} 115 ^a
S ^f	29	75	115 ^a
U	49	75	75 ^a

For SI: 1 foot = 304.8 mm, 1 square foot = 0.0929 m².

NP = Not permitted.

- a. Buildings equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1.
- b. Group H occupancies equipped throughout with an *automatic sprinkler system* in accordance with Section 903.2.5.
- c. For a room or space used for assembly purposes having fixed seating, see Section 1029.8.
- d. For the travel distance limitations in Group I-2, see Section 407.4.
- e. The *common path of egress travel* distance and occupant load limitations shall not apply to a building exclusively containing a single *dwelling unit* and no other occupancy or within a *townhouse*.
- f. The *common path of egress travel* distance in a Group S-2 parking garage shall be not more than 100 feet.
- g. For the travel distance limitations in Groups R-3, R-4 and R-5 equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.3, see Section 1006.2.2.6.
- h. 20 in buildings equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1. For spaces in Group R occupancies required by Section 1004 to use an occupant load factor more intensive than 125 gross, the maximum occupant load is 49.
- i. Buildings equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.2. See Section 903 for occupancies where *automatic sprinkler systems* are permitted in accordance with Section 903.3.1.2.

5. [Reserved.]

6. Revise Section 1006.2.2.1 by replacing "one is permitted to be a fixed ladder" with "one is allowed to be served by a fixed ladder" and replacing "equal to one-half" with "equal to not less than one-half."

6.1. Revise Section 1006.2.2.2 by replacing "equal to one-half" with "equal to not less than one-half."

7. Revise Section 1006.2.2.3 by deleting "approved."

8. Revise Section 1006.2.2.4 by replacing "2 1/2 years of age or less" with "less than 2 years of age."

9. Revise Section 1006.2.2.6 to read:

"1006.2.2.6 Groups R-3, R-4 and R-5.

Where Group R-3 and R-5 occupancies are permitted by Section 903.2.8 to be protected by an *automatic sprinkler system* installed in accordance with Section 903.3.1.3, the *exit access* travel distance for Group R-3 and R-5 shall be not more than 75 feet (22 860 mm). Where Group R-4 occupancies are permitted by Section 903.2.8 to be protected by an *automatic sprinkler system* installed in accordance with Section 903.3.1.3, the *exit access* travel distance for Group R-4 shall be not more than 75 feet (22 860 mm)."

10. Add a new Section 1006.2.2.7 to read:

"1006.2.2.7 Electrical rooms.

Two *exit access doorways* are required in electrical rooms containing equipment rated 3,000 amperes or greater. *Exit access doorways* shall be separated by a horizontal distance equal to not less than one-half the length of the maximum overall diagonal dimension of the room."

11. Revise Section 1006.3 to read:

"1006.3 Egress from stories or occupiable rooftops.

The *means of egress* system serving any *story* or *occupiable rooftop* shall be provided with the number of separate and distinct *exits* or access to *exits* based on the aggregate *occupant load* served in accordance with this section. Where *stairways* serve more than one *story* or *occupiable rooftop*, only the *occupant load* of each *story* or *occupiable rooftop* considered individually shall be used in calculating the required number of *exits* or access to *exits* serving that *story* or *occupiable rooftop*."

12. Revise exception 1 to Section 1006.3.1 by replacing "Group R-1, R-2 or R-2" with "Group R."

13. Delete exception 2 to Section 1006.3.1.

14. Revise Section 1006.3.2 to read:

"1006.3.2 Egress based on occupant load.

Each *story* and *occupiable rooftop* shall have the minimum number of separate and distinct *exits*, or access to *exits*, as specified in Table 1006.3.2. A single *exit* or access to a single *exit* shall be permitted in accordance with Section 1006.3.3. The required number of *exits*, or *exit access stairways* or *ramps* providing access to *exits*, from any *story* or *occupiable rooftop* shall be maintained until arrival at the *exit discharge* or a *public way*.

Exception: Two *exits* shall be permitted from a *story* separated by *fire barriers* into two areas of approximately equal size that are connected by a *horizontal exit* where the *occupant load* of each area does not exceed 300."

15. Revise Section 1006.3.3 to read:

"1006.3.3 Single exits.

A single *exit* or access to a single *exit* shall be permitted from any *story* or *occupiable rooftop* where one of the following conditions exists:

1. The *occupant load*, number of *dwelling units* and *common path of egress travel* distance do not exceed the values in Table 1006.3.3(1).
2. Rooms, areas and spaces in the first *story above grade plane* complying with Section 1006.2 with *exits* that discharge directly to the exterior at the *level of exit discharge*, are permitted to have one *exit*.
3. [Reserved.]
4. [Reserved.]
5. [Reserved.]
6. A level within or serving a single *dwelling unit*, provided all the following criteria are met:
 - 6.1. The level is either:
 - 6.1.1. A *story* not more than one story above or below the first *story above grade plane*.
 - 6.1.2. A *mezzanine* in accordance with Section 505.2.
 - 6.1.3. A *loft* in accordance with Section 1207.5.
 - 6.2. The floor area of the level does not exceed:
 - 6.2.1. 1,600 square feet (148.6 m²).
 - 6.2.2. 2,000 square feet (185.8 m²) where the building is protected throughout with an *automatic sprinkler system*.
 - 6.3. The *dwelling unit* has an *exit* door that discharges directly to the exterior at the *level of exit discharge*.
7. The third *story above grade plane* within a *dwelling unit* provided all the following criteria are met:
 - 7.1. The *building* is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1, 903.3.1.2 or 903.3.1.3.

- 7.2. The floor area of the *story* does not exceed 1,600 square feet (148.6 m²).
- 7.3. The *dwelling unit* has an *exit* that discharges directly to the exterior at the *level of exit discharge*.
8. The third *story above grade plane* within a *dwelling unit* provided that all of the following criteria are met:
- 8.1. The *building* does not contain more than three *stories above grade plane*.
- 8.2. The *building* does not contain more than three *dwelling units*.
- 8.3. The *net floor area* of the *story* does not exceed 800 square feet (74.3 m²).
- 8.4. Each *habitable space* on the *story* shall have a window with a minimum net clear opening of 5.7 square feet (0.53 m²), the bottom of which is not more than 44 inches (1118 mm) above the floor.
- 8.5. Each room within the *dwelling unit* used for sleeping purposes shall be provided with solid wood doors not less than 1 3/4 inches (45 mm) in thickness in a rabbeted frame, or doors in compliance with Section 716.2.2.1 with a fire protection rating of not less than 20 minutes.
- 8.6. In addition to an interior *exit access stairway*, the second *story above grade plane* shall have either a second interior *exit access stairway* leading directly to an *exit* door that discharges to the exterior at the *level of exit discharge* or access to an exterior *means of egress* leading directly to grade.
- 8.7. Either of the following shall be provided at the third *story above grade plane*:
- 8.7.1. The *exit access stairway* is enclosed with construction providing a *fire-resistance rating* of at least one hour and a door meeting criterion 8.5.
- 8.7.2. An *exterior balcony* is provided with a depth of at least 3 feet (914 mm).
9. The third and fourth *story above grade plane* in a *townhouse*, provided all the following criteria are met:
- 9.1. The *townhouse* construction type is I, IIA, IIIA or IV.
- 9.2. The floor area of the *story* does not exceed 1,500 square feet (139.4 m²).
- 9.3. At least one *exit access stairway* is contained within the *townhouse* and leads directly to the *exterior* at the *level of exit discharge*.
- 9.4. An egress balcony complying with Section 1021 shall connect at least 3 *townhouses* at the highest *story* to an *interior exit stairway*.
- 9.5. Each *habitable space* more than two stories above *grade plane* shall have a window with a minimum net clear opening of 5.7 square feet (0.53 m²), the bottom of which is not more than 44 inches (1118 mm) above the floor.
- 9.6. Each room in the *townhouse* used for sleeping purposes shall be provided with solid wood doors not less than 1-3/4 inches (45 mm) in thickness in a rabbeted frame, or doors in compliance with Section 716.2.2.1 with a fire protection rating of not less than 20 minutes.
10. An *occupiable rooftop* provided all the following criteria are met:
- 10.1. The *occupiable rooftop* is adjacent to a *public way*.
- 10.2. The *occupiable rooftop* is located no more than 12 feet (3658 mm) above the level of the *public way*.
- 10.3. The *net floor area* does not exceed 800 square feet (74.3 m²).
11. An *occupiable rooftop* provided both of the following criteria are met:
- 11.1. The *occupiable rooftop* is located no more than 40 feet (12.2 m) above *grade plane*.
- 11.2. The *net floor area* does not exceed 300 square feet (27.9 m²).
12. A *basement* with a floor area that does not exceed 2,000 square feet (185.8 m²) that is used exclusively for storage or mechanical equipment, with only incidental human occupancy.
13. An outdoor area as provided in the exceptions to Section 1004.7."

16. Replace Table 1006.3.3(1) as shown:

TABLE 1006.3.3(1)

STORIES WITH ONE EXIT OR ACCESS TO ONE EXIT FOR R-2, R-3, R-4 AND R-5 OCCUPANCIES

STORY	OCCUPANCY	MAXIMUM NUMBER OF DWELLING UNITS PER STORY	MAXIMUM COMMON PATH OF EGRESS TRAVEL DISTANCE
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First story below <i>grade plane</i> ; first or second story above <i>grade plane</i>	R-2, R-3, R-4, R-5	4 dwelling units	75 feet ^a
Third story above <i>grade plane</i> and higher	NP	NA	NA

For SI: 1 foot = 3048 mm.

NP = Not permitted.

NA = Not applicable.

- a. The maximum *common path of egress travel* distance shall be 125 feet (38.1 m) in *buildings* equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2 and *buildings* of Group R-3, R-4 or R-5 occupancy equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.3.

17. Delete Table 1006.3.3(2).

18. Delete Section 1006.3.3.1.

(Amend Coun. J. 7-24-19, p. 3646, § 31; Amend Coun. J. 2-19-20, p. 14473, Art. II, § 42; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 22; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 17; Amend Coun. J. 9-14-21, p. 35949, Art. XV, § 8; Amend Coun. J. 1-26-22, p. 43756, Art. II, § 8)

14B-10-1007 Exit and exit access doorway configuration.

The provisions of Section 1007 of IBC are adopted by reference with the following modifications:

1. Revise Section 1007.1 by adding the following sentence at the end:

"Interlocking or *scissor stairways* shall be counted as one *exit stairway*."

2. Revise Section 1007.1.1 to read:

"1007.1.1 Two exits or exit access doorways.

Where two *exits*, *exit access doorways*, *exit access stairways* or *exit access ramps*, or any combination thereof, are required from any portion of the *exit access*, they shall be placed a distance apart equal to not less than one-half of the length of the maximum overall diagonal dimension of the building or area to be served measured in a straight line between them.

Exceptions:

1. Where interior *exit stairways* or *exit access ramps* are interconnected by a 1-hour fire-resistance-rated *corridor* conforming to the requirements of Section 1020, the required separation distance shall be measured along the shortest direct line of travel within the *corridor*.
2. Where a *building* is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2, the separation distance shall be not less than one-third of the length of the maximum overall diagonal dimension of the area served.
3. In Group R-5 occupancies, the *exits*, *exit access doorways*, *exit access stairways* or *exit access ramps* shall be arranged remote from one another so as to minimize the chance of both being blocked by smoke or fire.
4. Where two *exits*, *exit access doorways*, *exit access stairways* or *exit access ramps* are required from or within a *dwelling unit*, they shall be arranged remote from one another so as to minimize the chance of both being blocked by smoke or fire.
5. Where two *exits*, *exit access doorways*, *exit access stairways* or *exit access ramps* are required from a Group B or Group M tenant space (other than an *ambulatory care facility*), a reduction in the required separation distance may be *approved* by the *building official* where the tenant space is separated from all other areas of the *building* by construction with a *fire-resistance rating* of not less than 1 hour and the separation distance is sufficient to minimize the chance of both being blocked by smoke or fire.
6. In Group R-2 occupancies where the *building* is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1 or 903.3.1.2 and dead-end *corridors* do not exceed 20 feet (6096 mm), the separation distance shall be not less than 15 feet (4572 mm)."

14B-10-1008 Means of egress illumination.

The provisions of Section 1008 of IBC are adopted by reference with the following modifications:

1. Revise Section 1008.2 to read:

"1008.2 Illumination required.

The *means of egress* serving a room or space shall be artificially illuminated at all times that the room or space served by the *means of egress* is occupied and the intensity of lighting required by Section 1008.2.1 is not provided by means of natural light.

Exceptions:

1. Occupancies in Group U.
2. *Aisle accessways* in Group A.
3. *Dwelling units* and *sleeping units* in Groups R.
4. *Sleeping units* of Group I occupancies."

2. Revise the exception to Section 1008.2.1 to read:

"Exception: For auditoriums, theaters, concert or opera halls, galleries and similar assembly occupancies, the illumination at the walking surface is permitted to be reduced during performances or exhibits to not less than 0.1 footcandle (1 lux) at the walking surface, provided that the required illumination is automatically restored upon failure of normal power or activation of the premises' fire alarm system."

3. Delete Section 1008.2.3.

4. Revise Sections 1008.3.1, 1008.3.2, and 1008.3.3 by replacing "power supply failure" with "normal power supply failure."

5. Revise item 2 in Section 1008.3.3 to read:

"Rooms or spaces containing *fire command centers*."

6. Revise Section 1008.3.4 to read:

"1008.3.4 Operation.

The emergency power system shall be in accordance with Article 700 of the *Chicago Electrical Code*."

14B-10-1009 Accessible means of egress.

The provisions of Section 1009 of IBC are adopted by reference with the following modifications:

1. Revise the exceptions to Section 1009.2.1 by replacing "an *automatic sprinkler system*" with "a supervised *automatic sprinkler system*."
2. [Reserved.]
3. Revise exception 2 to Section 1009.3.3 by replacing "an *automatic sprinkler system*" with "a supervised *automatic sprinkler system*."
4. Revise exception 2 to Section 1009.4.2 by replacing "an *automatic sprinkler system*" with "a supervised *automatic sprinkler system*."
5. Delete the exception to Section 1009.7.2.
6. Delete exception 6 to Section 1009.8.
7. Revise Section 1009.8.1 to read:

"1009.8.1 System requirements.

Two-way communication systems shall provide communication between each required location and the *fire command center* or a central control point location acceptable to the *fire code official*. Where the central control point is not a *constantly attended location*, a two-way communication system shall have a timed automatic telephone dial-out capability to a 24/7 monitoring service. The two-way communication system shall include both audible and visible signals. If a handset is provided, the handset cords shall be at least 29 inches (735 mm) long."

8. Revise Section 1009.10 by deleting "approved."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 43)

14B-10-1010 Doors, gates and turnstiles.

The provisions of Section 1010 of IBC are adopted by reference with the following modifications:

1. Revise exception 1 to Section 1010.1.1 to read:
 - "1. Within Group R-2, R-3 and R-5 *dwelling* and *sleeping units* that are not required to be an *Accessible unit*, *Type A unit* or *Type B unit*, the minimum clear opening width of an *exit access doorway*, other than the primary entrance to the unit, shall be 28 inches (711 mm), and the minimum and maximum width shall not apply to door openings that are not part of the required *means of egress*."
2. Delete exception 8 to Section 1010.1.1.

3. Revise exceptions 11 and 12 to Section 1010.1.1 to read:

"11. The minimum clear opening width shall not apply to doors for shower or sauna compartments not required to be *accessible*.

12. The minimum clear opening width shall not apply to the doors for toilet stalls or single-user toilet rooms not required to be *accessible*."

4. Revise exception 4 to Section 1010.1.2 to read:

"4. Doors within or serving a single *dwelling unit* in Groups R-2, R-3 and R-5."

5. Delete exceptions 6 and 8 to Section 1010.1.2.

6. Revise exception 9 to Section 1010.1.2 by replacing "Group H" with "Groups E and H."

7. Revise Section 1010.1.3 to read:

"1010.1.3 Door opening force.

The force for pushing or pulling open interior swinging egress doors, other than fire doors, shall not exceed 5 pounds (22 N). The force for pushing or pulling open exterior hinged doors and gates shall not exceed 8.5 pounds (38 N). These forces do not apply to the force required to retract latch bolts or disengage other devices that hold the door in a closed position. For other swinging doors, as well as sliding and folding doors, the door latch shall release when subjected to a 15-pound (67 N) force. The door shall be set in motion when subjected to a 30-pound (133 N) force. The door shall swing to a full-open position when subjected to a 15-pound (67 N) force.

Exceptions:

1. Group R-5.
2. Within *Type B units*.
3. Within or serving only *dwelling units* that are not required to be *Type B units*."

8. Revise the exception to Section 1010.1.4.1.2, item 4, by replacing "in an approved location and clearly identified" with "within sight of the revolving door."

9. Delete Section 1010.1.4.3.

10. Delete Section 1010.1.4.4.1.

11. Revise exception 1 to Section 1010.1.5, exclusive of subparts, to read:

"1. Doors serving individual *dwelling units* in Groups R-2, R-3 and R-5 occupancies where the following apply:"

12. Revise exception 3 to Section 1010.1.5 to read:

"3. In Group R-2, R-3 and R-5 occupancies not required to be *Accessible units*, *Type A units* or *Type B units*, the landing at an exterior storm or screen door, does not swing over the landing."

13. Revise exception 5 to Section 1010.1.5 to read:

"5. Exterior *decks*, patios or *exterior balconies* that are part of *Type B dwelling units*, have impervious surfaces and that are not more than 4 inches (102 mm) below the finished floor level of the adjacent interior space of the *dwelling unit*."

14. Revise Section 1010.1.6 to read:

"1010.1.6 Landings at doors.

Landings shall have a width not less than the width of the *stairway* or the door, whichever is greater. Doors in the fully open position shall not reduce a required dimension by more than one-quarter its required width. Landings shall have a length measured in the direction of travel of not less than 44 inches (1118 mm).

Exception: Landing length in the direction of travel in Groups R-3, R-5 and U and within individual units of Group R-2 need not exceed 36 inches (914 mm)."

15. Revise the exceptions to Section 1010.1.7 to read:

"Exceptions:

1. In occupancy Group R-2, R-3 or R-5, threshold heights for sliding and side-hinged exterior doors shall be permitted to be up to 8 inches (203 mm) in height if all of the following apply:

- 1.1. [Reserved.]
- 1.2. The door is not part of an *accessible route* as required by Chapter 11.
- 1.3. The door is not part of an *Accessible unit*, *Type A unit* or *Type B unit*.

2. In *Type B units*, where Exception 5 to Section 1010.1.5 permits a 4-inch (102 mm) elevation change at the door, the threshold height on the exterior side of the door shall not exceed 4 3/4 inches (120 mm) in height above the exterior *deck*, patio or *exterior balcony* for sliding doors or 4 1/2 inches (114 mm) above the exterior *deck*, patio or *exterior balcony* for other doors."

16. Revise the exceptions to Section 1010.1.8 to read:

"Exceptions:

1. The minimum distance between horizontal sliding power-operated doors in a series shall be 48 inches (1219 mm).
2. Storm and screen doors serving individual *dwelling units* in Groups R-2, R-3 and R-5 need not be spaced 48 inches (1219 mm) from the other door.
3. Doors within individual *dwelling units* in Groups R-2, R-3 and R-5 other than within *Accessible units* and *Type A units*."

17. Revise Section 1010.1.9.4 to read:

"1010.1.9.4 Locks and latches.

Locks and latches shall be permitted to prevent operation of doors where any of the following exist:

1. Places of detention or restraint.
2. Where *approved* by the *fire code official*, in buildings in occupancy Group A having an *occupant load* of 300 or less, Groups B, F, M and S, and in *places of religious worship*, the main door or doors are permitted to be equipped with key-operated locking devices from the egress side provided:
 - 2.1. The locking device is readily distinguishable as locked.
 - 2.2. A readily visible durable sign is posted on the egress side on or adjacent to the door stating: THIS DOOR TO REMAIN UNLOCKED WHEN THIS SPACE IS OCCUPIED. The sign shall be in letters 1 inch (25 mm) high on a contrasting background.
 - 2.3. The use of the key-operated locking device is revocable by the *fire code official* for due cause.
3. Where egress doors are used in pairs, automatic flush bolts shall be permitted to be used, provided that the door leaf having the automatic flush bolts does not have a doorknob or surface-mounted hardware.
4. Doors from individual *dwelling* or *sleeping units* of Group R occupancies having an *occupant load* of 10 or less are permitted to be equipped with a night latch, dead bolt or security chain, provided such devices are openable from the inside without the use of a key, tool or special knowledge.
5. *Fire doors* after the minimum elevated temperature has disabled the unlatching mechanism in accordance with *listed fire door* test procedures.
6. Doors serving roofs not intended to be occupied shall be permitted to be locked preventing entry to the *building* from the roof."

18. Delete exception 3 to Section 1010.1.9.5.

19. Revise exception 4 to Section 1010.1.9.5 to read:

"4. Where a pair of doors serves a Group B, F or S occupancy, manually operated edge- or surface-mounted bolts are permitted on the inactive leaf provided that such inactive leaf is not needed to meet egress capacity requirements. The inactive leaf shall not contain doorknobs, panic bars or similar operating hardware."

20. Revise item 5 in Section 1010.1.9.7 to read:

"5. The procedures for unlocking the doors shall be described and accepted as part of the emergency planning and preparedness required by the *Chicago Fire Prevention Code*."

20.1. Revise Section 1010.1.9.8 by inserting an item 3 to read:

"3. Group A occupancies in airport terminals."

21. Revise items 3 and 4 (including the exception) in Section 1010.1.9.8.1 to read:

"3. The delayed egress locking system shall have the capability of being deactivated at the *fire command panel*. Nothing in this section shall prohibit installation of deactivation means at additional locations.

4. An attempt to egress shall initiate an irreversible process that shall allow such egress not more than 15 seconds after pressure is applied to the egress-side door hardware for not more than 1 second. Initiation of the irreversible process shall activate an audible signal in the vicinity of the door. Once the delay electronics have been deactivated, rearming the delay electronics shall be by manual means only. The time delay shall not be field adjustable."

22. Delete item 6.2 in Section 1010.1.9.8.1.

23. Add new item 9 in Section 1010.1.9.8.1 to read:

"9. The door must swing in the direction of egress travel."

24. Revise Section 1010.1.9.11 by replacing "an *approved*" with "a."

Revise exceptions 3 through 5 to Section 1010.1.9.12 to read:

"3. In *interior exit stairways* serving more than 4 *stories*, *exit* doors are permitted to be locked from the side opposite the egress side, provided that they are openable from the egress side and capable of being unlocked simultaneously without unlatching both automatically, by smoke detectors or waterflow devices, and manually, upon a signal from the *fire command panel*, if present, or a signal by building management or emergency personnel from a single location inside the main entrance to the building. A telephone or other two-way communication system connected to a constantly monitored location shall be provided in the *exit stairway* at not less than every fifth *story*.

4. In *interior exit stairways* serving 4 or fewer *stories* in other than Group A, E, H, and I occupancies, *exit* doors are permitted to be locked from the side opposite the egress side.

5. *Stairway exit* doors shall be openable from the egress side and shall only be locked from the opposite side in Group R-2 or R-5 occupancies where the only interior access to the *dwelling unit* is from a single *interior exit stairway*."

25. Revise the exception to Section 1010.2 to read:

"**Exception:** Horizontal sliding or swinging gates exceeding the 4-foot (1219 mm) maximum leaf width limitation are permitted in exterior fences and walls."

26. Insert a new Section 1010.2.2 to read:

1010.2.2 Residential buildings.

Exterior gates which are part of the *exit discharge* for a Group R-2, R-3, R-4 or R-5 occupancy shall conform to the applicable requirements of this section.

1010.2.2.1 Small residential buildings.

On a *lot* containing *buildings* with not more than 4 *stories above grade plane* and not more than 4 *dwelling units* or *sleeping units*, an exterior gate is allowed to require a key or special tool to operate in either or both directions.

1010.2.2.2 Large residential buildings.

On a *lot* containing *buildings* with 5 or more *stories above grade plane* or containing 5 or more *dwelling units* or *sleeping units*, an exterior gate shall not require a key or special tool to operate in the direction of egress."

27. Revise Section 1010.3.2 by replacing "an *approved*" with "a."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 44; Amend Coun. J. 12-14-22, p. 58278, Art. III, § 3; Amend Coun. J. 1-24-24, p. 8634, Art. IV, § 3)

14B-10-1011 Stairways.

The provisions of Section 1011 of IBC are adopted by reference with the following modifications:

1. Revise exception 1 to Section 1011.2 by replacing "an *occupant load*" with "a *story* or *mezzanine* with an *occupant load*".

1.1. Revise exception 3 to Section 1011.2 by replacing "Group R-3" with "Group R-3 or R-5."

2. Revise Section 1011.3 by replacing "80 inches (2032 mm)" with "7 feet (2134 mm)."

3. Revise exception 2 to Section 1011.3 to read:

"2. The required headroom clearance shall be not less than 80 inches (2032 mm), and where the *nosings* of treads at the side of a *flight* extend under the edge of a floor opening through which the *stair* passes, the floor opening shall be allowed to project horizontally into the required headroom not more than 4 3/4 inches (121 mm) in any of the following conditions:

2.1. In Group R-3 and R-5 occupancies.

2.2. Within *dwelling units* in Group R-2 occupancies.

2.3. In Group U occupancies that are accessory to a Group R-3 occupancy or accessory to individual *dwelling units* in Group R-2 occupancies."

4. Revise Section 1011.4 to read:

"1011.4 Walkline.

Where a *stair* includes *winder* treads, the walkline shall comply with Section 1011.4.1 or 1011.4.2.

1011.4.1 Other than Group R-5 occupancies and within dwelling units.

The walkline across *winder* treads in other than R-5 occupancies and within individual *dwelling units* in Group R-2

occupancies shall be concentric to the direction of travel through the turn and located 12 inches (305 mm) from the side where the *winders* are narrower. The 12-inch (305 mm) dimension shall be measured from the widest point of the clear *stair* width at the walking surface of the *winder*. Where *winders* are adjacent within the *flight*, the point of the widest clear *stair* width of the adjacent *winders* shall be used.

1011.4.2 Group R-5 occupancies and within dwelling units.

The walkline across *winder* treads in R-5 occupancies and within individual *dwelling units* in Group R-2 occupancies shall be concentric to the direction of travel through the turn and located 18 inches (457 mm) from the side where the *winders* are narrower."

5. Revise exception 3 to Section 1011.5.2 to read:

"3. The maximum riser height shall be 8 inches (203 mm), the minimum tread depth shall be 9 inches (229 mm), the minimum *winder* tread depth at the walkline shall be 9 inches (229 mm) and a *nosing* projection not less than 1 inch (25 mm) but not more than 1 1/4 inches (32 mm) shall be provided where the tread depth is less than 10 inches (254 mm) in any of the following conditions:

- 3.1. In Group R-3 and R-5 occupancies.
- 3.2. Within *dwelling units* in Group R-2 occupancies.
- 3.3. In interior and exterior *stairways* in Group R-2 occupancies serving only units that are not required to be *Accessible units* or *Type A units*.
- 3.4. In Group U occupancies that are accessory to a Group R-3 occupancy or accessory to individual *dwelling units* in Group R-2 occupancies."

6. Revise Section 1011.5.3 to read:

"1011.5.3 Winder treads.

Winder treads are not permitted in *means of egress stairways*.

Exceptions:

1. Curved *stairways* in accordance with Section 1011.9.
2. *Spiral stairways* in accordance with Section 1011.10.
3. Within or serving a building of Group R-3 or R-5 occupancy.
4. Within a *dwelling unit* in a Group R occupancy."

- 6.1. Revise Section 1011.5.5 by deleting the second sentence.

7. Revise Section 1011.5.5.3 to read:

"1011.5.5.3 Riser profile.

Risers shall be solid and vertical or sloped under the tread above from the underside of the *nosing* above at an angle not more than 30 degrees (0.52 rad) from the vertical.

Exceptions:

1. Interior *stairways* that are not required to comply with Section 1009.3, provided that the opening between treads does not permit the passage of a sphere with a diameter of 6 inches (152 mm).
2. Occupancies in Group I-3 or in Group F, H and S occupancies other than areas accessible to the public. The size of the opening in the riser is not restricted.
3. *Spiral stairways* constructed in accordance with Section 1011.10.
4. Exterior *stairways* that are not required to comply with Section 1009.3."

8. [Reserved.]

9. Revise Section 1011.6 to read:

"1011.6 Stairway landings.

There shall be a floor or landing at the top and bottom of each *stairway*. The width of landings, measured perpendicularly to the direction of travel, shall be not less than the width of *stairways* served. Every landing shall have a minimum depth, measured parallel to the direction of travel, equal to the width of the *stairway* or 48 inches (1219 mm), whichever is less. Doors opening onto a landing shall not reduce the landing by more than one-quarter the required width. Where *wheelchair spaces* are required on the *stairway* landing in accordance with Section 1009.6.3, the *wheelchair space* shall not be located in the required width of the landing and doors shall not swing over the *wheelchair spaces*.

Exceptions:

1. Where *stairways* connect stepped *aisles* to cross *aisles* or concourses, *stairway* landings are not required

at the transition between *stairways* and stepped *aisles* constructed in accordance with Section 1029.

2. Where doors are allowed by Section 1010.1.5."

9.1. Revise Section 1011.7 to read:

"1011.7 Stairway construction.

Stairways shall be built of materials consistent with the type of construction of the building.

Exceptions:

1. *Stairways* connecting five or more *stories* shall be built of and supported by noncombustible materials. Shaft enclosures for *stairways* connecting five or more *stories* shall be built of noncombustible materials.
2. Wood *handrails* shall be permitted for all types of construction."

10. Revise the exception to Section 1011.7.3 by replacing "Group R-2 or R-3" with "Group R-2, R-3 or R-5."

11. Revise Section 1011.8 to read:

"1011.8 Vertical rise.

A *flight of stairs* shall not have a vertical rise greater than 12 feet (3658 mm) between floor levels or landings.

Exceptions:

1. *Spiral stairways* used as a means of egress from technical production areas.
 2. In Group R-3 and R-5 occupancies a *flight of stairs* may have a vertical rise not greater than 151 inches (3835 mm) between floor levels or landings."
12. Revise the exception to Section 1011.9 by replacing "Group R-3" with "Group R-3 and R 5."
13. Revise the first sentence of Section 1011.10 to read:

"*Spiral stairways* are permitted to be used as a component in the *means of egress* in the following locations only:

1. Within a *dwelling unit*.
2. Serving a single *dwelling unit*.
3. From a space not more than 250 square feet (23 m²) in area and serving not more than five occupants.
4. From a *technical production area* in accordance with Section 410.5 ."

14. Revise Section 1011.11 to read:

"1011.11 Handrails.

Flights of stairs shall have *handrails* on each side and shall comply with Section 1014. Where glass is used to provide or support the *handrail*, the *handrail* or support shall comply with Section 2407.

Exceptions:

1. *Flights of stairs* within *dwelling units* and *spiral stairways* are permitted to have a *handrail* on one side only.
 2. *Decks*, patios and walkways accessory to Group R-5 occupancies or individual *dwelling units* that have a single change in elevation where the landing depth on each side of the change of elevation is greater than what is required for a landing do not require *handrails*.
 3. Where allowed, a change in elevation consisting of a single riser at an entrance or egress door does not require *handrails*.
 4. Changes in elevation of three or fewer risers within or accessory to individual *dwelling units* and *sleeping units* in Group R-2, R-3 and R-5 do not require *handrails*.
 5. *Flights of stairs* that serve as stepped *aisles* for seating areas shall have *handrails* either at one or both sides of the *aisle* or within the *aisle* width."
15. Revise Section 1011.12 to read:

"1011.12 Stairway to roof.

In *buildings* with four or more *stories above grade plane*, one *stairway* shall extend to the roof surface unless the roof has a slope greater than or equal to three units vertical in 12 units horizontal (25-percent slope).

Exception: Other than where required by Section 1011.12.1, in *buildings* without an *occupiable rooftop*, access to the roof from the top *story* shall be permitted to be by an *alternating tread device*, a ship's ladder or a permanent ladder."

16. Revise Section 1011.12.2 to read:

"1011.12.2 Roof access.

Where a *stairway* is provided to a roof, access to the roof shall be provided through a penthouse complying with Section 1510.2.

Exception: In buildings without an *occupiable rooftop*, access to the roof shall be permitted to be a roof hatch or trap door not less than 16 square feet (1.5 m²) in area and having a minimum dimension of 2 feet (610 mm)."

17. Revise Section 1011.14 by adding the following after the last sentence:

"The minimum clear width at and below the *handrails* shall be 20 inches (508 mm)."

18. Delete the exception to Section 1011.14.2.

19. Revise the second sentence of Section 1011.16 to read:

"Permanent ladders shall be constructed in accordance with the *Chicago Mechanical Code*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 45; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 23; Amend Coun. J. 1-26-22, p. 43756, Art. II, § 9)

14B-10-1012 Ramps.

The provisions of Section 1012 of IBC are adopted by reference with the following modifications:

1. Revise exception 1 to Section 1012.6.3 by replacing "Group R-2 and R-3" with "Group R-2, R-3 and R-5."
2. Revise the exception to Section 1012.6.4 by replacing "Group R-2 and R-3" with "Group R-2, R-3 and R-5."

14B-10-1013 Exit signs.

The provisions of Section 1013 of IBC are adopted by reference with the following modifications:

1. Revise Section 1013.1 by deleting "*approved*."
2. Revise the exceptions to Section 1013.1 to read:

"Exceptions:

1. Exit signs are not required in rooms or areas that require only one *exit* or *exit access*.
 2. Main exterior *exit* doors or gates that are obviously and clearly identifiable as *exits* need not have exit signs where *approved* by the *fire code official*.
 3. Exit signs are not required in occupancies in Group U and individual *sleeping units* or *dwelling units* in Group R-1, R-2, R-3 or R-5.
 4. Exit signs are not required in dayrooms, sleeping rooms or dormitories in occupancies in Group I-3."
3. Delete Section 1013.2.
 4. Revise Section 1013.3 to read:

"1013.3 Illumination.

Exit signs shall be internally illuminated.

Exception: Visual character signs required by Section 1013.14 shall be internally or externally illuminated at all times. Tactile signs required by Section 1013.4 need not be provided with illumination."

5. Revise Section 1013.4 to read:

"1013.4 Raised character and braille signs.

Signs stating "AREA OF REFUGE," "EXTERIOR AREA FOR ASSISTED RESCUE," "EXIT," "STAIR" or "FIRE ESCAPE," as required by Sections 1009.9 and 1013.6.1, in visual characters, raised characters and braille and complying with ICC A117.1 shall be provided adjacent to each door to an *area of refuge*, or providing direct access to an *exit stairway*, an exterior area for assisted rescue, an *exit ramp*, an *exit passageway* or the *exit discharge*."

6. Delete Section 1013.5.
7. Revise Section 1013.6 and its subsections to read:

"1013.6 Internally illuminated exit signs.

Internally illuminated exit signs shall comply with Sections 1013.6.1 through 1013.6.3.

1013.6.1 Graphics.

Every exit sign and directional exit sign shall have plainly legible letters not less than 3 3/8 inches (86 mm) high with the principal strokes of the letters not less than 9/16 inch (14 mm) wide. Directional exit signs shall have a horizontal arrow

below the lettering not less than 1/2 inch (13 mm) wide, running the full length of the lettering. Signs larger than the minimum established in this section shall have letter widths, strokes and spacing in proportion to their height.

The lettering and arrow shall be red on a white translucent background and shall be clearly discernible when the means of exit sign illumination is or is not energized.

1013.6.1.1 Exit sign.

Exit signs, other than those subject to Section 1013.6.1.2 or 1013.6.1.3 shall read "EXIT."

1013.6.1.2 Stair sign.

Exit signs located immediately over an *exit access stairway* or *exit access door* leading to an *exit stairway* shall read "STAIR" or "EXIT STAIR." In *existing buildings*, new exit signs may also read "STAIRS" or "STAIRWAY" if consistent with existing exit signs.

1013.6.1.3 Fire escape sign.

In *existing buildings*, exit signs and directional exit signs leading to a fire escape shall read "FIRE ESCAPE."

1013.6.2 Exit sign illumination.

The face of an exit sign shall be illuminated to meet the luminance measurement test in UL 924.

1013.6.3 Power source.

Exit signs shall be illuminated at all times. To ensure continued illumination for a duration of not less than 90 minutes in case of primary power loss, the sign illumination means shall be connected to an emergency power system provided from storage batteries, unit equipment or an on-site generator. The installation of the emergency power system shall be in accordance with Article 700 of the *Chicago Electrical Code*."

14B-10-1014 Handrails.

The provisions of Section 1014 of IBC are adopted by reference with the following modifications:

1. Revise exception 2 to Section 1014.2 by replacing "Group R-3" with "Group R-3 or R-5."
2. Revise the exception to Section 1014.3 by replacing "Group R-3" with "Group R-3 or R-5."

14B-10-1015 Guards.

The provisions of Section 1015 of IBC are adopted by reference with the following modifications:

1. Add a new exception 8 to Section 1015.2 to read:

"8. On the loading side of passenger platforms for fixed guideway transit and passenger rail systems."

2. Revise item 1 of Section 1015.3 to read:

"1. From adjacent walking surfaces, excluding any horizontal surface 12 inches (305 mm) or less in depth measured perpendicular to the *guard*."

3. Revise exceptions 1, 2 and 3 to Section 1015.3 to read:

"1. For occupancies in Group R-3 and R-5 not more than three *stories above grade plane* in height and within individual *dwelling units* in occupancies in Group R-2, and outdoor areas of individual *dwelling units* located not more than three stories above *grade plane*, required *guards* shall be not less than 36 inches (914 mm) in height measured vertically above the adjacent walking surfaces.

2. For occupancies in Group R-3 and R-5, and within individual *dwelling units* in occupancies in Group R-2, *guards* on the open sides of *stairs* shall have a height not less than 34 inches (864 mm) measured vertically from a line connecting the leading edges of the treads.

3. For occupancies in Group R-3 and R-5, and within individual *dwelling units* in occupancies in Group R-2, where the top of the *guard* serves as a *handrail* on the open sides of *stairs*, the top of the *guard* shall be not less than 34 inches (864 mm) and not more than 38 inches (965 mm) measured vertically from a line connecting the leading edges of the treads."

4. Revise the exceptions to Section 1015.4 to read:

"Exceptions:

1. From a height of 36 inches (914 mm) to 42 inches (1067 mm), *guards* shall not have openings that allow passage of a sphere 6 inches (152 mm) in diameter.
2. The triangular openings at the open sides of a *stair*, formed by the riser, tread and bottom rail shall not allow passage of a sphere 6 inches (152 mm) in diameter.
3. At elevated walking surfaces for access to and use of electrical, mechanical or plumbing systems or equipment,

guards shall not have openings that allow passage of a sphere 21 inches (533 mm) in diameter.

4. In areas that are not open to the public within occupancies in Group I-3, F, H, M or S; for *alternating tread devices* and ship's ladders; and in similar locations where children under the age of 5 are unlikely to be present that are *approved* by the *building official*, *guards* shall not have openings that allow passage of a sphere 21 inches (533 mm) in diameter.

5. In assembly seating areas, *guards* required at the end of aisles in accordance with Section 1029.17.4 shall not have openings that allow passage of a sphere 4 inches (102 mm) in diameter up to a height of 26 inches (660 mm). From a height of 26 inches (660 mm) to 42 inches (1067 mm) above the adjacent walking surfaces, *guards* shall not have openings that allow passage of a sphere 8 inches (203 mm) in diameter.

6. Within individual *dwelling units* and *sleeping units* in Group R-2, R-3 and R-5 occupancies, *guards* shall not have openings that allow passage of a sphere 4 3/8 (111 mm) inches in diameter."

5. Revise Section 1015.5 to read:

"1015.5 Screen porches.

Porches and *decks* that are enclosed with insect screening shall be provided with *guards* where the walking surface is located more than 30 inches (762 mm) above the floor or grade below at any point within 36 inches (914 mm) horizontally to the edge of the screening."

6. Revise Section 1015.8 and its subparts to read:

"1015.8 Window openings.

Where the bottom of the clear opening of an operable window in a *dwelling unit*, *sleeping unit* or common area in a Group R occupancy is located less than 36 inches (914 mm) above the finished floor and more than 72 inches (1829 mm) above the finished grade or other surface below on the exterior of the building, the window shall comply with one of the following:

1. Operable windows where the bottom of the clear opening is located more than 75 feet (22 860 mm) above the finished grade or other surface below and that are provided with window fall prevention devices that comply with ASTM F2006.
2. Operable windows where the openings will not allow a 4-inch-diameter (102 mm) sphere within 36 inches (914 mm) of the finished floor or a 6-inch-diameter (152 mm) sphere within 42 inches (1067 mm) of the finished floor to pass through the opening when the window is in its largest opened position.
3. Operable windows where the openings are provided with window fall prevention devices that comply with ASTM F2090.
4. Operable windows that are provided with window opening control devices that comply with ASTM F2090.

Exception: In Group R-2 and R-5 occupancies this section only applies where either:

1. The bottom of the clear opening of an operable window is located less than 24 inches (610 mm) above the finished floor and at least 72 inches (1829 mm) but not more than 20 feet (6096 mm) above the finished grade or other surface below on the exterior of the *building*.
2. The bottom of the clear opening of an operable window is located less than 36 inches (914 mm) above the finished floor and more than 20 feet (6096 mm) above the finished grade or other surface below on the exterior of the *building*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 46; Amend Coun. J. 12-14-22, p. 58278, Art. III, § 4)

14B-10-1016 Exit access.

The provisions of Section 1016 of IBC are adopted by reference with the following modification:

1. Revise item 1 of Section 1016.2 by deleting the second sentence.
2. Revise the exception to item 5 of Section 1016.2 by replacing "kitchen" with "kitchen, laundry or open storage area."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 47; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 18)

14B-10-1017 Exit access travel distance.

The provisions of Section 1017 of IBC are adopted by reference with the following modifications:

1. Revise note a to Table 1017.2 to delete "Section 402: For the distance limitation in malls."
2. Revise note e to Table 1017.2 to replace "Group R-3 and R-4" with "Group R-3, R-4 and R-5."
3. Delete the exception to Section 1017.3.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 48)

14B-10-1018 Aisles.

The provisions of Section 1018 of IBC are adopted by reference without modification.

14B-10-1019 Exit access stairways and ramps.

The provisions of Section 1019 of IBC are adopted by reference with the following modifications:

1. Revise items 2, 3 and 4 in Section 1019.3 to read:

"2. In Group R-1, R-2, R-3 or R-5 occupancies, *exit access stairways* and *ramps* connecting four stories or less serving and contained within an individual *dwelling unit* or *sleeping unit* or *live/work unit*.

3. *Exit access stairways* connecting three stories or less serving and contained within a Group R-3 *congregate living facility* or a Group R-4 facility are not required to be enclosed.

4. [Reserved.]"
2. Revise item 7 in Section 1019.3 to delete "*smoke-protected* or."

14B-10-1020 Corridors.

The provisions of Section 1020 of IBC are adopted by reference with the following modifications:

1. Delete exception 1 to Section 1020.1.
2. Replace Table 1020.1 as shown:

TABLE 1020.1

CORRIDOR FIRE-RESISTANCE RATING

OCCUPANCY	OCCUPANT LOAD SERVED BY CORRIDOR	REQUIRED FIRE-RESISTANCE RATING (hours)	
		Without sprinkler system	With sprinkler system
OCCUPANCY	OCCUPANT LOAD SERVED BY CORRIDOR	REQUIRED FIRE-RESISTANCE RATING (hours)	
		Without sprinkler system	With sprinkler system
H-1, H-2, H-3	All	Not Permitted	1
H-4, H-5	Greater than 30	Not Permitted	1
A, B, F, M, S, U	Greater than 30	1	0 ^e
E	All	1	1
R	Greater than 10	1	1
I-2 ^a	All	Not Permitted	0
I-1, I-3	All	Not Permitted	1 ^b
I-4	All	1	0

- a. For requirements for occupancies in Group I-2, see Sections 407.2 and 407.3.
- b. For a reduction in the *fire-resistance rating* for occupancies in Group I-3, see Section 408.8.
- c. [Reserved.]
- d. [Reserved.]
- e. In Groups A-1, A-2, A-3, and A-4 occupancies, *corridors* serving and visually separated from a room or suite of rooms with an *occupant load* of 300 or more shall have a *fire-resistance rating* of not less than 1 hour.

2.1. Revise Section 1020.1.1 by replacing "Section 3006.2.1" with "Section 3002.1.1."
3. Replace Table 1020.2 as shown:

TABLE 1020.2

MINIMUM CORRIDOR WIDTH

OCCUPANCY	MINIMUM WIDTH (inches)
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OCCUPANCY	MINIMUM WIDTH (inches)
Any facility not listed in this table	44
Access to and utilization of mechanical, plumbing or electrical systems or equipment	24
With an <i>occupant load</i> of less than 50	36
Within or serving a single <i>dwelling unit</i>	36
In Group E-1 <i>corridors</i> serving classrooms, study rooms or assembly rooms	60 ^a
In <i>corridors</i> and areas serving stretcher traffic in <i>ambulatory care facilities</i>	72
Group I-2 in areas where required for bed movement	96

For SI: 1 inch = 25.4 mm.

a. When lockers are installed in corridors, the full required clear width of the corridor shall be provided between locker doors when open.

4. Revise exception 2 to Section 1020.4 to read:

"2. In occupancies in Groups B, E-2, F, I-1, M, R, S and U, where the building is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1, the length of the dead-end *corridors* shall not exceed 50 feet (15 240 mm)."

5. Revise exception 1 to Section 1020.5 by deleting "smoking lounges."

6. Add new exception 5 to Section 1020.5 to read:

"5. Use of a *corridor* as a source of makeup air for exhaust systems in Group R-2 *dwelling units* and *sleeping units* that open directly onto such *corridors* shall be permitted, provided that each such *corridor* is directly supplied with outdoor air at a rate greater than the rate of makeup air taken from the *corridor* and the *dwelling units* and *sleeping units* are provided with an independent source of air for *ventilation*."

7. Delete item 5 in Section 1020.5.1.

8. Add a new Section 1020.7 to read:

"1020.7 Toilet rooms.

Separation shall not be required between a toilet room and a fire-resistance-rated *corridor* where the toilet room meets the following criteria:

1. The toilet room shall be separated from the remainder of the building by fire-resistance-rated construction meeting the requirements of Section 1020.1.
2. The toilet room shall not directly connect to any *habitable space*.
3. The toilet room shall not contain any fuel-fired equipment or appliances."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 49; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 24; Amend Coun. J. 1-24-24, p. 8634, Art. IV, § 4)

14B-10-1021 Egress balconies.

The provisions of Section 1021 of IBC are adopted by reference with the following modification:

1. Revise Section 1021.4 to read:

"1021.4 Location.

Exterior egress balconies shall have a minimum *fire separation distance* of 10 feet (3048 mm) measured at right angles from the exterior edge of the egress balcony to:

1. The closest *abutting property line*.
2. The far side of a *public way*.
3. The *exterior walls* of other buildings on the same lot with openings that are not protected in accordance with Section 705 based on *fire separation distance*."

14B-10-1022 Exits.

The provisions of Section 1022 of IBC are adopted by reference without modification.

14B-10-1023 Interior exit stairways and ramps.

The provisions of Section 1023 of IBC are adopted by reference with the following modifications:

1. Revise section 1023.2 by replacing "*basements*" with "*basements and occupiable rooftops* (even if not counted as *stories* for other purposes)."
2. Delete exception 3 to Section 1023.3.1.
3. Revise item 3 in Section 1023.5 to read:
"3. Security systems covering the *interior exit stairway* or *ramp*."
4. Delete the exception to Section 1023.5.
5. Revise Section 1023.8 by replacing "an *approved*" with "a."
- 5.1. Insert a new Section 1023.9.2 to read:

"1023.9.2 Designation.

Stairways shall be designated using letters indicating the compass direction of the location of the *stairway* within the building.

Exception: Letter-based designation schemes approved by the *fire code official*."

6. Delete Section 1023.10.
7. Delete Section 1023.11 and its subsections.

(Amend Coun. J. 6-25-21, p. 32180, Art. II, § 19)

14B-10-1024 Exit passageways.

The provisions of Section 1024 of IBC are adopted by reference with the following modifications:

1. Revise item 3 in Section 1024.6 to read:
"3. Security systems covering the *exit passageway*."
2. Delete the exception to Section 1024.6.

14B-10-1025 IBC Section 1025.

The provisions of Section 1025 of IBC are not adopted.

14B-10-1026 Horizontal exits.

The provisions of Section 1026 of IBC are adopted by reference with the following modification:

1. Delete the exception to Section 1026.2.
2. Revise the exception to Section 1026.4.2 by replacing the initial "The" with "In Group I-3, the".

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 50)

14B-10-1027 Exterior exit stairways and ramps.

The provisions of Section 1027 of IBC are adopted by reference with the following modifications:

1. Revise Section 1027.2 to read:

"1027.2 Use in a means of egress.

Exterior exit stairways shall not be used as an element of a required *means of egress* for Group I-2 occupancies. For occupancies in other than Group I-2, *exterior exit stairways* and *exterior exit ramps* shall be permitted as an element of a required *means of egress* for levels not more than 45 feet (13.7 m) above grade adjacent to the *exterior exit stairway*.

1027.2.1 Limitation.

Not more than 50 percent of the number and minimum width or required capacity of *means of egress* components shall be provided using *exterior exit stairways* or *exterior exit ramps*.

Exceptions:

1. *Means of egress* components only serving an *occupiable rooftop*.
2. *Means of egress* components only serving occupancies that are open to the atmosphere, such as *open parking garages* or Group A-5 occupancies.
3. *Means of egress* components serving a single *basement*, the first *story above grade plane*, or the second *story above grade plane*."

2. Revise Section 1027.5 to read:

"1027.5 Location.

Exterior exit stairways and exterior exit ramps shall have a minimum separation distance of 10 feet (3048 mm) measured at right angles from the exterior edge of the stairway or ramps, including landings, to:

1. The closest *abutting property line*.
2. The far side of a *public way*.
3. The *exterior walls* of other buildings on the same *lot* with openings that are not protected in accordance with Section 705 based on *fire separation distance*.

Exceptions:

1. *Exterior exit stairways, porches and exterior exit ramps serving not more than 4 stories or occupiable rooftops in Group R-2, R-3 and R-5 occupancies shall have a minimum separation distance of 3 feet (914 mm) if constructed of combustible materials; a separation distance is not required for exterior exit stairways, porches and exterior exit ramps constructed of entirely noncombustible materials.*
2. *On any side of an exterior exit stairway or exterior exit ramp protected by a solid wall with a fire-resistance rating of at least one hour that extends at least 3 feet (914 mm) beyond the exterior edge of adjacent sides and 8 feet (2438 mm) above the highest landing, the separation distance may be reduced to 3 feet (914 mm).*
3. *Exterior exit stairways and exterior exit ramps serving occupiable rooftops above portions of buildings not more than two stories above grade plane shall have a minimum separation distance of 5 feet (1525 mm).*
4. *Exterior exit stairways and ramps serving occupiable rooftops above portions of buildings not more than one story above grade plane and not more than 15 feet (4572 mm) above adjoining grade are not required to have a separation distance."*

3. Revise exception 1 to Section 1027.6 to read:

"1. Separation from the interior of the *building* is not required for occupancies, other than Group R-1, in *buildings* that are not more than two *stories above grade plane* where a *level of exit discharge* serving such occupancies is the first *story above grade plane*.

4. Revise exception 2 to Section 1027.6 by deleting "*approved*."

5. Revise exception 4 to Section 1027.6 to read:

"4. In Group R-2, R-3, and R-5 occupancies not more than four *stories above grade plane*, *exterior exit stairways and exterior exit ramps* are not required to be separated from the interior of the *building* and openings are not limited where the *exterior exit stairway or exterior exit ramp* discharges directly to grade."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 51; Amend Coun. J. 1-26-22, p. 43756, Art. II, § 10)

14B-10-1028 Exit discharge.

The provisions of Section 1028 of IBC are adopted by reference with the following modifications:

1. Revise exception 1 to Section 1028.1 by replacing "through areas on the level of discharge" with "through a lobby or similar common area on the *level of exit discharge*."

1.1. Revise exception 1.1 to Section 1028.1 by replacing "from the point of termination of the enclosure." with "within 25 feet (7620 mm) and no more than two turns, each not exceeding 90 degrees (1.57 rad), from the point of termination of the enclosure. There shall be no dead-end *corridors* or similar conditions exceeding 10 feet (3048 mm) in length adjacent to the point of termination of the enclosure."

1.2. Revise exception 1.3 to Section 1028.1 by deleting "*approved*."

2. Add a new exception 4 to Section 1028.1 to read:

"4. Up to 100 percent of the number, minimum width, and required capacity of *interior exit stairways and exterior exit ramps* is permitted to egress through a lobby or similar common area on the *level of exit discharge* provided that all of the following conditions are met:

4.1. Discharge of *interior exit stairways and exterior exit ramps* shall be provided with a free and unobstructed path of travel to at least two distinct exterior *exit doors* and each exterior *exit door* is readily visible and identifiable within 25 feet (7620 mm) and no more than two turns, each not exceeding 90 degrees (1.57 rad), from the point of termination of the enclosure. There shall be no dead-end *corridors* or similar conditions exceeding 10 feet (3048 mm) in length adjacent to the point of termination of the enclosure.

4.2. The entire area of the *level of exit discharge* is separated from areas below by construction conforming to the *fire-resistance rating* for the enclosure.

4.3. The building is protected throughout by an *automatic sprinkler system* installed in accordance with Section 903.3.1.1 or 903.3.1.2.

4.4. *Means of egress* components are separated in accordance with Exception 1.4."

3. Revise Section 1028.3 to read:

"1028.3 Exit discharge components.

Exterior *exit discharge* components shall be sufficiently open to the atmosphere so as to minimize the accumulation of smoke and toxic gases."

4. Revise Section 1028.4.1 by replacing "Group R-3 and U occupancies" with "Group R-3, R-5 and U occupancies and Group R-2 occupancies in buildings with no more than 4 *stories above grade plane*."

5. Revise exception 2 to Section 1028.4.2 by replacing "Group R-3" with "Group R-3 or R 5."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 52; Amend Coun. J. 1-26-22, p. 43756, Art. II, § 11; Amend Coun. J. 12-14-22, p. 58278, Art. III, § 5)

14B-10-1029 Assembly.

The provisions of Section 1029 of IBC are adopted by reference with the following modifications:

1. Delete the exceptions to Section 1029.1.1.1.
2. Revise Section 1029.2 by replacing "10 feet (3048 mm)" with "20 feet (6096 mm)."
3. Delete Section 1029.3.
4. Revise Section 1029.6 to read:

"1029.6 Capacity of aisle for assembly.

The required capacity of *aisles* shall be not less than that determined in accordance with Section 1029.6.1 where indoor assembly seating is provided and Section 1029.6.3 where *open-air assembly seating* is provided."

5. Revise the title and first sentence of Section 1029.6.1 to read:

"1029.6.1 Indoor assembly seating.

The required capacity in inches (mm) of the *aisles* for indoor assembly seating shall be not less than the *occupant load* served by the egress element in accordance with all of the following, as applicable:"

6. Delete Section 1029.6.2 and its subsections.
7. Delete Table 1029.6.2.
8. [Reserved.]
9. [Reserved.]
10. Delete exception 1 (and its subparts) to Section 1029.7.
11. Revise exception 2 to Section 1029.8 by deleting "*smoke-protected* or."
12. Revise the exception to Section 1029.8.1 by deleting "*smoke-protected* or."
13. Revise exceptions 3 and 4 to Section 1029.9.5 by deleting "*smoke-protected* or."
14. Revise the exception to Section 1029.13.2.1 by deleting "*smoke-protected* or."
15. Revise Table 1029.13.2.1 by deleting "SMOKE-PROTECTED OR."
16. Revise the exception to Section 1029.13.2.2 by deleting "*smoke-protected* or."
17. Revise exception 3 to Section 1029.15 by replacing "three" with "five."
18. Revise exception 4 to Section 1029.15 to read:

"4. In a *building*, room or space used for assembly purposes where flexibility of the seating arrangement is an integral part of the design and function of the space and seating is on tiered levels, up to 200 seats shall not be required to be fastened to the floor. Plans showing seating, tiers and *aisles* shall be submitted to the *fire code official* for review."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 53)

14B-10-1030 IBC Section 1030.

The provisions of Section 1030 of IBC are not adopted.

The following language is adopted as a new Section 1031:

"1031. FRONTAGE AND FIRE DEPARTMENT ACCESS REQUIREMENTS FOR GROUP E-1

1031.1 Frontage requirement.

The portion of a building containing a Group E-1 occupancy shall abut the number and type of *public ways* or *yards* specified in Table 1031.1. For purposes of this section, each side shall be at least 12 percent of the perimeter of the occupancy. The width of required *public ways* and *yards* shall be measured as provided in Section 506.3.

TABLE 1031.1

FRONTAGE REQUIREMENTS

Occupant Load of Group E-1 Occupancy	Minimum Number of Sides Abutting a Public Way or Yard	
	Public Way at Least 30 Feet Wide	Public Way or Yard at Least 10 Feet Wide
Up to 100	1	0
101 to 500	1	1
501 to 1,000	1	2
1,001 to 2,500	2	1
More than 2,500	3	1

For SI: 1 foot = 304.8 mm.

1031.2 Fire department access panels.

Where the *exterior wall* on any *story* of a Group E-1 occupancy exceeds 100 linear feet (30.5 m) without windows, doorways or similar openings, fire department access panels shall be provided at intervals not exceeding 50 feet (15.2 m).

Exception: *Exterior walls* of auditoriums, assembly halls, field houses, gymnasiums, swimming pools, and theater areas.

1031.2.1 Dimensions.

Fire department access panels shall be not less than 32 inches (812.8 mm) wide and 48 inches (1219.2 mm) high. Fire department access panels shall be located not more than 32 inches (812.8 mm) above the floor.

1031.2.2 Construction.

Fire department access panels shall be *approved* by the *fire code official* and constructed of materials and installed in such a manner to be readily removed by the fire department.

(Added Coun. J. 10-7-20, p. 21791, Art. II, § 25)

CHAPTER 14B-11

ACCESSIBILITY

- 14B-11-1100 Defined terms.
- 14B-11-1101 General.
- 14B-11-1102 Compliance.
- 14B-11-1103 Scoping requirements.
- 14B-11-1104 Accessible route.
- 14B-11-1105 Accessible entrances.
- 14B-11-1106 Parking and passenger loading facilities.
- 14B-11-1107 Dwelling units and sleeping units.
- 14B-11-1108 Special occupancies.
- 14B-11-1109 Other features and facilities.

14B-11-1110 Recreational facilities.

14B-11-1111 Signage.

14B-11-1100 Defined terms.

Where provisions of Chapter 11 of IBC are adopted by reference, the following terms shall be reset in italic type: "accessible"; "building"; "disability"; "element"; "entrance"; "facility"; "space"; "structure"; "temporary" and "vehicular way."

14B-11-1101 General.

The provisions of Section 1101 of IBC are not adopted. The following language is adopted as Section 1101:

"1101. GENERAL

1101.1 Scope.

The provisions of this chapter and Appendix E shall control the design and construction of *facilities* for *accessibility* for individuals with *disabilities*.

1101.2 Rulemaking.

The Commissioner of the Mayor's Office for People with Disabilities and the *building official* may jointly promulgate rules necessary to implement the requirements of this chapter. Rules promulgated under this section shall comply with the public notice and comment procedures in Section 104.4 of the *Chicago Construction Codes Administrative Provisions* and be posted on the *City's* public website.

1101.2.1 Illinois Accessibility Code.

The Commissioner of the Mayor's Office for People with Disabilities may promulgate rules to enforce a specific provision of the *Illinois Accessibility Code* if the Commissioner determines that the provision requires a greater degree of access and usability for individuals with *disabilities* than the corresponding provision of this code.

1101.2.2 Projects owned or financed by a governmental unit.

The Commissioner of the Mayor's Office for People with Disabilities and the Commissioner of Housing may jointly promulgate rules for housing that is owned or financed by a *governmental unit*. Such rules may require a greater degree of access and usability for individuals with *disabilities* than is provided for in this chapter.

1101.3 Notice and warning.

While efforts have been made to coordinate the *accessibility* requirements of this code with similar laws, this code has not been certified by the federal government or State of Illinois as a safe-harbor under the Americans with Disabilities Act ("ADA"), the Rehabilitation Act, the Fair Housing Act or any State of Illinois accessibility laws, including but not limited to the Environmental Barriers Act, and may not conform in all respects with those laws. Therefore, compliance with this code does not assure compliance with the ADA, the Rehabilitation Act, the Fair Housing Act or any other federal or state laws, or any rules promulgated with respect to such laws. The City of Chicago is not responsible for enforcement of the ADA, Rehabilitation Act, Fair Housing Act or such other federal or state laws, except as specifically provided in this code. *Permit* applications submitted under this code will be reviewed and permitted work will be inspected for compliance with the *Chicago Construction Codes* and will not be evaluated for compliance with the requirements of the ADA, Rehabilitation Act, Fair Housing Act or other federal or state laws, except as specifically provided in this code. Therefore, it is the responsibility of the developer or *owner* to have plans or *facilities* independently evaluated by knowledgeable professionals in order to meet or exceed the applicable requirements of applicable federal and state accessibility laws. Similarly, compliance with federal and state laws does not establish compliance with this code.

Exception: Where work is done by, on behalf of or with financial assistance provided by the *City*, the Commissioner of the Mayor's Office for People with Disabilities shall prescribe procedures to ensure the work complies with the *City's* obligations under federal and state law and Section 1101.2.2.

1101.4 Modifications to ICC A117.1.

The text of ICC A117.1 shall be modified as indicated in Sections 1101.4.1 through 1101.4.8.

1101.4.1 ICC A117.1 Section 302.1.

Modify Section 302.1 by adding exceptions to read as follows:

- "1. Animal containment areas shall not be required to comply with this section or with Section 303.
2. Areas of sport activity shall not be required to comply with this section or with Section 303."

1101.4.2 ICC A117.1 Section 404.2.8.

Modify Section 404.2.8 by adding Item 3 to read as follows:

- "3. Exterior hinged doors and gates: 8.5 pounds (37.8N) maximum"

1101.4.3 ICC A117.1 Section 502.

Modify Section 502 by replacing it in its entirety with Section 502 of the *Illinois Accessibility Code*.

1101.4.4 ICC A117.1 Section 604.3.

Modify Section 604.3 by adding a new Section 604.3.4 to read as follows:

"604.3.4 Changes in Level. Floor or ground surfaces within the required clearance shall comply with Section 302. Changes in level are not permitted.

EXCEPTION: Slopes not steeper than 1:48 shall be permitted."

1101.4.5 ICC A117.1 Section 607.2.

Modify Section 607.2 by adding the following sentence and exception at the end:

"Floor or ground surfaces within the required clearance shall comply with Section 302. Changes in level are not permitted.

EXCEPTION: Slopes not steeper than 1:48 shall be permitted."

1101.4.6 ICC A117.1 Section 607.6.

Modify Section 607.6 to read:

"A hand shower with a hose 59 inches (1500 mm) minimum in length that can be used as both a fixed shower head and as a hand shower shall be provided. The shower spray unit shall be located within one or more of the reach ranges specified in Section 308. The hand shower shall have a control with a nonpositive shut-off feature. Where provided, an adjustable-height hand shower mounted on a vertical bar shall be installed so as to not obstruct the use of grab bars. Shower spray units shall deliver water that does not exceed the maximum temperature allowed by the *Chicago Plumbing Code*.

EXCEPTION: Slopes not steeper than 1:48 shall be permitted."

1101.4.7 ICC A117.1 Section 608.2.

Modify Section 608.2 by adding the following sentence and exception at the end:

"Changes in level are not permitted.

EXCEPTION: Slopes not steeper than 1:48 shall be permitted."

1101.4.8 ICC A117.1 Section 802.

Modify Section 802.1 to read as follows:

"802.1 General. Wheelchair spaces and wheelchair space locations in assembly areas with spectator seating shall comply with Section 802. Where tiered seating includes dining surfaces or work surfaces, wheelchair spaces and wheelchair space locations shall comply with Section 802.6, 802.7, 802.9, 802.10 and 902. Team and player seating shall comply with Sections 802.2 through 802.7."

Modify Section 802.7.2 to read as follows:

"802.7.2 Companion Seat Alignment. In row seating, the companion seat shall be located to provide shoulder alignment with the wheelchair space occupant. The shoulder of the wheelchair space occupant is considered to be 36 inches (915 mm) or more from the front and 12 inches (305 mm) or more from the rear of the wheelchair space. The floor surface for the companion seat shall be at the same elevation as the wheelchair space floor surface.

EXCEPTION: Companion seat alignment shall not be required in tiered seating that includes dining surfaces or work surfaces."

Modify Section 802.8.2 to read as follows:

"802.8.2 Identification. Each designated aisle seat shall be identified by a sign or marker."

Modify Section 802.10.1, excluding its exception, to read as follows:

"802.10.1 Horizontal Dispersion. Wheelchair space locations shall be dispersed horizontally to provide viewing options. Where seating encircles the stage or field, in whole or in part, horizontal dispersion shall include the entire seating area. Two wheelchair spaces shall be permitted to be located side-by-side."

Modify Section 802.10.4.1 to read as follows:

"802.10.4.1 Spaces with Seating on Risers. Where tiered seating is provided, wheelchair space locations shall be integrated into the tiered seating area on a riser or a cross-aisle."

Modify Section 802.10.4.2 to read as follows:

"802.10.4.2 Distance from the Screen. Wheelchair space locations shall be located in accordance with one of the following:

1. Within the rear 60 percent of the seats provided: or
2. Located within the area of an auditorium in which the vertical viewing angles, as measured to the top of the screen, are from the 40th to the 100th percentile of vertical viewing angles for all seats as ranked from the seats in the first row (1st percentile) to seats in the back row (100th percentile)."

1101.4.9 ICC A117.1 Section 803.

Modify Section 803 by adding a new Section 803.6 to read as follows:

"803.6 Mirrors. Mirrors provided within the room shall be installed with the bottom edge of the reflecting surface 35 inches (1220 mm) maximum above the finish floor or ground."

1101.4.10 IAC A117.1 Section 804.5.2.

Modify Section 804.5.2 by adding a new Exception 3 to read as follows:

"3. Filters in dishwashers and refrigerators and other parts or controls used only for periodic maintenance of appliances shall not be required to comply with Section 309."

1101.4.11 ICC A117.1 Section 1003.11.2.

Modify Section 1003.11.2 by adding a new Section 1003.11.2.6 to read as follows:

"1003.11.2.6 Toilet Paper Dispenser. Toilet paper dispensers shall comply with Section 604.7."

1101.4.12 ICC A117.1 Sections 1002.14 and 1003.14.

Modify Sections 1002.14 and 1003.14 by revising the exception to each section to read as follows:

"EXCEPTIONS:

1. In kitchens, at least 5 percent, but no less than one of each type of storage facility provided, shall comply with Section 905. Types of storage facilities include: drawers, open shelves, and shelves in cabinets.
2. In kitchens, the lowest shelf of a storage facility mounted above a work surface complies with Section 905 if it can be relocated so that at least that shelf is 48 inches (1220 mm) maximum above the floor without damaging the storage facility or adjacent cabinetry, walls, wall finish, doors and structural *elements*. The wall behind the relocatable storage facility shall be finished."

1101.4.13 ICC A117.1 Section 1005.4.

Modify Section 1005.4, including the exception, to read as follows:

"1005.4 Interior Spaces. The *entrance* level shall include a toilet room or bathroom complying with Section 1005.6, one food preparation area complying with Section 1005.7, and a habitable space, other than the food preparation area, with a floor area at least 70 square feet (6.5 m²)."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 54; Amend Coun. J. 1-26-22, p. 43756, Art. II, § 12)

14B-11-1102 Compliance.

The provisions of Section 1102 of IBC are adopted by reference with the following modification:

1. Add a new Section 1102.1.1 to read:

"1102.1.1 Equivalent facilitation.

Nothing in this chapter prevents the use of designs, products or technologies as alternatives to those prescribed, provided they result in substantially equivalent or greater access and usability for individuals with *disabilities*."

14B-11-1103 Scoping requirements.

The provisions of Section 1103 of IBC are adopted by reference with the following modifications:

1. Add a new Section 1103.1.1 to read:

"1103.1.1 Multiple purposes.

An area, room or *space* that is intended to be occupied at different times for different purposes shall comply with all requirements that are applicable to each of the purposes for which the area, room or *space* will be occupied."

2. Revise Section 1103.2.3 to read:

"1103.2.3 Group R-5.

Group R-5 occupancies, their accessory *structures* and their associated *sites* and *facilities* are not required to comply with this

chapter, except as required by Sections 1107.8 and 1107.9."

3. Revise Section 1103.2.5 to read:

"1103.2.5 Construction sites.

Structures, sites and equipment directly associated with the actual processes of construction including, but not limited to, scaffolding, bridging, materials hoists, materials storage or construction trailers are not required to comply with this chapter. Toilet or bathing facilities provided for use exclusively by construction personnel on a construction *site* are not required to comply with this chapter. *Temporary* pedestrian circulation routes around and adjacent to a construction *site* shall comply with Chapter 33."

4. Delete Section 1103.2.8.

14B-11-1104 Accessible route.

The provisions of Section 1104 of IBC are adopted by reference with the following modifications:

1. Revise the exception to Section 1104.1 to read:

"Exception: Other than in *buildings* or *facilities* containing or serving *Type B units*, an *accessible route* shall not be required between *site* arrival points and the *building* or *facility entrance* if the only means of access between them is a *vehicular way* not providing for pedestrian access. Access from *site* arrival points is permitted to include *vehicular ways*. Where a *vehicular way* or a portion of a *vehicular way* is provided for pedestrian travel, such as within a shopping center or shopping mall parking lot, this exception shall not apply."

2. Revise exception 1 to Section 1104.2 to read:

"1. An *accessible route* is not required between *accessible buildings*, *accessible facilities*, *accessible elements* and *accessible spaces* where all of the following conditions are met:

1.1. The only means of access between them is a *vehicular way* not providing for pedestrian access.

1.2. Due to circumstances outside the control of the *owner*, either the slope of the finished ground level between *accessible facilities* and *buildings* exceeds 1:12 or physical or legal restrictions prevent the installation of an *accessible route*.

1.3. Parking that complies with Section 1106 is provided at each *accessible building*, *facility*, *element* or *space*."

3. Revise exception 2 to Section 1104.3.1 to read:

"2. *Common use circulation paths*, located within *employee work areas*, that are an integral component of *work area equipment*, shall not be required to be *accessible routes*."

4. Revise Section 1104.4 to read:

"1104.4 Multistory buildings and facilities.

At least one *accessible route* shall connect each *accessible story*, *mezzanine* and *occupiable rooftops* in multilevel *buildings* and *facilities*.

Exceptions:

1. An *accessible route* is not required to a *basement*, *second story above grade plane* or *mezzanine* that has an area of not more than 1,000 square feet (93 m²) and is either the *second story* of a two- *story building*, the *mezzanine* of a one-*story building*, or the *basement* of a one- or two-*story building*. This exception shall not apply to:

1.1. Shopping centers or shopping malls.

1.2. *Stories* or *mezzanines* containing offices of health care providers (Group B or I).

1.3. Passenger transportation *facilities* and airports (Group A-3 or B).

1.4. *Publicly-owned buildings*.

2. *Stories*, *mezzanines* or *occupiable rooftops* that do not contain *accessible elements* or other *spaces* as determined by Section 1107 or 1108 are not required to be served by an *accessible route* from an *accessible level*.

3. In air traffic control towers, an *accessible route* is not required to serve the cab and the floor immediately below the cab.

4. *Spaces* greater than 1,000 square feet (93 m²) but less than 3,000 square feet (280 m²) in areas that are used exclusively for archival storage or for product storage in a business or mercantile occupancy are not required to be on an *accessible route*."

(Amend Coun. J. 1-26-22, p. 43756, Art. II, § 13)

14B-11-1105 Accessible entrances.

The provisions of Section 1105 of IBC are adopted by reference with the following modifications:

1. Add a new exception 3 to Section 1105.1 to read:

"3. In *multi-story housing*, at least 50 percent of all *public entrances* shall be *accessible*."

2. Add a new Section 1105.2 to read:

"1105.2 Entrances to rooms or spaces.

Within a *building* or *facility*, all required doors, doorways and gates on an *accessible route* which serve *accessible* rooms or spaces shall be *accessible*."

3. Add a new Section 1105.3 to read:

"1105.3 Group I-2 and ambulatory care facilities.

In Group I-2 occupancies and Group B *ambulatory care facilities* at least one *accessible entrance* shall be protected from the weather by a canopy or roof overhang. Such *entrance* shall incorporate a passenger loading zone complying with Section 1106.7."

4. Add a new Section 1105.4 to read:

"1105.4 Power-operated doors.

At least one *accessible entrance* door to the following *facilities*, or each door where the *entrance* consists of doors in series, shall be a *power-operated door* complying with Section 1010.1.4.2:

1. Governmental *facilities* which contain or provide administrative services available to the general public.
2. Museums with a *gross floor area* of 5,000 square feet (465 m²) or more.
3. Passenger terminals.
4. *High-rise buildings* containing *Accessible units* or *Type A units*.
5. Group E-1 occupancies."

14B-11-1106 Parking and passenger loading facilities.

The provisions of Section 1106 of IBC are adopted by reference with the following modifications:

1. Revise Table 1106.1 by replacing "2% of total" with "2% of total (rounded up to nearest whole number)."
2. Revise Section 1106.2 to read:

"1106.2 Groups R-2, R-3 and R-4.

Accessible parking spaces shall be provided in Group R-2, R-3 and R-4 occupancies in accordance with Items 1 through 4 as applicable.

1. In Group R-2, R-3 and R-4 occupancies that are required to have *Accessible*, *Type A* or *Type B dwelling units* or *sleeping units*, *accessible* parking spaces shall be provided in compliance with Table 1106.2.
2. [Reserved.]
3. [Reserved.]
4. Where parking is provided within or beneath a building, *accessible* parking spaces shall be provided within or beneath the building.

TABLE 1106.2

ACCESSIBLE PARKING SPACES FOR R-2, R-3 AND R-4 OCCUPANCIES

TOTAL PARKING SPACES PROVIDED IN PARKING FACILITIES	REQUIRED MINIMUM NUMBER OF ACCESSIBLE SPACES
TOTAL PARKING SPACES PROVIDED IN PARKING FACILITIES	REQUIRED MINIMUM NUMBER OF ACCESSIBLE SPACES
1 to 50	1
51 to 100	2
101 to 150	3
151 to 200	4
201 to 250	5
251 to 300	6

301 to 350	7
351 to 400	8
401 to 450	9
451 to 500	10
501 and over	2% of total (rounded up to nearest whole number)

3. Revise the exception to Section 1106.5 to read:

"Exceptions:

1. In *private garages* that serve Group R-2, R-3 and R-4 occupancies, van-accessible parking spaces shall be permitted to have vehicular routes, *entrances*, parking spaces and access aisles with a minimum vertical clearance of 7 feet (2134 mm).
2. Van-accessible spaces are not required in a parking *facility* exclusively serving the residents of a Group R-2, R-3 or R-4 occupancy that contains zero *Accessible units* and no more than 19 *dwelling units* or *sleeping units* that are either *Type A units* or *Type B units*."

4. Revise the exceptions to Section 1106.6 to read:

"Exceptions:

1. In multilevel parking *structures*, van-accessible parking spaces required by Section 1106.5 are permitted on one level.
2. *Accessible* parking spaces shall be permitted to be located in different parking *facilities* if substantially equivalent or greater access and usability to people with *disabilities* is provided in terms of distance from an *accessible entrance* or *entrances*, parking fee and user convenience."

5. Revise Section 1106.7.3 by adding the following sentence at the end:

"If at-grade parking is provided, at least one *accessible* space for self-parking of a vehicle shall be provided."

(Amend Coun. J. 7-20-22, p. 50878, § 2)

14B-11-1107 Dwelling units and sleeping units.

The provisions of Section 1107 of IBC are adopted by reference with the following modifications:

1. Revise Section 1107.2 to read:

"1107.2 Design.

Dwelling units and *sleeping units* that are required to be *Accessible units*, *Type A units*, *Type B units* or *Type C units* shall comply with the applicable provisions of Chapter 10 of ICC A117.1. Units required to be *Type A units* may be designed and constructed as *Accessible units*. Units required to be *Type B units* may be designed and constructed as *Accessible units* or *Type A units*. Units required to be *Type C units* may be designed and constructed as *Accessible units*, *Type A units*, or *Type B units*."

1.1. Revise Section 1107.3, excluding the exceptions, to read:

"1107.3 Accessible spaces.

Rooms and *spaces* available to the general public or available for use by residents and serving *Accessible units*, *Type A units*, *Type B units* or *Type C units* shall be *accessible*. *Accessible spaces* shall include toilet and bathing rooms, kitchen, living and dining areas and any exterior *spaces*, including patios, terraces and balconies."

2. Revise Section 1107.4 to read:

"1107.4 Accessible route.

Not fewer than one *accessible route* shall connect *accessible building* or *facility entrances* with the primary *entrance* of each *Accessible unit*, *Type A unit*, *Type B unit* and *Type C unit* within the *building* or *facility* and with those exterior and interior *spaces* and *facilities* that serve the units.

Exceptions:

1. [Reserved.]
2. In Group I-3 facilities, an *accessible route* is not required to connect *stories* or *mezzanines* where *Accessible units*, all *common use* areas serving *Accessible units* and all *public use* areas are on an *accessible route*.
3. In Group R-2 facilities with *Type A units* complying with Section 1107.6.2.2.1, an *accessible route* is not required to connect *stories* or *mezzanines* where *Type A units*, all *common use* areas serving *Type A units* and all *public use* areas are on an *accessible route*. This exception does not apply to *multi-story housing*.

4. In other than Group R-2 housing at places of education, in Group R-2 facilities with *Accessible units* complying with Section 1107.6.2.3.1, an *accessible route* is not required to connect *stories* or *mezzanines* where *Accessible units*, all *common use* areas serving *Accessible units* and all *public use* areas are on an *accessible route*. This exception does not apply to *multi-story housing*.
5. In Group R-1, an *accessible route* is not required to connect *stories* or *mezzanines* within individual units, provided the *accessible* level meets the provisions for *Accessible units* and sleeping accommodations for two persons minimum and toilet and bathing facilities are provided on that level.
6. In *congregate living facilities* in Groups R-3 and R-4, an *accessible route* is not required to connect *stories* or *mezzanines* where *Accessible units* or *Type B units*, all *common use* areas serving *Accessible units* and *Type B units* and all *public use* areas serving *Accessible units* and *Type B units* are on an *accessible route*. This exception does not apply to *multi-story housing*.
7. An *accessible route* between *stories* is not required where *Type B units* are exempted by Section 1107.7."

3. Revise Section 1107.5 to read:

"1107.5 Group I.

Accessible units, *Type B units* and units with communication features shall be provided in Group I occupancies in accordance with Sections 1107.5.1 through 1107.5.5 and Appendix E."

4. Revise Section 1107.5.1.1 to read:

"1107.5.1.1 Accessible units.

At least 10 percent, but not less than one, of the *dwelling units* and *sleeping units* shall be *Accessible units*."

5. Add a new Section 1107.5.1.1.1 to read:

"1107.5.1.1.1 Beds.

In *sleeping units* having more than 25 beds, at least 5 percent of the beds shall be *accessible*."

6. Revise Section 1107.5.2.2 to read:

"1107.5.2.2 Type B units.

In *structures* with four or more *dwelling units* or *sleeping units*, every *dwelling unit* and *sleeping unit* shall be a *Type B unit*."

7. Revise Section 1107.5.5 to read:

"1107.5.5 Group I-3.

Accessible units and units with communication features shall be provided in Group I-3 occupancies in accordance with Sections 1107.5.5.1 through 1107.5.5.3 and Appendix E."

8. Add a new Section 1107.5.5.1.1 to read:

"1107.5.5.1.1 Beds.

In *sleeping units* having more than 25 beds, at least 5 percent of the beds shall be *accessible*."

9. Revise Section 1107.6 to read:

"1107.6 Group R.

Accessible units, *Type A units* and *Type B units* and units with communication features shall be provided in Group R occupancies in accordance with Sections 1107.6.1 through 1107.6.4 and Appendix E."

10. Revise Section 1107.6.1.1 to read:

"1107.6.1.1 Accessible units.

Accessible units shall be provided in accordance with Table 1107.6.1.1. On a *multiple-building site*, Each *structure* that contains more than 50 *dwelling units* or *sleeping units* shall be considered separately for determining the minimum number and type of *Accessible units* required. On a *multiple-building site*, all *structures* that individually contain 50 or fewer *dwelling units* or *sleeping units*, shall be considered together to determine the minimum number and type of *Accessible units* required. *Accessible units* shall be dispersed among the various classes of units."

11. Revise the title of Table 1107.6.1.1 to "ACCESSIBLE UNITS."

12. Add a new Section 1107.6.1.1.1 to read:

"1107.6.1.1.1 Beds.

In *sleeping units* having more than 25 beds, at least 5 percent of the beds shall be *accessible*."

13. Add a new Section 1107.6.1.3 to read:

"1107.6.1.3 Doors.

Entrances, doors and doorways providing user passage into and within all *dwelling units* and *sleeping units* shall comply with the clear width requirements for *accessible* doors. Door handles and pulls shall be *accessible*.

Exception: Shower and sauna doors in units that are not required to be *Accessible units*.

14. Revise Section 1107.6.2 and its subsections to read:

"1107.6.2 Groups R-2 and R-3.

Accessible units, *Type A units* and *Type B units* shall be provided in Group R-2 and R-3 occupancies in accordance with Sections 1107.6.2.1 through 1107.6.2.3.

1107.6.2.1 Live/work units.

In *live/work units* constructed in accordance with Section 419, the nonresidential portion is required to be *accessible*. In *multi-story housing*, the residential portions of at least 20 percent of the *live/work units* shall be *Type A units*. In a *structure* where there are four or more *live/work units*, the residential portions of the *live/work units* shall be *Type B units*.

Exception: The number of *Type B units* is permitted to be reduced in accordance with Section 1107.7.

1107.6.2.2 Apartment houses and congregate living facilities.

Type A units and *Type B units* shall be provided in apartment houses and *congregate living facilities*, other than housing at a place of education required to comply with Section 1107.6.2.3, in accordance with Sections 1107.6.2.2.1 and 1107.6.2.2.2. Bedrooms in *congregate living facilities* shall be counted as *sleeping units* for the purpose of determining the number of units. Where the bedrooms are grouped into *dwelling units* or *sleeping units*, only one bedroom in each *dwelling unit* or *sleeping unit* shall count toward the number of required *Type A units*.

Exception: Where the number of required *Type A units* exceeds the number of *dwelling units* and *sleeping units* and at least one bedroom in each *dwelling unit* and *sleeping unit* is a *Type A unit*, more than one bedroom in each *dwelling unit* and *sleeping unit* may count toward the number of required *Type A units*.

1107.6.2.2.1 Type A units.

1107.6.2.2.1.1 Multi-story housing

In *multi-story housing*, at least 20 percent of the *dwelling units* and *sleeping units* shall be *Type A units*.

1107.6.2.2.1.2 Other than multi-story housing.

In *buildings* three *stories* or less in height containing more than 20 *dwelling units* or *sleeping units*, at least 2 percent but not less than one of the units shall be a *Type A unit*.

Exceptions:

1. The number of *Type A units* is permitted to be reduced in accordance with Section 1107.7 exceptions allowed for *Type B units*.
2. In *buildings* three *stories* or less in height, *townhouses* are not required to be *Type A units*.

1107.6.2.2.1.3 Dispersion.

Type A units shall be distributed proportionally throughout the *building* to provide a variety of sizes and locations and of *affordable units* and market rate units.

1107.6.2.2.2 Type B units.

Where there are four or more *dwelling units* or *sleeping units* in a single *structure*, every *dwelling unit* and *sleeping unit* shall be a *Type B unit*.

Exception: The number of *Type B units* is permitted to be reduced in accordance with Section 1107.7.

1107.6.2.3 Housing at a place of education.

In housing at a place of education, *Accessible units* and *Type B units* shall be provided in accordance with Sections 1107.6.2.3.1 and 1107.6.2.3.2. Bedrooms within *congregate living facilities* shall be counted as *sleeping units* for the purpose of determining the number of units. Where the *bedrooms* are grouped into *dwelling units* or *sleeping units*, only one *bedroom* in each *dwelling unit* or *sleeping unit* shall be permitted to count toward the number of required *Accessible units*.

Exceptions:

1. Where the number of required *Type A units* exceeds the number of *dwelling units* and *sleeping units* and at least one bedroom in each *dwelling unit* and *sleeping unit* is a *Type A unit*, more than one bedroom in each *dwelling unit* and *sleeping unit* may count toward the number of required *Type A units*.
2. Housing at a place of education that is leased on a year-round basis exclusively to graduate students or faculty and that does not contain any *public use areas* or *common use areas* available for

educational programming shall comply with Section 1107.6.2.2.

1107.6.2.3.1 Accessible units and units with communication features.

Accessible units and units with communication features shall be provided in accordance with Section 1107.6.1.1 and Appendix E.

1107.6.2.3.2 Type B units.

Where there are four or more *dwelling units* or *sleeping units* in a single *structure*, every *dwelling unit* and every *sleeping unit* shall be a *Type B unit*.

Exception: The number of *Type B units* is permitted to be reduced in accordance with Section 1107.7."

15. Revise Section 1107.6.3, including the exception, to read:

"1107.6.3 Attached residential buildings.

Where there are four or more Group R *dwelling units* or *sleeping units* in a single *structure*, even if that *structure* is classified as multiple *buildings* under other provisions of this code, the *structure* shall comply with Section 1107.6.1 or 1107.6.2. Bedrooms within Group R *congregate living facilities* shall be counted as *sleeping units* for the purpose of determining the number of units."

16. Revise Section 1107.6.4.1 to read:

"1107.6.4.1 Accessible units.

In Group R-4, Condition 1, at least one of the *sleeping units* shall be an *Accessible unit*. In Group R-4, Condition 2, at least two of the *sleeping units* shall be *Accessible units*.

In Group R-4, Condition 1 or Condition 2, owned or financed by a *governmental unit*, at least 20 percent of the *sleeping units* shall be *Accessible units*."

17. Revise Section 1107.6.4.2, excluding the exception, to read:

"1107.6.4.2 Type B units.

In *structures* with four or more *sleeping units*, every *sleeping unit* shall be a *Type B unit*."

18. Revise Section 1107.7 to read:

"1107.7 General exceptions.

Where specifically permitted by Section 1107.5 or 1107.6, the required number of *Type B units* is permitted to be reduced in accordance with Sections 1107.7.1 through 1107.7.5. The distribution and features of *Type A units* may be altered in accordance with Sections 1107.7.6 and 1107.7.7."

19. Revise Section 1107.7.1 to read:

"1107.7.1 Structures without elevator service.

Where elevator service is not provided in a *structure*, only the *dwelling units* and *sleeping units* that are located on *stories* indicated in Sections 1107.7.1.1 and 1107.7.1.2 are required to be *Type B units*."

20. Revise the title of Section 1107.7.2 to "Multistory Type B units."

21. Delete Section 1107.7.5.

22. Insert new Sections 1107.7.6 and 1107.7.7 to read:

"1107.7.6 Multistory Type A units.

A *multistory unit* is not required to be a *Type A unit* if the required number of *Type A units* is met by other types of units distributed throughout the *building*.

1107.7.7 Multistory Type A units with provision for future elevator.

A *multistory unit* required to be a *Type A unit* may be constructed so that an *accessible route* to levels above or below the *entrance level* can be provided by the *owner* by the installation of a limited-use/limited application elevator, private residence elevator or platform lift when appropriate. Provisions for the future elevator or platform lift shall comply with Section 1107.8.3.1.3, method 1."

23. Add a new Section 1107.8 to read:

"1107.8 Residential planned developments.

Dwelling units and *sleeping units* in a *planned development* shall be *accessible* in accordance with Sections 1107.8.1 through 1107.8.6.

Exceptions:

1. *Dwelling units* and *sleeping units* in Group I occupancies shall comply with Section 1107.5 and shall not

be considered in determining the number of units in a *planned development* for purposes of this section.

2. *Dwelling units* and *sleeping units* in Group R-1 occupancies shall comply with Section 1107.6.1 and shall not be considered in determining the number of units in a *planned development* for purposes of this section.

3. *Dwelling units* and *sleeping units* in *multi-story housing* shall comply with Section 1107.6.2 and shall not be considered in determining the number of units in a *planned development* for purposes of this section.

Affordable units shall also comply with Section 1107.9.

1107.8.1 Construction documents.

The following information shall be included in the *construction documents*:

1. A *site plan* of the entire *planned development* identifying the location of units required to comply with Section 1107.8.
2. Alternative elevations for kitchens in units required to comply with Section 1107.8.2 showing *accessible* features after adaptation.
3. Alternative plans for the visitable detached single-family dwellings required by Section 1107.8.4, item 2, showing *accessible* features that comply with the requirements for a *Type A unit*.

1107.8.2 Type A units.

At least 10 percent of units, but not less than one, in a *planned development* shall be *Type A units*. The required *Type A units* shall be distributed proportionally among the various types and classes of units provided in the *planned development*.

Exceptions:

1. Detached single-family dwellings shall comply with Section 1107.8.4.
2. Instead of meeting the requirements for *Type A units*, attached *multistory units* with separate *means of egress* may be designed and constructed in accordance with Section 1107.8.3.
3. In a *building* where a *multistory unit* is located above another unit, an *accessible route* to the upper unit is not required if a shaft or unenclosed open well is provided to accommodate the future installation of a residential elevator complying with the *Chicago Conveyance Device Code*, or if a limited-use limited-application elevator complying with the *Chicago Conveyance Device Code* or a wheelchair lift complying with the *Chicago Conveyance Device Code* is provided.

Where a shaft or unenclosed open well is provided, the construction shall be structurally sufficient to allow future installation of the elevator or lift without any additional reinforcement or alteration of structural *elements*. Conduit for future electrical wiring and power for operation of the elevator or lift shall be provided. Where a shaft is provided, the space provided for the shaft is not restricted from being used for other purposes before the elevator or wheelchair lift is installed. The floor openings within the shaft are not restricted from being in-filled with flooring provided that the in-fill flooring can be removed at the time of installation of the elevator or lift without additional reinforcement or alteration of other structural *elements*.

4. A *multistory unit* is not required to be *accessible* if a single-story *Type A unit* of equivalent total floor area is provided as part of the *planned development*. For purposes of this section, "equivalent total floor area" means a *net floor area* within 70 square feet (6.5 m²).

1107.8.3 Attached multistory units with separate means of egress.

Attached *multistory units* with separate *means of egress* required to be *Type A units* may be designed and constructed as *Accessible units* or in accordance with Section 1107.8.3.1.

1107.8.3.1 Multistory unit standard.

Units shall be constructed as *Type C units* and shall comply with Sections 1107.8.3.1.1 through 1107.8.3.1.3.

1107.8.3.1.1 Toilet and bathing facility.

All toilet and bathing facilities shall comply with the requirements for toilet and bathing facilities in *Type A units*, however toilet and bathing facilities other than those required for *Type C units* shall be allowed to be located on levels not initially provided with vertical access. This provision is not intended to require more than one toilet and bathing facility within a unit.

1107.8.3.1.2 Kitchen.

At least one kitchen complying with the kitchen requirements for a *Type A unit* shall be provided, however the kitchen shall be allowed to be located on a level not initially provided with vertical access.

1107.8.3.1.3 Provision for vertical access.

Provisions for future access within the unit to all levels above and below the *entrance* level shall be provided by one of the following methods:

1. A shaft or unenclosed open well is provided to accommodate the future installation of a residential elevator or lift complying with the *Chicago Conveyance Device Code*. The construction shall be structurally sufficient to allow future installation of the elevator or lift without any additional reinforcement or alteration of structural *elements*. Conduit for future electrical wiring and power for operation of the elevator or lift shall be provided at the shaft or unenclosed open well. If a shaft is provided, the space occupied for a shaft is not restricted from being used for other purposes before the elevator or wheelchair lift is installed. The floor openings within the shaft or unenclosed open well may be in-filled with flooring if the in-fill flooring can be removed at the time of installation of the elevator or lift without additional reinforcement or alteration of other structural *elements*. Maneuvering clearances at landings where *entrances* to the elevator or lift occur shall not be required if the width of the *corridor*, hallway or other space in front of the shaft is not less than 36 inches (914 mm).
2. A *stairway* is designed to accommodate the future installation of a wheelchair lift that complies with the *Chicago Conveyance Device Code*. Maneuvering clearances at the top and bottom of the wheelchair lift shall not be required if the width of the *corridor*, hallway or other space in front of the *stairway* is not less than 36 inches (914 mm). Conduit for future electrical wiring and power for operation of a wheelchair lift shall be provided from the power source to a location at the *stairway* where the connection would be made when the lift is installed.
3. A limited-use limited-application elevator or a wheelchair lift complying with the *Chicago Conveyance Device Code* is provided.

1107.8.4 Type B units.

Where there are four or more *dwelling units* or *sleeping units* in a single *structure*, every *dwelling unit* and every *sleeping unit* shall be a *Type B unit*.

Exception: The number of *Type B units* is permitted to be reduced in accordance with Section 1107.7.

1107.8.5 Type C (visitable) units.

Type C units shall be provided in accordance with Section 1107.8.5.1 or 1107.8.5.2.

1107.8.5.1 Multi-unit buildings.

At least 10 percent of units, but not less than one, in multi-unit buildings in *planned developments* shall be *Type C units*. Units provided to satisfy Section 1107.8.2 or 1107.8.3 shall not be counted toward the number of *Type C units* required by this section.

Exception: In a *building* in which the minimum number of required *Type A units* has been provided, the number of *Type C units* is not required to exceed the remaining number of units, if any, on the levels that contain *Type A units*.

1107.8.5.2 Detached single-family dwellings.

At least 20 percent, but not less than one, of all detached single-family dwellings in a *planned development* shall be *Type C units*.

1107.8.6 Units with visible alarm notification appliances.

In *buildings* required to have a fire alarm system of a type that sends an alarm signal to occupants of more than just an individual *dwelling unit* or *sleeping unit*, not less than 20 percent of the *dwelling units* and *sleeping units* shall be provided with visible alarm notification appliances or provided with connections to allow the future installation of visible alarm notification appliances. These *dwelling units* and *sleeping units* shall be distributed proportionally among the various types and classes of units provided."

24. Insert a new Section 1107.9 to read:

"1107.9 Affordable units.

Affordable units shall be *accessible* in accordance with Sections 1107.9.1 through 1107.9.4.

1107.9.1 Type A units.

At least 20 percent of *affordable units*, but not less than one, shall be *Type A units*. The required *Type A units* shall be distributed proportionally among the various types and classes of units provided in the development.

1107.9.2 Type B units.

Where there are four or more *dwelling units* or *sleeping units* in a single *structure*, every *affordable unit* shall be a *Type B unit*.

Exception: The number of *Type B units* is permitted to be reduced in accordance with Section 1107.7.

1107.9.3 Type C (visitable) units.

At least 10 percent of *affordable units*, but not less than one, shall be *Type C units*. *Type A units* provided to satisfy Section 1107.9.1 shall not be counted toward the number of *Type C units* required by this section.

Exception: In a *building* in which the minimum number of required *Type A units* has been provided, the number of *Type C units* is not required to exceed the remaining number of units, if any, on the levels that contain *Type A units*.

1107.9.4 Units with visible alarm notification appliances.

In *buildings* required to have a fire alarm system of a type that sends an alarm signal to occupants of more than just an individual *dwelling unit* or *sleeping unit*, not less than 20 percent of the *affordable units* shall be provided with visible alarm notification appliances or provided with connections to allow the future installation of visible alarm notification appliances. These units shall be distributed proportionally among the various types and classes of *affordable units* provided."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 55; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 20; Amend Coun. J. 9-14-21, p. 35949, Art. XV, § 9)

14B-11-1108 Special occupancies.

The provisions of Section 1108 of IBC are adopted by reference with the following modification:

1. Add a new Section 1180.2.9.2 to read:

"1108.2.9.2 Raised platform.

In banquet rooms or *spaces* where a head table or speaker's lectern is located on a raised platform, an *accessible route* shall be provided to the platform. Open edges of a raised platform shall be protected by placement of tables or by a curb."

14B-11-1109 Other features and facilities.

The provisions of Section 1109 of IBC are adopted by reference with the following modifications:

1. Revise the exception to Section 1109.1 to read:

"Exception: *Accessible units, Type A units, Type B units and Type C units* shall comply with Chapter 10 of ICC A117.1."

2. Add a new Section 1109.5.3 to read:

"1109.5.3 Bottle fillers.

Where a bottle filler is provided, the bottle filler shall be *accessible* and located on an *accessible route*."

3. Revise item 1 in Section 1109.8 to read:

"1. An *accessible route* to a performing area, head tables in banquet rooms and *spaces* and speaker platforms."

4. Add a new Section 1109.7.1 to read:

"1109.7.1 Limited use/limited application elevators.

Limited use/limited application elevators shall be allowed for the following applications:

1. For applications listed in Section 1109.8.
2. For applications subject to an exception to Section 1104.4.
3. To provide access to the *mezzanine* or *second story above grade plane* of a two-story building or to the *basement* or *mezzanine* of a one-story building, other than a shopping mall, shopping center, or the professional office of a health care provider, where each *story* is not more than 3,000 square feet (279 m²)."

5. Revise Section 1109.9 to read:

"1109.9 Storage.

Where fixed or built-in storage *elements* such as cabinets, coat hooks, shelves, medicine cabinets, lockers, closets and drawers are provided in required *accessible spaces* or for *Type A units* or *Accessible units*, at least 5 percent, but not less than one of each type shall be *accessible*.

Exception: Archival storage areas and product storage areas described in Section 1104.4, exception 4 are not required to be *accessible*."

6. Revise Section 1109.9.2 to read:

"1109.9.2 Shelving and display units.

Self-service shelves and display units shall be located on an *accessible route*. Such shelving and display units shall not be required to comply with reach-range provisions. The clear width of aisles between stacks in libraries shall be 42 inches (1065 mm) minimum.

Exception: In *existing buildings* and *alterations*, the minimum width of aisles between stacks in libraries shall be 36 inches (915 mm)."

7. Add new Sections 1109.12.1.1 and 1109.12.1.2 to read:

"1109.12.1.1 Coat hooks, shelves and mirrors.

Where coat hooks, shelves or mirrors are provided in dressing, fitting or locker rooms without individual compartments, at least one of each type shall be *accessible*. Where coat hooks, shelves or mirrors are provided in individual compartments required to be *accessible*, at least one of each type shall be *accessible* within the compartment.

1109.12.1.2 Mirrors.

Where mirrors are provided in dressing rooms of the same use, then at least one *accessible* mirror shall be provided in an *accessible* dressing room."

8. Add a new Section 1109.13.1 to read:

"1109.13.1 Operable windows.

Where operable windows are provided for operation by occupants, at least one operable window in each room shall be *accessible* and each required operable window shall be *accessible*.

Exceptions:

1. Windows in bathrooms and kitchens.
2. In Group R-1 occupancies, only windows in *Accessible units* are required to be *accessible*."

14B-11-1110 Recreational facilities.

The provisions of Section 1110 of IBC are adopted by reference with the following modifications:

1. Revise Section 1110.4.7 to read:

"1110.4.7 Animal containment areas.

Animal cages, pens, corrals, and similar areas in which animals are contained shall be on an *accessible route* but shall not otherwise be required to comply with these requirements. Public circulation routes where animals may travel, such as in petting zoos and passageways alongside animal pens in state fairs, are not eligible for this exception."

2. Add a new Section 1110.4.16 to read:

"1110.4.16 Other recreational facilities.

The following recreational facilities shall comply with the *Illinois Accessibility Code*:

1. Golf *facilities*.
2. Camping *facilities*.
3. Picnic *facilities*.
4. Outdoor viewing areas.
5. Trails.
6. Beach access routes.
7. Benches in outdoor locations."

14B-11-1111 Signage.

The provisions of Section 1111 of IBC are adopted by reference with the following modifications:

1. Delete the exception to item 1 in Section 1111.1.
2. Revise item 5 in Section 1111.2 by deleting "*approved*."

CHAPTER 14B-12

INTERIOR ENVIRONMENT

14B-12-1201 General.

14B-12-1202 Ventilation.

14B-12-1203 Temperature control.

14B-12-1204 Lighting.

14B-12-1205 Yards and courts.

14B-12-1206 IBC Section 1206.

14B-12-1207 Interior space dimensions.

14B-12-1208 Access to unoccupied spaces.

14B-12-1209 Toilet and bathroom requirements.

14B-12-1210 Rodent protection.

14B-12-1211 Insect screens.

14B-12-1212 Residential security devices.

14B-12-1201 General.

The provisions of Section 1201 of IBC are adopted by reference with the following modification:

1. Revise Section 1201.1 by deleting "sound transmission."

14B-12-1202 Ventilation.

The provisions of Section 1202 of IBC are adopted by reference with the following modifications:

1. Revise Section 1202.1 to read:

"1202.1 General.

Every space intended for human occupancy shall be provided with natural *ventilation* in accordance with Section 1202.5, or mechanical *ventilation* in accordance with the *Chicago Mechanical Code*.

Exceptions:

1. Where the air infiltration rate in a *dwelling unit* is less than 5 air changes per hour where tested with a blower door at a pressure 0.2 inch w.c. (50 Pa) in accordance with the *Chicago Energy Conservation Code*, the *dwelling unit* shall be ventilated by mechanical means in accordance with Section 403 of the *Chicago Mechanical Code*.
2. In *dwelling units* in Group I-1, R-2, R-3, and R-5 occupancies, natural *ventilation* in accordance with Section 1202.5 shall be provided in all living, dining and sleeping rooms with a floor area of 70 square feet (6.5 m²) or more.
3. Rooms containing bathtubs, showers, spas and similar bathing fixtures shall be mechanically ventilated in accordance with the *Chicago Mechanical Code*.
4. *Ambulatory care facilities* and Group I-2 occupancies shall be ventilated by mechanical means in accordance with the *Chicago Mechanical Code*.
5. Natural *ventilation* shall not be substituted for mechanical *ventilation* in rooms or spaces where mechanical *ventilation* is specifically required by the *Chicago Mechanical Code* or *Chicago Energy Conservation Code*.
6. *Ventilation* and exhaust systems for occupancies and operations involving flammable or combustible hazards or other contaminant sources as covered in the *Chicago Mechanical Code* or the *Chicago Fire Prevention Code* shall be provided as required by both codes."

2. Revise the exceptions to Section 1202.2.1 to read:

"Exception: The net free cross-ventilation area shall be permitted to be reduced to 1/300 provided at least 40 percent and not more than 50 percent of the required venting area is provided by ventilators located in the upper portion of the *attic* or rafter space. Upper ventilators shall be located not more than 3 feet (914 mm) below the ridge or highest point of the space, measured vertically, with the balance of the *ventilation* provided by eave or cornice vents. Where the location of wall or roof framing members conflicts with the installation of upper ventilators, installation more than 3 feet (914 mm) below the ridge or highest point of the space shall be permitted."

3. Revise item 4 in Section 1202.3 to read:

"4. Any *air-impermeable insulation* shall be a Class II vapor retarder or shall have a Class II vapor retarder coating or covering in direct contact with the underside of the insulation."

4. Revise item 5.1.3 in Section 1202.3 to read:

"5.1.3. Where both *air-impermeable insulation* and air-permeable insulation are provided, the *air-impermeable insulation* shall be applied in direct contact with the underside of the structural roof sheathing in accordance with Item 5.1.1 and shall be in accordance with the *R-values* in Table 1202.3 for condensation control. The *air-permeable insulation* shall be installed

directly under the *air-impermeable insulation*."

5. Delete item 5.1.4 in Section 1202.3.

6. Revise exception 2 to Section 1202.3 to read:

"2. Section 1202.3 does not apply to enclosures that are humidified beyond 35 percent during winter months."

7. Revise Table 1202.3 by deleting all rows except for Climate Zone 5.

8. Revise note a to Table 1202.3 to read:

"a. Contributes to, but does not supersede, thermal resistance requirements for attic and roof assemblies in the *Chicago Energy Conservation Code*."

9. Revise Section 1202.4 by deleting "1202.4.2."

10. Revise Section 1202.4.1 to read:

"1202.4.1 Ventilation openings.

Ventilation openings through foundation walls shall be provided. The openings shall be placed so as to provide cross-ventilation of the under-floor space. The net area of *ventilation* openings shall be in accordance with Section 1202.4.1.1 or 1202.4.1.2."

11. Delete Section 1202.4.2.

12. Delete Section 1202.4.4.

13. Revise Section 1202.5 and its subsections to read:

"1202.5 Natural ventilation.

Where required, natural *ventilation* shall be through windows, doors, louvers or other openings to the outdoors that open upon a *public way* or a *yard* or *court* complying with Section 1205. The operating mechanism for such openings shall be readily controllable by the occupants of the room or space served.

1202.5.1 Ventilation area required.

The net openable area of the openings to the outdoors, without regard for window opening control devices or limiters, shall be at least 4 percent of the floor area being ventilated.

1202.5.1.1 Adjoining spaces.

For the purpose of natural *ventilation*, any space is permitted to be considered as a portion of an adjoining room where at least one-half of the area of the common wall is open and unobstructed and provides an opening of not less than 10 percent of the floor area of the interior space or 25 square feet (2.32 m²), whichever is greater. The net openable area shall be at least 4 percent of the aggregate floor area.

1202.5.1.2 Remote rooms.

Where rooms and spaces without openings to the outdoors are ventilated through an adjoining room, the opening to the adjoining room shall either:

1. be unobstructed and shall have an area of not less than 8 percent of the floor area of the remote room or space, but not less than 25 square feet (2.3 m²).
2. be equipped with doors, movable panels, or louvers that are operable from within the remote room and that when fully open provide a net open area of not less than 10 percent of the floor area of the remote room or space, but not less than 30 square feet (2.8 m²).

The openable area of the openings to the outdoors shall be based on the total floor area being ventilated.

1202.5.1.3 Residential basements.

Where at least 40 percent of the net floor area of a Group R *dwelling unit*, not used as *congregate living facilities*, is located entirely above grade and provided with natural *ventilation* as required in this section, *basement-level occupiable spaces* may be provided with mechanical *ventilation* and exhaust providing at least 0.35 air changes per hour in lieu of natural *ventilation* openings.

1202.5.1.4 Openings below grade.

Where openings below grade provide required natural *ventilation*, the outside horizontal clear space measured perpendicular to the opening shall be one and one-half times the depth of the opening. The depth of the opening shall be measured from the average adjoining ground level to the bottom of the opening.

1202.5.2 Contaminants exhausted.

Contaminant sources in naturally ventilated spaces shall be removed in accordance with the *Chicago Mechanical Code* and the *Chicago Fire Prevention Code*."

14. Delete Section 1202.6.

14B-12-1203 Temperature control.

The provisions of Section 1203 of IBC are adopted by reference with the following modifications:

1. Revise Section 1203.1, excluding the exceptions, to read:

"1203.1 Heating equipment and systems.

Interior spaces intended for human occupancy shall be provided with active or passive space heating systems capable of maintaining an indoor temperature of not less than 68°F (20°C) at a point 3 feet (914 mm) above the floor when the outdoor temperature is -7°F (-22°C)."

2. Add a new Section 1203.2 to read:

"1203.2 Cooling and dehumidification equipment and systems.

Permanent cooling and dehumidification equipment or systems shall be provided in accordance with Sections 1203.2.1 and 1203.2.2.

1203.2.1 Group I-1, I-2, I-4 and R-4 occupancies.

In Group I-1, I-2, I-4 and R-4 occupancies, indoor spaces intended for human occupancy shall be provided with active or passive space cooling and dehumidification equipment or systems capable of maintaining an indoor temperature of 75°F (24°C) and 50 percent relative humidity at a point 3 feet (914 mm) above the floor when the outdoor temperature is 100°F (38°C).

1203.2.2 New Group E, I-3, R-1, R-2, R-3 and R-5 occupancies.

In Group E, I-3, R-1, R-2, R-3 and R-5 occupancies constructed or established after the effective date of the amendatory ordinance of 2022, indoor spaces intended for human occupancy shall be provided with active or passive space cooling and dehumidification equipment or systems capable of maintaining an indoor temperature of 75°F (24°C) and 50 percent relative humidity at a point 3 feet (914 mm) above the floor when the outdoor temperature is 92°F (33°C) and the mean coincident wet bulb temperature is 75°F (24°C)."

14B-12-1204 Lighting.

The provisions of Section 1204 of IBC are adopted by reference with the following modifications:

1. Revise Section 1204.1 to read:

"1204.1 General.

Every space intended for human occupancy shall be provided with natural light in accordance with Section 1204.2 or shall be provided with artificial light in accordance with Section 1204.3.

Exceptions:

1. In *dwelling units* and *sleeping units* in Group R, I-1, and I-2 occupancies, all living, dining, and sleeping rooms with a floor area of 70 square feet (6.5 m²) or more shall be provided with natural light in accordance with Section 1204.2 and artificial light in accordance with Section 1204.3.
2. In Group I-4 occupancies all living and sleeping rooms for persons receiving *custodial care* shall be provided with natural light in accordance with Section 1204.2 and artificial light in accordance with Section 1204.3.
3. In Group E occupancies, all regular classrooms shall be provided with natural light in accordance with Section 1204.2 and artificial light in accordance with Section 1204.3."

2. Revise Section 1204.2 and its subsections to read:

"1204.2 Natural light.

Where required, natural light shall be provided in accordance with Section 1204.2.1, 1204.2.2, 1204.2.3 or 1204.2.4. Openings shall comply with Sections 1204.2.5 and 1204.2.6.

1204.2.1 Direct natural light.

The room shall be provided with exterior openings with a net glazed area of at least 8 percent of the floor area of the room.

1204.2.2 Adjoining spaces.

For the purpose of natural lighting, any room is permitted to be considered as a portion of an adjoining room where at least one-half of the area of the common wall is open and unobstructed and provides an opening of not less than 10 percent of the floor area of the interior room or 25 square feet (2.32 m²), whichever is greater. The net glazed area of exterior openings

shall be at least 8 percent of the aggregate floor area.

1204.2.3 Remote rooms.

A room without sufficient exterior openings is permitted to borrow natural lighting from an adjoining room that meets the requirements of Section 1204.2.1 if the remote room, adjoining room and exterior openings meet all of the following criteria:

1. The ceiling height in both the remote room and adjoining room shall be no less than 9 feet 4 inches (2844 mm) at the lowest point.
2. The top of the exterior glazing shall be no less than 9 feet (2743 mm) above the finished floor.
3. The opening between the remote room and the adjoining room shall be parallel to the plane of the exterior glazed openings, as near as possible to the ceiling and either:
 - 3.1. If open and unobstructed, shall have an area at least 8 percent of the floor area of the remote room or 25 square feet (2.32 m²), whichever is greater.
 - 3.2. If equipped with transparent or translucent glazing, shall have a net glazed area at least 10 percent of the floor area of the remote room or 30 square feet (2.8 m²), whichever is greater.
4. The remote room shall be immediately adjacent to the adjoining room.
5. The net glazed area of exterior openings shall be at least 8 percent of the aggregate floor area of the adjoining room and remote room.

1204.2.4 Residential basements.

Where at least 40 percent of the net floor area of a Group R *dwelling unit*, not used as *congregate living facilities*, is located entirely above grade and provided with natural light in accordance with Section 1204.2.1 or 1204.2.2, *basement* spaces may be provided with natural light as follows:

1. Bedrooms shall have exterior openings with a net glazed area of at least 6 percent of the floor area of the room.
2. For *habitable spaces* other than bedrooms, the *basement* level, as a whole, shall be provided with exterior openings with a net glazed area of at least 4 percent of the aggregate floor area of all *habitable spaces* in the *basement*.

1204.2.5 Exterior openings.

Exterior openings required by Section 1204.2 for natural light shall open directly onto a *public way* or a *yard* or *court* complying with Section 1205.

Exceptions:

1. Required exterior openings are permitted to open into an *open porch* or *exterior balcony* where the *open porch* or *exterior balcony* meets all of the following criteria:
 - 1.1. Abuts a *public way*, or a *yard* or *court* complying with Section 1205.
 - 1.2. Has a ceiling height of not less than 7 feet (2134 mm).
 - 1.3. Has a longer side, parallel to the plane of the openings, at least 55 percent open and unobstructed from finished floor to underside of surface above.
2. Skylights that are unobstructed to the sky are not required to open directly onto a *public way*, *yard* or *court*.

1204.2.6 Openings below grade.

Where openings below grade are used to provide required natural light, the outside horizontal clear space measured perpendicular to the opening shall be one and one-half times the depth of the opening. The depth of the opening shall be measured from the average adjoining ground level to the bottom of the opening."

3. Revise Section 1204.3 to read:

"1204.3 Artificial light.

Artificial light shall be provided that is adequate to provide an average illumination of 10 footcandles (107 lux) over the area of the room at a height of 30 inches (762 mm) above the floor level. Lighting outlets shall be provided as required in Section 210.70 of the *Chicago Electrical Code*.

Exception: Within *dwelling units*, wall switch-controlled receptacles shall be permitted in lieu of lighting outlets in accordance with Section 210.70(A)(1), exception 1 of the *Chicago Electrical Code*."

4. Revise Section 1204.4 by replacing "*dwelling unit*" with "*single dwelling unit*."

(Amend Coun. J. 6-25-21, p. 32180, Art. II, § 21; Amend Coun. J. 1-26-22, p. 43756, Art. II, § 15)

The provisions of Section 1205 of IBC are not adopted. The following language is adopted as Section 1205:

"1205. YARDS AND COURTS

1205.1 General.

This section shall apply to *yards* and *courts* adjacent to exterior openings that provide natural light or natural *ventilation*. Such *yards* and *courts* shall be on the same *lot* as the building. For purposes of this section, width shall be measured perpendicular to the plane of openings and length shall be measured parallel to the plane of openings.

1205.2 Yards.

Yards shall be not less than 3 feet (914 mm) in width for an *exterior wall* not more than 40 feet (12.2 m) in height above the lowest floor receiving natural light or natural *ventilation* or *grade plane*, whichever is higher. Where the *exterior wall* exceeds 40 feet (12.2 m) in height, the minimum width of the *yard* shall be increased 2 inches (76 mm) for each 1 foot (305 mm) or fraction thereof in height up to a maximum width of 15 feet (4572 mm).

1205.3 Courts.

The length of a *court* shall be at least 150% of the horizontal dimension of required openings. *Outer courts* shall comply with Section 1205.3.2. *Inner courts* shall comply with Section 1205.3.3.

1205.3.1 [Reserved.]

1205.3.2 Outer courts.

Outer courts shall be not less than 3 feet (914 mm) in width for an *exterior wall* not more than 40 feet (12.2 m) in height above the lowest floor receiving natural light or natural *ventilation* or *grade plane*, whichever is higher. Where the *exterior wall* exceeds 40 feet (12.2 m) in height, the minimum width of the *outer court* shall be increased 2 inches (76 mm) for each 1 foot (305 mm) or fraction thereof in height up to a maximum width of 15 feet (4572 mm). Openings on opposite sides of an *outer court* shall be separated by not less than 6 feet (1829 mm).

1205.3.3 Inner courts.

Inner courts shall be not less than 6 feet (1829 mm) in width for an *exterior wall* not more than 40 feet (12.2 m) in height above the lowest floor receiving natural light or natural *ventilation* or *grade plane*, whichever is higher. Where the *exterior wall* exceeds 40 feet (12.2 m) in height, the minimum width of the *inner court* shall be increased 4 inches (102 mm) for each 1 foot (305 mm) or fraction thereof in height up to a maximum width of 15 feet (4572 mm). For irregularly-shaped *inner courts*, the average width shall be at least as great as the minimum width required by this section and the width shall not be less than 5 feet (1524 mm) at any point.

1205.3.3.1 Court access.

Access shall be provided to the bottom of *inner courts* for cleaning purposes.

1205.3.3.2 Air intake.

Inner courts more than two *stories* in height shall be provided with a horizontal air intake at the bottom with a cross-sectional area not less than 10 square feet (0.93 m²) and leading to a *yard* or *public way*. Air intakes shall be of construction having a *fire-resistance rating* of not less than two hours and shall be open at both ends and unobstructed for their full size and length, except that grilles of noncombustible construction are permitted at both ends.

Exception: In buildings equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1, the *fire-resistance rating* shall be not less than one hour.

1205.3.3.3 Court drainage.

The bottom of every *inner court* shall be properly graded and drained to a public sewer or other *approved* drainage system complying with the *Chicago Plumbing Code*.

1205.4 Easements.

The *building official* may adopt rules for considering permanent and perpetual easements or restrictive declarations on adjoining *lots* as a means of creating spaces equivalent to *courts* or *yards*."

(Amend Coun. J. 6-25-21, p. 32180, Art. II, § 22; Amend Coun. J. 1-26-22, p. 43756, Art. II, § 16)

The provisions of Section 1206 of IBC are not adopted.

The provisions of Section 1207 of IBC are not adopted. The following language is adopted as Section 1207:

"1207. INTERIOR SPACE DIMENSIONS

1207.1 Minimum room dimensions.

Kitchens shall have a clear passageway of not less than 3 feet (914 mm) between counter fronts and appliances or counter fronts and walls.

Exception: Within a *dwelling unit* having a *net floor area* not exceeding 700 square feet (65 m²) and not required to be an *Accessible unit*, *Type A unit*, *Type B unit* or *Type C unit*, a clear working space of not less than 30 inches (762 mm) shall be provided in front of kitchen counters and kitchen appliances.

1207.2 Minimum ceiling heights.

Occupiable spaces and *corridors* shall have a ceiling height of not less than 7 feet 6 inches (2286 mm) above the finished floor.

Exceptions:

1. Bathrooms, toilet rooms, kitchens, storage rooms, laundry rooms and spaces above and below a *mezzanine* shall have a ceiling height of not less than 7 feet (2134 mm) above the finished floor.
2. Within a *dwelling unit* or *sleeping unit* in a Group R occupancy, *habitable spaces* and hallways shall have a ceiling height of not less than 7 feet (2134 mm) above the finished floor. Bathrooms, toilet rooms and laundry rooms shall have a ceiling height of not less than 6 feet 8 inches (2032 mm) above the finished floor.
3. If any room in a building has a sloped ceiling, the prescribed ceiling height for the room is required in one-half the area thereof. Any portion of the room measuring less than 5 feet (1524 mm) from the finished floor to the ceiling shall not be included in any computation of the minimum area thereof.
4. *Lofts* in accordance with Section 1207.5.

1207.2.1 Furred ceiling or projecting elements.

Any room with a furred ceiling or projecting elements, such as beams, girders, ducts, or pipes, shall be required to have the minimum ceiling height in three-quarters of the area thereof, and in no case shall the clearance below a furred ceiling or projecting element be less than 80 inches (2032 mm) above the finished floor.

Exception: Within a *dwelling unit* or *sleeping unit* in a Group R occupancy, the clearance below a furred ceiling or projecting element shall not be less than 6 feet 4 inches (1931 mm).

1207.3 Room area.

Every *dwelling unit* shall have at least one room that shall have not less than 120 square feet (11.2 m²) of *net floor area*. Dining spaces shall have a *net floor area* of not less than 60 square feet (5.6 m²). Other *habitable spaces* shall have a *net floor area* of not less than 70 square feet (6.5 m²). Where a single room is provided for cooking, dining and living purposes, it shall have a *net floor area* of not less than 180 square feet (16.7 m²).

1207.4 [Reserved.]

1207.5 Lofts.

Lofts used for sleeping or living space within a *dwelling unit* or *sleeping unit* in a Group R occupancy, and not containing or providing exclusive means of access to any plumbing fixture, shall conform to the minimum requirements of this code except as follows:

1. The *loft* shall have a floor area of not less than 35 square feet (3.25 m²) and not more than 150 square feet (13.9 m²).
2. The *loft* shall not be less than 5 feet (1524 mm) in any horizontal dimension.
3. A portion of every *loft* must have a ceiling height of at least 5 feet (1524 mm), measured from the finished floor to the finished ceiling. Portions of a *loft* with a sloped ceiling measuring less than 3 feet (914 mm) from the finished floor to the finished ceiling shall not be considered as contributing to the minimum required or maximum permitted area of the *loft*.
4. The *loft* shall be accessed by one of the following means, which shall count as the *means of egress*:
 - 4.1. A *stairway* complying with Section 1011.
 - 4.2. A *stairway* with a clear width of not less than 20 inches (508 mm), with risers not less than 7 inches (178 mm) and not more than 10 inches (254 mm) in height and a tread depth equal to 20 inches (508 mm) minus four-thirds of the riser height and with a *handrail* on one side. The *stairway* shall have headroom of at least 6 feet 2 inches (1880 mm). Where the ceiling height of the *loft* is less than 6 feet 2 inches (1880 mm) at the location where the *stairway* meets the *loft*, the top of the *stairway* shall be a landing platform, not less than 20 inches (508 mm) in length and width, located not more than 18 inches (457 mm) vertically below the finished floor of the *loft*.
 - 4.3. A permanent ladder with a rung width of not less than 12 inches (305 mm) and 10-inch (254 mm) to 14-inch (356 mm) spacing between rungs installed at an incline between 70 to 80 degrees from horizontal.

Ladders shall be capable of supporting a 300-pound (136 kg) load on any rung. Rung spacing shall be uniform within 3/8 inch (9.5 mm).

4.4. An *alternating tread device* complying with Section 1011.14.

4.5. A ship's ladder with a clear width of 20 inches (508 mm) complying with Section 1011.15.

5. *Guards* shall be provided at the open side of *lofts*. *Loft guards* shall be not less than 36 inches (914 mm) in height or one-half of the clear height to the ceiling, whichever is less.

Exception: *Lofts* with guardrails that meet the requirements of the Consumer Products Safety Commission for bunk beds.

6. The floor level of the *loft* shall be at least 30 inches (762 mm) and no more than 151 inches (3835 mm) above the main floor level of the unit.

1207.5.1 Accessibility.

Lofts shall not be allowed within *Accessible units* and *Type A units*. Within *Type B units*, *lofts* shall not contain the primary living or sleeping space, and all other spaces within the unit shall be on an *accessible route*.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 58; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 26; Amend Coun. J. 12-14-22, p. 58278, Art. III, § 6)

14B-12-1208 Access to unoccupied spaces.

The provisions of Section 1208 of IBC are adopted by reference with the following modifications:

1. Revise Section 1208.1 to read:

"1208.1 Crawl spaces.

Crawl spaces shall be provided with not less than one access opening that shall be not less than 24 inches by 24 inches (610 mm by 610 mm)."

2. Delete Section 1208.2.

14B-12-1209 Toilet and bathroom requirements.

The provisions of Section 1209 of IBC are adopted by reference with the following modifications:

1. Revise Section 1209.1 by replacing "Chapter 29" with "the *Chicago Plumbing Code*."
2. Revise Sections 1209.2.1, 1209.2.2 and 1209.2.3 by replacing "a smooth, hard, nonabsorbent" with "an easily-cleaned, nonabsorbent."
3. Revise Section 1209.3 and its subsections to read:

"1209.3 Privacy.

Privacy at water closets, urinals and bathing areas shall be provided in accordance with the *Chicago Plumbing Code*."

(Amend Coun. J. 10-7-20, p. 21791, Art. II, § 27)

14B-12-1210 Rodent protection.

The following language is adopted as a new Section 1210:

"1210. RODENT PROTECTION

1210.1 General.

Buildings containing rooms and spaces in which people live, sleep or work, or in which feed, food or foodstuffs are stored, prepared, processed, served or sold, shall be constructed in accordance with the provisions of this section.

1210.2 Ventilation openings.

Ventilation openings in *exterior walls* located less than 2 feet (610 mm) above the adjacent ground level shall be covered for their entire height and width with perforated sheet metal plates not less than 0.070 inch (1.8 mm) thick, expanded sheet metal plates not less than 0.047 inch (1.2 mm) thick, cast-iron grills or grating, extruded aluminum load-bearing vents, hardware cloth of 0.035 inch (0.89 mm) wire or heavier, or other *approved* material providing equivalent protection against the entry of rodents. The openings therein shall not exceed 1/4 inch (6.4 mm).

1210.3 Windows.

When closed, the maximum clearance between any exterior window, window jambs and sills, where the window sill is located less than 2 feet (610 mm) above the adjacent ground level shall be not greater than 3/8 inch (9.5 mm).

1210.4 Exterior wall penetrations.

Annular spaces around pipes, electric cables, conduits and similar openings in *exterior walls* shall be protected against the passage of rodents by closing such openings with cement mortar, concrete masonry or noncorrosive metal.

1210.5 Doors.

When closed, the maximum clearance between any exterior door, door jambs and sills shall be not greater than 3/8 inch (9.5 mm)."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 59)

14B-12-1211 Insect screens.

The following language is adopted as a new Section 1211:

"1211. INSECT SCREENS

1211.1 General.

Every exterior door, operable window and other outside opening used to provide *ventilation for habitable spaces*, food preparation areas, food service areas, or areas where products to be included or utilized in food for human consumption are processed, manufactured, packed or stored shall be supplied with tightly fitting screens of minimum 16 mesh per inch (16 mesh per 25 mm). Every screen door provided for insect control shall be self-closing.

Exceptions:

1. Insect screens shall not be required where other effective means of insect protection, such as air curtains or insect repellent fans are provided.
2. Insect screens shall not be required for openings more than 50 feet (15.2 m) above the ground unless the *building official* determines that unusual insect prevalence exists at the premises."

(Added Coun. J. 2-19-20, p. 14473, Art. II, § 60)

14B-12-1212 Residential security devices.

The following language is adopted as a new Section 1212:

"1212. RESIDENTIAL SECURITY DEVICES

1212.1 General.

Building entrances and each *dwelling unit* in buildings containing more than one Group I-1, R-2, R-3, R-4 or R-5 *dwelling unit* shall be provided with security devices in accordance with Section 403 of the *Chicago Minimum Requirements for Existing Buildings*."

(Added Coun. J. 2-19-20, p. 14473, Art. II, § 61)

CHAPTER 14B-13

ENERGY EFFICIENCY

14B-13-1301 General.

14B-13-1301 General.

The provisions of Section 1301 of IBC are adopted by reference without modification.

CHAPTER 14B-14

EXTERIOR WALLS

14B-14-1401 General.

14B-14-1402 Performance requirements.

14B-14-1403 Materials.

14B-14-1404 Installation of wall coverings.

14B-14-1405 Combustible materials on the exterior side of exterior walls.

14B-14-1406 Metal composite materials (MCM).

14B-14-1407 Exterior insulation and finish systems (EIFS).

14B-14-1408 High-pressure decorative exterior-grade compact laminates (HPL).

14B-14-1409 Plastic composite decking.

14B-14-1401 General.

The provisions of Section 1401 of IBC are adopted by reference without modification.

14B-14-1402 Performance requirements.

The provisions of Section 1402 of IBC are adopted by reference with the following modifications:

1. Delete exception 2 and its subparts to Section 1402.2.
2. Revise Section 1402.4 by replacing "*Exterior walls*" with "*Exterior walls, including fire-resistant joint systems,*".
3. Revise Section 1402.5 to read:

"1402.5 Vertical and lateral flame propagation.

Exterior walls on buildings of Type I, II, III or IV construction that are greater than 40 feet (12 192 mm) in height above *grade plane* and contain a combustible *water-resistive barrier* or air barrier shall be tested in accordance with and comply with the acceptance criteria of NFPA 285. For the purposes of this section, *fenestration* products, flashing of fenestration products and flashing and accessories at other locations, including through-wall flashings, shall not be considered part of the *water-resistive barrier*.

Exceptions:

1. Walls in which the *water-resistive barrier* or air barrier required by the *Chicago Energy Conservation Code* is the only combustible component and the *exterior wall* has an *exterior wall covering* of brick, concrete, stone, terracotta, stucco or steel with minimum thicknesses in accordance with Table 1404.2.
 2. Walls in which the *water-resistive barrier* or air barrier required by the *Chicago Energy Conservation Code* is the only combustible component and the *water-resistive barrier* or air barrier required by the *Chicago Energy Conservation Code* has a peak heat release rate of less than 150 kW/m², a total heat release of less than 20 MJ/m² and an effective heat of combustion of less than 18 MJ/kg as determined in accordance with ASTM E1354 and has a flame spread index of 25 or less and a smoke-developed index of 450 or less as determined in accordance with ASTM E84 or UL 723. The ASTM E1354 test shall be conducted on specimens at the thickness intended for use, in the horizontal orientation and at an incident radiant heat flux of 50 kW/m²."
4. Delete Sections 1402.6 and 1402.7.
 5. Add a new Section 1402.8 to read:

"1402.8 Energy conservation.

Exterior walls shall meet additional performance requirements found in the *Chicago Energy Conservation Code*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 62)

14B-14-1403 Materials.

The provisions of Section 1403 of IBC are adopted by reference with the following modifications:

1. Delete Section 1403.2.
2. Add a new Section 1403.3.3 to read:

"1403.3.3 Wood shingles and shakes.

Wood shingles and shakes shall comply with Section 1507.8.5."

3. Revise Section 1403.4 by replacing "steel framing" with "metal framing."
4. Delete Section 1403.5.3.

5. Add a new Section 1403.5.4 to read:

"1403.5.4 Other metals.

Other metals, used as siding, shall conform to the applicable standard listed in Table 1507.4.3(1)."

6. Revise Section 1403.8 by deleting "*approved*."

7. Revise Section 1403.9 by replacing "*approved* quality control agency" with "*approved agency*."

8. Revise Section 1403.10 by replacing "*approved* quality control agency" with "*approved agency*."

9. Delete Section 1403.12 and its subsections.

14B-14-1404 Installation of wall coverings.

The provisions of Section 1404 of IBC are adopted by reference with the following modifications:

1. Revise Section 1404.2 by deleting "*approved*."

2. Revise Table 1404.2 by deleting the rows for: asbestos cement board, asbestos shingles, lead-coated copper, and lead-coated high-yield copper; and replacing "Steel (approved corrosion resistant)" with "Steel (corrosion resistant)."

3. Revise note d to Table 1404.2 to read:

"d. 16 ounces per square foot for cold-rolled copper, 12 ounces per square foot for copper shingles and high- yield copper."

4. Revise Section 1404.3 by replacing "an *approved* design" with "a design."

5. Revise Section 1404.3.1, excluding the exceptions, to read:

"1404.3.1 Class I and II vapor retarders.

Class I or II vapor retarders shall be provided on the interior side of frame walls."

6. Revise Table 1404.3.2 by deleting all rows except Zone 5.

7. Revise Section 1404.3.4 to read:

"1404.3.4 Vented cladding.

For the purposes of this section, vented cladding shall include the following:

1. Vinyl, or horizontal aluminum siding applied over a weather-resistive barrier as specified in this chapter.
2. Brick veneer with a clear airspace as specified in this code.
3. Other *approved* vented claddings."

8. Revise the first sentence of Section 1404.4 to read:

"Flashing, including end dams, shall be installed in such a manner so as to prevent moisture from entering the wall or to redirect that moisture to the exterior."

9. Revise Section 1404.5 to read:

"1404.5 Wood veneers.

Wood veneers on *exterior walls* of buildings of Type I, II, III and IV construction shall be not less than 1 inch (25 mm) nominal thickness and shall conform to the following:

1. The veneer shall not exceed 40 feet (12 190 mm) in height above grade.
2. The veneer is attached to or furred from a noncombustible backing that is fire-resistance rated as required by other provisions of this code.
3. Where open or spaced wood veneers (without concealed spaces) are used, they shall not project more than 24 inches (610 mm) from the building wall."

10. Revise Section 1404.7 to read:

"1404.7 Anchored stone veneer.

Anchored stone veneer units not exceeding 10 inches (254 mm) in thickness shall be anchored directly to masonry, concrete or to metal stud construction by corrosion-resistant clips or anchors located in the edge or back of the units. Veneer anchors shall be capable of resisting a force equal to two times the weight of the attached veneer in tension and compression."

11. Delete Section 1404.8.

12. Revise Section 1404.9 to read:

"1404.9 Terra cotta.

Terra cotta or ceramic units not less than 1 5/8 inches (41 mm) thick shall be anchored directly to masonry, concrete or metal stud construction. The facing shall be tied to the backing with corrosion-resistant metal anchors. The veneer anchors shall have sufficient strength to support twice the weight of the veneer in tension."

13. Revise Section 1404.10.1.2.1 to read:

"1404.10.1.2.1 Flashing at foundation.

A corrosion-resistant screed or flashing shall be installed to extend below the foundation plate line on exterior stud walls in accordance with Section 1404.4. The *water-resistive barrier* shall lap over the exterior of the attachment flange of the screed or flashing."

14. Revise Section 1404.10.2 to read:

"1404.10.2 Exterior adhered masonry veneers- porcelain tile.

Adhered *porcelain tile* units shall not exceed 5/8 inch (15.8 mm) thickness and 24 inches (610 mm) in any face dimension nor more than 3 square feet (0.28 m²) in total face area and shall not weigh more than 9 pounds psf (0.43 kN/m²). *Porcelain tile* shall be adhered to a backing system."

15. Revise Section 1404.11 to read:

"1404.11 Metal veneers.

Veneers of metal shall be fabricated from corrosion-resistant materials or shall be protected front and back with porcelain enamel, or otherwise be treated to render the metal resistant to corrosion."

16. Revise Section 1404.11.1 to read:

"1404.11.1 Attachment.

Exterior metal veneer shall be securely attached to the supporting masonry or framing members with corrosion-resistant fastenings, metal ties or by other *approved* devices or methods. Such attachments and their supports shall be designed and constructed to resist the wind loads as specified in Section 1609 for components and cladding."

17. Revise Section 1404.11.2 to read:

"1404.11.2 Weather protection.

Metal supports for exterior metal veneer shall be corrosion-resistant. Wood studs, furring strips or other wood supports for exterior metal veneer shall be pressure-treated wood or protected as required in Section 1402.2. Joints and edges exposed to the weather shall be caulked with durable waterproofing material or by other *approved* means to prevent penetration of moisture."

18. Delete Section 1404.12.3.

19. Revise Section 1404.12.4.1 to read:

"1404.12.4.1 Installation above sidewalk level.

Where thin exterior structural glass veneer is installed above the level of the top of a bulkhead facing, or at a level more than 36 inches (914 mm) above the sidewalk level, the mastic cement binding shall be supplemented with nonferrous metal shelf angles located in the horizontal joints in every course. Such shelf angles shall be not less than 0.0478-inch (1.2 mm) thick and not less than 2 inches (51 mm) long, with not less than two angles for each glass unit. Shelf angles shall be secured to the wall or backing with expansion bolts, toggle bolts or by other *approved* methods."

20. Revise Section 1404.12.5 to read:

"1404.12.5 Joints.

Unless otherwise specifically *approved* by the *building official*, abutting edges of thin exterior structural glass veneer shall be ground square. Horizontal joints shall be held to not less than 0.063 inch (1.6 mm) by a nonrigid substance or device. Where thin exterior structural glass veneer abuts nonresilient material at sides or top, expansion joints not less than 1/4 inch (6.4 mm) wide shall be provided."

21. Revise Section 1404.12.6 by replacing "sidewalk" with "ground."

22. Revise Section 1404.13.1 to read:

"1404.13.1 Installation.

Windows and doors shall be installed in accordance with manufacturer's written instructions or an engineering analysis. Fastener size and spacing shall be provided in such instructions and shall be calculated based on maximum loads and spacing used in the test or analysis."

23. Revise Section 1404.14 and its subsections to read:

"1404.14 Vinyl siding.

Vinyl siding conforming to the requirements of this section and complying with ASTM D3679 shall be permitted on *exterior walls* of buildings where the *building height* is less than or equal to 45 feet (13 716 mm). Vinyl siding shall be secured to the building so

as to provide weather protection for the *exterior walls* of the building.

1404.14.1 Application.

The siding shall be applied over sheathing or materials listed in Section 2304.6. Siding shall be applied to conform to the *water-resistive barrier* requirements in Section 1402. Siding and accessories shall be installed in accordance with manufacturer's written instructions. Unless otherwise specified in the manufacturer's instructions, nails used to fasten the siding and accessories shall have a minimum 0.313-inch (7.9 mm) head diameter and 1/8-inch (3.18 mm) shank diameter. The nails shall be corrosion resistant and shall be long enough to penetrate the studs or nailing strip not less than 3/4 inch (19 mm). For cold-formed steel *light-frame construction*, corrosion-resistant fasteners shall be used. Screw fasteners shall penetrate the cold-formed steel framing not fewer than three exposed threads. Other fasteners shall be installed in accordance with the *permitted construction documents* and manufacturer's instructions. Where the siding is installed horizontally, the fastener spacing shall not exceed 16 inches (406 mm) horizontally and 12 inches (305 mm) vertically. Where the siding is installed vertically, the fastener spacing shall not exceed 12 inches (305 mm) horizontally and 12 inches (305 mm) vertically."

24. Revise Section 1404.16 to read:

"1404.16 Fiber-cement siding.

Fiber-cement siding complying with Section 1403.10 shall be allowed on *exterior walls* of Type I, II, III, IV and V construction for wind pressure resistance or wind speed exposures as indicated by the manufacturer's listing and *label* and installation instructions. Where specified, the siding shall be installed over sheathing or materials listed in Section 2304.6 and shall be installed to conform to the *water-resistive barrier* requirements in Section 1402. Siding and accessories shall be installed in accordance with manufacturer's instructions. Unless otherwise specified in the manufacturer's instructions, nails used to fasten the siding to wood studs shall be corrosion-resistant round head smooth shank and shall be long enough to penetrate the studs not less than 1 inch (25 mm). For cold-formed steel *light-frame construction*, corrosion-resistant fasteners shall be used. Screw fasteners shall penetrate the cold-formed steel framing not fewer than three exposed full threads. Other fasteners shall be installed in accordance with the *permitted construction documents* and manufacturer's instructions."

25. Revise Sections 1404.16.1 and 1404.16.2 by deleting "*approved*."

26. Revise Section 1404.17 by deleting "*approved*."

27. Delete Section 1404.18.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 63)

14B-14-1405 Combustible materials on the exterior side of exterior walls.

The provisions of Section 1405 of IBC are adopted by reference with the following modifications:

1. Revise Section 1405.1.1 to read:

"1405.1.1 Types I, II, III and IV construction.

On buildings of Types I, II, III and IV construction, *exterior wall coverings* shall be permitted to be constructed of combustible materials, complying with the following limitations:

1. Combustible *exterior wall coverings* shall not exceed 10 percent of an *exterior wall* surface area where the *fire separation distance* is 15 feet (4572 mm) or less.
2. Combustible *exterior wall coverings* shall be limited to 40 feet (12 192 mm) in height above *grade plane*.

Exceptions:

1. *Metal composite material (MCM) systems* complying with Section 1406.
2. *Exterior insulation and finish systems (EIFS)* complying with Section 1407.
3. *High-pressure decorative exterior-grade compact laminate (HPL)* systems complying with Section 1408.
4. *Exterior wall coverings* containing *foam plastic insulation* complying with Section 2603.
3. [Reserved.]
4. Wood veneers shall comply with this section and Section 1404.5."

2. Revise Sections 1405.1.1.1.1 and 1405.1.1.1.2 to read:

"1405.1.1.1.1 Fire separation 15 feet or less.

Where installed on *exterior walls* having a *fire separation distance* of 15 feet (4572 mm) or less, combustible *exterior wall coverings* shall not exhibit sustained flaming as defined in NFPA 268.

1405.1.1.1.2 Fire separation greater than 15 feet.

For *fire separation distances* greater than 15 feet (4572 mm), any *exterior wall covering* shall be allowed that has been exposed to a reduced level of incident radiant heat flux in accordance with the NFPA 268 test method without exhibiting

sustained flaming. The minimum *fire separation distance* required for the *exterior wall covering* shall be determined from Table 1405.1.1.1.2 based on the maximum tolerable level of incident radiant heat flux that does not cause sustained flaming of the *exterior wall covering*."

3. Revise Table 1405.1.1.1.2 by deleting all entries for a fire separation distance of 15 feet or less, and revising the entry for "16" feet to read "greater than 15."

(Amend Coun. J. 12-14-22, p. 58278, Art. III, § 7)

14B-14-1406 Metal composite materials (MCM).

The provisions of Section 1406 of IBC are adopted by reference with the following modifications:

1. In provisions adopted by reference, reset the following term in italic type: "MCM system."
2. Revise Section 1406.5 to read:

"1406.5 Approval.

Results of tests or an engineering analysis shall be submitted to the *building official* as part of the *construction documents* to demonstrate compliance with the requirements of Chapter 16 for wind loads."

3. Revise Section 1406.7 by deleting "*approved*."
4. Revise Section 1406.10.2 by replacing "an approved" with "a."
5. Revise item 1 in Section 1406.10.3 to read:

"1. The *MCM system* has passed tests conducted in accordance with NFPA 286 and the acceptance criteria of Section 803.1.1.1, UL 1040 or UL 1715. Such testing shall be performed with the MCM in the maximum thickness intended for use. The *MCM system* shall include seams, joints and other typical details used in the installation and shall be tested in the manner intended for use."

6. Revise Section 1406.11 and its subsections to read:

"1406.11 Alternate conditions.

Where *approved* by the *building official*, MCM and *MCM systems* shall not be required to comply with Sections 1406.10.1 through 1406.10.4 provided that such systems comply with Section 1406.11.1.

1406.11.1 Installations up to 40 feet in height.

MCM shall not be installed more than 40 feet (12 190 mm) in height above *grade plane* where installed in accordance with Sections 1406.11.1.1 and 1406.11.1.2.

1406.11.1.1 Fire separation distance of 15 feet or less.

Where the *fire separation distance* is 15 feet (4572 mm) or less, the area of MCM shall not exceed 10 percent of the *exterior wall surface*.

1406.11.1.2 Fire separation distance greater than 15 feet.

Where the *fire separation distance* is greater than 15 feet (4572 mm), the area of *exterior wall surface coverage* using MCM shall not be limited."

7. Delete Section 1406.11.2 and its subsections.
8. Delete Section 1406.11.3 and its subsections.
9. Delete Table 1406.11.3.4.
10. Delete Section 1406.11.4 and its subsections.

14B-14-1407 Exterior insulation and finish systems (EIFS).

The provisions of Section 1407 of IBC are adopted by reference with the following modifications:

1. Revise Section 1407.4.1 and its subsections to read:

"1407.4.1 Drainage.

EIFS without drainage (barrier-type) shall not be permitted. EIFS shall have an average minimum drainage efficiency of 90 percent when tested in accordance with the requirements of ASTM E2273. Details of the required drainage mechanisms shall be provided on the *construction documents*.

1407.4.1.1 Water-resistive barrier.

The *water-resistive barrier* shall comply with Section 1403.2 or ASTM E2570."

2. Revise Section 1407.5 to read:

"1407.5 Installation.

Installation of the EIFS shall be in accordance with the manufacturer's instructions."

3. Delete Section 1407.6.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 64; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 23)

14B-14-1408 High-pressure decorative exterior-grade compact laminates (HPL).

The provisions of Section 1408 of IBC are adopted by reference with the following modifications:

1. Revise Section 1408.1 to read:

"1408.1 General.

The provisions of this section shall govern the materials, construction and quality of *High-Pressure Decorative Exterior-Grade Compact Laminates (HPL)* for use as *exterior wall coverings* in addition to other applicable requirements of Chapters 14 and 16."

2. Revise Section 1408.5 to read:

"1408.5 Approval.

Results of tests or an engineering analysis shall be submitted to the *building official* with the *construction documents* to demonstrate compliance with the requirements of Chapter 16 for wind loads."

3. Revise Section 1408.7 by deleting "approved."

4. Revise Section 1408.10.2 by replacing "an approved" with "a."

5. Revise Section 1408.10.3 by replacing "is specifically approved based on" with "has passed."

6. Revise Section 1408.11 and its subsections to read:

"1408.11 Alternate conditions.

Where *approved* by the *building official*, HPL and *HPL systems* shall not be required to comply with Section 1408.10.4 provided that such systems comply with Section 1408.11.1.

1408.11.1 Installations up to 40 feet in height.

HPL shall not be installed more than 40 feet (12 190 mm) in height above *grade plane* where installed in accordance with Sections 1408.11.1.1 and 1408.11.1.2.

1408.11.1.1 Fire separation distance of 15 feet or less.

Where the *fire separation distance* is 15 feet (4572 mm) or less, the area of HPL shall not exceed 10 percent of the *exterior wall* surface.

1408.11.1.2 Fire separation distance greater than 15 feet.

Where the *fire separation distance* is greater than 15 feet (4572 mm), the area of *exterior wall* surface coverage using HPL shall not be limited."

7. Delete Section 1408.11.2 and its subsections.

8. Revise Section 1408.13 to read:

"1408.13 Foam plastic insulation.

HPL systems containing *foam plastic insulation* shall comply with the requirements of Section 2603."

14B-14-1409 Plastic composite decking.

The provisions of Section 1409 of IBC are adopted by reference without modification.

CHAPTER 14B-15

ROOF ASSEMBLIES AND ROOFTOP STRUCTURES

14B-15-1500 Defined terms.

14B-15-1501 General.

14B-15-1502 Roof drainage.

- 14B-15-1503 Weather protection.**
- 14B-15-1504 Performance requirements.**
- 14B-15-1505 Fire classification.**
- 14B-15-1506 Materials.**
- 14B-15-1507 Requirements for roof coverings.**
- 14B-15-1508 Roof insulation.**
- 14B-15-1509 Radiant barriers installed above deck.**
- 14B-15-1510 Unoccupied rooftop structures.**
- 14B-15-1511 Reroofing.**
- 14B-15-1512 IBC Section 1512.**
- 14B-15-1513 Occupiable rooftops.**
- 14B-15-1514 Vegetative roofs, roof gardens and landscaped roofs.**
- 14B-15-1515 Solar reflectance.**
- 14B-15-1516 Interior roof edge for shafts and fully enclosed courts.**

14B-15-1500 Defined terms.

Where provisions of Chapter 15 of IBC are adopted by reference, the following terms shall be reset in italic type: "built-up roof covering"; "interlayment"; "grade plane"; "metal roof panel"; "metal roof shingle"; "roof assembly"; "roof covering"; "roof covering system"; "roof deck"; "rooftop structures"; "single-ply membrane"; "skylights and sloped glazing" and "underlayment."

14B-15-1501 General.

The provisions of Section 1501 of IBC are adopted by reference with the following modification:

1. Add a new Section 1501.1.1 to read:

"1501.1.1 Dormers.

Dormers shall either be of the same type of construction as required for the *roof assembly* in which such dormers are located or the same type of construction as required for the *exterior walls* of the *building*."

14B-15-1502 Roof drainage.

The provisions of Section 1502 of IBC are adopted by reference with the following modifications:

1. Revise Section 1502.1 to read:

"1502.1 General.

Design and installation of roof drainage systems shall comply with Section 1502 of this code and Section 1106 of the *Chicago Plumbing Code*."

2. Delete Section 1502.2.
3. Delete Section 1502.3.
4. Revise Section 1502.4 to read:

"1502.4 Gutters.

Gutters and leaders placed on the exterior of *buildings* shall be of noncombustible material.

Exceptions:

1. *One-story buildings.*
2. *Buildings* of Group R occupancy with no more than 4 *stories above grade plane.*
3. *Buildings* of Type V construction."

(Amend Coun. J. 7-24-19, p. 3646, § 32; Amend Coun. J. 2-19-20, p. 14473, Art. II, § 65)

14B-15-1503 Weather protection.

The provisions of Section 1503 of IBC are adopted by reference with the following modification:

- 1. Revise Section 1503.1 by deleting "approved."

14B-15-1504 Performance requirements.

The provisions of Section 1504 of IBC are adopted by reference with the following modifications:

- 1. Revise Section 1504.1.1 to read:

"1504.1.1 Wind resistance of asphalt shingles.

Asphalt shingles shall be tested in accordance with ASTM D7158 and have a classification of D, G or H. Asphalt shingle packaging shall bear a label to indicate compliance with ASTM D7158 and the required classification.

Exception: Asphalt shingles not included in the scope of ASTM D7158 shall be tested and labeled in accordance with ASTM D3161 and have a classification of A, D or F. Asphalt shingle packaging shall bear a label to indicate compliance with ASTM D3161 and the required classification."

- 1.1. Delete Table 1504.1.1.
- 1.2. Revise Section 1504.4 to read:

"1504.4 Ballasted low-sloped roof systems.

Ballasted *low-sloped roof* single-ply roof system coverings installed in accordance with Sections 1507.12 and 1507.13 shall be designed in accordance with Section 1504.8 and ANSI/SPRI RP-4."

- 2. Revise Section 1504.5 to read:

"1504.5 Edge securement for low-sloped roofs.

Low-sloped roof built-up, modified bitumen and single-ply roof system metal edge securement, except gutters, shall be designed and installed for wind loads in accordance with Chapter 16 and tested for resistance in accordance with Test Methods RE-1, RE-2 and RE-3 of ANSI/SPRI ES-1, except basic design wind speed, *V*, shall be determined from Section 1609.3."

- 3. Revise Section 1504.6 to read:

"1504.6 Physical properties.

Roof coverings installed on *low-sloped roofs* in accordance with Section 1507 shall demonstrate physical integrity over the working life of the roof based on 2,000 hours of exposure to accelerated weathering tests conducted in accordance with ASTM G152, ASTM G154 or ASTM G155. Those *roof coverings* that are subject to cyclical flexural response due to wind loads shall not demonstrate any significant loss of tensile strength for unreinforced membranes or breaking strength for reinforced membranes when tested as herein required."

- 4. Revise Section 1504.7 to read:

"1504.7 Impact resistance.

Roof coverings installed on *low-sloped roofs* in accordance with Section 1507 shall resist impact damage based on the results of tests conducted in accordance with ASTM D3746, ASTM D4272 or the "Resistance to Foot Traffic Test" in Section 5.5 of FM 4470."

- 5. Revise Section 1504.8 to read:

"1504.8 Surfacing and ballast materials.

For a building with a *mean roof height* exceeding that permitted by Table 1504.8 based on the exposure category, the following materials shall not be used on the roof:

- 1. *Aggregate* used as surfacing for *roof coverings*.
- 2. *Aggregate*, gravel or stone used as *ballast*."

- 6. Replace Table 1504.8 as shown:

TABLE 1504.8

MAXIMUM ALLOWABLE MEAN ROOF HEIGHT PERMITTED FOR BUILDINGS WITH AGGREGATE ON THE ROOF

RISK CATEGORY ^b	MEAN ROOF HEIGHT (ft) ^a		
	Exposure category ^c		
	B	C	D

I or II	55	55	30
III	55	40	20
IV	55	25	NP

For SI: 1 foot = 304.8 mm

NP = gravel and stone not permitted for any roof height.

- a. *Mean roof height* as defined in ASCE 7.
- b. *Risk category* shall be determined in accordance with Section 1604.5.
- c. Exposure category shall be determined in accordance with Section 1609.4.3.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 66)

14B-15-1505 Fire classification.

The provisions of Section 1505 of IBC are adopted by reference with the following modifications:

1. Revise Section 1505.2 by replacing "*approved testing agency*" with "*approved agency*."
2. Revise the exceptions to 1505.2 by replacing "decks" with "*roof decks*."
3. Revise Table 1505.1 by changing the value for IIB from "C ^C" to "B."
4. Revise the notes to Table 1505.1 to read:
 - "a. Unless otherwise required due to the location of the building within a fire district in accordance with Appendix D.
 - b. Class C *roof coverings* shall be allowed on *buildings* of Group R-3, R-4, R-5 and Group U occupancies, where there is a minimum *fire-separation distance* of 6 feet measured from the leading edge of the roof.
 - c. *Buildings* that are not more than two *stories above grade plane* and having not more than 6,000 square feet of projected roof area and where there is a minimum 10-foot *fire separation distance* from the leading edge of the roof to an *abutting property line* on all sides of the *building*, shall be allowed to have roofs of *fire-retardant-treated wood* shingles and shakes constructed in accordance with Section 1505.6 or No. 1 cedar or redwood shakes and No. 1 shingles constructed in accordance with Section 1505.7."
5. Revise Section 1505.3 by replacing "*approved testing agency*" with "*approved agency*."
6. Revise Section 1505.4 by replacing "*approved testing agency*" with "*approved agency*."
7. Delete Section 1505.5.
8. Revise Section 1505.10 to read:

"1505.10 Vegetative roofs, roof gardens and landscaped roofs.

Vegetative roofs, roof gardens and landscaped roofs shall comply with Sections 1507.16 and 1514."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 67)

14B-15-1506 Materials.

The provisions of Section 1506 of IBC are adopted by reference with the following modification:

1. Revise Section 1506.3 by replacing "*approved testing agency*" with "*approved agency*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 68)

14B-15-1507 Requirements for roof coverings.

The provisions of Section 1507 of IBC are adopted by reference with the following modifications:

1. Revise Tables 1507.1.1(1), 1507.1.1(2) and 1507.1.1(3) by changing the title of the third column to "INSTALLATION STANDARD" and deleting the fourth column.
2. Revise the first sentence of Section 1507.1.2 to read:

"At eaves, an ice barrier shall be installed for asphalt shingles, *metal roof shingles*, mineral-surfaced roll roofing, slate and slate-type shingles, wood shingles and wood shakes."
3. Revise Table 1507.2.8.2 to delete the rows for lead, lead-coated copper, lead-coated high-yield copper, and painted terne.
4. Revise Section 1507.4.2 by deleting "approved."

5. Revise Table 1507.4.3(1) to delete the rows for hard lead, lead-coated copper, soft lead, and terne and terne coated stainless.
6. Revise Section 1507.4.4 to replace "approved manufacturer's" with "manufacturer's recommended."
7. Revise the title of Section 1507.4.5 to "Underlayment."
8. Revise Section 1507.5.6 by deleting "approved."
9. Revise Table 1507.8 by deleting "(Type 316 for coastal areas)."
10. Revise Section 1507.8.9 by replacing "approved grading or inspection bureau or agency" with "approved agency."
11. Revise Section 1507.9.10 by replacing "approved grading or inspection bureau or agency" with "approved agency."
12. Revise Section 1507.11 and its subsections by replacing each instance of "modified bitumen roofing" with "modified bitumen roof covering."
13. Revise Section 1507.12.2 by deleting "ASTM D5019."
14. Revise Section 1507.12.3 to read:

"1507.12.3 Ballasted thermoset low-sloped roofs.

Ballasted thermoset *low-sloped roofs* shall be installed in accordance with this section and Section 1504.4. Stone used as *ballast* shall comply with ASTM D448 or ASTM D7655."

15. Revise Section 1507.13.3 to read:

"1507.13.3 Ballasted thermoplastic low-sloped roofs.

Ballasted thermoplastic *low-sloped roofs* shall be installed in accordance with this section and Section 1504.4. Stone used as *ballast* shall comply with ASTM D448 or ASTM D7655."

16. Revise Section 1507.15 to read:

"1507.15 Liquid-applied roofing.

The installation of cold and hot applied liquid-applied roofing shall comply with the provisions of this section."

17. Revise Section 1507.15.2 to read:

"1507.15.2 Material standards.

Liquid-applied roofing shall comply with ASTM C836, ASTM C957, ASTM D1227 or ASTM D3468, ASTM D6083, ASTM D6694, ASTM D6947 or other *approved* material standard."

18. Revise Section 1507.16 to read:

"1507.16 Vegetative roofs, roof gardens and landscaped roofs.

Vegetative roofs, roof gardens and landscaped roofs shall comply with the requirements of this chapter and Section 1607.13.3."

19. Revise Section 1507.17.2 to read:

"1507.17.2 Deck slope.

Photovoltaic shingles shall not be installed on *low-sloped roofs*."

20. Revise Section 1507.18.2 to read:

"1507.18.2 Deck slope.

BIPV roof panels shall not be used on *low-sloped roofs*."

21. Revise Section 1507.18.4.2, excluding the exception, to read:

"1507.18.4.2 Ice barrier.

An ice barrier consisting of not fewer than two layers of *underlayment* cemented together or of a self-adhering polymer/modified bitumen sheet shall be used instead of normal *underlayment* and extend from the lowest edges of all roof surfaces to a point not less than 24 inches (610 mm) inside the exterior wall line of the building."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 69)

14B-15-1508 Roof insulation.

The provisions of Section 1508 of IBC are adopted by reference with the following modification:

1. Revise Section 1508.1 to read:

"1508.1 General.

The use of above-deck thermal insulation shall be permitted provided that such insulation is covered with a *roof covering* and passes the tests of NFPA 276 or UL 1256 when tested as an assembly.

Exceptions:

1. *Foam plastic insulation* shall conform to the material and installation requirements of Chapter 26.
2. Where a concrete *roof deck* is used and the above-deck thermal insulation is covered with a *roof covering*."

14B-15-1509 Radiant barriers installed above deck.

The provisions of Section 1509 of IBC are adopted by reference with the following modifications:

1. Revise Section 1509.1 to replace "deck" with "*roof deck*."
2. Revise Section 1509.2 to replace "decks" with "*roof decks*" and "an approved" with "a."
3. Revise Section 1509.4 to replace "'deck" with "*roof deck*."

14B-15-1510 Unoccupied rooftop structures.

The provisions of Section 1510 of IBC are adopted by reference with the following modifications:

1. Revise Section 1510.1 and its subsections to read:

"1510.1 General.

The provisions of this section shall govern the construction of unoccupied *rooftop structures*.

1510.1.1 Area limitation.

The combined area of parapet walls, unoccupied *rooftop structures*, rooftop access *penthouses* in accordance with Section 1510.2 and structures accessory to an *occupiable rooftop* in accordance with Section 1513.1.1 shall not exceed one-third the area of the supporting *roof deck*. Such *rooftop structures* shall not be required to be included in determining *building height*, number of *stories above grade plane* or *building area* in accordance with Chapter 2, nor shall the area of such *rooftop structures* be included in determining the *fire areas* specified in Section 901.7.

Exception: If the combined area of *rooftop structures* exceeds one-third the horizontal area of the supporting *roof deck*, the height and area of the *rooftop structures* shall be considered in determining *building height*, number of *stories above grade plane*, *building area* and *fire area*."

2. Revise Section 1510.2 and its subsections to read:

"1510.2 Mechanical penthouses.

Mechanical *penthouses* in compliance with Sections 1510.2.1 through 1510.2.5 shall be considered as a portion of the *story* directly below the *roof deck* on which such *penthouses* are located. Other *penthouses* shall be considered as an additional *story* of the *building*.

1510.2.1 Height above roof deck.

Mechanical *penthouses* shall not exceed 18 feet (5486 mm) in height above the *roof deck* as measured to the average height of the roof of the *penthouse*.

Exception: On *buildings* of Type I construction, mechanical *penthouses* shall be permitted to have a maximum height of 28 feet (8534 mm) above the *roof deck*.

1510.2.2 Use limitations.

Mechanical *penthouses* shall not be used for purposes other than the shelter of mechanical or electrical equipment, tanks, elevators and related machinery, or vertical shaft openings in the *roof assembly*.

1510.2.3 Weather protection.

Provisions such as louvers, louver blades or flashing shall be made to protect the mechanical and electrical equipment and the building interior from the elements.

1510.2.4 Type of construction.

Mechanical *penthouses* shall be constructed with walls, floors and roofs as required for the type of construction of the building on which such *penthouses* are built.

Exceptions:

1. On buildings of Type I and II-A construction, the *exterior walls* and *roof decks* of mechanical *penthouses* with a *fire separation distance* greater than 5 feet (1524 mm) and less than 20 feet (6096 mm) shall be noncombustible construction and permitted to have not less than a 1-hour *fire-resistance rating*. The

exterior walls and roofs of mechanical *penthouses* with a *fire separation distance* of 20 feet (6096 mm) or greater shall be noncombustible construction and not be required to have a *fire-resistance rating*.

2. [Reserved.]

3. On *buildings* of Type III, IV or V construction, the *exterior walls* of mechanical *penthouses* with a *fire separation distance* greater than 5 feet (1524 mm) and less than 20 feet (6096 mm) shall be permitted to have not less than a 1-hour *fire-resistance rating* or a lesser *fire-resistance rating* as required by Table 602. On *buildings* of Type III, IV or VA construction, the *exterior walls* of mechanical *penthouses* with a *fire separation distance* of 20 feet (6096 mm) or greater shall be permitted to be of heavy timber construction complying with Sections 602.4 and 2304.11 or noncombustible construction and shall not be required to have a *fire-resistance rating*."

3. Revise Section 1510.3 to read:

"1510.3 Tanks.

Tanks having a capacity of more than 500 gallons (1893 L) located on the *roof deck* of a building shall be supported on masonry, reinforced concrete, steel or heavy timber construction complying with Section 2304.11 provided that, where such supports are located in the *building* above the lowest *story*, the support shall have a *fire-resistance rating* as required for Type IA construction."

4. Revise Section 1510.5 and its subsections to read:

"1510.5 Towers, spires, minarets, domes and cupolas.

Towers, spires, minarets, domes, cupolas, and similar features shall be of a type of construction having *fire-resistance ratings* not less than required for the *building* on top of which such feature is built. Such features greater than 85 feet (25 908 mm) in height above *grade plane* as measured to the highest point on such feature, and either greater than 200 square feet (18.6 m²) in horizontal area or used for any purpose other than a belfry or an architectural embellishment, shall be constructed of and supported on Type I or IIA construction.

1510.5.1 Noncombustible construction required.

Features greater than 60 feet (18 288 mm) in height above the highest point at which such feature contacts the roof as measured to the highest point on such feature, or that exceeds 200 square feet (18.6 m²) in area at any horizontal section, or which is intended to be used for any purpose other than a belfry or architectural embellishment, or is located on the top of a *building* greater than 50 feet (1524 mm) in *building height* shall be constructed of and supported by noncombustible materials and shall be separated from the *building* below by construction having a *fire-resistance rating* of not less than 1.5 hours with openings protected in accordance with Section 711. Such features located on the top of a *building* greater than 50 feet (15 240 mm) in *building height* shall be supported by noncombustible construction.

1510.5.2 Enclosed features

Enclosed features shall have *exterior walls* constructed as required for the *building* on top of which such features are built. The *roof covering* of features shall be not less than the same class of *roof covering* required for the *building* on top of which the feature is located."

5. Revise Section 1510.6 by replacing "5 feet (1524 mm)" with "15 feet (4572 mm)."

6. Revise the exception to Section 1510.6.1 by replacing "Type IA" with "Type I."

7. Revise Section 1510.6.2 to read:

"1510.6.2 Type I, II, III or IV construction.

Regardless of the requirements in Section 1510.6, *mechanical equipment screens* that are located on the *roof decks* of buildings of Type I, II, III or IV construction shall be permitted to be constructed of combustible materials in accordance with the following limitation: the *fire separation distance* shall be not less than 20 feet (6096 mm) and the height of the *mechanical equipment screen* above the roof deck shall not exceed 4 feet (1219 mm) as measured to the highest point on the *mechanical equipment screen*."

8. Revise Section 1510.6.3 to read:

"1510.6.3 Type V construction.

The height of *mechanical equipment screens* located on the *roof decks* of buildings of Type V construction, as measured from *grade plane* to the highest point on the *mechanical equipment screen*, shall not exceed the maximum *building height* allowed for the building by other provisions of this code."

9. Revise Section 1510.7 to read:

"1510.7 Photovoltaic panels and modules.

Rooftop-mounted *photovoltaic panels* and *modules* shall be designed in accordance with this section and the *Chicago Fire Prevention Code*."

10. Revise Section 1510.8 by replacing "Sections 1510.8.1 through 1510.8.5, as applicable" with "Section 1510.8.1."

11. Delete Sections 1510.8.2, 1510.8.3, 1510.8.4, and 1510.8.5.

14B-15-1511 Reroofing.

The provisions of Section 1511 of IBC are not adopted. The following language is adopted as Section 1511:

"1511. REROOFING

1511.1 General.

Roof repairs and roof replacement shall comply with Section 306 of the *Chicago Building Rehabilitation Code*."

14B-15-1512 IBC Section 1512.

The provisions of Section 1512 of IBC are not adopted.

14B-15-1513 Occupiable rooftops.

The following language is adopted as a new Section 1513:

"1513. OCCUPIABLE ROOFTOPS

1513.1 General.

The provisions of this section shall govern the construction of *occupiable rooftops* and associated *rooftop structures*.

1513.1.1 Area limitation.

The combined area of *rooftop structures* associated with an *occupiable rooftop* and exceeding 48 inches (1219 mm) in height above the lowest walking surface of the *occupiable rooftop* shall comply with Section 1510.1.1.

1513.2 Rooftop access penthouses.

Where the *occupied rooftop* is located above the highest *story*, *penthouses* providing access to or egress from an *occupiable rooftop* in compliance with Sections 1513.2.1 through 1513.2.5 shall be considered as a portion of the *story* directly below the *roof deck* on which such *penthouses* are located. Other fully-enclosed *occupiable space* at the roof level shall be considered as an additional *story* of the *building*.

1513.2.1 Height above roof deck.

Rooftop access *penthouses* shall not exceed 12 feet (3658 mm) in height above the *roof deck* as measured to the average height of the roof of the *penthouse*.

Exception: Where used to enclose elevators that travel to the roof level, rooftop access *penthouses* shall be permitted to have a maximum height of 20 feet (6096 mm) above the *roof deck*.

1513.2.2 Use limitations.

Rooftop access *penthouses* shall not contain *habitable space* or any *occupiable space* other than *circulation paths* necessary to access *means of egress* and elevators serving an *occupiable rooftop*. Rooftop access *penthouses* shall not contain toilet rooms or bathrooms.

1513.2.3 Type of construction.

Rooftop access *penthouses* shall be constructed with walls, floors and roofs as required for the type of construction of the *building* on which such *penthouses* are built.

1513.3 Outdoor elements.

The walking surface and other outdoor elements of an *occupiable rooftop* shall comply with the requirements of Sections 1513.3.1 through 1513.3.2.

1513.3.1 Buildings not exceeding 55 feet in building height.

The outdoor elements of *occupiable rooftops* on *buildings* not exceeding 55 feet (16.8 m) in *building height* shall be of noncombustible materials or combustible materials as allowed by Table 1513.3.1 or Table 705.2.1 or 705.2.2 for *decks*. Outdoor elements not listed in the tables shall conform to the type of construction and *fire-resistance rating* required for the *building* to which they are attached.

TABLE 1513.3.1

REQUIRED FIRE SEPARATION DISTANCE FOR OUTDOOR ELEMENTS OF OCCUPIABLE ROOFTOPS ON BUILDINGS UP TO 55 FEET IN BUILDING HEIGHT ^a

	Material	Fire Separation Distance (feet)

Type of Element	Type	0 to less than 3	3 to less than 5	5 to less than 10	10 or greater
Type of Element	Material Type	Fire Separation Distance (feet)			
		0 to less than 3	3 to less than 5	5 to less than 10	10 or greater
Walking surfaces covering no more than one third of the roof area or 500 square feet, whichever is greater, on side protected by a noncombustible parapet wall providing a 2-hour <i>fire-resistance rating</i> , at least 36 inches above the walking surface	U	Yes	Yes	Yes	Yes
Walking surfaces on <i>buildings</i> of Group R-5 occupancy covering more than one third of the roof area, with or without parapet wall	FRTW	No	Yes	Yes	Yes
Walking surfaces covering more than one third of the roof area or on side without a parapet wall	U	No	No	Yes	Yes
	UNC	Yes	Yes	Yes	Yes
Pickets, handrails and similar vertical elements not more than 48 inches in height above the adjacent walking surface	U	No	Yes	Yes	Yes
	UNC	Yes	Yes	Yes	Yes
Vertical elements of pergolas, trellises, arbors, privacy screens and similar structures greater than 42 inches in height above the adjacent walking surface, with 50 percent or more of surface area consisting of regularly distributed openings	U ^b	No	Yes	Yes	Yes
	UNC	Yes	Yes	Yes	Yes
Vertical elements of pergolas, trellises, arbors, privacy screens and similar structures greater than 42 inches in height above the adjacent walking surface, solid or with less than 50 percent of surface area consisting of regularly distributed openings	U ^b	No	No	Yes	Yes
	UNC	Yes	Yes	Yes	Yes
Horizontal elements of pergolas, trellises, arbors and similar structures greater than 42 inches in height above the adjacent walking surface	U ^b	No	Yes	Yes	Yes
	UNC	Yes	Yes	Yes	Yes

For SI: 1 foot = 304.8 mm; 1 inch = 25.4 mm; 1 square foot = 0.0929 m².

U = any material not less fire resistive than untreated wood or plastic composites that comply with ASTM D7032 and Section 2612, unprotected

FRTW = *Fire-retardant treated wood*

P = any material, protected with construction providing *fire-resistance rating* as required for *exterior wall*

HT = heavy timber construction, complying with Section 2304.11

UNC = noncombustible, unprotected

PNC = noncombustible, protected with construction providing *fire-resistance rating* as required for *exterior wall*

- a. Where material type U is allowed, material types P, HT, UNC and PNC shall also be allowed; where material type UNC is allowed, material type PNC shall also be allowed.
- b. Combustible pergolas, arbors, trellises, privacy screens and similar structures shall not exceed 12 feet in height above the highest adjacent walking surface.

1513.3.2 Buildings greater than 55 feet in height.

The outdoor elements of *occupiable rooftops* on buildings greater than 55 feet (16.8 m) in height shall be of noncombustible materials.

Exceptions:

1. Pickets, handrails and similar vertical elements not more than 4 feet (1219 mm) in height above the adjacent walking surface of material not less fire resistive than untreated wood or plastic composites that comply with ASTM D7032 and Section 2612 if the *fire separation distance* is greater than or equal to 20 feet (6096 mm).
2. Walking surfaces *listed* as a Class A *roof assembly* or a Class A *roof covering* installed over a noncombustible *roof assembly*.
3. Limited non-structural elements of material not less fire resistive than untreated wood or plastic composites that comply with ASTM D7032 where *approved* by the *building official*.

1513.4 Structures providing weather protection.

Structures above an *occupiable rooftop* that provide weather protection equivalent to a roof and project more than 24 inches (610 mm) horizontally from a wall or other support shall be treated as part of the *building* to which they are attached for purposes of determining *building height* and number of *stories above grade plane*.

1513.5 Means of egress.

Occupiable rooftops shall be provided with *means of egress* as required in Chapter 10.

1513.6 Construction type.

Occupiable rooftops shall not be installed on buildings of Type VB construction. The *roof assembly* below an *occupiable rooftop* shall have a *fire-resistance rating* of not less than 1 hour."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 71)

14B-15-1514 Vegetative roofs, roof gardens and landscaped roofs.

The following language is adopted as a new Section 1514:

"1514. VEGETATIVE ROOFS, ROOF GARDENS AND LANDSCAPED ROOFS

1514.1 General.

Vegetative roofs, roof gardens and landscaped roofs shall be installed and maintained in accordance with Sections 1505, 1507.16, and 1514.2 through 1514.6.

1514.1.1 Intended for occupancy.

Vegetative roofs, roof gardens and landscaped roofs that are intended for occupancy shall also comply with Section 1513.

1514.2 Standard.

Vegetative roofs, roof gardens and landscaped roofs shall be installed in accordance with ANSI/SPRI VF-1.

1514.3 Maximum area.

Vegetative roof, roof garden or landscaped roof areas shall not exceed 15,625 square feet (1450 m²) for any single area with a maximum dimension of 125 feet (39 m) in length and width. A minimum 6 foot (1.8 m) wide clearance consisting of Class A *roof covering* shall be provided between adjacent *vegetative roof*, roof garden or landscaped roof areas.

1514.4 Clearances.

Vegetative roof, roof garden and landscaped roof areas shall be separated from combustible vertical surfaces, intake and exhaust openings and rooftop equipment by a 6 foot (1.8 m) wide surface consisting of Class A *roof covering*.

Exception: Separation from plumbing exhaust openings is not required.

1514.5 Maintenance.

Vegetation shall be maintained in accordance with Sections 1514.5.1 through 1514.5.3.

1514.5.1 Maintenance plan.

The *fire code official* is authorized to require a maintenance plan for vegetation placed on roofs.

1514.5.2 Water source.

A water source shall be provided to maintain levels of hydration necessary to keep vegetation alive and to keep dry foliage to a minimum.

1514.5.3 Dead foliage.

Excess biomass, such as overgrown vegetation, leaves and other dead and decaying material shall be removed at regular intervals not less than two times per year.

1514.6 Maintenance equipment.

Fueled equipment used for the care and maintenance of vegetation on roofs shall be stored in accordance with the *Chicago Fire Prevention Code*."

14B-15-1515 Solar reflectance.

The following language is adopted as a new Section 1515:

"1515. SOLAR REFLECTANCE

1515.1 Scope.

Roof coverings shall comply with the solar reflectance requirements in Section 1515.2.

Exceptions:

1. Walking surfaces of *occupiable rooftops*.
2. *Vegetative roofs*, roof gardens and landscaped roofs.
3. Photovoltaic and solar thermal equipment.

1515.2 Solar reflectance.

All *roof coverings* shall have a minimum solar reflectance as specified in Sections 1515.2.1 or 1515.2.2 as tested in accordance with ASTM C1549, ASTM E903, ASTM E1918 or CRRC-S100.

1515.2.1 Low-sloped roofs.

Roof coverings on *low-sloped roofs* shall have an initial reflectance value of 0.72 or a three-year-aged solar reflectance value of 0.5 or greater.

Exceptions:

1. Where more than 50 percent of the roof area is a *vegetative roof* or roof garden, the remainder of the roof area shall have a three-year-aged solar reflectance value of 0.3 or greater.
2. Roofs with a minimum of 15 pounds per square foot (0.72 kN/m^2) of *ballast* installed over the entire roof area to resist wind uplift shall have a three-year-aged solar reflectance value of 0.3 or greater.

1515.2.2 Other than low-sloped roofs.

Roof coverings on other than *low-sloped roofs* shall have an initial reflectance value of 0.15 or greater.

Exceptions:

1. *Roof coverings* installed with a slope of 5:12 (41 percent) or greater.
2. *Historic buildings.*"

(Amend Coun. J. 9-21-22, p. 52080, Art. II, § 6)

14B-15-1516 Interior roof edge for shafts and fully enclosed courts.

The following language is adopted as a new Section 1516:

“1516. LOW-SLOPED ROOF EDGES AND OPENINGS

1516.1 Scope.

Low-sloped roofs shall comply with the applicable requirements in Sections 1516.2 and 1516.3.

1516.2 Roof edges adjoining shafts and fully enclosed courts.

The edges of a *low-sloped roof* that adjoin a shaft or a *court* that is enclosed on all sides shall be provided with a parapet, extended masonry, or guard, or any combination thereof, that meets the requirements of Section 1015.

1516.3 Skylights and other openings.

All skylights and other openings located in the plane of a *low-sloped roof* that are not otherwise required to remain open and unobstructed by law shall be either: (i) glazed with wired glass, plain glass, or glass block that is designed to meet the applicable live load requirements for the roof under Section 1607; or (ii) provided with a parapet, extended masonry, or guard, or any combination thereof, that meets the requirements of Section 1015.”

(Added Coun. J. 10-9-24, p. 18818, § 1)

CHAPTER 14B-16

STRUCTURAL DESIGN

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14B-16-1602 Notations.

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14B-16-1605 Load combinations.

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14B-16-1608 Snow loads.

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14B-16-1611 Rain loads.

14B-16-1612 Flood design.

14B-16-1613 Earthquake loads.

14B-16-1614 Atmospheric ice loads.

14B-16-1615 IBC Section 1615.

14B-16-1616 Structural integrity.

14B-16-1601 General.

The provisions of Section 1601 of IBC are adopted by reference without modification.

14B-16-1602 Notations.

The provisions of Section 1602 of IBC are adopted by reference without modification.

14B-16-1603 Construction documents.

The provisions of Section 1603 of IBC are adopted by reference with the following modifications:

1. Revise the exception to Section 1603.1 by deleting item 5.

2. Revise the first two sentences of Section 1603.1.3 to read:

"The ground snow load, P_g , shall be indicated. The following additional information shall also be provided, regardless of whether snow loads govern the design of the roof:"

3. Revise item 6 in Section 1603.1.3 to read:

"6. Drift surcharge load(s), P_d ."

4. Revise item 1 in Section 1603.1.4 to read:

"1. Basic design wind speed, V , miles per hour or allowable stress design wind speed, V_{asd} , as determined in accordance with Section 1609.3.1."

5. Add a new item 6 to Section 1603.1.4 to read:

"6. Design base shear(s)."

6. Revise Section 1603.1.5 by inserting the following between items 6 and 7:

"The following information shall be shown for *structures* assigned to *seismic design category* B, C or D:"

7. Delete Section 1603.1.7.

8. Delete Section 1603.1.9.

14B-16-1604 General design requirements.

The provisions of Section 1604 of IBC are adopted by reference with the following modifications:

1. Revise Section 1604.2 to read:

"1604.2 Strength.

Buildings and other *structures*, and parts thereof, shall be designed and constructed to support the factored loads in load combinations defined in this code without exceeding the appropriate strength limit states for the materials of construction. Alternatively, *buildings* and other *structures*, and parts thereof, shall be designed and constructed to support the *nominal loads* in load combinations defined in this code without exceeding the appropriate specified allowable stresses for the materials of construction.

Loads and forces for occupancies or uses not covered in this chapter shall be *approved* by the *building official*."

2. Revise Section 1604.5, including the exception, to read:

"1604.5 Risk category.

Each *building* and *structure* shall be assigned a *risk category* in accordance with Table 1604.5. Where a referenced standard specifies an occupancy category, the *risk category* shall not be taken as lower than the occupancy category specified therein. Where

a referenced standard specifies that the assignment of a *risk category* be in accordance with ASCE 7, Table 1.5-1, Table 1604.5 shall be used in lieu of ASCE 7, Table 1.5-1."

3. Replace Table 1604.5 as shown:

TABLE 1604.5

RISK CATEGORY OF BUILDINGS AND OTHER STRUCTURES

RISK CATEGORY	NATURE OF OCCUPANCY
I	<i>Structures</i> that represent a low hazard to human life in the event of failure, including but not limited to: • Agricultural <i>buildings</i>. • Minor storage <i>structures</i>.
II	<i>Structures</i> except those listed in <i>Risk Categories</i> I, III and IV.
III	<i>Structures</i> that represent a substantial hazard to human life in the event of failure, including but not limited to: • <i>Buildings</i> containing one or more Group A occupancies with a combined <i>occupant load</i> in Group A occupancies greater than 1,000^a • <i>Buildings</i> containing Group E occupancies with an <i>occupant load</i> greater than 500.^a • <i>Buildings</i> containing Group I-2, Condition 1 occupancies with 50 or more care recipients. • <i>Buildings</i> containing Group I-2, Condition 2 occupancies not providing emergency surgery or emergency treatment. • <i>Buildings</i> containing Group I-3 occupancies. • Any other <i>building</i> with an <i>occupant load</i> greater than 5,000.^a • Power-generating stations, water treatment facilities for potable water, wastewater treatment facilities and other public utility <i>structures</i> not included in <i>Risk Category</i> IV. • <i>Structures</i> not included in <i>Risk Category</i> IV containing quantities of <i>toxic</i> or <i>explosive</i> materials that exceed maximum allowable quantities per <i>control area</i> as given in Table 307.1(1) or 307.1(2) or per outdoor <i>control area</i> in accordance with the <i>Chicago Fire Prevention Code</i>; and are sufficient to pose a threat to the public if released.^b
IV	<i>Structures</i> designated as <i>essential facilities</i>, including but not limited to: • <i>Buildings</i> containing Group I-2, Condition 2 occupancies providing emergency surgery or emergency treatment. • <i>Buildings</i> containing fire, rescue, ambulance and police stations and emergency vehicle garages. • <i>Buildings</i> containing earthquake, hurricane or other emergency shelters. • Designated emergency preparedness, communications and operations centers and other <i>structures</i> required for emergency response. • Power-generating stations and other utility <i>structures</i> providing emergency backup for <i>Risk Category</i> IV <i>structures</i>. • <i>Structures</i> containing quantities of <i>highly toxic</i> materials that: exceed maximum allowable quantities per <i>control area</i> as given in Table 307.1(2) or per outdoor <i>control area</i> in accordance with the <i>Chicago Fire Prevention Code</i>; and are sufficient to pose a threat to the public if released.^b • Aviation control towers, air traffic control centers and emergency aircraft hangars. • <i>Structures</i> having critical public safety or national defense functions. • Water storage and pump <i>structures</i> required to maintain water pressure for fire suppression.

a. *Occupant load* shall be determined in accordance with Table 1004.5.

b. Where *approved* by the *fire code official*, the classification of *structures* as *Risk Category* III or IV based on their quantities of *toxic*, *highly toxic* or *explosive* materials is permitted to be reduced to *Risk Category* II, provided that it can be demonstrated by a hazard assessment in accordance with Section 1.5.3 of ASCE 7 that a release of the *hazardous materials* is not sufficient to pose a threat to the public.

4. Delete the exception to Section 1604.5.1.

5. Revise Section 1604.7 by replacing "*approved*" with "*accepted*."

6. Revise Section 1604.8.3 to read:

"1604.8.3 Decks, porches, exterior balconies and similar structures.

Where supported by attachment to an *exterior wall*, *decks*, *porches*, *exterior balconies* and similar *structures* shall be positively anchored to the primary building structure and designed for both vertical and lateral loads as applicable. Such attachment shall not be accomplished by the use of toenails or nails subject to withdrawal. Where positive connection to the primary building structure cannot be verified, *decks*, *porches*, *exterior balconies* and similar *structures* shall be self-supporting. Connections of *decks*, *porches*, *exterior balconies* and similar *structures* with cantilevered framing members to *exterior walls* or other framing members shall be designed for both of the following:

1. The reactions resulting from the dead load and live load specified in Table 1607.1, or the snow load specified in

Section 1608, in accordance with Section 1605, acting on all portions of the *structure*.

2. The reactions resulting from the dead load and live load specified in Table 1607.1, or the snow load specified in Section 1608, in accordance with Section 1605, acting on the cantilevered portion of the *structure*, and no live load or snow load on the remaining portion of the *structure*."

7. Delete Section 1604.10.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 72)

14B-16-1605 Load combinations.

The provisions of Section 1605 of IBC are adopted by reference with the following modifications:

1. Revise item 1 in the first paragraph of Section 1605.1 to read:

"1. The load combinations specified in Section 1605.2 or 1605.3.1."

2. Delete item 3 in the third paragraph of Section 1605.1.

3. Revise Section 1605.1.1 by replacing "*registered design professional*" with "*registered geotechnical engineer*."

4. Delete the first sentence of Section 1605.2.1.

5. Revise Section 1605.3 by deleting "or 1605.3.2."

6. Delete the first sentence of Section 1605.3.1.2.

7. Delete Section 1605.3.2 and its subsections.

14B-16-1606 Dead loads.

The provisions of Section 1606 of IBC are adopted by reference with the following modification:

1. Revise Section 1606.2 to read:

"1606.2 Design dead load.

For purposes of design, the actual weights of materials of construction and fixed equipment shall be used. In the absence of definite information, values used shall be *approved* by the *building official*."

14B-16-1607 Live loads.

The provisions of Section 1607 of IBC are adopted by reference with the following modifications:

1. Revise Table 1607.1 by replacing "OCCUPANCY OR USE" with "FUNCTION OR USE" in the heading of the first column.

1.1. Revise row 2 of Table 1607.1 to read:

2. Access floor systems		
Office use	50	2,000
Telecommunication equipment area	100	2,000

2. Revise row 4 of Table 1607.1 by inserting an entry for "Dressing rooms" with a value of "40."

3. Revise row 5 of Table 1607.1 to read:

5. Porches, exterior balconies, decks and similar structures ^h		
Accessible from a single dwelling unit	60	300 ^f
All others	100	300 ^f

4. Revise row 9 of Table 1607.1 by adding "including associated kitchens" in the first column.

5. Revise row 13 of Table 1607.1 by deleting the entry for "on single family dwellings only."

6. Revise row 21 of Table 1607.1 to read:

21. Mechanical Rooms	125 ^p	—
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7. Revise row 22 of Table 1607.1 to read:

22. Offices			
Corridors above first floor			
File rooms other than high-density files	80	2,000	
<i>Telecommunication equipment areas</i>	100	—	
	125 ^P	—	
Lobbies and first-floor corridors	100	2,000	
Offices and associated work areas, meeting rooms, break rooms, reception areas, and similar nonpublic gathering areas	50	2,000	

8. Revise row 25 of Table 1607.1 to read:

25. Residential			
Group R-3, R-4 and R-5 occupancies			
Uninhabitable attics without storage ⁱ	10		
Uninhabitable attics with storage ^{i, j, k}	20		
All other areas	40	—	
	40		
Group R-1 and R-2 occupancies	100		
Private rooms and corridors serving them			
Public rooms ^m and corridors serving them			

9. Revise row 26 of Table 1607.1 by deleting "except one- and two-family dwellings" and replacing "Occupiable roofs" with "Occupiable rooftops."

10. Revise row 27 of Table 1607.1 by adding an entry for "Shops" with a uniform value of "100" and concentrated value of "2,000."

11. Revise row 30 of Table 1607.1 by replacing "One- and two-family" with "Group R-5 (interior)."

12. Revise row 31 of Table 1607.1 to read:

31. Storage (shall be designed for heavier loads if required for anticipated storage)			
Heavy	250 ⁿ		
Light	125 ⁿ	—	
Telecommunications equipment areas	100 ^P		

13. Revise note d to Table 1607.1 by replacing "an approved method" with "an accepted engineering practice."

14. Revise note k to Table 1607.1 by replacing "habitable attics and sleeping rooms" with "all other areas."

15. Revise note l to Table 1607.1 to read:

"l. Areas of *occupiable rooftops* accessible from a single *dwelling unit* shall be designed in accordance with Row 5. Areas of roofs with vegetation and not intended for occupancy shall be designed in accordance with Section 1607.13.3."

16. Add a new note p to Table 1607.1 to read:

"p. Where actual fixed equipment loads are known, permanent aisles shall be allowed to be designed for a live load of 40 pounds per square foot and equipment areas shall be allowed to be designed for the actual dead loads."

17. Revise Section 1607.2 to read:

"1607.2 Loads not specified.

For occupancies or uses not designated in Table 1607.1, the live load shall be *approved* by the *building official*."

18. Revise item 1 in Section 1607.7.2 to read:

"1. The actual operational loads, including outrigger reactions and contact areas of the vehicles."

19. Revise the exception to Section 1607.7.3 by deleting "and are approved by the building official."
20. Revise exception 1 to Section 1607.8.1 to read:
 - "1. For Group R-5 occupancies and within individual *dwelling units*, only the single concentrated load required by Section 1607.8.1.1 shall be applied."
21. Revise Section 1607.9 to replace "an *approved* method" with "an accepted engineering practice."
- 21.1. Revise Section 1607.10.4 to read:

"1607.10.4 Fall arrest, lifeline, and rope descent system anchorages.

In addition to any other applicable *live loads*, fall arrest, lifeline, and rope descent system anchorages and structural elements that support these anchorages shall be designed for a *live load* of not less than 3,100 pounds (13.8 kN) for each attached line, in any direction that the *load* can be applied. Anchorages of horizontal lifelines and the structural elements that support these anchorages shall be designed for the maximum tension that develops in the horizontal lifeline from these *live loads*."

22. Revise Section 1607.13 to delete "and marquees."
23. Revise Section 1607.13.2 to delete "and marquees."
24. Revise Section 1607.13.3 to read:

"1607.13.3 Occupiable rooftops.

Occupiable rooftops and *vegetative roofs* are permitted to have their uniformly distributed live loads reduced in accordance with Section 1607.11 for floors."

25. Delete Section 1607.15.2.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 73; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 28; Amend Coun. J. 1-26-22, p. 43756, Art. II, § 17; Amend Coun. J. 1-24-24, p. 8634, Art. IV, § 5)

14B-16-1608 Snow loads.

The provisions of Section 1608 of IBC are adopted by reference with the following modifications:

1. Revise Section 1608.2 to read:

"1608.2 Ground snow load.

The ground snow load to be used in determining the design snow loads for roofs shall be 25 pounds per square foot (1.20 kN/ m²)."

2. Delete all figures and tables included in Section 1608.

14B-16-1609 Wind loads.

The provisions of Section 1609 of IBC are adopted by reference with the following modifications:

1. Revise Section 1609.1.1 to read:

"1609.1.1 Determination of wind loads.

Wind loads on every *building* or *structure* shall be determined in accordance with Chapters 26 to 30 of ASCE 7. The basic design wind speed, *V*, and the exposure category for a site shall be determined in accordance with Section 1609. Wind shall be assumed to come from any horizontal direction and wind pressures shall be assumed to act normal to the surface considered.

Exceptions:

1. [Reserved.]
2. *Buildings* of Group R-3, R-4 or R-5 occupancy using the provisions of AWC WFCM (including *buildings* containing up to 3 *dwelling units*).
3. *Buildings* of Group R-3, R-4 or R-5 occupancy using the provisions of AISI S230 (including *buildings* containing up to 3 *dwelling units*).
4. Designs using NAAMM FP 1001.
5. Designs using TIA-222 for antenna-supporting *structures* and antennas, provided that the horizontal extent of Topographic Category 2 escarpments in Section 2.6.6.2 of TIA-222 shall be 16 times the height of the escarpment.
6. Wind tunnel tests in accordance with ASCE 49 and Sections 31.4 and 31.5 of ASCE 7.
7. Wind loads determined in accordance with Chapter 28 of ASCE 7 shall only be allowed for *buildings* containing only Group F, S or U occupancies.

8. Wind loads determined in accordance with Section 1609.6.

The wind speeds in Section 1609.3 are basic design wind speeds, V . Allowable stress design wind speeds, V_{asd} , from Section 1609.3.1 shall be used when the provisions of the standards referenced in Exceptions 4 and 5 are used."

2. Delete Section 1609.1.1.1.

3. Add a new Section 1609.1.2 to read:

"1609.1.2 Special investigations.

Special investigations shall be required to determine wind loads for *buildings* and other *structures* with a *mean roof height* exceeding 400 feet (122 m). Special investigations shall be required to determine wind loads for *buildings* and other *structures* with a *mean roof height* of 400 feet (122 m) or less and having irregular shapes, response characteristics or site locations with shielding or channeling effects that warrant special consideration."

4. Delete Section 1609.2 and its subsections and tables.

5. Revise Section 1609.3 to read:

"1609.3 Basic design wind speed.

The basic design wind speed, V , for nominal design 3-second gust wind speeds at 33 feet (10 m) above ground for exposure category C for the determination of the wind loads shall be as follows:

1. The basic design wind speed, V , for use in the design of *Risk Category I structures* shall be 101 mph (45 m/s).
2. The basic design wind speed, V , for use in the design of *Risk Category II structures* shall be 107 mph (48 m/s).
3. The basic design wind speed, V , for use in the design of *Risk Category III structures* shall be 114 mph (51 m/s).
4. The basic design wind speed, V , for use in the design of *Risk Category IV structures* shall be 119 mph (53 m/s)."

6. Delete all figures and tables included in Section 1609.3.

7. Revise Section 1609.3.1 to read:

"1609.3.1 Wind speed conversion.

Where required, the basic design wind speeds of Section 1609.3 shall be converted to allowable stress design wind speeds, V_{asd} , using Equation 16-33."

8. Revise the Description of V in Equation 16-33 to read:

" V = Basic design wind speed determined from Section 1609.3."

9. Delete Table 1609.3.1.

10. Revise Section 1609.4.2 by deleting the second sentence describing Surface Roughness D.

11. Insert a new Section 1609.6 to read:

"1609.6 Alternative wind loads for enclosed structure with mean roof height of 60 feet or less.

Where the *mean roof height* of a *structure* is 60 feet (18.3 m) or less, wind loads on the *main windforce-resisting system* and on components and cladding shall be permitted to be determined in accordance with Section 1609.6, provided the following conditions are met:

1. The *structure* is classified as "enclosed" in accordance with Chapter 26 of ASCE 7.
2. Both the windward and leeward wind loads are transmitted by roof and vertically spanning wall assemblies, through continuous floor and roof diaphragms, to the *main windforce-resisting system*.
3. The longest horizontal dimension of the *structure* is no more than five times the shortest horizontal dimension of the *structure*.
4. The *structure* has no unusual geometrical irregularity in spatial form.
5. The *structure* does not have response characteristics that make it subject to across-wind loading, vortex shedding, or instability caused by galloping or flutter.
6. The *structure* does not have a site location for which channeling effects or buffeting in the wake of upwind obstructions warrant special consideration.

1609.6.1 Main windforce-resisting system.

The load due to wind pressure, W , on the *main windforce-resisting system* shall be based on exposure category and *risk category* as indicated in Table 1609.6.1.

TABLE 1609.6.1

LOAD DUE TO WIND PRESSURE ON MAIN WINDFORCE-RESISTING SYSTEM ^{c, d, e, f, g}

EXPOSURE CATEGORY	LOAD DUE TO WIND PRESSURE (pounds per square foot)		
	Risk Category		
	I or II	III	IV
B	25	28	31
C	32	37	40
D	38	43	46

For SI: 1 pound per square foot = 0.0479 kN/ m²

- Exposure category* shall be determined in accordance with Section 1609.4.3.
- Risk category* shall be determined in accordance with Section 1604.5.
- Wind loads shall use the following load cases (see ASCE 7, Chapter 27):
 - Case 1:** 100 percent of the design wind pressure acting on the projected area perpendicular to each principal axis of the structure, considered separately for each principal axis.
 - Case 2:** 75 percent of the design wind pressure acting on the projected area perpendicular to each principal axis of the structure, in conjunction with a torsional moment resulting from this wind pressure acting at an eccentricity equal to 15 percent of the building width, considered separately for each principal axis.
 - Case 3:** Wind loading as defined in Case 1 but considered to act simultaneously on each principal axis at 75 percent of the value specified in Case 1.
 - Case 4:** Wind loading as defined in Case 2 but considered to act simultaneously on each principal axis at 75 percent of the value specified in Case 2.
- For buildings that include parapets, additional parapet load shall be applied in accordance with ASCE 7, Section 27.3.4.
- For the *main windforce-resisting system* of a roof, the values shall be multiplied by 1.0 for negative (uplift) pressure and 0.3 for positive (downward) pressure. (See Section 1609.6.2 for components and cladding.)
- All wind loads are ultimate wind loads (W) and shall be utilized in load combinations using strength or load and resistance factor design per Section 1605.2. Where required, service level loads may be obtained by dividing the values in the table by 1.6, however these service level loads shall not be used for either method of load combination prescribed in Section 1605.2 or 1605.3.1.
- Projecting elements such as flag poles, sign structures, tanks and towers shall be designed in accordance with ASCE 7.

1609.6.2 Components and cladding.

The load due to wind pressure, *W*, on components and cladding shall be the values in Table 1609.6.1 multiplied by the multiplier provided in Table 1609.6.2.

TABLE 1609.6.2

MULTIPLIERS TO DETERMINE COMPONENTS AND CLADDING WIND PRESSURE

CONDITION	MULTIPLIER ^k	
	Flat Roof (less than 7° slope)	Sloped Gable Roof ^j
Wall (field) ^a	1.2	1.2
Wall (corner) ^{a, c}	1.4	1.4
Roof (field) ^b	1.0 ^h	2.0
Roof (edge) ^b	2.3 ^{d, h}	3.0 ^{f, i}
Roof (corner) ^b	3.2 ^{e, h}	3.6 ^{g, i}

For SI: 1 foot = 304.8 mm, 1 degree = 0.017 rad.

- Wall pressures shall act toward (positive) or away (negative) from the surface.
- Roof pressures shall be considered as acting away (negative) from the surface.
- Wall corners shall be defined as a distance from the corner equal to the lesser of 0.1 times the least horizontal dimension or 0.4 times the height in both directions (minimum 3 feet).
- Flat roof edges shall be defined as a distance from the roof edge of 1.2 times the height.
- Flat roof corners shall be defined as a distance from each roof corner of 0.6 times the height (for a width of 0.2 times the height).
- Sloped gable roof edges shall be defined as a distance from the roof edge equal to the lesser of 0.1 times the least horizontal dimension or 0.4 times the height (minimum 3 feet) from the roof edge.

- g. Sloped gable roof corners shall be defined as a distance from any roof corner or peak equal to the lesser of 0.1 times the least horizontal dimension or 0.4 times the height (minimum 3 feet) in both directions.
- h. For overhangs on flat roofs utilize the flat roof multipliers above (includes pressure contribution from both upper and lower surface).
- i. For overhangs on sloped gable roofs, increase the sloped gable roof multipliers above by 30 percent (includes pressure contribution from both upper and lower surface).
- j. Loads above shall not apply to hip roofs (Refer to Chapter 30 of ASCE 7).
- k. Resulting wind loads are ultimate wind loads and shall be utilized in load combinations using strength or load and resistance factor design per Section 1605.2 or converted in accordance with Section 1609.3.1.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 74; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 24)

14B-16-1610 Soil lateral loads.

The provisions of Section 1610 of IBC are adopted by reference with the following modifications:

1. Revise Section 1610.1, excluding the exception, to read:

"1610.1 General.

Foundation walls and retaining walls shall be designed to resist lateral soil loads. Soil loads specified in Table 1610.1 shall be used as the minimum design lateral soil loads unless determined otherwise by a geotechnical investigation in accordance with Section 1803. Foundation walls and other walls in which horizontal movement is restricted at the top shall be designed for at-rest pressure. Retaining walls free to move and rotate at the top shall be permitted to be designed for active pressure. Design lateral pressure from surcharge loads shall be added to the lateral earth pressure load. Foundation walls shall be designed to support the weight of the full hydrostatic pressure of undrained backfill unless a drainage system is installed in accordance with Sections 1805.4.2 and 1805.4.3. Where a portion or the whole of the adjacent soil is below a free-water surface, computations shall be based on the weight of the soil diminished by buoyancy, plus full hydrostatic pressure."

2. Replace Table 1610.1 as follows:

TABLE 1610.1

LATERAL SOIL LOAD

LATERAL PRESSURE	DESIGN LATERAL SOIL LOAD ^a (pounds per square foot per foot of depth)
Active pressure	45
At rest pressure	60

For SI: 1 pound per square foot per foot of depth = 0.157 kPa/m, 1 foot = 304.8 mm.

14B-16-1611 Rain loads.

The provisions of Section 1611 of IBC are adopted by reference with the following modifications:

1. Revise the last sentence of Section 1611.1 to read:
"The design rainfall shall be based on a 100-year hourly rainfall rate of 4.03 inches (102 mm)."
2. Delete figures accompanying Section 1611.1.
3. Delete Equation 16-35.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 75)

14B-16-1612 Flood design.

The provisions of Section 1612 of IBC are not adopted. The following language is adopted as Section 1612:

"1612. FLOOD DESIGN.

1612.1 General.

Development in special flood hazard areas shall comply with Chapter 16-6 of the *Municipal Code*."

14B-16-1613 Earthquake loads.

The provisions of Section 1613 of IBC are adopted by reference with the following modifications:

1. Revise the exceptions to Section 1613.1 to read:

"Exceptions:

1. *Buildings and structures* assigned to *Seismic Design Category A* shall comply with Section 1613.4.
2. Group R-3 and R-5 occupancies.
3. The *seismic force-resisting system* of wood-frame *buildings* of *light-frame construction* that conform to the provisions of Section 2308 are not required to be analyzed as specified in this section.
4. Agricultural and industrial storage *structures* intended only for incidental human occupancy.
5. *Structures* that require special consideration of their response characteristics and environment that are not addressed by this code or ASCE 7 and for which other regulations provide seismic criteria, such as vehicular bridges, electrical transmission towers, hydraulic *structures*, buried utility lines and their appurtenances and nuclear reactors.
6. References within ASCE 7 to Chapter 14 shall not apply, except as specifically required herein."

2. Revise Section 1613.2 to read:

"1613.2 Seismic ground motion values.

Seismic ground motion values shall be determined in accordance with this section.

1613.2.1 Acceleration parameters.

The acceleration parameters shall be: mapped 0.2 sec. spectral response acceleration (5% of critical damping, *Site Class B*) $S_s = 0.125$ g and mapped 1.0 sec. spectral response acceleration (5% of critical damping, *Site Class B*) $S_1 = 0.066$.

1613.2.2 Site class definitions.

Based on the site soil properties, the site shall be classified as *Site Class A*, *B*, *C*, *D*, *E* or *F* in accordance with Chapter 20 of ASCE 7. The *site class* shall reflect the soil conditions that affect the ground motion input to the *structure*.

Where the soil properties are not known in sufficient detail to determine the *site class*, *Site Class D*, subjected to the requirements of Section 1613.2.3, shall be used unless the *building official* or a *registered geotechnical engineer* determines that *Site Class E* or *F* soils are present at the *site*.

Where site investigations that are performed in accordance with Chapter 20 of ASCE 7 reveal rock conditions consistent with *Site Class B*, but site-specific velocity measurements are not made, the site coefficients F_a and F_v shall be taken at unity (1.0).

1613.2.3 Site coefficients and adjusted maximum considered earthquake spectral response acceleration parameters.

The maximum considered earthquake spectral response acceleration for short periods, S_{MS} , and at 1-second period, S_{M1} , adjusted for *site class* effects and five-percent damped design spectral response acceleration at short periods, S_{DS} , and at 1-second period, S_{D1} shall be as provided in Table 1613.2.3.

Where the simplified design procedure of ASCE 7 Section 12.14 is used, these values shall be determined in accordance with ASCE 7 Section 12.14.8.1.

TABLE 1613.2.3

SEISMIC DESIGN PARAMETERS

Site Class	Maximum Spectral Response Accelerations		Design Spectral Response Accelerations		Seismic Design Category	
					Risk Category	
	S_{MS}	S_{M1}	S_{DS}	S_{D1}	I, II or III	IV
Site Class	Maximum Spectral Response Accelerations		Design Spectral Response Accelerations		Seismic Design Category	
					Risk Category	
	S_{MS}	S_{M1}	S_{DS}	S_{D1}	I, II or III	IV
A	0.100	0.053	0.067	0.035	A	A
B	0.113	0.053	0.075	0.035	A	A
C	0.163	0.099	0.108	0.066	A	A
D	0.200	0.155	0.133	0.103	B ^a	C
E	0.300	0.277	0.200	0.185	C	D
F	note b	note b	note b	note b	note c	

- a. The *structure* shall be assigned to *Seismic Design Category A* where the *mean roof height* does not exceed 60 feet (18.3 m) and the horizontal distance between vertical elements of the *seismic force-resisting system* does not exceed 40 feet (12.2 m).
- b. Values shall be determined in accordance with Section 11.4.8 of ASCE 7.
- c. Determine in accordance with ASCE 7.

1613.2.4 [Reserved.]

1613.2.5 Determination of seismic design category.

Structures shall be assigned to a *Seismic Design Category* in accordance with Table 1613.2.3.

1613.2.5.1 [Reserved.]

1613.2.5.2 Simplified design procedure.

Where the alternate simplified design procedure of ASCE 7 is used, the *seismic design category* shall be determined in accordance with ASCE 7."

3. Revise the last sentence of Section 1613.3 to read:

"In *structures* assigned to *Seismic Design Category C* or *D*, ballasted nonpenetrating systems shall be designed to accommodate seismic displacement determined by nonlinear response-history or other engineering analysis or shake-table testing, using input motions consistent with ASCE 7 lateral and vertical seismic forces for nonstructural components on roofs."

4. Insert a new Section 1613.4 to read:

"1613.4 Seismic design requirements for buildings assigned to Seismic Design Category A.

Buildings and other *structures* assigned to *Seismic Design Category A* need only comply with the requirements of this section. Nonstructural components of *buildings* and *structures* in *Seismic Design Category A* are exempt from seismic design requirements.

Exception: Tanks assigned to *Risk Category IV* shall satisfy the freeboard requirements of Section 15.6.5.1 of ASCE 7.

1613.4.1 General.

All *structures* shall be provided with a continuous load path in accordance with the requirements of Section 1613.4.2 and shall have a complete lateral force-resisting system with adequate strength to resist the forces indicated in Section 1613.4.3. All members of the structural system shall be connected to their supporting members in accordance with Section 1613.4.4. Structural walls shall be anchored to diaphragms and supports in accordance with Section 1613.4.5. The effects on the *structure* and its components caused by the forces stipulated in this section shall be taken as the notional load, N , and combined with the effects of other loads in accordance with the load combinations of Section 1605. Where material resistance depends on load duration, notional loads are permitted to be taken as having a duration of 10 minutes.

1613.4.2 Load path connections.

All parts of the *structure* between separation joints shall be interconnected to form a continuous path to the lateral force-resisting system, and the connections shall be capable of transmitting the lateral forces induced by the parts being connected. Any smaller portion of the *structure* shall be tied to the remainder of the *structure* with elements having the strength to resist a force of not less than 5% of the portion's weight.

1613.4.3 Lateral forces.

Each *structure* shall be analyzed for the effects of static lateral forces applied independently in each of two orthogonal directions. In each direction, the static lateral forces at all levels shall be applied simultaneously. For purposes of analysis, the force at each level shall be determined using Equation 16-36a.

$$F_x = 0.01 W_x \quad \text{(Equation 16-36a)}$$

where:

F_x = The design lateral force applied at level x .

W_x = the portion of the total dead load of the structure, D , located or assigned to level x .

1613.4.4 Connection to supports.

A positive connection for resisting a horizontal force acting parallel to the member shall be provided for each beam, girder or truss either directly to its supporting elements or to slabs designed to act as diaphragms. Where the connection is through a diaphragm, the member's supporting element shall also be connected to the diaphragm. The connection shall have the strength to resist a force of 5% of the unfactored dead load plus live load reaction imposed by the supported member on the supporting member.

1613.4.5 Anchorage of structural walls.

Walls that provide vertical load bearing or lateral shear resistance for a portion of the *structure* shall be anchored to the roof and all floors and members that provide lateral support for the wall or that are supported by the wall. The anchorage shall provide a direct connection between the walls and the roof or floor construction. The connections shall be capable of resisting a strength level horizontal force perpendicular to the plane of the wall equal to 0.2 times the weight of the wall

tributary to the connection, but not less than 5 psf (0.24 kN/m²)."

(Amend Coun. J. 7-24-19, p. 3646, § 33; Amend Coun. J. 2-19-20, p. 14473, Art. II, § 76)

14B-16-1614 Atmospheric ice loads.

The provisions of Section 1614 of IBC are adopted by reference with the following modification:

1. Revise Section 1614.1 to read:

"1614.1 General.

Ice-sensitive structures shall be designed for atmospheric ice loads in accordance with Chapter 10 of ASCE 7."

14B-16-1615 IBC Section 1615.

The provisions of Section 1615 of IBC are not adopted.

14B-16-1616 Structural integrity.

The provisions of Section 1616 of IBC are adopted by reference without modification.

CHAPTER 14B-17

SPECIAL INSPECTIONS AND TESTS

14B-17-1700 Defined terms.

14B-17-1701 General.

14B-17-1702 IBC Section 1702.

14B-17-1703 Products, materials and assemblies.

14B-17-1704 Special inspections and tests, contractor responsibility and structural observation.

14B-17-1705 Required special inspections and tests.

14B-17-1706 Design strength of materials.

14B-17-1707 Alternative test procedure.

14B-17-1708 In-situ load tests.

14B-17-1709 Preconstruction load tests.

14B-17-1700 Defined terms.

Where provisions of Chapter 17 of IBC are adopted by reference, the following terms shall be reset in italic type: "approved agency"; "building"; "designated seismic system"; "seismic force-resisting system"; "special inspection"; "special inspector"; "sprayed fire-resistant materials"; "structure" and "owner."

14B-17-1701 General.

The provisions of Section 1701 of IBC are adopted by reference with the following modification:

1. Revise Section 1701.1 to read:

"1701.1 Scope.

The provisions of this chapter shall govern the quality, workmanship and requirements for materials covered. Construction materials and test procedures shall conform to the applicable standards listed in this code and the other *Chicago Construction Codes*."

14B-17-1702 IBC Section 1702.

The provisions of Section 1702 of IBC are not adopted.

14B-17-1703 Products, materials and assemblies.

The provisions of Section 1703 of IBC are adopted by reference with the following modifications:

1. Delete Section 1703.1 and its subsections.
2. Delete Section 1703.2.
3. Delete Section 1703.3.
4. Revise Section 1703.4 to read:

"1703.4 Performance.

Specific information consisting of test reports conducted by an *approved agency* in accordance with the appropriate referenced standards, or other such information as necessary, shall be provided for the *building official* to confirm that a product, material or assembly meets applicable code requirements."

5. Revise Section 1703.4.1 to read:

"1703.4.1 Research and investigation.

Sufficient technical data shall be submitted to the *building official* to substantiate the proposed use of any product, material or assembly. If it is determined that the evidence submitted is satisfactory proof of performance for the use intended, the *building official* shall allow the use of the product, material or assembly subject to the requirements of this code. The costs, reports and investigations required under these provisions shall be paid by the *owner*."

6. Revise Section 1703.6 and its subsections to read:

"1703.6 Off-site fabrication.

Where structural components or other items regulated by this code are not visible for inspection after completion of a prefabricated assembly, the *owner* shall maintain a report of each prefabricated assembly. The report shall indicate the complete details of the assembly, including a description of the assembly and its components, the basis upon which the assembly is being evaluated, test results and similar information and other data as necessary for the *building official* to confirm conformance to this code.

1703.6.1 Follow-up inspection.

The *owner* shall provide for *special inspections* of *fabricated items* in accordance with Section 1704.2.5.

1703.6.2 Test and inspection records.

Copies of necessary test and *special inspection* records shall be maintained by the *owner* and made available to the *building official*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 77)

14B-17-1704 Special inspections and tests, contractor responsibility and structural observation.

The provisions of Section 1704 of IBC are adopted by reference with the following modifications:

1. Revise Section 1704.2 to read:

"1704.2 Special inspections and tests.

Where application is made to the *building official* for construction as specified in Chapter 4 of the *Chicago Construction Codes Administrative Provisions*, the *owner* shall employ one or more *special inspection agencies* to provide *special inspections* and tests during construction on the types of work specified in Section 1705 and identify the *special inspection agencies* to the *building official*. These *special inspections* and tests are in addition to the inspections by the *building official* that are identified in Section 502 of the *Chicago Construction Codes Administrative Provisions*.

Exceptions:

1. *Special inspections* and tests are not required for construction of a minor nature or as warranted by conditions *approved* by the *building official*.
2. [Reserved.]
3. *Special inspections* and tests are not required for portions of *structures* designed and constructed in accordance with the cold-formed steel *light-frame construction* provisions of Section 2211.1.2 or the conventional wood *light-frame construction* provisions of Section 2308.
4. The contractor is permitted to employ the *special inspection agencies*."

2. Revise Section 1704.2.1 to read:

"1704.2.1 Special inspection agency qualifications.

Prior to the *start of construction*, the *special inspection agencies* shall provide written documentation to the *owner* demonstrating the competence and relevant experience or training of the *special inspectors* who will perform the *special inspections* and tests during construction. Experience or training shall be considered to be relevant where the documented experience or training is

related in complexity to the same type of *special inspection* or testing activities for projects of similar complexity and material qualities. These qualifications are in addition to qualifications specified in other sections of the *Chicago Construction Codes*. The *owner* shall make this documentation available to the *building official* upon request.

The *registered design professional in responsible charge* and engineers of record involved in the design of the project are allowed to act as the *special inspection agency* and their personnel are permitted to act as *special inspectors* for the work designed by them, provided they qualify as *special inspectors*.

1704.2.1.1 Evidence of qualifications.

A *special inspection agency* shall provide all information as necessary for the *owner* to determine that the agency meets the applicable requirements specified in Sections 1704.2.1.1.1 through 1704.2.1.1.3.

1704.2.1.1.1 Independence.

A *special inspection agency* shall be objective, competent and independent from the contractor responsible for the work being inspected. The agency shall disclose to the *owner* and the *registered design professional in responsible charge* possible conflicts of interest so that objectivity can be confirmed.

1704.2.1.1.2 Equipment.

A *special inspection agency* shall have adequate equipment to perform required tests. The equipment shall be periodically calibrated.

1704.2.1.1.3 Personnel.

A *special inspection agency* shall employ experienced personnel educated in conducting, supervising and evaluating tests and *special inspections*.

3. Revise Section 1704.2.3 by replacing "Section 107.1" with "Section 411.1 of the *Chicago Construction Codes Administrative Provisions*."

4. Revise Section 1704.2.4 to read:

"1704.2.4 Report requirement.

Special inspection agencies shall keep records of *special inspections* and tests. The *special inspection agency* shall submit reports of *special inspections* and tests to the *owner* and to the *registered design professional in responsible charge*. Reports shall indicate that work inspected or tested was or was not completed in conformance to *permitted construction documents*. Discrepancies shall be brought to the immediate attention of the contractor for correction. If they are not corrected, the discrepancies shall be brought to the attention of the *building official* and to the *registered design professional in responsible charge* prior to the completion of that phase of the work. The *building official* may require that a final report documenting required *special inspections* and tests, and correction of any discrepancies noted in the inspections or tests, be submitted to the *building official* prior to occupancy."

5. Revise the last sentence of Section 1704.2.5.1 to read:

"At completion of fabrication, the fabricator shall submit a *certificate of compliance* to the *owner* stating that the work was performed in accordance with the *permitted construction documents*. The *certificate of compliance* shall be made available to the *building official*."

6. Delete the exception to Section 1704.3.

7. Revise Section 1704.3.1, item 4, by deleting "1705.11."

8. Delete Section 1704.3.3.

9. Revise Section 1704.4 to read:

"1704.4 Contractor responsibility.

Each contractor responsible for the construction of a *seismic force-resisting system*, *designated seismic system* or a seismic force-resisting component listed in the statement of *special inspections* shall submit a written statement of responsibility to the *owner* prior to the commencement of work on the system or component. The contractor's statement of responsibility shall contain acknowledgement of awareness of the special requirements contained in the statement of *special inspections*."

10. Revise the title and first sentence of Section 1704.5 to read:

"1704.5 Documents made available to the building official.

In addition to the reports of *special inspections* and tests in accordance with Section 1704.2.4, reports and certificates shall be submitted to the *owner* and made available to the *building official* for each of the following:"

11. Revise Section 1704.5, item 7, by replacing "Seismic Design Category B, C, D, E or F" with "Seismic Design Category C or D."

12. Revise Section 1704.6 and its subsections to read:

"1704.6 Structural observations.

Where required by the provisions of Section 1704.6.1 or 1704.6.2, the *owner* shall employ a *registered design professional* to

perform *structural observations*. *Structural observation* does not include or waive the responsibility for the inspections required by Chapter 5 of the *Chicago Construction Codes Administrative Provisions* or the *special inspections* in Section 1705 or other sections of this code.

Prior to the commencement of observations, the structural observer shall submit to the *owner* a written statement identifying the frequency and extent of *structural observations*. The statement shall be made available to the *building official*.

At the conclusion of the work included in the *permit*, the structural observer shall submit to the *owner* a written statement that the site visits have been made and identify any reported deficiencies that, to the best of the structural observer's knowledge, have not been resolved. The statement shall be made available to the *building official*. The structural observer shall notify the *building official* in writing of any deficiencies that, to the best of the structural observer's knowledge, have not been resolved by the *owner* at the conclusion of work.

1704.6.1 Structural observations for structures.

Structural observations shall be provided for those *structures* where the work involves construction or *alteration* of the *primary structural frame* and one or more of the following conditions exist:

1. The *structure* is classified as *Risk Category IV*.
2. The *structure* is a *high-rise building*.
3. Such observation is required by the *registered design professional* responsible for the structural design.
4. Such observation is specifically required by the *building official*.

1704.6.2 Structural observations for seismic resistance - Risk Category IV.

Structural observations shall be provided for those *structures* assigned to *Seismic Design Category D* where the structure is classified as *Risk Category IV*."

14B-17-1705 Required special inspections and tests.

The provisions of Section 1705 of IBC are adopted by reference with the following modifications:

1. Revise Section 1705.1.1 to read:

"1705.1.1 Special cases.

Special inspections and tests may be required by the *building official* or *registered design professional* for proposed work that is, in the opinion of the *building official* or *registered design professional*, unusual in its nature, such as, but not limited to, the following examples:

1. Construction materials and systems that are *approved* as alternatives to materials and systems prescribed by this code.
2. Unusual design applications of materials described in this code.
3. Materials and systems required to be installed in accordance with additional manufacturer's instructions that prescribe requirements not contained in this code or in standards referenced by this code."

2. Revise Section 1705.2 and its exception to read:

"1705.2 Steel construction.

The *special inspections* and nondestructive testing of steel construction shall be in accordance with this section.

Exception: *Special inspections* of the steel fabrication process shall not be required where the fabrication process for the entire *building* or *structure* does not include any welding, thermal cutting or heating operation of any kind. In such cases, the fabricator shall be required to maintain records so that, at any time during the fabrication process, the material specification and grade for the main stress-carrying elements are capable of being determined."

3. Revise Section 1705.2.1 by deleting "in buildings, structures and portions thereof."
4. Revise Section 1705.2.3 by deleting "in buildings, structures and portions thereof."
5. Revise Section 1705.2.4 by deleting "*approved*."
6. Revise the exception to Section 1705.3, items 1 and 2, to read:
 - "1. Isolated spread concrete footings of *buildings* four *stories* or less above *grade plane* that are fully supported on earth or rock.
 2. Continuous concrete footings supporting walls of *buildings* four *stories* or less above *grade plane* that are fully supported on earth or rock where:
 - 2.1. The footings support walls of *light-frame construction*.
 - 2.2. The footings are designed in accordance with Table 1809.7.

2.3. The structural design of the footing is based on a specified compressive strength, f'_c , not more than 2,500 pounds per square inch (psi) (17.2 MPa), regardless of the compressive strength specified in the *permitted construction documents* or used in the footing construction."

7. Revise the exception to Section 1705.4, item 1, by deleting "where they are part of a structure classified as *Risk Category I, II or III*."

8. Delete Section 1705.4.1.

9. Revise Section 1705.5.1 by replace both instances of "*approved*" with "*permitted*."

10. [Reserved.]

11. Revise Section 1705.6, excluding the exception, to read:

"1705.6 Soils.

Special inspections and tests of existing site soil conditions, fill placement and load-bearing requirements shall be performed in accordance with this section and Table 1705.6. The geotechnical report prepared by the *registered geotechnical engineer* and the *permitted construction documents* shall be used to determine compliance. During fill placement, the *special inspector* shall verify that proper materials and procedures are used in accordance with the provisions of the geotechnical report."

12. Revise the second sentence of Section 1705.7 to read:

"The geotechnical report prepared by the *registered geotechnical engineer* and the *permitted construction documents* shall be used to determine compliance."

13. Revise the second sentence of Section 1705.8 to read:

"The geotechnical report prepared by the *registered geotechnical engineer* and the *permitted construction documents* shall be used to determine compliance."

14. Revise Section the second sentence of 1705.9 to read:

"The geotechnical report prepared by the *registered geotechnical engineer* and the *permitted construction documents* shall be used to determine compliance."

15. Delete Section 1705.11 and its subsections.

16. Revise Section 1705.12 and its subsections to read:

"1705.12 Special inspections for seismic resistance for Risk Category IV buildings.

Special inspections for seismic resistance for *Risk Category IV buildings* shall be required as specified in Sections 1705.12.1 through 1705.12.9, unless exempted by the exceptions of Section 1704.2.

Exception: The *special inspections* specified in Sections 1705.12.1 through 1705.12.9 are not required for *structures* designed and constructed in accordance with one of the following:

1. The *structure* consists of *light-frame construction*; and the *building height* of the *structure* does not exceed 35 feet (10 668 mm).
2. The *seismic force-resisting system* of the *structure* consists of reinforced masonry or reinforced concrete and the *building height* of the *structure* does not exceed 25 feet (7620 mm).

1705.12.1 Structural steel.

Special inspections for seismic resistance shall be in accordance with Section 1705.12.1.1 or 1705.12.1.2, as applicable.

1705.12.1.1 Seismic force-resisting systems.

Special inspections of structural steel in the *seismic force-resisting systems* in *buildings* and *structures* assigned to *Seismic Design Category C, D, E or F* shall be performed in accordance with the quality assurance requirements of AISC 341.

Exceptions:

1. In *buildings* and *structures* assigned to *Seismic Design Category C*, *special inspections* are not required for structural steel *seismic force-resisting systems* where the response modification coefficient, R , designated for "Steel systems not specifically detailed for seismic resistance, excluding cantilever column systems" in ASCE 7, Table 12.2-1, has been used for design and detailing.
2. In *structures* assigned to *Seismic Design Category D*, *special inspections* are not required for structural steel *seismic force-resisting systems* where design and detailing in accordance with AISC 360 is permitted by ASCE 7, Table 15.4-1.

1705.12.1.2 Structural steel elements.

Special inspections of structural steel elements in the *seismic force-resisting systems* of *buildings* and *structures*

assigned to *Seismic Design Category C* or *D* other than those covered in Section 1705.12.1.1, including struts, collectors, chords and foundation elements, shall be performed in accordance with the quality assurance requirements of AISC 341.

Exceptions:

1. In *buildings* and *structures* assigned to *Seismic Design Category C*, *special inspections* of *structural steel elements* are not required for *seismic force-resisting systems* with a response modification coefficient, R , of 3 or less.
2. In *structures* assigned to *Seismic Design Category D*, *special inspections* of *structural steel elements* are not required for *seismic force-resisting systems* where design and detailing other than AISC 341 is permitted by ASCE 7, Table 15.4-1. *Special inspection* shall be in accordance with the applicable referenced standard listed in ASCE 7, Table 15.4-1.

1705.12.2 Structural wood.

For structural wood elements in the *seismic force-resisting systems* of *structures* assigned to *Seismic Design Category C* or *D*:

1. *Continuous special inspection* shall be required during field gluing operations of elements of the *seismic force-resisting system*.
2. *Periodic special inspection* shall be required for nailing, bolting, anchoring and other fastening of elements of the *seismic force-resisting system*, including wood shear walls, wood diaphragms, drag struts, braces, shear panels and hold-downs.

Exception: *Special inspections* are not required for wood shear walls, shear panels and diaphragms, including nailing, bolting, anchoring and other fastening to other elements of the *seismic force-resisting system*, where the fastener spacing of the sheathing is more than 4 inches (102 mm) on center.

1705.12.3 Cold-formed steel light-frame construction.

For the *seismic force-resisting systems* of *structures* assigned to *Seismic Design Category C* or *D*, *periodic special inspection* shall be required for both:

1. Welding operations of elements of the *seismic force-resisting system*.
2. Screw attachment, bolting, anchoring and other fastening of elements of the *seismic force-resisting system*, including shear walls, braces, diaphragms, collectors (drag struts) and hold-downs.

Exception: *Special inspections* are not required for cold-formed steel light-frame shear walls and diaphragms, including screw installation, bolting, anchoring and other fastening to components of the *seismic force-resisting system*, where either of the following applies:

1. The sheathing is gypsum board or fiberboard.
2. The sheathing is wood structural panel or steel sheets on only one side of the shear wall, shear panel or diaphragm assembly and the fastener spacing of the sheathing is more than 4 inches (102 mm) on center.

1705.12.4 Designated seismic systems.

For *structures* assigned to *Seismic Design Category C* or *D*, the *special inspector* shall examine *designated seismic systems* requiring seismic qualification in accordance with Section 13.2.2 of ASCE 7 and verify that the label, anchorage and mounting conform to the *certificate of compliance*.

1705.12.5 Architectural components.

Periodic special inspection is required for the erection and fastening of exterior cladding, interior and exterior nonbearing walls and interior and exterior veneer in *structures* assigned to *Seismic Design Category D*.

Exception: *Periodic special inspection* is not required for the following:

1. Exterior cladding, interior and exterior nonbearing walls and interior and exterior veneer 30 feet (9144 mm) or less in height above grade or walking surface.
2. Exterior cladding and interior and exterior veneer weighing 5 psf (24.5 N/ m²) or less.
3. Interior nonbearing walls weighing 15 psf (73.5 N/ m²) or less.

1705.12.5.1 Access floors.

Periodic special inspection is required for the anchorage of access floors in *structures* assigned to *Seismic Design Category D*.

1705.12.6 Plumbing, mechanical and electrical components.

Periodic special inspection of plumbing, mechanical and electrical components shall be required for the following:

1. Anchorage of electrical equipment for *emergency power systems* and *standby power systems* in structures assigned to *Seismic Design Category C* or *D*.
2. [Reserved.]
3. Installation and anchorage of piping systems designed to carry hazardous materials and their associated mechanical units in *structures* assigned to *Seismic Design Category C* or *D*.
4. Installation and anchorage of ductwork designed to carry hazardous materials in *structures* assigned to *Seismic Design Category C* or *D*.
5. Installation and anchorage of vibration isolation systems in *structures* assigned to *Seismic Design Category C* or *D* where the *permitted construction documents* require a nominal clearance of 1/4 inch (6.4 mm) or less between the equipment support frame and restraint.
6. Installation of mechanical and electrical equipment, including duct work, piping systems and their structural supports, where *automatic sprinkler systems* are installed in *structures* assigned to *Seismic Design Category C* or *D* to verify one of the following:
 - 6.1. Minimum clearances have been provided as required by Section 13.2.3 ASCE/SEI 7.
 - 6.2. A nominal clearance of not less than 3 inches (76 mm) has been provided between fire protection sprinkler system drops and sprigs and: structural members not used collectively or independently to support the sprinklers; equipment attached to the building structure; and other systems' piping.

Where flexible sprinkler hose fittings are used, *special inspection* of minimum clearances is not required.

1705.12.7 Storage racks.

Periodic special inspection is required for the anchorage of storage racks that are 8 feet (2438 mm) or greater in height in *structures* assigned to *Seismic Design Category D*.

1705.12.8 Seismic isolation systems.

Periodic special inspection shall be provided for seismic isolation systems in seismically isolated *structures* assigned to *Seismic Design Category C* or *D* during the fabrication and installation of isolator units and energy dissipation devices.

1705.12.9 Cold-formed steel special bolted moment frames.

Periodic special inspection shall be provided for the installation of cold-formed steel special bolted moment frames in the *seismic force-resisting systems* of *structures* assigned to *Seismic Design Category D*.

17. Add a new exception to Section 1705.13 to read:

"Exception: *Buildings and structures* assigned to *Seismic Design Category A* or *B*."

18. Revise Section 1705.13.1.1 by replacing "*Seismic Design Category B, C, D, E or F*" with "*Seismic Design Category C or D*."
19. Revise exception 1 to Section 1705.13.1.1 by replacing "*B or C*" with "*C*."
20. Revise exception 2 to Section 1705.13.1.1 by replacing "*D, E or F*" with "*D*."
21. Revise Section 1705.13.1.2 by replacing "*Seismic Design Category B, C, D, E or F*" with "*Seismic Design Category C or D*."
22. Revise exception 1 to Section 1705.13.1.2 by replacing "*B or C*" with "*C*."
23. Revise exception 2 to Section 1705.13.1.2 by replacing "*D, E or F*" with "*D*."
24. Revise Section 1705.13.2 to read:

"1705.13.2 Nonstructural components.

For *structures* assigned to *Seismic Design Category C* or *D*, where the requirements of Section 13.2.1 of ASCE 7 for nonstructural components, supports or attachments are met by seismic qualification as specified in Item 2 therein, the *registered design professional* shall specify on the *permitted construction documents* the requirements for seismic qualification by analysis, testing or experience data. *Certificates of compliance* for the seismic qualification shall be made available to the *building official* as specified in Section 1704.5."

25. Revise Section 1705.13.3 to read:

"1705.13.3 Designated seismic systems.

For *structures* assigned to *Seismic Design Category C* or *D* and with *designated seismic systems* that are subject to the requirements of Section 13.2.2 of ASCE 7 for certification, the *registered design professional* shall specify on the *permitted construction documents* the requirements to be met by analysis, testing or experience data as specified therein. *Certificates of compliance* documenting that the requirements are met shall be made available to the *building official* as specified in Section 1704.5."

26. Revise Section 1705.13.4 by replacing "*Seismic Design Category B, C, D, E or F*" with "*Seismic Design Category C or D*."

27. Revise the second sentence of Section 1705.14 to read:

"*Special inspections* shall be based on the *permitted construction documents*."

28. Revise Section 1705.14.2 to read:

"1705.14.2 Structural member surface conditions.

The surfaces shall be prepared in accordance with the *permitted construction documents* and the written instructions of the manufacturer. The prepared surface of structural members to be sprayed shall be inspected by the *special inspector* before the application of the *sprayed fire-resistant material*."

29. Revise Section 1705.14.3 to read:

"1705.14.3 Application.

The substrate shall have a minimum ambient temperature before and after application as specified in the written instructions of the manufacturer. The area for application shall be ventilated during and after application as required by the written instructions of the manufacturer."

30. Revise Section 1705.14.4 by replacing "*approved* fire-resistance design" with "*permitted construction documents*."

31. Revise Section 1705.14.5 by replacing "*approved* fire-resistance design" with "*permitted construction documents*."

32. Revise Section 1705.14.6.3 to read:

"1705.14.6.3 Primer, paint and encapsulant bond tests.

Bond tests to qualify a primer, paint or encapsulant shall be conducted where the *sprayed fire-resistant material* is applied to a primed, painted or encapsulated surface for which acceptable bond-strength performance between these coatings and the *sprayed fire-resistant material* has not been determined. A bonding agent *approved* by the *sprayed fire-resistant material* manufacturer shall be applied to a primed, painted or encapsulated surface where the bond strengths are found to be less than required values."

33. Revise Section 1705.15 to read:

"1705.15 Mastic and intumescent fire-resistant coatings.

Special inspections and tests for *mastic fire-resistant coatings* and *intumescent fire-resistant coatings* applied to structural elements and decks shall be performed in accordance with AWCI 12-B. *Special inspections* and tests shall be based on the *permitted construction documents* and manufacturer's instructions."

34. Delete Section 1705.16 and its subsections.

35. Revise Section 1705.18 to read:

"1705.18 Testing for smoke control.

Smoke control systems shall be tested by a *special inspector* prior to acceptance testing by or in the presence of the *fire code official*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 78)

14B-17-1706 Design strength of materials.

The provisions of Section 1706 of IBC are adopted by reference with the following modifications:

1. Revise Section 1706.1 to read:

"1706.1 Conformance to standards.

The design strengths and permissible stresses of any structural material that are identified by a *manufacturer's designation* as to manufacture and grade by mill tests, or the strength and stress grade is otherwise confirmed to the satisfaction of the *building official*, shall conform to the specifications and methods of design of accepted engineering practice or be *approved* in accordance with Chapter 10 of the *Chicago Construction Codes Administrative Provisions*."

2. Delete Section 1706.2.

14B-17-1707 Alternative test procedure.

The provisions of Section 1707 of IBC are adopted by reference with the following modification:

1. Revise Section 1707.1 to read:

"1707.1 General.

New materials shall be evaluated in accordance with Chapter 10 of the *Chicago Construction Codes Administrative Provisions*."

14B-17-1708 In-situ load tests.

The provisions of Section 1708 of IBC are adopted by reference with the following modifications:

1. Revise Section 1708.1 to read:

"1708.1 General.

Whenever there is a reasonable doubt as to the strength, stability or load-bearing capacity of a completed *building, structure* or portion thereof for the expected loads, an engineering assessment shall be required. The engineering assessment shall involve either a structural analysis or an in-situ load test, or both, as determined by the *building official*. A structural analysis shall be based on actual material properties and other as-built conditions that affect strength, stability or load-bearing capacity, and shall be conducted in accordance with the applicable design standard. In-situ load tests shall be conducted in accordance with Section 1708.2. If the *building, structure* or portion thereof is found to have inadequate strength, stability or load-bearing capacity for the expected loads, modifications to ensure structural adequacy or the removal of the inadequate construction shall be required."

2. Revise the first sentence of Section 1708.2.2 to read:

"In the absence of applicable load test procedures contained within a material standard referenced by this code or acceptance criteria for a specific material or method of construction, the *existing structure* shall be subjected to a test procedure developed by a *registered design professional* that simulates applicable loading and deformation conditions. The test procedure shall be *approved* by the *building official*."

14B-17-1709 Preconstruction load tests.

The provisions of Section 1709 of IBC are adopted by reference with the following modifications:

1. Revise Section 1709.1 to read:

"1709.1 General.

Where proposed construction is not capable of being designed by accepted engineering practices, or where the proposed construction design method does not comply with the applicable material design standard, the system of construction or the structural unit and the connections shall be subjected to the tests prescribed in Section 1709. The *building official* shall accept certified reports of such tests conducted by an *approved agency*, provided that such tests meet the requirements of this code and specified procedures."

2. Revise Section 1709.5.1 by replacing "approved labeling agency" with "*approved agency*."

CHAPTER 14B-18

SOILS AND FOUNDATIONS

14B-18-1800 Defined terms.

14B-18-1801 General.

14B-18-1802 Design basis.

14B-18-1803 Geotechnical investigations.

14B-18-1804 Excavation, grading and fill.

14B-18-1805 Dampproofing and waterproofing.

14B-18-1806 Presumptive load-bearing values of soils.

14B-18-1807 Foundation walls, retaining walls and embedded posts and poles.

14B-18-1808 Foundations.

14B-18-1809 Shallow foundations.

14B-18-1810 Deep foundations.

14B-18-1800 Defined terms.

Where provisions of Chapter 18 of IBC are adopted by reference, the following terms shall be reset in italic type: "deep foundation"; "drilled shaft"; "helical pile"; "light-frame construction"; "micropile"; "plain masonry"; "shallow foundation"; "site" and "socketed drilled shaft."

14B-18-1801 General.

The provisions of Section 1801 of IBC are adopted by reference without modification.

14B-18-1802 Design basis.

The provisions of Section 1802 of IBC are adopted by reference without modification.

14B-18-1803 Geotechnical investigations.

The provisions of Section 1803 of IBC are adopted by reference with the following modifications:

1. Revise Section 1803.1 by replacing "*registered design professional*" with "*registered geotechnical engineer*."
2. Add new exceptions following Section 1803.1 to read:

"Exceptions:

1. Geotechnical investigations are not required for single-story storage structures without *basements* and detached *private garages* not exceeding 2,000 square feet (185.8 m²).
2. Geotechnical investigations are not required for *alterations* and *additions* that do not require new foundations and do not result in an increase in loads exceeding 5 percent of the existing foundation design capacity.
3. Full geotechnical investigations by a *registered geotechnical engineer* are not required where excavation more than 8 feet (2438 mm) below existing grade is not required, a test pit is used for soil classification and to locate the ground-water table, and the values used as the basis for design do not exceed the presumptive values in Table 1806.2(1)."
3. Revise Section 1803.3.1 by replacing "*registered design professional*" with "*registered geotechnical engineer*."
4. Revise Section 1803.4 by replacing "*registered design professional*" with "*registered geotechnical engineer*."
5. Revise Section 1803.5.2 to read:

"1803.5.2 Questionable soil or fill.

Where the classification, strength or compressibility of the soil or fill is in doubt or where a load-bearing value superior to that specified in this code is claimed, the *building official* shall require that a geotechnical investigation be conducted."

6. Revise Section 1803.5.3 to read:

"1803.5.3 Expansive soil or fill.

In areas likely to have expansive soil or fill, the *registered geotechnical engineer* shall perform soil tests to determine where such soils or fills do exist.

Soils or fills meeting all four of the following provisions shall be considered to be expansive, except that tests to show compliance with Items 1, 2 and 3 shall not be required if the test prescribed in Item 4 is conducted:

1. Plasticity index (PI) of 15 or greater, determined in accordance with ASTM D4318.
2. More than 10 percent of the soil particles pass a No. 200 sieve (75 µm), determined in accordance with ASTM D422.
3. More than 10 percent of the soil particles are less than 5 micrometers in size, determined in accordance with ASTM D422.
4. Expansion index greater than 20, determined in accordance with ASTM D4829.

Steel slag shall be tested for expansive properties in accordance with ASTM D4792 prior to use as a fill material."

7. Delete the exception to Section 1803.5.4.
8. Revise Section 1803.5.5 to read:

"1803.5.5 Deep foundations.

Where *deep foundations* will be used, a geotechnical investigation shall be conducted and shall include all of the following, as applicable:

1. Recommended *deep foundation* types and installed capacities.
2. Recommended center-to-center spacing of *deep foundation* elements.
3. Recommended driving installation acceptance criteria.
4. Recommended installation procedures.
5. Field inspection and reporting procedures (to include procedures for verification of the installed bearing capacity where required).

6. Load test requirements.
 7. Suitability of *deep foundation* materials for the intended environment.
 8. Designation of bearing stratum or strata.
 9. Reductions for group action, where necessary."
9. Revise items 1 and 2 in Section 1803.5.8 to read:
- "1. Recommended specifications for the preparation of the *site* prior to placement of compacted fill material.
 2. Recommended specifications for material to be used as compacted fill."
10. Revise items 1 and 2 in Section 1803.5.9 to read:
- "1. Recommended specifications for the preparation of the *site* prior to placement of the CLSM.
 2. Recommended specifications for the CLSM."
11. Delete Section 1803.5.10.
12. Revise the title and first sentence of Section 1803.5.11 to read:

"1803.5.11 Seismic Design Category C or D.

For *structures* assigned to *Seismic Design Category C* or *D* a geotechnical investigation shall be conducted, and shall include an evaluation of all of the following potential geologic and seismic hazards:"

13. Revise the title and first sentence of Section 1803.5.12 to read:

"1803.5.12 Seismic Design Category D.

For *structures* assigned to *Seismic Design Category D*, the geotechnical investigation required by Section 1803.5.11 shall include all of the following as applicable:"

14. Revise item 5 in Section 1803.6 to replace "expansive" with "questionable."
15. Revise Section 1803.6 to add new items to read:
- "11. Construction considerations: earth retention systems, underpinning, excavation slopes, effects on adjacent structures, utilities and improvements, corrosivity, drainage systems and dewatering, as necessary.
 12. *Site class* in accordance with Chapter 20 of ASCE 7 and the horizontal seismic forces to be considered in slope stability analysis."
16. Add a new exception following Section 1803.6 to read:

"Exception: For construction not exceeding four *stories above grade plane*, not requiring excavation more than 8 feet (2438 mm) below existing grade, and not covering more than 16,000 square feet (1486.4 m²) of ground area, a written report of geotechnical investigations shall not be required at the time of *permit* application but shall be completed prior to the actual start of construction to confirm presumed conditions used as the basis for design. The report shall be made available to the *building official* upon request. Where the written report of geotechnical investigations, completed after *permit* application, does not confirm presumed conditions used as the basis for design, a revised *permit* application and revised *construction documents* shall be submitted to the *building official* and construction shall not commence until the revised *permit* has been issued."

14B-18-1804 Excavation, grading and fill.

The provisions of Section 1804 of IBC are adopted by reference with the following modifications:

1. Revise Section 1804.2.1 by replacing "*approved*" with "*permitted*."
2. Delete Section 1804.4.
3. Delete Section 1804.5.
4. Revise Section 1804.6 to read:

"1804.6 Compacted controlled fill material.

Where *shallow foundations* will bear on compacted controlled fill material, the compacted controlled fill shall comply with the provisions of a geotechnical report acceptable to the *building official*, as set forth in Section 1803.

Exception: Compacted controlled fill material 12 inches (305 mm) in depth or less need not comply with a geotechnical report, provided that the in-place dry density is not less than 90 percent of the maximum dry density at optimum moisture content determined in accordance with ASTM D1557. The compaction shall be verified by *special inspection* in accordance with Section 1705.6."

5. Revise Section 1804.7 by replacing "an *approved*" with "a."

The provisions of Section 1805 of IBC are adopted by reference with the following modifications:

1. Revise Section 1805.1 to read:

"1805.1 General.

Walls or portions thereof that retain earth and enclose interior spaces and floors below grade shall be waterproofed and dampproofed in accordance with this section.

Exception: Spaces containing occupancies other than Group A, E, I or R where such omission is not detrimental to the *building* or occupancy."

2. Delete Section 1805.1.1.
3. Revise Section 1805.1.2 by replacing "an *approved*" with "a".
4. Delete Section 1805.1.2.1.
5. Revise Section 1805.2 to read:

"1805.2 Dampproofing.

Where hydrostatic pressure will not occur as determined by Section 1803.5.4, floors and walls shall be dampproofed in accordance with this section."

6. Revise Section 1805.2.1 by replacing "*approved*" with "equivalent."
7. Revise Section 1805.2.2 by replacing "*approved*" with "equivalent."
8. Revise Section 1805.2.2.1 to read:

"1805.2.2.1 Surface preparation of walls.

Prior to application of dampproofing materials on concrete walls, holes and recesses resulting from the removal of form ties shall be sealed with a bituminous material or other equivalent methods or materials. Unit masonry walls shall be parged on the exterior surface below ground level with not less than 3/8 inch (9.5 mm) of Portland cement mortar. The parging shall be coved at the footing.

Exception: Parging of unit masonry walls is not required where a material is intended for direct application to the masonry."

9. Revise Section 1805.3.1 by replacing "*approved*" with "equivalent."
10. Revise Section 1805.3.2 by replacing "*approved*" with "equivalent."
11. Revise Section 1805.3.3 by replacing "*approved*" with "equivalent."
12. Revise Section 1805.4 to read:

"1805.4 Subsoil drainage system.

Where a permanent hydrostatic pressure condition does not exist, dampproofing shall be provided and a granular base course shall be installed under the floor and a drain installed around the foundation perimeter. A subsoil drainage system designed and constructed in accordance with Section 1805.1.3 shall be deemed adequate for lowering the ground-water table."

13. Revise Section 1805.4.2 to read:

"1805.4.2 Foundation drain.

A drain shall be placed at the perimeter of the foundation."

14. Revise Section 1805.4.3, excluding the exception, to read:

"1805.4.3 Drainage discharge.

The floor base course and foundation perimeter drain shall discharge by gravity or mechanical means into a drainage system that complies with the *Chicago Plumbing Code*."

The provisions of Section 1806 of IBC are adopted by reference with the following modifications:

1. Revise Section 1806.1 by replacing "Table 1806.2" with "Table 1806.2(1) or 1806.2(2), as applicable" and deleting the second sentence.
2. Revise Section 1806.2 to read:

"1806.2 Presumptive load-bearing values.

Where only a limited geotechnical investigation in accordance with Section 1803.1, exception 3, is performed, the load-bearing values used in design for supporting soils near the surface shall not exceed the values specified in Table 1806.2(1). Where the *building official* has reason to doubt the classification, strength or compressibility of the soil, the requirements of Section 1803.5.2 shall be satisfied.

Where a full geotechnical investigation is performed in accordance with Section 1803, the load-bearing values used in design for supporting soils near the surface shall not exceed the values specified in Table 1806.2(2), unless the use of higher values is recommended by the *registered geotechnical engineer*.

Presumptive load-bearing values shall apply to materials with similar physical characteristics and dispositions. Mud, organic silt, organic clays, peat or unprepared fill shall not be assumed to have a presumptive load-bearing capacity unless data to substantiate the use of such a value are submitted."

3. Delete Table 1806.2.

4. Insert Tables 1806.2(1), 1806.2(2) and 1806.2(3) to read:

TABLE 1806.2(1)

PRESUMPTIVE LOAD-BEARING VALUES WITHOUT FULL GEOTECHNICAL INVESTIGATION

CLASS OF MATERIALS	VERTICAL FOUNDATION PRESSURE (psf)	LATERAL BEARING PRESSURE (psf/ft below natural grade)	LATERAL SLIDING RESISTANCE	
			Coefficient of friction ^a	Cohesion (psf) ^b
Sandy gravel and gravel (GW and GP)	3,000	200	0.35	—
Sand, silty sand, clayey sand, silty gravel and clayey gravel (SW, SP, SM, SC, GM and GC)	2,000	150	0.25	—
Clay, sandy clay, silty clay, clayey silt, silt and sandy silt (CL, ML, MH and CH)	1,500	100	—	130
Non-engineered fill	500	50	0.1	—

For SI: 1 pound per square foot = 0.0479 kPa, 1 pound per square foot per foot = 0.157 kPa/m.

a. Coefficient to be multiplied by the dead load.

b. Cohesion value to be multiplied by the contact area, as limited by Section 1806.3.2.

TABLE 1806.2(2)

PRESUMPTIVE VERTICAL FOUNDATION PRESSURE VALUES WITH GEOTECHNICAL INVESTIGATION

CLASS	DESCRIPTION	NOTES	CONSISTENCY OR RELATIVE DENSITY ^a	ALLOWABLE NET BEARING PRESSURE (psf)
CLASS	DESCRIPTION	NOTES	CONSISTENCY OR RELATIVE DENSITY ^a	ALLOWABLE NET BEARING PRESSURE (psf)
1	DOLOMITE bedrock	c	Hard, sound rock, minor jointing	200,000
2a	Fractured DOLOMITE surface	d, h	Blocky limestone with soil filled seams and fractures	60,000
2b	Fractured DOLOMITE surface	e, h	Blocky limestone with soil filled seams and fractures	100,000
2c	DOLOMITE surface	f, h	Rock with minor fractures and jointing	120,000
2d	DOLOMITE surface	g, h	Rock with minor fractures and jointing	150,000
3	HARDPAN, hard silty clay, or very dense clayey silt, (CL, CL-ML)	i	Very hard or very dense	20,000
		j	Hard or dense	12,000
4	SAND, GRAVEL or SILT, silty sand, clayey sand, silty gravel or	k	Dense	3,000
			Medium dense	2,000

	clayey gravel (SW, SP, GW, GP, SC, SM, GC, GM, ML)		Loose	1,000
			Very loose	Note b
5	Silty CLAY, Clayey SILT (CH, CL, ML-CL, MH)	–	Hard	4,000
			Very stiff	3,000
			Stiff	2,000
			Medium	1,500
			Soft	Note b
			Very soft	Note b
6	Organic SILT, CLAY or PEAT	m	–	Note b
7	Non-Engineered FILL	n	–	500
8	CONTROLLED FILL	o	95% Modified Proctor	3,000
9	Improved In-Situ Soil or Fill	p	–	Note b

For SI: 1 foot = 304.8 mm, 1 pound per square foot = 0.0479 kPa.

- a. Classified in accordance with Table 1806.2(3).
- b. Requires engineering analysis and testing by a *registered geotechnical engineer*.
- c. 1 foot into sound rock (recovery greater than 95% and RQD greater than 70%) with 20% increase per foot of additional penetration up to a maximum of 400,000 pounds per square foot.
- d. Surface of refusal to standard caisson rig rock auger, correlated to borings with recovery greater than 90% and RQD greater than 40%, rock unconfined compressive strength greater than 2,000 pounds per square inch.
- e. Surface of refusal to standard caisson rig rock auger, correlated to borings with recovery greater than 90% and RQD greater than 50%, rock unconfined compressive strength greater than 2,500 pounds per square inch.
- f. Surface of refusal to standard caisson rig rock auger, correlated to borings with recovery greater than 95% and RQD greater than 60%, rock unconfined compressive strength greater than 3,000 pounds per square inch.
- g. Surface of refusal to standard caisson rig rock auger, correlated to borings with recovery greater than 95% and RQD greater than 70%, rock unconfined compressive strength greater than 3,500 pounds per square inch.
- h. No load tests are required up to presumptive bearing pressures. If higher bearing pressures are required than shown in table or RQD or rock compressive strength does not meet values shown, load tests are required to confirm allowable bearing pressure.
- i. Where q_u is greater than 14,000 pounds per square foot, or N is greater than or equal to 50.
- j. Where q_u is greater than or equal to 8,000 pounds per square foot, or N is greater than or equal to 30.
- k. Allowable net bearing value assumes minimum depth of embedment is 2.5 feet and a minimum width of 1.5 feet. Allowable bearing values shall be reduced proportionally for less depth of embedment, or the effect of reduced confinement from a rise in the water table above the foundation level.
- l. [Reserved.]
- m. Organic means more than 10% organic matter as determined in accordance with ASTM D2974.
- n. Presumptive value for light or temporary structure. Otherwise engineering analysis required.
- o. Controlled fill shall be placed in specified lifts with moisture control, under full time observation of a *registered geotechnical engineer* or their qualified representative in accordance with ASTM D1557.
- p. Improved fill or soil in-situ by dynamic compaction, compaction grouting, or other methods.

TABLE 1806.2(3)

STANDARDS FOR CLASSIFYING SOIL CONSISTENCY AND RELATIVE DENSITY

Clay		Sand and Gravel	
DESCRIPTIVE TERM	UNCONFINED STRENGTH (psf)	DESCRIPTIVE TERM	SPT N VALUE
Clay		Sand and Gravel	
DESCRIPTIVE TERM	UNCONFINED STRENGTH (psf)	DESCRIPTIVE TERM	SPT N VALUE
Very soft	< 500	Very loose	0 - 3
Soft	500 - 1,000	Loose	4 - 9
Medium	1,000 - 2,000	Medium	10 - 29
Stiff	2,000 - 4,000	Dense	30 - 49
Very stiff	4,000 - 8,000	Very Dense	≥ 50

Hard	> 8,000	
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For SI: 1 pound per square foot = 0.0479 kPa.

5. Revise Section 1806.3 to read:

"1806.3 Lateral load resistance.

Where the presumptive values of Table 1806.2(1) are used to determine resistance to lateral loads, the calculations shall be in accordance with Sections 1806.3.1 through 1806.3.4. Where the presumptive values of Table 1806.2(2) are used, resistance to lateral loads shall be as provided in the geotechnical report."

6. Revise Section 1806.3.1 by replacing "Table 1806.2" with "Table 1806.2(1)."
7. Revise Section 1806.3.3 by replacing "Table 1806.2" with "Table 1806.2(1)."

(Amend Coun. J. 10-7-20, p. 21791, Art. II, § 29)

14B-18-1807 Foundation walls, retaining walls and embedded posts and poles.

The provisions of Section 1807 of IBC are adopted by reference with the following modifications:

1. Revise Section 1807.1.3 by replacing "*Seismic Design Category C, D, E or F*" with "*Seismic Design Category C or D.*"
2. Delete Section 1807.1.4.
3. Revise Table 1807.1.6.2 by:
 - a. Deleting all columns for 30 and 45 psf per foot;
 - b. Deleting note d.
4. Revise Section 1807.1.6.2.1 by replacing "*Seismic Design Categories C, D, E and F*" with "*Seismic Design Categories C and D.*"
5. Revise Table 1807.1.6.3(1) by:
 - a. Deleting the columns for 30 and 45 psf per foot;
 - b. Deleting all rows for maximum unbalanced backfill height of 7 feet or greater.
 - c. Deleting note f.
6. Revise Table 1807.1.6.3(2) by:
 - a. Deleting the columns for 30 and 45 psf per foot;
 - b. Deleting note e.
7. Revise Table 1807.1.6.3(3) by:
 - a. Deleting the columns for 30 and 45 psf per foot;
 - b. Deleting note e.
8. Revise Table 1807.1.6.3(4) by:
 - a. Deleting the columns for 30 and 45 psf per foot;
 - b. Deleting note e.
9. Delete item 4 in Section 1807.1.6.3.2.
10. Revise Section 1807.2.2 by replacing "*Seismic Design Category D, E or F*" with "*Seismic Design Category D.*"
11. Revise Section 1807.2.3 by replacing "1.5 in each case" with "1.5 and 2.0 respectively."
12. Revise the exception to Section 1807.2.3 by adding a sentence at the end to read:

"Earthquake loads shall not be required to be considered for the design of temporary earth retention systems during construction."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 79)

14B-18-1808 Foundations.

The provisions of Section 1808 of IBC are adopted by reference with the following modifications:

1. Revise Section 1808.1 to read:

"1808.1 General.

Foundations shall be designed and constructed in accordance with Sections 1808.2 through 1808.9. *Shallow foundations* shall also satisfy the requirements of Section 1809. *Deep foundations* shall also satisfy the requirements of Section 1810."

2. Revise Section 1808.2 to read:

"1808.2 Design for capacity and settlement.

Foundations shall be so designed that the allowable bearing capacity of the soil is not exceeded, and that differential settlement is minimized. Foundations in areas with expansive soils or fills shall be designed in accordance with the provisions of Section 1808.6. Analysis shall consider soil layers underlying the designated load-bearing layer."

3. Add a new Section 1808.2.1 to read:

"1808.2.1 Factor of safety.

Foundation elements shall develop a minimum factor of safety against bearing capacity of 3.0 when routine testing is used to determine soil properties. Foundation elements shall develop a minimum factor of safety of 2.5 against bearing capacity failure when in-situ testing is used to determine soil properties."

4. Revise Section 1808.6 to read:

"1808.6 Design for expansive soils or fills.

Expansive soils or fills shall be removed in accordance with Section 1808.6.3."

5. Delete Section 1808.6.1.
6. Delete Section 1808.6.2.
7. Revise Section 1808.6.3 to read:

"1808.6.3 Removal of expansive soil or fill.

Where expansive soil or fill is removed, the soil shall be removed to a depth sufficient to ensure a constant moisture content in the remaining soil or fill. Newly placed fill material shall not contain expansive soils or fill and shall comply with Section 1804.5 or 1804.6.

Exception: Expansive soil or fill need not be removed to the depth of constant moisture, provided that the confining pressure in the expansive soil or fill created by the newly placed fill material and supported structure exceeds the swell pressure."

8. Revise Section 1808.6.4 to read:

"1808.6.4 Stabilization.

Where the active zone of expansive soils or fill is stabilized in lieu of designing foundations in accordance with Section 1808.6.1 or 1808.6.2, the soil or fill shall be stabilized by chemical, dewatering, presaturation or equivalent techniques."

9. Delete Section 1808.7 and its subsections.
10. Delete Figure 1808.7.1.
11. Revise Table 1808.8.1 by replacing "Seismic Design Category D, E or F" with "Seismic Design Category D."
12. Revise row 4 of Table 1808.8.1 to read "4. Drilled shafts and auger cast piles."
13. Revise the second and third sentences of Section 1808.8.3 to read:

"1808.8.3 Placement of concrete.

Concrete shall not be placed through water unless a tremie or other *approved* method is used. Where placed under or in the presence of water, the concrete shall be deposited by means to ensure minimum segregation of the mix and negligible turbulence of the water."

14. Revise Section 1808.8.6 by replacing "*Seismic Design Category C, D, E or F*" with "*Seismic Design Category C or D*"; "*Seismic Design Category D, E or F*" with "*Seismic Design Category D*" and "*Detached one- and two- family dwellings*" with "*Group R-5 occupancies*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 80)

14B-18-1809 Shallow foundations.

The provisions of Section 1809 of IBC are adopted by reference with the following modifications:

1. Revise item 1 of Section 1809.5 to read:
"1. Extending at least 42 inches (1067 mm) below final grade."

2. Revise Section 1809.6 by deleting "in an *approved* manner."
 3. Revise the notes to Table 1809.7 by replacing "Seismic Design Category C, D, E or F" with "*Seismic Design Category C or D*" and "Group R-3" with "Group R-3, R-4 and R-5."
- Revise the exception to Section 1809.8 by replacing "Group R-3" with "Group R-3, R-4 or R-5."
4. Revise Section 1809.10 by replacing "*Seismic Design Categories D, E and F*" with "*Seismic Design Category D*."
 5. Revise Section 1809.11 by deleting "*approved*."
 6. Delete Section 1809.12.
 7. Revise Section 1809.13 by replacing "*Seismic Design Category D, E or F*" with "*Seismic Design Category D*."

14B-18-1810 Deep foundations.

The provisions of Section 1810 of IBC are adopted by reference with the following modifications:

1. Revise Section 1810.1.2 to read:

"1810.1.2 Use of existing deep foundation elements.

Deep foundation elements left in place where a structure has been demolished shall not be used for the support of new construction unless satisfactory evidence is submitted to the *building official*, which indicates that the elements are sound and meet the requirements of this code. Such elements shall be analyzed by a *registered geotechnical engineer* using non-destructive testing or material testing, load tested or redriven to verify their capacities. The design load applied to such elements shall be the lowest allowable load as determined by analysis, tests or redriving data."

2. Revise Section 1810.2.1 by replacing "*registered design professional*" with "*registered geotechnical engineer*."

- 2.1 Revise the exceptions to Section 1810.2.2 to read:

"Exceptions:

1. Isolated cast-in-place *deep foundation* elements without lateral bracing shall be permitted where the least horizontal dimension is not less than 30 inches (762 mm), adequate lateral support in accordance with Section 1810.2.1 is provided for the entire height and a reinforcing cage is provided for the full depth unless an engineering analysis shows that a full depth cage is unnecessary.
2. A single row of *deep foundation* elements without lateral bracing is permitted for Group R-5 occupancies and *light-frame construction* not exceeding two *stories above grade plane* or 35 feet (10 668 mm) in *building height*, provided that the centers of the elements are located within the width of the supported wall."

3. Revise Section 1810.2.3 to read:

"1810.2.3 Settlement.

The settlement of a single *deep foundation* element or group thereof shall be estimated based on accepted engineering analysis. The predicted settlement shall cause neither harmful distortion of, nor instability in, the structure, nor cause any element to be loaded beyond its capacity."

4. Revise Section 1810.2.4 by replacing "*registered design professional*" with "*registered geotechnical engineer*."
5. Revise the title and first sentence of Section 1810.2.4.1 to read:

"1810.2.4.1 Seismic Design Category D.

For *structures* assigned to *Seismic Design Category D deep foundation* elements on *Site Class E or F sites*, as determined in Section 1613.2.2, shall be designed and constructed to withstand maximum imposed curvatures from earthquake ground motions and structure response."

6. Revise Section 1810.3.1.1 by deleting "*approved*."
7. Revise Section 1810.3.1.3 to read:

"1810.3.1.3 Mislocation.

The foundation or superstructure shall be designed to resist the effects of the mislocation of any *deep foundation* element (individual pier or centroid of a group) by not less than 3 inches (76 mm) and not less than any larger tolerance identified in the report of geotechnical investigations. To resist the effects of mislocation, compressive overload of *deep foundation* elements to 110 percent of the allowable design load shall be permitted."

8. Revise Section 1810.3.2.1.1 by replacing "*Seismic Design Category C, D, E or F*" with "*Seismic Design Category C or D*."
9. Revise Section 1810.3.2.5 by replacing "*approved by the building official*" with "*recommended by the registered geotechnical engineer*."
10. Add a new Section 1810.3.2.5.1 to read:

"1810.3.2.5.1 Steel piles.

For steel piles subjected to a corrosive environment, 1/32 inch (0.8 mm) of steel shall be deducted from the exposed surface in computing allowable load. The *registered geotechnical engineer* shall provide testing to establish corrosivity of fill, or it will be assumed to be corrosive. Natural soils below the water table shall be assumed to be non-corrosive."

11. Revise Table 1810.3.2.6, excluding the notes, as follows:

MATERIAL TYPE AND CONDITION	MAXIMUM ALLOWABLE STRESS ^a
MATERIAL TYPE AND CONDITION	MAXIMUM ALLOWABLE STRESS ^a
1. Concrete or grout in compression ^b	
Cast-in-place with a permanent casing in accordance with Section 1810.3.2.7, Section 1810.3.5.3.4 or bedrock	$0.4 f'_c$
Cast-in-place in a pipe, tube, other permanent casing	$0.33 f'_c$
Cast-in-place concrete without a permanent casing	$0.3 f'_c$
Cast-in-place grout without a permanent casing	$0.25 f'_c$
Precast nonprestressed	$0.33 f'_c$
Precast prestressed	$0.33 f'_c - 0.27 f'_{pc}$
2. Nonprestressed reinforcement in compression	$0.4 f_y \leq 30,000 \text{ psi}$
3. Steel in compression	
Cores within concrete-filled pipes or tubes	$0.5 F_y \leq 32,000 \text{ psi}$
Pipes, tubes, helicals or H-piles, where justified in accordance with Section 1810.3.2.8	$0.5 F_y \leq 32,000 \text{ psi}$
Pipes or tubes for micropiles	$0.4 F_y \leq 32,000 \text{ psi}$
Other pipes, tubes, helicals or H-piles	$0.35 F_y \leq 24,000 \text{ psi}$
4. Nonprestressed reinforcement in tension	
Within micropiles	$0.6 f_y \leq 40,000 \text{ psi}$
Other conditions	$0.5 f_y \leq 40,000 \text{ psi}$
5. Steel in tension	
Pipes, tubes, helicals or H-piles, where justified in accordance with Section 1810.3.2.8	$0.5 F_y \leq 32,000 \text{ psi}$
Other pipes, tubes, helicals or H-piles	$0.35 F_y \leq 24,000 \text{ psi}$
6. Timber	In accordance with the ANSI/AWC NDS

12. Revise the last sentence of Section 1810.3.2.8 to read:

"The design and installation of the *deep foundation* elements shall be under the direct supervision of a *registered geotechnical engineer* knowledgeable in the field of soil mechanics and *deep foundations* or their qualified representative who shall submit a report to the *owner* stating that the elements as installed satisfy the design criteria. The report shall be made available to the *building official*."

13. Revise Section 1810.3.3 to read:

"1810.3.3 Determination of allowable loads.

The allowable axial and lateral loads on *deep foundation* elements shall be determined by an accepted engineering analysis or load tests."

14. Revise the first two sentences of Section 1810.3.3.1.1 to read:

"The allowable compressive load on any driven *deep foundation* element where determined by the application of a driving formula shall not exceed 60 tons (598 kN) on soil and 100 tons (996 kN) on rock. For allowable loads above those values, the wave equation method of analysis shall be used to estimate driveability for both driving stresses and net displacement per blow at the ultimate load."

15. Revise Section 1810.3.3.1.2 to read:

"1810.3.3.1.2 Load tests.

Where design compressive loads are greater than those determined using the allowable stresses specified in Section 1810.3.2.6, where the design load for any *deep foundation* element is in doubt or exceeds the value in Table 1810.3.3.1.2, or where cast-in-place *deep foundation* elements have an enlarged base formed either by compacting concrete or by driving a precast base, control

test elements shall be tested in accordance with ASTM D1143, ASTM D4945 or ASTM D8169. When using ASTM D1143 or ASTM D8169, loading procedure B shall be used in soil and loading procedure A shall be allowed to be used in rock. One element or more shall be load tested in each area of uniform subsoil conditions. Where required by the *building official*, additional elements shall be load tested where necessary to establish the safe design capacity. The resulting allowable loads shall not be more than one-half of the ultimate axial load capacity of the test element as assessed by one of the published methods listed in Section 1810.3.3.1.3 with consideration for the test type, duration and subsoil. The ultimate axial load capacity shall be determined by a *registered geotechnical engineer* with consideration given to tolerable total and differential settlements at design load in accordance with Section 1810.2.3. In subsequent installation of the balance of *deep foundation* elements, all elements shall be deemed to have a supporting capacity equal to that of the control element where such elements are of similar type, size and relative length as the test element; are installed using the same or comparable methods and equipment as the test element; are installed in similar subsoil conditions as the test element; and, for driven elements, where the rate of penetration (for example, net displacement per blow) of such elements is equal to or less than that of the test element driven with the same hammer through a comparable driving distance."

16. Insert new Table 1810.3.3.1.2 to read:

TABLE 1810.3.3.1.2

MAXIMUM AXIAL LOADS ALLOWED WITHOUT LOAD TEST

TYPE OF PILE	MAXIMUM ALLOWABLE AXIAL LOAD WITHOUT LOAD TEST (tons)
Helical piles	10
Micropiles and non-driven piles	15 ^a
Timber piles	30
Driven structural steel piles in soil	60
Driven structural steel piles to bedrock	100

For SI: 1 ton = 907 kg

a. Nondestructive tests are required for non-driven piles.

17. Revise Section 1810.3.3.1.3 to read:

"1810.3.3.1.3 Load test evaluation methods.

Load test results must be accepted by the *registered geotechnical engineer* who prepared the geotechnical investigation report, unless otherwise *approved* by the *building official*. It shall be permitted to evaluate load tests of *deep foundation* elements using any of the following methods:

1. Davisson Offset Limit.
2. Brinch-Hansen 90-percent Criterion.
3. Butler-Hoy Criterion.
4. The failure load corresponding to a settlement of $0.2 \sqrt{D + PL/AE}$

where D = the diameter or diagonal dimension in inches, P = Load, L = Length, A = Area, E = Modulus of Elasticity.

5. Other methods *approved* by the *building official*."

18. Revise Section 1810.3.3.1.4 to read:

"1810.3.3.1.4 Allowable shaft resistance.

The assumed shaft resistance developed by any uncased cast-in-place *deep foundation* element shall not exceed one-sixth of the bearing value of the soil material at minimum depth as set forth in Table 1806.2, up to 500 psf (24 kPa), unless a greater value is recommended by the *registered geotechnical engineer* and allowed by the *building official* on the basis of a geotechnical investigation as specified in Section 1803 or a greater value is substantiated by a load test in accordance with Section 1810.3.3.1.2. Shaft resistance and end-bearing resistance shall not be assumed to act simultaneously unless determined by a geotechnical investigation in accordance with Section 1803."

19. Revise Section 1810.3.3.1.5 to read:

"1810.3.3.1.5 Uplift capacity of a single deep foundation element.

Where required by the design, the uplift capacity of a single *deep foundation* element shall be determined by accepted engineering analysis based on a minimum factor of safety of three or by load tests conducted in accordance with ASTM D3689 or ASTM D8169. The maximum allowable uplift load shall not exceed the ultimate load capacity as determined in Section 1810.3.3.1.2, using the results of load tests conducted in accordance with ASTM D3689 or ASTM D8169, divided by a factor of safety of two."

20. Revise Section 1810.3.3.1.7 to read:

"1810.3.3.1.7 Load-bearing capacity.

Ultimate load capacities shall be determined in accordance with Section 1810.3.3.1.7.1, 1810.3.3.1.7.2 or 1810.3.3.1.7.3.

1810.3.3.1.7.1 Load testing.

Deep foundation elements shall develop ultimate load capacities of not less than 2 times the design working loads in the designated load-bearing layers when load testing is used.

1810.3.3.1.7.2 Routine soil testing.

Deep foundation elements shall develop ultimate load capacities of not less than 3 times the design working loads in the designated load-bearing layers when routine soil testing is used to determine soil properties. Analysis shall show that soil layers underlying the designated load-bearing layers do not cause the load-bearing safety factor to be less than 3.

1810.3.3.1.7.3 In-situ soil testing.

Deep foundation elements shall develop ultimate load capacities of not less than 2.5 times the design working loads in the designated load-bearing layers when in-situ soil testing is used to determine soil properties. Analysis shall show that soil layers underlying the designated load-bearing layers do not cause the load-bearing safety factor to be less than 2.5."

21. Revise Section 1810.3.3.2 by replacing "an *approved* method of analysis" with "accepted engineering analysis."

22. Revise the exception to Section 1810.3.5.2.2 to read:

"Exception: The length of the element is permitted to exceed 30 times the specified diameter, provided that the design and installation of the *deep foundations* are under the direct supervision of a *registered geotechnical engineer* knowledgeable in the field of soil mechanics and *deep foundations* or their qualified representative. The *registered geotechnical engineer* shall submit a report to the *owner* stating that the elements were installed in compliance with the *permitted construction documents*."

23. Delete the second sentence of Section 1810.3.5.2.3.

24. Revise Section 1810.3.5.3.4 to read:

"1810.3.5.3.4 Steel pipes and tubes.

Steel pipes and tubes used as *deep foundation* elements shall have a nominal outside diameter of not less than 7 inches (178 mm).

Where steel pipes or tubes are driven open ended, they shall have not less than 0.34 square inch (219 mm²) of steel in cross section to resist each 1,000 foot-pounds (1356 Nm) of pile hammer energy, or shall have the equivalent strength for steels having a yield strength greater than 35,000 psi (241 MPa) or the wave equation analysis shall be permitted to be used to assess compression stresses induced by driving to evaluate if the pile section is appropriate for the selected hammer. Where a pipe or tube with wall thickness less than 0.179 inch (4.6 mm) is driven open ended, a suitable cutting shoe shall be provided. Concrete-filled steel pipes or tubes in structures assigned to *Seismic Design Category C* or *D* shall have a wall thickness of not less than 3/16 inch (5 mm). The pipe or tube casing for *socketed drilled shafts* shall have a nominal outside diameter of not less than 18 inches (457 mm), a wall thickness of not less than 3/8 inch (9.5 mm) and a suitable steel driving shoe welded to the bottom; the diameter of the rock socket shall be approximately equal to the inside diameter of the casing.

Exceptions:

1. The minimum diameter for drilled *micropiles* shall be 4.5 inches (114 mm) and for jacked piles shall be 3 inches (76 mm).
2. For mandrel-driven pipes or tubes, the minimum wall thickness shall be 1/10 inch (2.5 mm)."

25. Revise Section 1810.3.5.3.5 to read:

"1810.3.5.3.5 Helical piles.

Dimensions of the central shaft and the number, size and thickness of helical bearing plates shall be sufficient to support the design loads. The minimum pipe shaft shall have a diameter of not less than 3.5 inches (89 mm) and wall thickness not less than 1/4 inch (6.4 mm)."

26. Revise the first paragraph of Section 1810.3.6 to read:

"1810.3.6 Splices.

Splices shall be constructed so as to provide and maintain true alignment and position of the component parts of the *deep foundation* element during installation and subsequent thereto and shall be designed to resist the axial and shear forces and moments occurring at the location of the splice during driving and for design load combinations. Where *deep foundation* elements of different materials or different types are being spliced, splices shall develop the full compressive strength. Where structural steel cores are to be spliced, the ends shall be milled or ground to provide full contact and shall be full-depth welded."

27. Revise the title and first sentence of Section 1810.3.6.1 to read:

"1810.3.6.1 Seismic Design Categories C and D.

For *structures* assigned to *Seismic Design Category C* or *D* splices of *deep foundation* elements shall develop the lesser of the

following:"

28. Revise the title and first sentence of Section 1810.3.8.2.2 to read:

"1810.3.8.2.2 Seismic reinforcement in Seismic Design Categories C and D.

For *structures* assigned to *Seismic Design Category* C or D, precast nonprestressed piles shall be reinforced as specified in this section."

29. Revise Section 1810.3.8.2.3 to read:

"1810.3.8.2.3 Additional seismic reinforcement in Seismic Design Category D.

For *structures* assigned to *Seismic Design Category* D, transverse reinforcement shall be in accordance with Section 1810.3.9.4.2."

30. Revise the title and first sentence of Section 1810.3.8.3.3 to read:

"1810.3.8.3.3 Seismic reinforcement in Seismic Design Category D.

For *structures* assigned to *Seismic Design Category* D, precast prestressed piles shall have transverse reinforcement in accordance with the following:"

31. Revise the title and first sentence of Section 1810.3.8.3.4 to read:

"1810.3.8.3.4 Axial load limit in Seismic Design Categories C and D.

For *structures* assigned to *Seismic Design Category* C or D, the maximum factored axial load on precast prestressed piles subjected to a combination of seismic lateral force and axial load shall not exceed the following values:"

32. Revise exception 3 to Section 1810.3.9 to read:

"3. For Group R-3, R-5 and U occupancies not exceeding two *stories* of *light-frame construction*, reinforcement is permitted to be placed after concreting, while the concrete is still in a semifluid state, and the concrete cover requirement is permitted to be reduced to 2 inches (51 mm), provided that the construction method can be demonstrated to the satisfaction of the *building official*."

33. Revise Section 1810.3.9.4 to read:

"1810.3.9.4 Seismic reinforcement.

Where a *structure* is assigned to *Seismic Design Category* C, reinforcement shall be provided in accordance with Section 1810.3.9.4.1. Where a *structure* is assigned to *Seismic Design Category* D, reinforcement shall be provided in accordance with Section 1810.3.9.4.2.

Exceptions:

1. Isolated *deep foundation* elements supporting posts of Group R-3, R-4, R-5 and U occupancies not exceeding two stories of *light-frame construction* shall be permitted to be reinforced as required by rational analysis but with not less than one No. 4 bar, without ties or spirals, where detailed so the element is not subject to lateral loads and the soil provides adequate lateral support in accordance with Section 1810.2.1.
2. Isolated *deep foundation* elements supporting posts and bracing from decks and patios appurtenant to Group R-3, R-4, R-5 and U occupancies not exceeding two *stories* of *light-frame construction* shall be permitted to be reinforced as required by rational analysis but with not less than one No. 4 bar, without ties or spirals, where the lateral load, *E*, to the top of the element does not exceed 200 pounds (890 N) and the soil provides adequate lateral support in accordance with Section 1810.2.1.
3. *Deep foundation* elements supporting the concrete foundation wall of Group R-3, R-4, R-5 and U occupancies not exceeding two *stories* of *light-frame construction* shall be permitted to be reinforced as required by rational analysis but with not less than two No. 4 bars, without ties or spirals, where the design cracking moment determined in accordance with Section 1810.3.9.1 exceeds the required moment strength determined using the load combinations with overstrength factor in Section 2.3.6 or 2.4.5 of ASCE 7 and the soil provides adequate lateral support in accordance with Section 1810.2.1.
4. Closed ties or spirals where required by Section 1810.3.9.4.2 shall be permitted to be limited to the top 3 feet (914 mm) of *deep foundation* elements 10 feet (3048 mm) or less in depth supporting Group R-3, R-4, R-5 and U occupancies of *Seismic Design Category* D, not exceeding two *stories* of *light-frame construction*."

34. Revise the title and first sentence of Section 1810.3.9.4.2 to read:

"1810.3.9.4.2 Seismic reinforcement in Seismic Design Category D.

For *structures* assigned to *Seismic Design Category* D, cast-in-place *deep foundation* elements shall be reinforced as specified in this section. Reinforcement shall be provided where required by analysis."

35. Revise Section 1810.3.9.5 to read:

"1810.3.9.5 Belled drilled shafts.

Where *drilled shafts* are belled at the bottom, the edge thickness of the bell shall be not less than that required for the edge of footings. Where the sides of the bell slope at an angle less than 60 degrees (1 rad) from the horizontal, the effects of vertical shear shall be considered. The bell diameter shall not exceed three times the shaft diameter.

The design of the belled drilled shaft shall be based on the predicted settlement and on the allowable load bearing pressure on the bottom of the bell; side resistance shall be ignored.

Continuous special inspections of belled *drilled shafts* using an explosion proof camera, defined by Sections 1810.4.12, 1705.7 and 1705.8, shall be required to confirm the suitable bearing soil and bottom of bell cleanup to remove excess cuttings. Alternatively, where belled caissons are used, the caisson bell diameter shall be allowed to be oversized by 15 percent, for a minimum diameter increase of 6 inches (152 mm) up to a maximum diameter increase of 1 foot (305 mm)."

36. Revise Section 1810.3.9.6 to read:

"1810.3.9.6 Socketed drilled shafts.

Socketed drilled shafts shall have a permanent pipe or tube casing that extends down to bedrock and an uncased socket drilled into the bedrock, both filled with concrete. *Socketed drilled shafts* shall have reinforcement or a structural steel core for the length as indicated by accepted engineering analysis.

The depth of the rock socket shall be sufficient to develop the full load-bearing capacity of the element with a minimum safety factor in accordance with Section 1810.3.3.1.7, but the depth shall be not less than one foot (305 mm)."

37. Revise Section 1810.3.10.2 to read:

"1810.3.10.2 Materials.

Reinforcement shall consist of deformed reinforcing bars in accordance with ASTM A615 Grade 60, 75 or 80 or ASTM A722 Grade 150.

The steel pipe or tube shall have a minimum wall thickness of 1/4 inch (6.3 mm). Splices shall comply with Section 1810.3.6. The steel pipe or tube shall have a minimum yield strength of 45,000 psi (310 MPa) and a minimum elongation of 15 percent as shown by mill certifications or two coupon test samples per 40,000 pounds (18 160 kg) of pipe or tube."

38. Revise Section 1810.3.10.4 to replace "*Seismic Design Category D, E or F*" with "*Seismic Design Category D.*"

39. Revise the title and first sentence of Section 1810.3.11.1 to read:

"1810.3.11.1 Seismic Design Categories C and D.

For *structures* assigned to *Seismic Design Category C* or *D*, concrete *deep foundation* elements shall be connected to the pile cap by embedding the element reinforcement or field-placed dowels anchored in the element into the pile cap for a distance equal to their development length in accordance with ACI 318."

40. Revise the title and first sentence of Section 1810.3.11.2 to read:

"1810.3.11.2 Seismic Design Category D.

For *structures* assigned to *Seismic Design Category D*, *deep foundation* element resistance to uplift forces or rotational restraint shall be provided by anchorage into the pile cap, designed considering the combined effect of axial forces due to uplift and bending moments due to fixity to the pile cap."

41. Revise Section 1810.3.12 to replace "*Seismic Design Category D, E or F*" with "*Seismic Design Category D.*"

42. Revise Section 1810.3.13 to replace "*Seismic Design Category C, D, E or F*" with "*Seismic Design Category C or D.*"

43. Revise the exception to Section 1810.3.13 to read:

"Exception: In Group R-3, R-4, R-5 and U occupancies of *light-frame construction*, *deep foundation* elements supporting foundation walls, isolated interior posts detailed so the element is not subject to lateral loads or exterior decks and patios are not subject to interconnection where the soils are of adequate stiffness."

44. Revise section 1810.4 by adding the following sentence at the end:

"Where installing *drilled shafts* through unstable soils or soft clay, the installation procedures shall include casing or drilling slurry as recommended by the *registered geotechnical engineer*. The *registered geotechnical engineer* shall analyze the soil for shaft squeeze-in potential in soft clay."

45. Revise Section 1810.4.4 to read:

"1810.4.4 Preexcavation.

The use of jetting, augering or other methods of preexcavation shall be allowed where recommended by the *registered geotechnical engineer*. Where permitted, preexcavation shall be carried out in the same manner as used for *deep foundation* elements subject to load tests and in such a manner that will not impair the carrying capacity of the elements already in place or damage adjacent structures. Element tips shall be advanced below the preexcavated depth until the required resistance or penetration is obtained."

46. Revise Section 1810.4.5 to read:

"1810.4.5 Vibratory driving.

Vibratory drivers shall only be used to install *deep foundation* elements where the element load capacity is verified by load tests in accordance with Section 1810.3.3.1.2, unless the pile installation is completed by driving with an impact hammer in accordance with Section 1810.3.3.1.1, or the pile is to be used only for lateral resistance. The installation of production elements shall be controlled according to power consumption, rate of penetration or other approved means that ensure element capacities equal or exceed those of the test elements."

47. Revise Section 1810.4.8 to read:

"1810.4.8 Hollow-stem augered, and drilled displacement cast-in-place elements.

Where concrete or grout is placed by pumping through a hollow-stem auger, the auger shall be permitted to rotate in a clockwise direction during withdrawal. As the auger is withdrawn at a steady rate or in increments not to exceed 1 foot (305 mm), concreting or grouting pumping pressures shall be measured and maintained high enough at all times to offset hydrostatic and lateral earth pressures. Concrete or grout volumes shall be measured to ensure that the volume of concrete or grout placed in each element is equal to or greater than 110 percent of the theoretical volume of the hole created by the auger. Where the installation process of any element is interrupted or a loss of concreting or grouting pressure occurs, the element shall be redrilled to 5 feet (1524 mm) below the elevation of the tip of the auger when the installation was interrupted or concrete or grout pressure was lost and reformed.

Augered cast-in-place elements shall not be installed within six diameters center to center of an element filled with concrete or grout less than 12 hours old, unless *approved* by the *building official*. If the concrete or grout level in any completed element drops due to installation of an adjacent element, the element shall be replaced. All piles shall have a full-length reinforcing bar with a centralizer at the tip."

48. Revise Section 1810.4.9 to read:

"1810.4.9 Socketed drilled shafts.

The rock socket and pipe or tube casing, if any, of *socketed drilled shafts* shall be thoroughly cleaned of foreign materials before filling with concrete. Steel cores shall be bedded in concrete or sand/cement grout at the base of the rock socket. Concrete or sand/cement grout shall be placed using a tremie method."

49. Delete item 3 to Section 1810.4.10.

50. Revise item 5 to Section 1810.4.10 to read:

"5. *Micropiles* shall be grouted as soon as practical after drilling is completed."

51. Revise Section 1810.4.11 by replacing "*registered design professional*" with "*registered geotechnical engineer*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 81; Amend Coun. J. 1-26-22, p. 43756, Art. II, § 18)

CHAPTER 14B-19

CONCRETE

14B-19-1901 General.

14B-19-1902 Definitions.

14B-19-1903 Specifications for tests and materials.

14B-19-1904 Durability requirements.

14B-19-1905 Modifications to ACI 318.

14B-19-1906 Structural plain concrete.

14B-19-1907 Minimum slab provisions.

14B-19-1908 Shotcrete.

14B-19-1901 General.

The provisions of Section 1901 of IBC are adopted by reference with the following modifications:

1. Revise Section 1901.2 by replacing "*Seismic Design Category C, D, E or F*" with "*Seismic Design Category C or D*."
2. Revise Section 1901.5 by replacing "*Seismic Design Category D, E or F*" with "*Seismic Design Category D*."

14B-19-1902 Definitions.

The provisions of Section 1902 of IBC are adopted by reference without modification.

14B-19-1903 Specifications for tests and materials.

The provisions of Section 1903 of IBC are adopted by reference without modification.

14B-19-1904 Durability requirements.

The provisions of Section 1904 of IBC are adopted by reference with the following modification:

1. Revise Section 1904.1 by replacing "*Group R-2 and R-3*" with "*Group R*."

14B-19-1905 Modifications to ACI 318.

The provisions of Section 1905 of IBC are adopted by reference with the following modifications:

1. Revise Section 1905.1.2 by replacing "*Seismic Design Category B, C, D, E or F*" with "*Seismic Design Category B, C or D*" and "*Seismic Design Category C, D, E or F*" with "*Seismic Design Category C or D*."
2. Revise Section 1905.1.7 by replacing "*Seismic Design Category C, D, E or F*" with "*Seismic Design Category C or D*"; "*one- and two-family*" with "*one-, two- and three-family*" and "*Seismic Design Category D or E*" with "*Seismic Design Category D*."

14B-19-1906 Structural plain concrete.

The provisions of Section 1906 of IBC are adopted by reference with the following modification:

1. Revise the exception to Section 1906.1 by replacing "*Group R-3*" with "*Groups R-3 and R-5*."

14B-19-1907 Minimum slab provisions.

The provisions of Section 1907 of IBC are adopted by reference with the following modifications:

1. Revise Section 1907.1 by deleting "*approved*" and replacing "*6 inches (152 mm)*" with "*12 inches (305 mm)*."
2. Revise the exceptions to Section 1907.1 by replacing "*Group R-3*" with "*Group R-3 or R-5*."

(Amend Coun. J. 6-25-21, p. 32180, Art. II, § 25)

14B-19-1908 Shotcrete.

The provisions of Section 1908 of IBC are adopted by reference with the following modifications:

1. Revise the exception to Section 1908.4.2 to read:

"Exception: Where *approved* by the *building official*, required clearances may be reduced where it is demonstrated by preconstruction tests that adequate encasement of the bars used in the design will be achieved."

2. Revise the last sentence of Section 1908.5 to read:

"Reports of preconstruction tests shall be made available to the *building official* as specified in Section 1704.5."

3. Revise Section 1908.7 by replacing "*approved*" with "*permitted*."

4. Revise Sections 1908.9.1 and 1908.9.2 by replacing "*an approved*" with "*a*."

CHAPTER 14B-20

ALUMINUM

14B-20-2001 General.

14B-20-2002 Materials.

14B-20-2001 General.

The provisions of Section 2001 of IBC are adopted by reference without modification.

14B-20-2002 Materials.

The provisions of Section 2002 of IBC are adopted by reference without modification.

CHAPTER 14B-21

MASONRY

- 14B-21-2100 Defined terms.**
- 14B-21-2101 General.**
- 14B-21-2102 Notations.**
- 14B-21-2103 Masonry construction materials.**
- 14B-21-2104 Construction.**
- 14B-21-2105 Quality assurance.**
- 14B-21-2106 Seismic design.**
- 14B-21-2107 Allowable stress design.**
- 14B-21-2108 Strength design of masonry.**
- 14B-21-2109 IBC Section 2109.**
- 14B-21-2110 Glass unit masonry.**
- 14B-21-2111 Masonry fireplaces.**
- 14B-21-2112 Masonry heaters.**
- 14B-21-2113 Masonry chimneys.**
- 14B-21-2114 Dry-stack masonry.**

14B-21-2100 Defined terms.

Where provisions of Chapter 21 of IBC are adopted by reference, the following terms shall be reset in italic type: "AAC masonry"; "adhered masonry veneer"; "bed joint"; "chimney"; "fireplace"; "fireplace throat"; "fire-resistance rating"; "hollow masonry unit"; "masonry unit"; "mortar"; "solid masonry unit"; "stone masonry"; "structural clay tile"; "surface-bonding mortar" and "wythe."

14B-21-2101 General.

The provisions of Section 2101 of IBC are adopted by reference with the following modification:

1. Add new Section 2101.2.2 to read:

"2101.2.2 Single-wythe exterior walls.

Masonry units used in single-wythe exterior walls of buildings containing a Group R occupancy shall be tested in accordance with ASTM E514 and shall have integral water repellant included in the manufacturing process.

Exception: Grade SW clay or shale *masonry units*.

Mortar used in single-wythe exterior walls of buildings containing a Group R occupancy shall have a compatible water repellant added to the mixture."

(Amend Coun. J. 10-7-20, p. 21791, Art. II, § 30)

14B-21-2102 Notations.

The provisions of Section 2102 of IBC are adopted by reference without modification.

14B-21-2103 Masonry construction materials.

The provisions of Section 2103 of IBC are adopted by reference with the following modifications:

1. Revise Section 2103.1, excluding the exception, to read:

"2103.1 Masonry units.

Concrete *masonry units*, clay or shale *masonry units*, stone *masonry units*, glass unit masonry and AAC masonry units shall comply with Article 2.3 of TMS 602. Architectural cast stone shall conform to ASTM C1364 and TMS 504. Calcium silicate-based *masonry units* shall conform to ASTM C73 or C744. Adhered manufactured stone masonry veneer units shall conform to ASTM

C1670."

2. Add a new exception following Section 2103.1.1 to read:

"Exception: Undamaged *masonry units* may be reused on the same *site* for the same purpose without needing to conform to the requirements for new *masonry units*."

14B-21-2104 Construction.

The provisions of Section 2104 of IBC are adopted by reference with the following modification:

1. Revise the last sentence of Section 2104.1.2 to read:

"Terra cotta and metal cornices shall be anchored to a structural frame of noncombustible material."

14B-21-2105 Quality assurance.

The provisions of Section 2105 of IBC are adopted by reference with the following modification:

1. Revise Section 2105.1 by replacing "*approved*" with "*permitted*."

14B-21-2106 Seismic design.

The provisions of Section 2106 of IBC are adopted by reference without modification.

14B-21-2107 Allowable stress design.

The provisions of Section 2107 of IBC are adopted by reference without modification.

14B-21-2108 Strength design of masonry.

The provisions of Section 2108 of IBC are adopted by reference without modification.

14B-21-2109 IBC Section 2109.

The provisions of Section 2109 of IBC are not adopted.

14B-21-2110 Glass unit masonry.

The provisions of Section 2110 of IBC are adopted by reference with the following modification:

1. Revise Section 2110.1.1, excluding the exception, to read:

"2110.1.1 Limitations.

Solid or hollow glass block shall not be used in *fire walls*, party walls, *fire barriers*, *fire partitions* or *smoke barriers*, or for load-bearing construction. Glass blocks shall be erected with mortar and reinforcement in metal channel-type frames, structural frames, masonry or concrete recesses, embedded panel anchors as provided for both exterior and interior walls or other *approved* joint materials. Wood strip framing shall not be used in walls required to have a *fire-resistance rating* by other provisions of this code."

14B-21-2111 Masonry fireplaces.

The provisions of Section 2111 of IBC are adopted by reference with the following modifications:

1. Delete the last sentence of Section 2111.3.

2. Delete the last sentence of Section 2111.4.

3. Revise Section 2111.14.3 to read:

"2111.14.3 Exterior air intake.

The exterior air intake shall be capable of providing all combustion air from the exterior of the *building*. The exterior air intake shall not be located within a garage, *attic*, basement or crawl space nor shall the air intake be located at an elevation higher than the firebox. The exterior air intake shall be covered with a corrosion-resistant screen of 1/4-inch (6.4 mm) mesh."

14B-21-2112 Masonry heaters.

The provisions of Section 2112 of IBC are adopted by reference with the following modification:

1. Revise Section 2112.4 by replacing "*Seismic Design Category D, E or F*" with "*Seismic Design Category D*."

14B-21-2113 Masonry chimneys.

The provisions of Section 2113 of IBC are adopted by reference with the following modifications:

1. Delete the last sentence of Section 2113.3.
2. Revise Section 2113.5 to read:

"2113.5 Corbeling.

Masonry chimneys shall not be corbeled more than half of the chimney's wall thickness from a wall or foundation, nor shall a *chimney* be corbeled from a wall or foundation that is less than 12 inches (305 mm) in thickness unless it projects equally on each side of the wall, except that on the second *story* of a two-story building of Group R-5 occupancy, corbeling of *chimneys* on the exterior of the enclosing walls is permitted to equal the wall thickness. The projection of a single course shall not exceed one-half the unit height or one-third of the unit bed depth, whichever is less."

3. Revise Section 2113.7 to read:

"2113.7 Offsets.

Where a *masonry chimney* is constructed with a fireclay flue liner surrounded by one *wythe* of *masonry*, the maximum offset shall be such that the centerline of the flue above the offset does not extend beyond the center of the *chimney* wall below the offset. Where the *chimney* offset is supported by *masonry* below the offset in accordance with accepted engineering practices, the maximum offset limitations shall not apply. Each individual corbeled *masonry* course of the offset shall not exceed the projection limitations specified in Section 2113.5."

4. Revise Section 2113.11.1.4 to read:

"2113.11.1.4 Oil-fired appliances listed for use with L-vent.

Flue lining and vent systems for use in *masonry chimneys* with oil-fired appliances *listed* for use with Type L vent shall be limited to flue lining systems complying with Section 2113.11.1 and *listed* chimney liners complying with UL 641 (see Section 2113.11.1.5 for marking)."

5. Revise Section 2113.11.2.2 by deleting "an *approved*."
6. Revise Section 2113.11.3.2 by deleting "an *approved*."
7. Revise Section 2113.11.3.4 by deleting "*approved*."

14B-21-2114 Dry-stack masonry.

The provisions of Section 2114 of IBC are adopted by reference without modification.

CHAPTER 14B-22

STEEL

14B-22-2200 Defined terms.

14B-22-2201 General.

14B-22-2202 Identification of steel for structural purposes.

14B-22-2203 Protection of steel for structural purposes.

14B-22-2204 Connections.

14B-22-2205 Structural steel.

14B-22-2206 Composite structural steel and concrete structures.

14B-22-2207 Steel joists.

14B-22-2208 Steel cable structures.

14B-22-2209 Steel storage racks.

14B-22-2210 Cold-formed steel.

14B-22-2211 Cold-formed steel light-frame construction.

14B-22-2212 Cast iron.

Where provisions of Chapter 22 of IBC are adopted by reference, the following terms shall be reset in italic type: "cold-formed steel construction"; "light-frame construction"; "seismic force-resisting system"; "steel joist" and "structural steel element."

14B-22-2201 General.

The provisions of Section 2201 of IBC are adopted by reference without modification.

14B-22-2202 Identification of steel for structural purposes.

The provisions of Section 2202 of IBC are adopted by reference without modification.

14B-22-2203 Protection of steel for structural purposes.

The provisions of Section 2203 of IBC are adopted by reference without modification.

14B-22-2204 Connections.

The provisions of Section 2204 of IBC are adopted by reference with the following modification:

1. Revise Section 2204.3 by replacing "*approved*" with "*permitted*."

14B-22-2205 Structural steel.

The provisions of Section 2205 of IBC are adopted by reference with the following modifications:

1. Revise Section 2205.2.1.2 to read:

"2205.2.1.2 Seismic Design Category D.

Structures assigned to *Seismic Design Category D* shall be designed and detailed in accordance with AISC 341, except as permitted in ASCE 7, Table 15.4-1."

2. Revise item 1 to Section 2205.2.2 to read:

"1. The structure is assigned to *Seismic Design Category D* except as permitted in ASCE 7, Table 15.4-1."

14B-22-2206 Composite structural steel and concrete structures.

The provisions of Section 2206 of IBC are adopted by reference without modification.

14B-22-2207 Steel joists.

The provisions of Section 2207 of IBC are adopted by reference with the following modifications:

1. Revise Section 2207.4 by replacing "*approved construction documents*" with "*permitted construction documents*."
2. Revise Section 2207.5 by replacing "*approved construction documents*" with "*permitted construction documents*."

14B-22-2208 Steel cable structures.

The provisions of Section 2208 of IBC are adopted by reference without modification.

14B-22-2209 Steel storage racks.

The provisions of Section 2209 of IBC are adopted by reference without modification.

14B-22-2210 Cold-formed steel.

The provisions of Section 2210 of IBC are adopted by reference without modification.

14B-22-2211 Cold-formed steel light-frame construction.

The provisions of Section 2211 of IBC are adopted by reference with the following modifications:

1. Revise Section 2211.1.1.2 to read:

"2211.1.1.2 Seismic Design Category D.

In cold-formed steel *light-frame construction* assigned to *Seismic Design Category D* the *seismic force-resisting system* shall be designed and detailed in accordance with AISI S400."

2. Revise Section 2211.1.2 to read:

"2211.1.2 Prescriptive framing.

Group R-5 occupancies, less than or equal to three *stories above grade plane*, shall be permitted to be constructed in accordance with AISI S230 subject to the limitations therein, provided that the *building* may contain up to three *dwelling units*."

3. Add a new Section 2211.3 to read:

"2211.3 Vertical shaft enclosures in buildings greater than 55 feet in height.

In *buildings* of cold-formed steel *light-frame construction* greater than 55 feet (19.8 m) in *building height*, shaft enclosures required by Section 713 to have a *fire-resistance rating* greater than 1 hour shall be constructed of structural masonry or concrete."

14B-22-2212 Cast iron.

The following language is adopted as a new Section 2212:

"2212. CAST IRON

2212.1 General.

The structural use of cast iron elements in *buildings*, *structures* and portions thereof shall be in accordance with ASTM A48.

2212.2 Minimum thickness.

The minimum thickness of cast iron shall be not less than the following:

1. For columns, one-twelfth the smallest dimension of cross section with a minimum thickness of 3/4 inch (19 mm).
2. For bases and flanges, 1 inch (25 mm) and reinforced with fillets and brackets.
3. For lintels, 3/4 inch (19 mm) and limited to a maximum span of six feet (1829 mm).

2212.3 Limitations.

Cast iron columns shall not be used where subject to eccentric loads that produce a net tension in the material, nor in any part of a structural frame which is required to resist stress due to wind.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 82)

CHAPTER 14B-23

WOOD

14B-23-2300 Defined terms.

14B-23-2301 General.

14B-23-2302 Design requirements.

14B-23-2303 Minimum standards and quality.

14B-23-2304 General construction requirements.

14B-23-2305 General design requirements for lateral force-resisting systems.

14B-23-2306 Allowable stress design.

14B-23-2307 Load and resistance factor design.

14B-23-2308 Conventional light-frame construction.

14B-23-2309 Wood frame construction manual.

14B-23-2300 Defined terms.

Where provisions of Chapter 23 of IBC are adopted by reference, the following terms shall be reset in italic type: "accreditation body"; "braced wall line"; "braced wall panel"; "cripple wall"; "diaphragm"; "drag strut"; engineered wood rim board"; "fiberboard"; "gable"; "hardboard"; "hold-down"; "naturally durable wood"; "particleboard"; "performance category"; "prefabricated wood I-joist"; "risk category"; "structural composite lumber"; "tie-down" and "wood structural panel."

14B-23-2301 General.

The provisions of Section 2301 of IBC are adopted by reference without modification.

14B-23-2302 Design requirements.

The provisions of Section 2302 of IBC are adopted by reference without modification.

14B-23-2303 Minimum standards and quality.

The provisions of Section 2303 of IBC are adopted by reference with the following modifications:

1. Revise Section 2303.1.1 and its subsections to read:

"2303.1.1 Sawn lumber.

Sawn lumber used for load-supporting purposes, including end-jointed or edge-glued lumber, machine stress-rated or machine-evaluated lumber, shall be identified by the *grade mark* of a lumber grading or inspection agency accredited by an *accreditation body* that complies with DOC PS 20 or *approved* equivalent. Grading practices and identification shall comply with rules published by an agency certified in accordance with the procedures of DOC PS 20 or *approved* equivalent procedures.

2303.1.1.1 Certificate of inspection.

In lieu of a *grade mark* on the material, a certificate of inspection as to species and *grade* issued by a lumber grading or inspection agency meeting the requirements of this section shall be accepted for precut, remanufactured or rough-sawn lumber and for sizes larger than 3 inches (76 mm) nominal thickness.

2303.1.1.2 End-jointed lumber.

Where *approved* for use, end-jointed lumber may be used interchangeably with solid-sawn members of the same species and *grade*. End-jointed lumber used in an assembly required to have a *fire-resistance rating* shall have the designation "Heat Resistant Adhesive" or "HRA" included in its *grade mark*."

2. Revise Section 2303.1.3 to read:

"2303.1.3 Structural glued-laminated timber.

Structural glued-laminated timbers shall be manufactured and identified as required in ANSI/AITC A 190.1 and ASTM D3737."

3. Revise Section 2303.1.4 to read:

"2303.1.4 Structural glued cross-laminated timber.

Where *approved* for use, *cross-laminated timber* shall be manufactured and identified in accordance with ANSI/APA PRG 320."

4. Revise Section 2303.1.5 to read:

"2303.1.5 Wood structural panels.

Wood structural panels, where used structurally (including those used for siding, roof and wall sheathing, subflooring, diaphragms and built-up members), shall conform to the requirements for their type in DOC PS 1, DOC PS 2 or ANSI/APA PRP 210. Each panel or member shall be identified for *grade*, bond classification, and *Performance Category* by the trademarks of an *approved agency*. The *Performance Category* value shall be used as the "nominal panel thickness" or "panel thickness" whenever referenced in this code. *Wood structural panel* components shall be designed and fabricated in accordance with the applicable standards listed in Section 2306.1 and identified by the trademarks of an *approved agency* indicating conformance to the applicable standard. In addition, *wood structural panels* where permanently exposed in outdoor applications shall be of exterior type, except that *wood structural panel* roof sheathing exposed to the outdoors on the underside is permitted to be Exposure 1 type."

5. Revise Section 2303.1.6.2 by replacing "an *approved* roof covering" with "a *roof covering*."

- 5.1. Revise Section 2303.1.6.3 by replacing "an *approved*" with "a".

6. Delete Section 2303.1.6.3.1.

7. Revise Section 2303.1.11 to read:

"2303.1.11 Structural log members.

Stress grading of structural log members of nonrectangular shape, as typically used in log buildings, shall be in accordance with ASTM D3957. Such structural log members shall be identified by the *grade mark* of an *approved agency*. In lieu of a *grade mark* on the material, a certificate of inspection as to species and *grade* issued by a lumber grading or inspection agency meeting the requirements of this section shall be permitted."

8. Revise Section 2303.2.2 by replacing "approved" with "acceptable."

9. Revise Section 2303.2.5 by replacing "an *approved* method" with "a method."

10. Revise Section 2303.2.9 to read:

"2303.2.9 Type I and II construction applications.

See Section 603.1 for allowed uses of *fire-retardant-treated wood* in buildings of Type I or II construction."

11. Revise the first sentence of Section 2303.4.1.1 to read:

"The written, graphic and pictorial depiction of each individual truss shall be provided to the *registered design professional in responsible charge* for approval prior to installation."

12. Delete Section 2303.4.1.4 and its subsections.

13. Revise Section 2303.4.6 by deleting the last sentence.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 83)

14B-23-2304 General construction requirements.

The provisions of Section 2304 of IBC are adopted by reference with the following modifications:

1. Revise note c to Table 2304.8(3) to delete "approved."
2. Revise note a to Table 2304.8(4) to delete "approved."
3. Delete Section 2304.10.5.2.
4. Revise Section 2304.10.6 to delete "*approved*."
5. Revise note b to Table 2304.11 to delete "approved."
6. Revise the first two sentences of Section 2304.11.1.1 to read:

"Minimum dimensions of columns shall be in accordance with Table 2304.11. Columns shall be continuous or superimposed throughout all *stories*."

7. Revise Section 2304.11.1.2 to read:

"2304.11.1.2 Floor framing.

Minimum dimensions of floor framing shall be in accordance with Table 2304.11. Wall plate boxes or hangers shall be provided where wood beams, girders or trusses rest on masonry or concrete walls. Where intermediate beams are used to support a floor, they shall rest on top of girders, or shall be supported by a metal hanger into which the ends of the beams shall be closely fitted. Where traditional heavy timber detailing is used, these connections shall be permitted to be supported by ledgers or blocks securely fastened to the sides of the girders."

8. Delete Section 2304.11.2.1.

9. Revise Section 2304.11.3.1 to read:

"2304.11.3.1 Cross-laminated timber floors.

Where *approved* for use, *cross-laminated timber* floors shall comply with this section. *Cross-laminated timber* floors shall be not less than 4 inches (102 mm) in actual thickness. *Cross-laminated timber* floors shall be continuous from support to support and individual panels shall be mechanically fastened to one another. *Cross-laminated timber* floors shall be permitted to be connected to walls without a shrinkage gap providing swelling or shrinking is considered in the design. Corbelling of masonry walls under the floor shall be permitted to be used."

10. Revise Section 2304.11.4 to replace "*approved* hardware" with "other *approved* hardware."

11. Revise Section 2304.11.4.1 to read:

"2304.11.4.1 Cross-laminated timber roofs.

Where *approved* for use, *cross-laminated timber* roofs shall comply with this section. *Cross-laminated timber* roofs shall be not less than 3 inches (76 mm) in actual thickness, and shall be continuous from support to support. Individual panels shall be mechanically fastened to one another."

12. Revise Sections 2304.12.1.1, 2304.12.1.2, 2304.12.1.3, 2304.12.1.4 and 2304.12.1.5 to replace "naturally durable or *preservative-treated wood*" with "*naturally durable wood* or *preservative-treated wood*."

13. Revise Sections 2304.12.2.1, 2304.12.2.2, 2304.12.2.3, 2304.12.2.4 and 2304.12.2.5 to replace "naturally durable or *preservative-treated wood*" with "*naturally durable wood* or *preservative-treated wood*."

14. Delete the exception to Section 2304.12.2.3.

15. Revise Section 2304.12.3, excluding the exception, to read:

"2304.12.3 Wood in contact with the ground or fresh water.

Wood used in contact with exposed earth shall be *naturally durable wood* for both decay and termite resistance or *preservative treated wood* in accordance with AWPA U1 for soil or fresh water use."

16. Delete Section 2304.12.4.

14B-23-2305 General design requirements for lateral force-resisting systems.

The provisions of Section 2305 of IBC are adopted by reference with the following modification:

1. Revise Section 2305.2 by replacing "approved method" with "accepted engineering practice."

14B-23-2306 Allowable stress design.

The provisions of Section 2306 of IBC are adopted by reference with the following modifications:

1. Revise Section 2306.1.3 by replacing "an *approved* method" with "a method."
2. Revise note g to Table 2306.3(1) by replacing "Seismic Design Category D, E or F" with "*Seismic Design Category D*."
3. Revise note e to Table 2306.3(2) by replacing "Seismic Design Category D, E or F" with "*Seismic Design Category D*."

14B-23-2307 Load and resistance factor design.

The provisions of Section 2307 of IBC are adopted by reference without modification.

14B-23-2308 Conventional light-frame construction.

The provisions of Section 2308 of IBC are adopted by reference with the following modifications:

1. Delete the last sentence of Section 2308.1.
2. Revise Table 2308.2.1 to read:

TABLE 2308.2.1

ALLOWABLE STORY HEIGHT

SEISMIC DESIGN CATEGORY	ALLOWABLE STORIES ABOVE GRADE PLANE ^a
A or B	Three stories
C	One story

a. Where *cripple walls* exceed 14 inches (356 mm) in height, the allowable number of *stories above grade plane* shall be reduced by one.

- 2.1. Revise Section 2308.2.3 by deleting item 3.
3. Delete Section 2308.2.4.
4. Revise Section 2308.2.6 by deleting "assigned to *Seismic Design Category B, C, D or F*."
5. Revise Section 2308.3.1 by replacing "*approved* anchors" with "other anchors."
6. Delete Section 2308.3.1.1.
7. Delete Section 2308.3.1.2.
8. Revise Table 2308.4.1.1(1) by deleting the columns for snow loads of 50 psf and 70 psf and the heading rows "GROUND SNOW LOAD (psf)" and "30".
- 8.1. Revise note d to Table 2308.4.1.1(1) by replacing "an approved framing anchor" with "a framing anchor."
- 8.2. Delete note e to Table 2308.4.1.1(1).
9. Revise note d to Table 2308.4.1.1(2) by replacing "an approved framing anchor" with "a framing anchor."
- 9.1. Delete Table 2308.4.2.1(1).
10. Revise Section 2308.4.2.3 by deleting "in an approved manner."
11. Revise Section 2308.4.4.1 to read:

"2308.4.4.1 Openings in floor diaphragms in Seismic Design Category B or C.

Openings in horizontal *diaphragms* in *Seismic Design Category B or C* with a dimension that is greater than 4 feet (1219 mm) shall be constructed with metal ties and blocking in accordance with this section and Figure 2308.4.4.1(1). Metal ties shall be not less than 0.058 inch [1.47 mm (16 galvanized gage)] in thickness by 1 1/2 inches (38 mm) in width and shall have a yield stress not less than 33,000 psi (227 Mpa). Blocking shall extend not less than the dimension of the opening in the direction of the tie and blocking. Ties shall be attached to blocking in accordance with the manufacturer's instructions but with not less than eight 16d common nails on each side of the header-joist intersection."

12. Delete Section 2308.4.4.2 and its exception.
13. Delete Section 2308.4.10.
14. Revise the first two sentences of Section 2308.5.5.1 to read:
 "Headers shall be provided over each opening in exterior bearing walls. The size and spans in Table 2308.4.1.1(1) are permitted to be used for Group R-5 occupancies."
15. Revise Table 2308.5.11 by deleting the row for "Hardboard panel siding" and adding a row to read:
 "Cement board sheathing | 1/2 inch | 16 inches on center"
- 15.1. Revise Section 2308.6.1 by deleting the last sentence.
- 15.2. Revise Table 2308.6.1 by deleting the row for a two-story building in Seismic Design Category C and all rows for Seismic Design Categories D and E.
16. Revise Table 2308.6.3(5) by deleting all rows for "direct to studs" siding.
17. Revise Section 2308.6.5.1 by deleting "approved."
18. Revise Section 2308.6.6.1 to read:
"2308.6.6.1 Cripple wall bracing in Seismic Design Categories A, B and C.
 For the purposes of this section, *cripple walls* in *Seismic Design Categories A, B and C* having a stud height exceeding 14 inches (356 mm) shall be considered to be a *story* and shall be braced in accordance with Table 2308.6.1. Spacing of *edge nailing* for required cripple wall bracing shall not exceed 6 inches (152 mm) on center along the foundation plate and the top plate of the cripple wall. Nail size, nail spacing for *field nailing* and more restrictive *boundary nailing* requirements shall be as required elsewhere in the code for the specific bracing material used."
19. Delete Section 2308.6.6.2.
20. Revise Section 2308.6.7.2 by deleting "approved."
21. Revise Section 2308.6.8.1 by deleting all text after the first exception, including the subsequent exceptions.
22. Delete Section 2308.6.8.2.
23. Revise Section 2308.6.8.3 by replacing "*Seismic Design Categories B, C, D and E*" with "*Seismic Design Categories B and C*."
24. Delete Section 2308.6.10.2.
- 24.1. Revise the subtitle of Table 2308.7.1 (2) by deleting "limited."
- 24.2. Revise the subtitle of Table 2308.7.2(3) by deleting "Ground snow load = 30 psf,".
- 24.3. Delete Table 2308.7.2(4).
- 24.4. Revise the subtitle of Table 2308.7.2(5) by deleting "Ground snow load = 30 psf,".
- 24.5. Delete Table 2308.7.2(6).
- 24.6. Revise Table 2308.7.3.1 by deleting the columns for no snow load, deleting the columns for ground snow load = 50 psf, deleting the heading rows for "GROUND SNOW LOAD (pounds per square foot)" and "30 pounds per square foot," and deleting note g.
25. Revise the title of Table 2308.7.5 by deleting "approved."
26. Revise Table 2308.7.5 by:
 - a. Revising the first column in the upper portion of the table to read:
 "RISK CATEGORY
 I or II
 III"
 - b. Deleting the rows for $V_{asd} = 100$ and 110.
 - c. Revising note h by deleting "approved."
 - d. Deleting note g.
27. Revise Section 2308.7.6.1 by replacing "*Seismic Design Categories B, C, D and E*" with "*Seismic Design Categories B or C*."
28. Revise Section 2308.7.11 by deleting "approved."

14B-23-2309 Wood frame construction manual.

The provisions of Section 2309 of IBC are adopted by reference without modification.

CHAPTER 14B-24

GLASS AND GLAZING

14B-24-2400 Defined terms.

14B-24-2401 General.

14B-24-2402 IBC Section 2402.

14B-24-2403 General requirements for glass.

14B-24-2404 Wind, snow, seismic and dead loads on glass.

14B-24-2405 Sloped glazing and skylights.

14B-24-2406 Safety glazing.

14B-24-2407 Glass in guards.

14B-24-2408 Glazing in athletic facilities.

14B-24-2409 Glass in walkways, elevator hoistways and elevator cars.

14B-24-2400 Defined terms.

Where provisions of Chapter 24 of IBC are adopted by reference, the following terms shall be reset in italic type: "decorative glazing" and "unit skylight."

14B-24-2401 General.

The provisions of Section 2401 of IBC are adopted by reference without modification.

14B-24-2402 IBC Section 2402.

The provisions of Section 2402 of IBC are not adopted.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 85)

14B-24-2403 General requirements for glass.

The provisions of Section 2403 of IBC are adopted by reference with the following modifications:

1. Revise Section 2403.1 by replacing "*approved construction documents*" with "*permitted construction documents*."
2. Revise the last sentence of Section 2403.5 to read:

"Where other glass types are used, the design shall be *approved* by the *building official*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 86)

14B-24-2404 Wind, snow, seismic and dead loads on glass.

The provisions of Section 2404 of IBC are adopted by reference without modification.

14B-24-2405 Sloped glazing and skylights.

The provisions of Section 2405 of IBC are adopted by reference with the following modifications:

1. Revise the first sentence of exception 4 to Section 2405.3 to read:

"Screens shall not be required in individual *dwelling units* in Groups R-2, R-3 and R-5 where fully tempered glass is used as single glazing or as both panes in an insulating glass unit, and the following conditions are met:"

2. Revise the first sentence of exception 5 to Section 2405.4 to read:

"Screens shall not be required for laminated glass with a 15-mil (0.38 mm) polyvinyl butyral (or equivalent) interlayer used in individual *dwelling units* in Groups R-2, R-3 and R-5 within the following limits:"

3. Revise Section 2405.4 to read:

"2405.4 Framing.

In Type I and II construction, sloped glazing and skylight frames shall be constructed of noncombustible materials. In structures where acid fumes deleterious to metal are incidental to the use of the buildings, pressure- treated wood or other noncorrosive materials may be *approved* for sashes and frames. Framing supporting sloped glazing and skylights shall be designed to resist the tributary roof loads in Chapter 16. Skylights set at an angle of less than 45 degrees (0.79 rad) from the horizontal plane shall be mounted not less than 8 inches (203 mm) above the plane of the roof on a curb constructed as required for the frame. Skylights shall not be installed in the plane of the roof where the roof pitch is less than 45 degrees (0.79 rad) from the horizontal.

Exception: Installation of a skylight without a curb shall be permitted on roofs with a minimum slope of 14 degrees (three units vertical in 12 units horizontal) in Group R-3 and R-5 occupancies. *Unit skylights* installed in a roof with a pitch flatter than 14 degrees (0.25 rad) shall be mounted not less than 4 inches (102 mm) above the plane of the roof on a curb constructed as required for the frame unless otherwise specified in the manufacturer's installation instructions."

4. Revise Section 2405.5 by replacing "*approved* labeling agency" with "*approved agency*."

14B-24-2406 Safety glazing.

The provisions of Section 2406 of IBC are adopted by reference with the following modifications:

1. Revise exception 1 to Section 2406.3 to read:

"1. For other than tempered glass, manufacturer's designations are not required where the use of a certificate, affidavit or other evidence confirming compliance with this code is *approved* by the *building official*."

2. Revise exception 4 to Section 2406.4.2 to read:

"4. Glazing in walls on the latch side of and perpendicular to the plane of the door in a closed position in buildings of Group R-5 occupancy or within *dwelling units* and *sleeping units* in other Group R occupancies."

3. Revise exception 1 to Section 2406.4.6 to read:

"1. The side of a *stairway*, landing or *ramp* that has a *guard* complying with the provisions of Sections 1015 and 1607.8."

4. Revise the exception to Section 2406.4.7 to read:

"Exception: Glazing that is protected by a *guard* complying with Sections 1015 and 1607.8."

14B-24-2407 Glass in guards.

The provisions of Section 2407 of IBC are adopted by reference with the following modifications:

1. Revise Section 2407.1 to read:

"2407.1 Materials.

Glass used in a *guard* or to support a *handrail* shall be laminated glass constructed of fully tempered or heat- strengthened glass and shall comply with Category II or CPSC 16 CFR Part 1201 or Class A of ANSI Z97.1. Glazing in *guard* in-fill panels shall be of a recognized safety glazing material that conforms to the provisions of Section 2406.1.1. For all glazing types, the minimum nominal thickness shall be 1/4 inch (6.4 mm).

Exception: Single fully tempered glass complying with Category II of CPSC 16 CFR Part 1201 or Class A of ANSI Z97.1 shall be permitted to be used in *guards* where there is no walking surface beneath them or the walking surface is permanently protected from the risk of falling glass."

2. Revise Section 2407.1.1 to read:

"2407.1.1 Loads.

The panels and their support system shall be designed to withstand the loads specified in Section 1607.8. Glass *guard* elements shall be designed using a factor of safety of four in combination with the mean modulus of rupture for the type of glass provided in Table 2407.1.1.

TABLE 2407.1.1

MODULUS OF RUPTURE

TYPE OF GLASS	MEAN MODULUS OF RUPTURE
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Annealed glass	6,000 psi
Heat-strengthened glass	12,000 psi
Fully tempered glass	24,000 psi

For SI: 1 pound per square inch = 6894.75 N/m²."

3. Revise Section 2407.1.2 to read:

"2407.1.2 Structural glass baluster panels.

Guards with structural glass baluster panels shall be installed with an attached top rail or *handrail*. The top rail or *handrail* shall be supported by not fewer than three glass baluster panels, or shall be otherwise supported to remain in place should one glass baluster panel fail.

Exception: An attached top rail or *handrail* is not required where the glass baluster panels are laminated glass with two or more glass plies of equal thickness and of the same glass type. The panels shall be tested to remain in place as a barrier following impact or glass breakage in accordance with ASTM E2353."

4. Revise Section 2407.1.3 to read:

"2407.1.3 Parking garages.

Glazing materials shall not be installed in *guards* in parking garages except for pedestrian areas not exposed to impact from vehicles."

5. Delete Section 2407.1.4 and its subsections.

14B-24-2408 Glazing in athletic facilities.

The provisions of Section 2408 of IBC are adopted by reference without modification.

14B-24-2409 Glass in walkways, elevator hoistways and elevator cars.

The provisions of Section 2409 of IBC are adopted by reference with the following modifications:

1. Delete Section 2409.2.2.
2. Delete Section 2409.4.2.

CHAPTER 14B-25

GYPSUM BOARD, GYPSUM PANEL PRODUCTS AND PLASTER

14B-25-2500 Defined terms.

14B-25-2501 General.

14B-25-2502 Performance.

14B-25-2503 IBC Section 2503.

14B-25-2504 Vertical and horizontal assemblies.

14B-25-2505 Shear wall construction.

14B-25-2506 Gypsum board and gypsum panel product materials.

14B-25-2507 Lathing and plastering.

14B-25-2508 Gypsum construction.

14B-25-2509 Wet areas.

14B-25-2510 Lathing and furring for cement plaster (stucco).

14B-25-2511 Interior plaster.

14B-25-2512 Exterior plaster.

14B-25-2513 Exposed aggregate plaster.

14B-25-2514 Reinforced gypsum concrete.

14B-25-2500 Defined terms.

Where provisions of Chapter 25 of IBC are adopted by reference, the following terms shall be reset in italic type: "cement plaster"; "exterior surfaces"; "gypsum board"; "gypsum panel products"; "gypsum plaster"; "gypsum veneer plaster"; "interior surfaces" and "wire backing."

14B-25-2501 General.

The provisions of Section 2501 of IBC are adopted by reference with the following modification:

1. Revise Section 2501.2 to read:

"2501.2 Other materials.

Other wall or ceiling coverings may be *approved* by the *building official* and, if so, shall be installed in accordance with the recommendations of the manufacturer and the conditions of approval."

14B-25-2502 Performance.

The provisions of Section 2502 of IBC are adopted by reference with the following modification:

1. Revise Section 2502.1 by adding the following at the end:

"Where walls or ceiling coverings of gypsum materials will be permanently or periodically exposed to the exterior conditions, they must be designed for exterior exposure."

14B-25-2503 IBC Section 2503.

The provisions of Section 2503 of IBC are not adopted.

14B-25-2504 Vertical and horizontal assemblies.

The provisions of Section 2504 of IBC are adopted by reference without modification.

14B-25-2505 Shear wall construction.

The provisions of Section 2505 of IBC are adopted by reference without modification.

14B-25-2506 Gypsum board and gypsum panel product materials.

The provisions of Section 2506 of IBC are adopted by reference with the following modifications:

1. Revise Table 2506.2 by inserting a row to read:
"Abuse-resistant gypsum panel products | ASTM C1629"
2. Delete Section 2506.2.1.

14B-25-2507 Lathing and plastering.

The provisions of Section 2507 of IBC are adopted by reference without modification.

14B-25-2508 Gypsum construction.

The provisions of Section 2508 of IBC are adopted by reference with the following modifications:

1. Revise Section 2508.4 to read:

"2508.4 Adhesives.

Gypsum board and *gypsum panel products* secured to framing with adhesives in ceiling assemblies shall be attached using the fastening schedule recommended by the manufacturer. Expandable foam adhesives for fastening gypsum wallboard shall conform to ASTM D6464. Other adhesives for the installation of gypsum wallboard shall conform to ASTM C557."

2. Revise note b to Table 2508.6 to read:

"Values shall be reduced 50 percent in *Seismic Design Category D*."

14B-25-2509 Wet areas.

The provisions of Section 2509 of IBC are adopted by reference with the following modifications:

1. Revise Section 2509.1 to read:

"2509.1 Finish materials.

Shower areas, bathing areas and public toilet walls shall conform to Section 1209.2."

2. Revise Table 2509.2 by adding a row to read:

"Fiber-reinforced gypsum panels | ASTM C1278"

14B-25-2510 Lathing and furring for cement plaster (stucco).

The provisions of Section 2510 of IBC are adopted by reference with the following modifications:

1. Delete exception 2 to Section 2510.6.
2. Revise Section 2510.7 by deleting "*approved*."

14B-25-2511 Interior plaster.

The provisions of Section 2511 of IBC are adopted by reference with the following modifications:

1. Revise the exception to Section 2511.1 to read:

"Exception: *Gypsum veneer plaster* and *cement plaster* specifically designed and recommended by the manufacturer for one-coat applications."

2. Revise the first two sentences of Section 2511.4 to read:

"Condition of surfaces shall be as specified in Section 2510.7. Specially prepared *gypsum plaster* designed for application to concrete surfaces or acoustical plaster shall be permitted."

3. Revise Section 2511.5 by replacing "*an approved*" with "*a*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 87)

14B-25-2512 Exterior plaster.

The provisions of Section 2512 of IBC are adopted by reference with the following modifications:

1. Revise the first sentence of Section 2512.2 to read:

"Only plasticity agents in amounts recommended by the manufacturer shall be added to Portland cement or blended cements."

2. Add an exception to Section 2512.9 to read:

"Exception: Acrylic-based or siliconized acrylic-based finish coat systems manufactured for application as a finish coat over Portland cement based exterior stucco or plaster assemblies shall be allowed as a finish coat."

14B-25-2513 Exposed aggregate plaster.

The provisions of Section 2513 of IBC are adopted by reference with the following modification:

1. Revise the first sentence of Section 2513.3 to read:

"The bedding coat for *interior surfaces* or *exterior surfaces* shall be composed of one part Portland cement and one part Type S lime; or one part blended cement and one part Type S lime; or masonry cement; or plastic cement and not more than three parts of graded white or natural sand by volume."

14B-25-2514 Reinforced gypsum concrete.

The provisions of Section 2514 of IBC are adopted by reference without modification.

CHAPTER 14B-26

PLASTIC

14B-26-2601 General.

14B-26-2602 Finish and trim.

- 14B-26-2603 Foam plastic insulation.**
- 14B-26-2604 Interior finish and trim.**
- 14B-26-2605 Plastic veneer.**
- 14B-26-2606 Light-transmitting plastics.**
- 14B-26-2607 Light-transmitting plastic wall panels.**
- 14B-26-2608 Light-transmitting plastic glazing.**
- 14B-26-2609 Light-transmitting plastic roof panels.**
- 14B-26-2610 Light-transmitting plastic skylight glazing.**
- 14B-26-2611 Light-transmitting plastic interior signs.**
- 14B-26-2612 Plastic composites.**
- 14B-26-2613 Fiber-reinforced polymer.**
- 14B-26-2614 IBC Section 2614.**

14B-26-2601 General.

The provisions of Section 2601 of IBC are adopted by reference with the following modification:

1. Revise Section 2601.1 by replacing "foam plastic insulation" with "*foam plastic insulation* (including insulated panels with *foam plastic insulation*)."

14B-26-2602 Finish and trim.

The provisions of Section 2602 of IBC are adopted by reference without modification.

14B-26-2603 Foam plastic insulation.

The provisions of Section 2603 of IBC are adopted by reference with the following modifications:

1. In provisions adopted by reference, reset the following term in italic type: "foam plastic insulation."
2. Revise Section 2603.1 by replacing "foam plastic insulation" with "*foam plastic insulation* (including insulated panels with *foam plastic insulation*)."
3. Revise exception 2 to Section 2603.3 to read:

"2. In cold storage buildings, ice plants, food plants, food processing rooms and similar areas, *foam plastic insulation* where tested in a thickness of 4 inches (102 mm) shall be permitted in a thickness up to 10 inches (254 mm) where the *building* is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1."
4. Revise exception 4 to Section 2603.3 by replacing "approved" with "*approved*."
5. Delete exception 5 to Section 2603.3.
6. Revise Section 2603.4 by replacing "an approved" with "a."
7. Revise item 1 of Section 2603.4.1.5 to read:

"1. The *roof assembly* is separated from the interior of the *building* by wood structural panel sheathing not less than 0.47 inch (11.9 mm) in thickness bonded with exterior glue, with edges supported by blocking, tongue-and-groove joints or other *approved* type of edge support or material."
8. Delete Section 2603.4.1.6.
9. Delete Section 2603.4.1.8.
10. Revise the exception to Section 2603.4.1.9 to read:

"Exception: Garage doors using *foam plastic insulation* complying with Section 2603.3 in detached and attached *private garages* associated with *buildings* of Group R-5 occupancy need not be provided with a thermal barrier."
11. Delete Section 2603.4.1.12.
12. Revise Section 2603.4.1.14 by replacing "approved" with "*approved*."
- 12.1. Revise the third sentence of Section 2603.5 to read:

"*Exterior walls* of buildings of Type V construction shall comply with Sections 2603.2, 2603.3, 2603.4 and 2603.5.7."

13. Revise Section 2603.7 to read:

"2603.7 Foam plastic as interior finish or interior trim in plenum-served spaces.

Foam plastic used as interior wall or ceiling finish, or interior trim in plenum-served spaces, shall exhibit a flame spread index of 25 or less and a smoke-developed index of 50 or less when tested in accordance with ASTM E84 or UL 723 at the maximum thickness and density intended for use, and shall be tested in accordance with NFPA 286 and meet the acceptance criteria of Section 803.1.1. As an alternative to testing to NFPA 286, the foam plastic may be *approved* based on tests conducted in accordance with Section 2603.9.

Exceptions:

1. Foam plastic used as interior wall or ceiling finish or interior trim in plenum-served spaces shall exhibit a flame spread index of 75 or less and a smoke-developed index of 450 or less when tested in accordance with ASTM E84 or UL 723 at the maximum thickness and density intended for use, where it is separated from the airflow in the plenum by a thermal barrier complying with Section 2603.4.
2. Foam plastic used as interior wall or ceiling finish or interior trim in plenum-served spaces shall exhibit a flame spread index of 75 or less and a smoke-developed index of 450 or less when tested in accordance with ASTM E84 or UL 723 at the maximum thickness and density intended for use, where it is separated from the airflow in the plenum by corrosion-resistant steel having a base metal thickness of not less than 0.0160 inch (0.4 mm).
3. Foam plastic used as interior wall or ceiling finish or interior trim in plenum-served spaces shall exhibit a flame spread index of 75 or less and a smoke-developed index of 450 or less when tested in accordance with ASTM E84 or UL 723 at the maximum thickness and density intended for use, where it is separated from the airflow in the plenum by not less than a 1-inch (25 mm) thickness of masonry or concrete."

14. Delete Section 2603.8 and Figure 2603.8.

15. Revise Section 2603.9 by replacing "specifically approved" with "*approved*."

16. Revise Section 2603.11 by replacing "approved" with "*engineered*."

17. Revise exception 1 to Section 2603.11 by deleting "approved."

Revise Section 2603.12 to read:

"2603.12 Cladding attachment over foam sheathing to cold-formed steel framing.

Cladding shall be specified and installed in accordance with Chapter 14 and the cladding manufacturer's installation instructions, including any limitations for use over foam plastic sheathing, or an engineered design. Where used, furring and furring attachments shall be designed to resist design loads determined in accordance with Chapter 16. In addition, the cladding or furring attachments through foam sheathing to cold-formed steel framing shall meet or exceed the minimum fastening requirements of Sections 2603.12.1 and 2603.12.2, or an engineered design for support of cladding weight.

Exceptions:

1. Where the cladding manufacturer has provided installation instructions for application over foam sheathing, those requirements shall apply.
2. For exterior insulation and finish systems, refer to Section 1407.
3. For anchored masonry or stone veneer installed over foam sheathing, refer to Section 1404."

18. Revise note c to Table 2603.12.2 by replacing "approved" with "*engineered*."

19. Revise Section 2603.13 to read:

"2603.13 Cladding attachment over foam sheathing to wood framing.

Cladding shall be specified and installed in accordance with Chapter 14 and the cladding manufacturer's installation instructions. Where used, furring and furring attachments shall be designed to resist design loads determined in accordance with Chapter 16. In addition, the cladding or furring attachments through foam sheathing to framing shall meet or exceed the minimum fastening requirements of Section 2603.13.1 or 2603.13.2, or an engineered design for support of cladding weight.

Exceptions:

1. Where the cladding manufacturer has provided installation instructions for application over foam sheathing, those requirements shall apply.
2. For *exterior insulation and finish systems*, refer to Section 1407.
3. For *anchored masonry veneer* or anchored stone veneer installed over foam sheathing, refer to Section 1404."

20. Revise note c to Table 2603.13.2 by replacing "approved" with "*engineered*."

14B-26-2604 Interior finish and trim.

The provisions of Section 2604 of IBC are adopted by reference with the following modification:

1. Revise Section 2604.1 by deleting "where approved."

14B-26-2605 Plastic veneer.

The provisions of Section 2605 of IBC are adopted by reference with the following modification:

1. Revise item 2 of Section 2605.2 by replacing "50 feet (15 240 mm)" with "40 feet (12 190 mm)" and "grade" with "*grade plane*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 89)

14B-26-2606 Light-transmitting plastics.

The provisions of Section 2606 of IBC are adopted by reference with the following modifications:

1. In provisions adopted by reference, reset the following terms in italic type: "approved"; "greenhouse" and "light- diffusing system."
2. Revise items 1 and 2 of Section 2606.7 to read:
 - "1. Group A with an *occupant load* of 300 or more.
 2. Group E."
3. Revise item 5 of Section 2606.7 to read:

"5. *Interior exit stairways, interior exit ramps and exit passageways.*

4. Revise Section 2606.7.4 to read:

"2606.7.4 Fire suppression system.

In *buildings* that are equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1, *light-diffusing systems* shall be protected both above and below unless the *automatic sprinkler system* has been *approved* for installation only above the *light-diffusing system*. Areas of *light-diffusing systems* that are protected in accordance with this section shall not be limited."

5. Revise Section 2606.7.5 by deleting "approved."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 90)

14B-26-2607 Light-transmitting plastic wall panels.

The provisions of Section 2607 of IBC are adopted by reference with the following modifications:

1. In provisions adopted by reference, reset the following terms in italic type: "greenhouse" and "light-transmitting plastic wall panels."
2. Revise Section 2607.1 to read:

"2607.1 General.

Light-transmitting plastics shall not be used as wall panels in *exterior walls* in occupancies in Groups A-1, A-2, A-3, A-4, E, H, I-2 and I-3. In other groups, light-transmitting plastics shall be permitted to be used as wall panels in *exterior walls*, provided that the walls are not required to have a *fire-resistance rating* and the installation conforms to the requirements of this section. Such panels shall be erected and anchored on a foundation, waterproofed or otherwise protected from moisture absorption and sealed with a coat of mastic or other *approved* waterproof coating. *Light-transmitting plastic wall panels* shall comply with Section 2606."

3. Revise Section 2607.3 to read:

"2607.3 Height limitation.

Light-transmitting plastics shall not be installed more than 40 feet (12 190 mm) above *grade plane*."

4. Revise the exceptions to Section 2607.4 by deleting "approved."

5. Revise Section 2607.5 to read:

"2607.5 Automatic sprinkler system.

Where the *building* is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1, the maximum percentage area of *exterior wall* in any *story* in *light-transmitting plastic wall panels* and the maximum square footage of a single area given in Table 2607.4 shall be increased 100 percent, but the area of *light-transmitting plastic wall panels* shall not exceed 50 percent of the *exterior wall* area in any *story*, or the area permitted by Section 705.8 for unprotected openings, whichever is

smaller."

14B-26-2608 Light-transmitting plastic glazing.

The provisions of Section 2608 of IBC are adopted by reference with the following modifications:

1. Revise item 2 to section 2608.2, excluding its exception, to read:
 - "2. Flame barriers extending 30 inches (762 mm) beyond the *exterior wall* in the plane of the floor, or vertical panels not less than 4 feet (1219 mm) in height, shall be installed between glazed units located in adjacent *stories*."
2. Revise item 3 to section 2608.2, including its exception, to read:
 - "3. Light-transmitting plastics shall not be installed more than 40 feet (12 190 mm) above *grade plane*."

14B-26-2609 Light-transmitting plastic roof panels.

The provisions of Section 2609 of IBC are adopted by reference with the following modifications:

1. In provisions adopted by reference, reset the following terms in italic type: "fire-resistance rating" and "light-transmitting plastic roof panels."
2. Revise exception 3 to Section 2609.4 to read:
 - "3. *Greenhouses* that are occupied for growing or maintaining plants, without public access, shall be exempt from the area limitations of Table 2609.4 provided that they have a minimum fire separation distance of 10 feet (3048 mm)."
3. Delete exception 4 to Section 2609.4.

14B-26-2610 Light-transmitting plastic skylight glazing.

The provisions of Section 2610 of IBC are adopted by reference with the following modifications:

1. In provisions adopted by reference, reset the following terms in italic type: "approved"; "light-transmitting plastic roof panels"; "roof covering" and "unit skylight."
2. Revise exception 1 to Section 2610.2 to read:
 - "1. Curbs shall not be required for skylights used on roofs having a minimum slope of three units vertical in 12 units horizontal (25-percent slope) in occupancies in Group R-3 or R-5 and on buildings with a nonclassified roof covering."
3. Revise the exception to Section 2610.4 to read:

"Exception: The area limitation shall not apply where the *building* is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1."
4. Revise the exception to Section 2610.5 to read:

"Exception: The aggregate area limitations of light-transmitting plastic skylights shall be increased 100 percent beyond the limitations set forth in this section where the *building* is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1."
5. Revise exception 2 to Section 2610.6 to read:
 - "2. In Group R-3 or R-5, multiple skylights located above the same room or space with a combined area not exceeding the limits set forth in Section 2610.4."

14B-26-2611 Light-transmitting plastic interior signs.

The provisions of Section 2611 of IBC are adopted by reference with the following modification:

1. Delete the exception to Section 2611.1.

14B-26-2612 Plastic composites.

The provisions of Section 2612 of IBC are adopted by reference with the following modification:

1. Delete Section 2612.4.

14B-26-2613 Fiber-reinforced polymer.

The provisions of Section 2613 of IBC are adopted by reference with the following modification:

1. In provisions adopted by reference, reset the following term in italic type: "approved."
2. Revise Section 2613.5, exception 2.2 by replacing "5 feet (1524 mm)" with "15 feet (4572 mm)."

14B-26-2614 IBC Section 2614.

The provisions of Section 2614 of IBC are not adopted.

CHAPTER 14B-27

ELECTRICAL

14B-27-2701 General.

14B-27-2702 Emergency and standby power systems.

14B-27-2701 General.

The provisions of Section 2701 of IBC are adopted by reference without modification.

14B-27-2702 Emergency and standby power systems.

The provisions of Section 2702 of IBC are adopted by reference with the following modifications:

1. In provisions adopted by reference, reset the following terms in italic type: "ambulatory care facility"; "emergency power system"; "high-rise building" and "standby power system."

2. Revise Section 2702.1.2 to read:

"2702.1.3 Installation.

Emergency power systems and standby power systems required by this code, the *Chicago Electrical Code* or the *Chicago Fire Prevention Code* shall be installed in accordance with the *Chicago Electrical Code*."

3. Revise Sections 2702.1.4 and 2702.1.5 by replacing "in this code" with "in this code or the *Chicago Electrical Code*."

4. Delete Section 2702.1.8.

5. Revise Section 2702.2 to read:

"2702.2 Where required.

Emergency power systems and standby power systems shall be provided where required by Sections 2702.2.1 through 2702.2.18 or Article 700 of the *Chicago Electrical Code*."

6. Revise Section 2702.2.2 to read:

"2702.2.2 Elevators and platform lifts.

Standby power shall be provided for elevators and platform lifts as required in Sections 1009.4.1, 1009.5, 3003.1 and 3007.8."

7. Delete Section 2702.2.3.

8. Revise Section 2702.2.4 to read:

"2702.2.4 Voice communication systems.

Emergency power shall be provided for voice communication systems as required in Section 907.2.12.2. The system shall be capable of powering the required load for a duration of not less than 24 hours, as required in NFPA 72."

9. Delete Section 2702.2.5.

10. Delete Section 2702.2.12.

11. Revise Section 2702.2.13 by replacing "90 minutes" with "required by the *Chicago Electrical Code*."

12. Revise Section 2702.2.14 by deleting the last sentence.

12.1. Revise Section 2702.15 by deleting "as required in Section 415.11.10."

13. Delete Section 2702.2.17.

14. Delete Section 2702.3.

15. Delete Section 2702.4.

CHAPTER 14B-28

MECHANICAL SYSTEMS

14B-28-2801 General.

14B-28-2801 General.

The provisions of Section 2801 of IBC are adopted by reference without modification.

CHAPTER 14B-29

PLUMBING SYSTEMS

14B-29-2900 Chapter 29.

14B-29-2900 Chapter 29.

The provisions of Chapter 29 of IBC are not adopted. The following language is adopted as Chapter 29:

"2901 GENERAL

2901.1 Scope.

The provisions of the *Chicago Plumbing Code* shall govern the design, construction, erection and installation of plumbing components, appliances, equipment and systems used in *buildings* and structures covered by this code. Toilet and bathing rooms shall be constructed in accordance with Section 1209. The *Chicago Fire Prevention Code*, the *Chicago Minimum Requirements for Existing Buildings* and the *Chicago Plumbing Code* shall govern the use and maintenance of plumbing components, appliances, equipment and systems. The *Chicago Building Rehabilitation Code* and the *Chicago Plumbing Code* shall govern the *alteration, repair, relocation, replacement and addition* of plumbing components, *appliances, equipment* and systems."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 92)

CHAPTER 14B-30

CONVEYANCE DEVICES

14B-30-3001 General.

14B-30-3002 Hoistway enclosures.

14B-30-3003 Elevator emergency operations.

14B-30-3004 Other conveyance devices.

14B-30-3005 Machine rooms.

14B-30-3006 IBC Section 3006.

14B-30-3007 Fire service access elevator.

14B-30-3008 IBC Section 3008.

14B-30-3001 General.

The provisions of Section 3001 of IBC are adopted by reference with the following modifications:

1. Revise Section 3001.1 to read:

"3001.1 Scope.

This chapter and the *Chicago Conveyance Device Code* govern the design, construction, installation, *alteration* and *repair* of *conveyance devices* and their components."

2. Delete Section 3001.2.

3. Revise Section 3001.3 to read:

"3001.3 Applicable standards.

The design, construction, installation, *alteration*, *repair* and maintenance of *conveyance devices* and their components shall conform to the *Chicago Conveyance Device Code*."

4. Delete Table 3001.3.

5. Revise Section 3001.5 to read:

"3001.5 Change in use.

A change in use of an elevator from freight to passenger, passenger to freight, or from one freight class to another freight class shall comply with Section 8.7 of Chapter 3 of the *Chicago Conveyance Device Code*."

14B-30-3002 Hoistway enclosures.

The provisions of Section 3002 of IBC are adopted by reference with the following modifications:

1. Revise Section 3002.1.2 to read:

"3002.1.2 Hardware.

Hardware on opening protectives shall be installed as tested, except that interlocks, mechanical locks and electric contacts, door and gate electric contacts and door-operating mechanisms shall be exempt from the fire test requirements."

2. Revise Section 3002.3, including its exception, to read:

"3002.3 Emergency signs.

A pictorial sign shall be posted adjacent to each elevator call station on all floors instructing occupants to use the *exit stairways* and not to use the elevators in case of fire."

3. Revise Section 3002.4 to read:

"3002.4 Elevator car to accommodate ambulance stretcher in new buildings.

Where elevators are provided in new *buildings* and *additions* with six or more *stories above grade plane* or two or more *basements*, not less than one elevator shall be provided for fire department emergency access to all *stories*. The elevator car shall be of such a size and arrangement to accommodate an ambulance stretcher 24 inches by 84 inches (610 mm by 2134 mm) with not less than 5-inch (127 mm) radius corners, in the horizontal, open position and shall be identified by the international symbol for emergency medical services (star of life). The symbol shall be not less than 3 inches (76 mm) in height and shall be placed inside on both sides of the hoistway door frame."

4. Delete Section 3002.5.

5. Delete the exception to Section 3002.7.

6. Revise Section 3002.9 to read:

"3002.9 Plumbing, gas and mechanical systems.

Plumbing, fuel gas and mechanical systems, other than elevator equipment, shall not be located in an elevator hoistway enclosure.

Exceptions:

1. Floor drains, sumps and sump pumps shall be permitted at the base of the hoistway enclosure provided that they are indirectly connected to the plumbing system.
2. Mechanical systems associated with hoistway pressurization or cooling."

7. Add new Sections 3002.10 and 3002.11 to read:

"3002.10 Electrical, fire protection and communications wiring.

Electrical, fire protection and communications wiring not essential to the elevator shall not be located in an elevator hoistway enclosure.

3002.11 Sprinkler locations.

Automatic sprinklers shall not be installed in elevator machine rooms, elevator machinery spaces, control rooms, control spaces and elevator hoistways, except sprinklers shall be installed to protect the bottom of the pit in newly-installed sprinkler systems.

Exception: Sprinkler locations required by state or federal law."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 93)

14B-30-3003 Elevator emergency operations.

The provisions of Section 3003 of IBC are adopted by reference with the following modifications:

1. Revise section 3003.2 to read:

"3003.2 Fire fighters' emergency operation.

Elevators shall be provided with Phase I emergency recall operation and Phase II emergency in-car operation in accordance with Chapters 3 and 4 of the *Chicago Conveyance Device Code*."

2. Revise section 3003.3 to read:

"3003.3 Standardized fire service elevator keys.

All elevators shall be equipped to operate with a standardized fire service elevator key in accordance with the *Chicago Conveyance Device Code*."

14B-30-3004 Other conveyance devices.

The provisions of Section 3004 of IBC are adopted by reference with the following modifications:

1. Revise Section 3004.1 to read:

"3004.1 General.

Conveyance devices, other than elevators, shall comply with the *Chicago Conveyance Device Code* and the provisions of Sections 3004.2 through 3004.4."

2. Delete the word "*approved*" in Section 3004.2.

3. Revise Section 3004.2.1 to read:

"3004.2.1 Floor openings.

Escalator floor openings shall comply with Section 712."

4. Revise Section 3004.3 to read:

"3004.3 Conveyors.

Conveyors and conveying systems shall comply with Chapter 10 of the *Chicago Conveyance Device Code*."

5. Revise Section 3004.3.1 to read:

"3004.3.1 Enclosure.

Conveyors and related equipment connecting successive floors or levels, other than on construction sites, shall be enclosed with *shaft enclosures* complying with Section 713."

6. Revise Section 3004.4 by replacing "an *approved*" with "a."

14B-30-3005 Machine rooms.

The provisions of Section 3005 of IBC are adopted by reference with the following modifications:

1. Revise Section 3005.1 by replacing "An *approved*" with "Acceptable."
2. Revise Section 3005.4 by replacing "*fire-protection rating*" with "*fire-resistance rating*."
3. Revise the exceptions to Section 3005.4 to read:

"Exceptions:

1. For other than fire service access elevators, where machine rooms, machinery spaces, control rooms and control spaces do not abut and do not have openings to the hoistway enclosure they serve, the *fire barriers* constructed in accordance with Section 707 or *horizontal assemblies* constructed in accordance with Section 711, or both, shall be permitted to be reduced to a 1-hour *fire-resistance rating*.
2. For other than fire service access elevators, in *buildings* four *stories* or less above *grade plane* equipped with an *automatic sprinkler system* where machine room, machinery spaces, control rooms and control spaces do not abut and do not have openings to the hoistway enclosure they serve, the machine room, machinery spaces, control rooms and control spaces are not required to be fire- resistance rated."

14B-30-3006 IBC Section 3006.

The provisions of Section 3006 of IBC are not adopted.

14B-30-3007 Fire service access elevator.

The provisions of Section 3007 of IBC are adopted by reference with the following modifications:

1. Revise Section 3007.1, including its exception, to read:

"3007.1 General.

Where required by Section 403.6.1, every area of every story of the *building* shall be served by a fire service access elevator complying with Sections 3007.1, 3007.5, 3007.6.5, 3007.7, and 3007.8. Except as modified in this section, fire service access elevators shall be installed in accordance with this chapter and the *Chicago Conveyance Device Code*."

2. Delete Section 3007.2 and its subsections.
3. Delete Section 3007.3.
4. Delete Section 3007.4.
5. Delete Section 3007.5.2.
6. Delete Section 3007.6 and its subsections except 3007.6.5.
7. Revise the first sentence of Section 3007.6.5 to read:

"A pictorial symbol of a standardized design designating which elevators are fire service access elevators shall be installed on each side of the hoistway door frame."

- 7.1. Delete Section 3007.7.
8. Delete item 2 in Section 3007.8.
9. Delete Section 3007.9 and its subsections.

(Amend Coun. J. 1-24-24, p. 8634, Art. IV, § 6)

14B-30-3008 IBC Section 3008.

The provisions of Section 3008 of IBC are not adopted.

CHAPTER 14B-31

SPECIAL CONSTRUCTION

14B-31-3101 General.

14B-31-3102 Membrane structures.

14B-31-3103 Temporary structures.

14B-31-3104 Pedestrian walkways and tunnels.

14B-31-3105 Awnings and canopies.

14B-31-3106 Marquees.

14B-31-3107 IBC Section 3107.

14B-31-3108 Telecommunication and broadcast towers.

14B-31-3109 Swimming pools, spas and hot tubs.

14B-31-3110 Automatic vehicular gates.

14B-31-3111 Solar energy systems.

14B-31-3112 Greenhouses.

14B-31-3113 IBC Section 3113.

14B-31-3114 Fences.

14B-31-3101 General.

The provisions of Section 3101 of IBC are adopted by reference with the following modification:

1. Revise Section 3101.1 to read:

"3101.1 Scope.

The provisions of this chapter shall govern special construction including membrane structures, temporary structures, *pedestrian walkways* and tunnels, automatic *vehicular gates*, *awnings* and *canopies*, *marquees*, towers, antennas swimming pool enclosures and safety devices, solar energy systems, *greenhouses* and fences."

14B-31-3102 Membrane structures.

The provisions of Section 3102 of IBC are adopted by reference with the following modifications:

1. Revise Section 3102.1 by replacing "the *International Fire Code*" with "Section 3103."
2. Revise Section 3102.3, excluding its exception, to read:

"3102.3 Type of construction.

Noncombustible membrane structures shall be classified as Type IIB construction. Noncombustible frame or cable-supported structures covered by a membrane in accordance with Section 3102.3.1 shall be classified as Type IIB construction. Other membrane structures shall be classified as Type VB construction."

3. Revise Section 3102.8.1.1 to read:

"3102.8.1.1 Auxiliary inflation system.

In addition to the primary inflation system an auxiliary inflation system shall be provided with sufficient capacity to maintain the inflation of the *structure* in case of primary system failure. The auxiliary inflation system shall operate automatically when there is a loss of internal pressure and when the primary blower system becomes inoperative."

4. Revise item 5 of Section 3102.8.1.2 to read:

"5. Blower inlets shall be located to provide protection from air contamination. The location of inlets shall be shown on the *construction documents*."

14B-31-3103 Temporary Structures.

The provisions of Section 3103 of IBC are not adopted. The following is adopted as Section 3103:

"3103.1 General.

The provisions of Sections 3103.1 through 3103.3 shall apply to *temporary structures*. *Structures* erected on the same *site* for more than 180 days within a 12-month period shall comply with provisions of this code applicable to permanent *structures*.

Exceptions:

1. *Temporary structures* associated with ongoing construction activity shall not be subject to the 180-day limit.
2. Membrane structures that are erected on *occupiable rooftops*, *exterior balconies*, *decks* or similar *structures* for any period of time shall be regulated as permanent membrane structures in accordance with Section 3102 unless otherwise *approved* by the *building official*.

3103.1.1 Conformance.

Temporary structures and uses shall conform to the structural strength, fire safety, *means of egress*, accessibility, light, ventilation and sanitary requirements of this code as necessary to ensure public health, safety and general welfare except as modified by this section.

3103.1.2 Fire protection.

Temporary structures and uses shall be protected from fire hazards in accordance with Sections 3103.1.2.1 and 3103.1.2.2.

3103.1.2.1 Portable fire extinguishers.

Temporary structures exceeding 600 square feet (56 m²) in floor area or where cooking will occur within 20 feet (6096 mm) of the *structure* shall be provided with portable fire extinguishers in accordance with Section 905, and no less than one.

3103.1.2.2 Emergency lighting.

Temporary structures exceeding 1,000 square feet (93 m²) in floor area that will be used for a period longer than 48 hours shall be provided with an emergency lighting system complying with Article 700 of the *Chicago Electrical Code*.

3103.1.3 Means of egress.

Temporary structures shall be provided with *means of egress* in accordance with Chapter 10. For *temporary structures* exceeding 600 square feet (56 m²) in floor area, a dimensioned *exit* plan, showing aisles, seating and other furnishings, shall be provided as part of the *permit* application. *Temporary structures* exceeding 1,000 square feet (93 m²) in floor area shall be provided with illuminated exit signs.

3103.1.4 Structural requirements.

Temporary structures shall not be required to be designed for seismic forces. *Temporary structures* installed for 6 weeks or less shall be designed to withstand at least 56 percent of the wind loads applicable to permanent structures. *Temporary structures* installed for more than 6 weeks shall be designed to withstand at least 64 percent of the wind loads applicable to permanent structures. *Temporary structures* shall be allowed to be designed for reduced environmental forces, such as snow loads and ice loads, based on limited seasonal exposure. Where *temporary structures* do not meet all structural requirements applicable to permanent structures, the *building official* may issue a conditional permit requiring monitoring of weather conditions and allowing the *structure* to be used up to a specified wind velocity or under other applicable climatic limitations.

3103.1.5 Sanitary requirements.

Temporary structures and uses shall be provided with *temporary* sanitary facilities in accordance with the provisions of the *Chicago Plumbing Code*.

3103.1.6 Electrical equipment and wiring.

Electrical equipment and wiring shall comply with the *Chicago Electrical Code*.

3103.2 Temporary tents and membrane structures.

Temporary tents and other membrane structures shall comply with Sections 3103.2.1 through 3103.2.4.

3103.2.1 Location.

Temporary tents and membrane structures shall not be erected within 12 feet (3658 mm) of *existing buildings* or *property lines*.

Exceptions:

1. A *temporary* covered passageway, no more than 8 feet (2438 mm) wide, with open or enclosed sides, may be provided to connect *temporary structures* to *existing buildings*.
2. *Temporary structures* need not be separated from *existing buildings* where the aggregate floor area of *temporary structures* does not exceed 5,000 square feet (465 m²) and the *temporary structure* will not impede required *means of egress* for the *existing building*.

3103.2.2 Flame propagation.

The *permit* application shall be accompanied by a certificate of fire resistance executed by an *approved agency* certifying that the *structure* meets the flame propagation performance requirements of NFPA 701. If the treatment indicated on the certificate is not of the permanent type, the certificate shall be dated within 360 days preceding the last date of intended use. If the tent fabric is certified as having been rendered permanently fire resistant during manufacture, or is made of an inherently fire resistant fabric, then the certificate shall be dated no earlier than seven years prior to the last date of intended use. The certificate shall indicate the type of fabric and method of compliance with NFPA 701.

3103.2.2.1 Label.

Membrane structures and their components shall have a permanently affixed label bearing the identification of size and fabric or material type.

3103.2.3 Certification.

The *permit* application shall be accompanied by a certificate providing all of the following information relative to the flame propagation performance criteria of the fabric:

1. The name and address for the *owner* of the *structure*.
2. Date the fabric was last treated with flame-retardant solution, if applicable.
3. Trade name or kind of chemical used in treatment.
4. Name of person or firm treating the material.
5. Name of testing agency and test standard by which the fabric was tested.

A copy of the *certificate* must be kept on *site* and made available to the *building official* or *fire code official*.

3103.2.4 Liquefied petroleum gas (LP gas).

Liquefied petroleum gas (LP gas) tanks shall not be permitted inside of or within 10 feet (3048 mm) of *temporary* tents and membrane structures.

3103.2.5 Air-supported and air-inflated structures.

Temporary air-supported structures and air-inflated structures shall comply with Sections 3103.2.5.1 through 3103.2.5.3.

3103.2.5.1 Design.

The design and construction of the fabric envelope and the method of anchoring shall be in accordance with Architectural Fabric Structures Institute FSAAS.

3103.2.5.2 Blowers.

An *air-supported structure* containing *occupiable space* shall be furnished with not less than two blowers, each of which has adequate capacity to maintain full inflation pressure with normal leakage. The design of the blower shall be so as to provide integral limiting pressure at the design pressure specified by the manufacturer.

3103.2.5.3 Inflation system.

Temporary structures with an *occupant load* greater than 100 shall also comply with Section 3102.8.

3103.3 Temporary structures other than tents and membrane structures.

Temporary structures, other than tents and other membrane structures, shall comply with Sections 3103.3.1 and 3103.2.

3103.3.1 Location.

Temporary structures of combustible construction shall not be erected within 12 feet (3658 mm) of *existing buildings* or *property lines*.

Exceptions:

1. A *temporary* covered passageway, no more than 8 feet (2438 mm) wide, with open or enclosed sides, may be provided to connect *temporary structures* to *existing buildings*.
2. *Temporary structures* need not be separated from *existing buildings* where the aggregate floor area of *temporary structures* does not exceed 5,000 square feet (465 m²) and the *temporary structure* will not impede required *means of egress* for the *existing building*.

3103.3.2 Liquefied petroleum gas (LP gas).

Liquefied petroleum gas (LP gas) tanks shall not be permitted inside of or within 10 feet (3048 mm) of *temporary structures* of combustible construction."

(Amend Coun. J. 7-24-19, p. 3646, § 34; Amend Coun. J. 2-19-20, p. 14473, Art. II, § 94; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 26)

14B-31-3104 Pedestrian walkways and tunnels.

The provisions of Section 3104 of IBC are adopted by reference with the following modifications:

1. Delete the exceptions to Section 3104.3.
2. Revise Section 3104.4 to read:

"3104.4 Contents.

No furnishings or decorations shall be located in the *pedestrian walkway* unless *approved by the fire code official*."

3. Delete the exception to Section 3104.5.3.

14B-31-3105 Awnings and canopies.

The provisions of Section 3105 of IBC are adopted by reference with the following modifications:

1. Add a new Section 3105.1.1 to read:

"3105.1.1 Awnings and canopies over the public way.

Awnings and *canopies* installed over the *public way* shall also comply with Chapter 32 of this code and Chapter 10-28 of the *Municipal Code*."

2. Revise the first sentence of Section 3105.3 to read:

"*Awnings* and *canopies* shall be provided with a covering that complies with one of the following:"

3. Delete the exception to Section 3105.3.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 95)

14B-31-3106 Marquees.

The provisions of Section 3106 of IBC are adopted by reference with the following modifications:

1. In provisions adopted by reference, reset the following term in italic type: "marquee."
2. Add a new Section 3106.1.1 to read:

"3106.1.1 Marquees over the public way.

Marquees installed over the *public way* shall also comply with Chapter 32 and Chapter 10-28 of the *Municipal Code*."

14B-31-3107 IBC Section 3107.

The provisions of Section 3107 of IBC are not adopted.

14B-31-3108 Telecommunication and broadcast towers.

The provisions of Section 3108 of IBC are adopted by reference without modification.

14B-31-3109 Swimming pools, spas and hot tubs.

The provisions of Section 3109 of IBC are adopted by reference with the following modification:

1. Revise Section 3109.1 by replacing "*International Swimming Pool and Spa Code*" with "*Chicago Plumbing Code*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 96)

14B-31-3110 Automatic vehicular gates.

The provisions of Section 3110 of IBC are adopted by reference without modification.

14B-31-3111 Solar energy systems.

The provisions of Section 3111 of IBC are adopted by reference with the following modification:

1. Revise Section 3111.3.5.1 by replacing "local jurisdiction" with "*fire code official*."

14B-31-3112 Greenhouses.

The provisions of Section 3112 of IBC are adopted by reference without modification.

14B-31-3113 IBC Section 3113.

The provisions of Section 3113 of IBC are not adopted.

14B-31-3114 Fences.

The following language is adopted as a new Section 3114:

"3114. FENCES

3114.1 General.

Fences shall comply with the provisions of Sections 3114.2 through 3114.4.

3114.2 Classification.

Fences shall be classified according to the general form of their construction as screen *fences* or solid *fences* and according to their combustibility in accordance with Sections 3114.2.1 through 3114.2.3.

3114.2.1 Screen fences.

A screen *fence* is a *fence* so constructed that at least 50 percent of the superficial area thereof consists of regularly distributed apertures.

3114.2.2 Solid fences.

A solid *fence* is a *fence* so constructed that less than 50 percent of the superficial area thereof consists of regularly distributed apertures.

3114.2.3 Combustible fences.

Any *fence* not constructed entirely of noncombustible materials shall be classed as a combustible *fence*.

3114.3 Height.

The height of *fences* shall not exceed the values in Table 3114.3.

TABLE 3114.3

MAXIMUM HEIGHT OF FENCES

MATERIAL	CLASSIFICATION	MAXIMUM HEIGHT (feet)	NOTES
MATERIAL	CLASSIFICATION	MAXIMUM HEIGHT (feet)	NOTES
Combustible	Screen <i>fence</i>	8	All
	Solid <i>fence</i>	6	Horizontal distance from <i>fence</i> to <i>public way</i> less than or equal to height of <i>fence</i>
		8	Horizontal distance from <i>fence</i> to <i>public way</i> exceeds height of <i>fence</i>
Noncombustible	Screen <i>fence</i>	10	All others
		15	Enclosing a Group A-5 occupancy; enclosing a <i>yard</i> accessory to a Group B, F, I, M or S occupancy
	Solid <i>fence</i>	8	All others
		12	Enclosing a Group A-5 occupancy; enclosing a <i>yard</i> accessory to a Group B, F, I, M or S occupancy

For SI: 1 foot = 304.8 mm.

3114.4 Wind load.

Fences shall be designed and constructed to resist a horizontal wind pressure of not less than 30 pounds per square foot (1436 kN/m²) in addition to all other forces to which they may be subjected.

3114.5 Barbed wire and razor wire.

The use of barbed wire or razor wire at a height less than 8 feet (2438 mm) above the ground shall be prohibited.

Exception: Where wholly located on or over private property, the use of barbed wire or razor wire more than 7 feet (2134 mm) above the ground shall be allowed."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 97)

CHAPTER 14B-32

ENCROACHMENTS INTO THE PUBLIC WAY

14B-32-3201 General.

14B-32-3202 Encroachments.

14B-32-3201 General.

The provisions of Section 3201 of IBC are adopted by reference with the following modifications:

1. Revise Section 3201.1 by replacing "public right-of-way" with "*public way*."
2. Revise Section 3201.2 by replacing "*lot line*" with "*property line*."
3. Revise Section 3201.4 to read:

"3201.4 Drainage.

Drainage water collected from a roof, *awning*, *exterior balcony*, *canopy* or *marquee*, and condensate from mechanical equipment shall not be directed to flow onto or over a public sidewalk."

14B-32-3202 Encroachments.

The provisions of Section 3202 of IBC are adopted by reference with the following modifications:

1. Revise Section 3202.1.1 to read:

"3202.1.1 Structural support.

A part of a *building* erected below grade that is necessary for structural support of the *building* or *structure* shall not project beyond the *property lines*, except as provided in Sections 3202.1.1.1 through 3202.1.1.4."

2. Add new Sections 3202.1.1.1 through 3202.1.1.4 to read:

"3202.1.1.1 Foundations.

The portions of foundations above a level 20 feet (6096 mm) below *city datum* may project into a *public way* four and one-half inches (114 mm) per foot (305 mm) of depth below sidewalk or alley grade but not more than 36 inches (914 mm).

3202.1.1.2 Centerline of public way.

Except as provided in Section 3202.1.1.4, in no case shall foundations extend within five feet (1524 mm) of the centerline of any *public way*.

3202.1.1.3 No usable floor area.

Except where sub-sidewalk space is permitted, no foundation, or any part thereof, shall project into a *public way* in such manner as to add to the floor area of any *building* or *structure*.

3202.1.1.4 Deep foundations.

Portions of foundations, constructed lower than 20 feet (6096 mm) below *city datum*, may project into a *public way* such distance as the *building official* may approve as necessary for the stability of the *building* or *structure* of which they are a part."

3. Revise Section 3202.1.2 to read:

"3202.1.2 Vaults and other enclosed spaces.

The construction and utilization of vaults and other enclosed spaces below grade shall be subject to Chapter 10-28 of the *Municipal Code*."

4. Revise Section 3202.2, including its subsections, to read:

"3202.2 Encroachments above grade and below 12 feet in height.

Encroachments into the *public way* above grade and below 12 feet (3658 mm) in clear height shall be prohibited. Doors and windows shall not open or project into the *public way*."

5. Revise Section 3202.3, including its subsections, to read:

"3202.3 Encroachments 12 feet or more above grade.

Encroachments 12 feet (3658 mm) or more above grade shall comply with Sections 3202.3.1 through 3202.3.4.

3202.3.1 Awnings, canopies, marquees and signs.

Awnings, canopies, marquees and signs shall be constructed so as to support applicable loads as specified in Chapter 16. *Awnings, canopies, marquees* and signs shall comply with Chapters 10-24 and 10-28 of the *Municipal Code*.

3202.3.2 Cornices, belt courses and similar projections.

Where the vertical clearance above grade to projecting cornices, rustications, quoins, moldings, belt courses, lintel, sills, oriel windows, pediments and similar projections of a decorative character is more than 12 feet (3658 mm), the maximum encroachment shall be 2 feet (610 mm) and shall not exceed 5 percent of the wall area.

3202.3.3 Fire escapes.

Fire escapes hereafter erected shall not project over a *public way* except when erected to replace an existing required fire escape which projects over the *public way*.

3202.3.4 Pedestrian walkways.

The installation of a pedestrian walkway over a *public way* shall be subject to Chapter 10-28 of the *Municipal Code*."

6. Revise Section 3202.4 to read:

"3202.4 Temporary vestibules.

Temporary vestibules shall not be erected for a period of time exceeding 6 months in any 1 year and shall not encroach more than 3 feet (914 mm) nor more than one-fourth of the width of the sidewalk beyond the *property line*. Such enclosures shall comply with Chapter 11 and shall not contain seating."

CHAPTER 14B-33

WORK SITE SAFETY AND OPERATIONS

- 14B-33-3301 General.
- 14B-33-3302 Construction safeguards.
- 14B-33-3303 Demolition.
- 14B-33-3304 Site work.
- 14B-33-3305 Sanitary.
- 14B-33-3306 Protection of pedestrians.
- 14B-33-3307 Protection of adjoining property.
- 14B-33-3308 Temporary use of streets, alleys and public property.
- 14B-33-3309 Fire extinguishers.
- 14B-33-3310 Means of egress.
- 14B-33-3311 Standpipes.
- 14B-33-3312 Automatic sprinkler system.
- 14B-33-3313 IBC Section 3313.
- 14B-33-3314 Owner's responsibility for fire protection.
- 14B-33-3315 Temporary heating equipment.
- 14B-33-3316 Precautions against fire.
- 14B-33-3317 Flammable and combustible liquids.
- 14B-33-3318 Flammable gases.
- 14B-33-3319 Explosive materials.
- 14B-33-3320 Fire reporting.
- 14B-33-3321 Motorized construction equipment.
- 14B-33-3322 Safeguarding roofing operations.
- 14B-33-3323 Scaffolding.
- 14B-33-3324 Construction site cleanliness.

14B-33-3301 General.

The provisions of Section 3301 of IBC are adopted by reference with the following modifications:

1. Revise Section 3301.1 to read:

"3301.1 Scope.

The provisions of this chapter shall govern safety during construction, demolition and *rehabilitation* work and the protection of adjacent public and private properties. Provisions of this chapter pertaining to dust control shall be enforced by the Chief Sustainability Officer in accordance with Chapter 11-4 of the *Municipal Code*."

2. Add a new Section 3301.3 to read:

"3301.3 Delivery and removal.

Delivery and removal of equipment and materials from the construction site shall be done in a manner that prevents injury or damage to persons, adjoining properties and the *public way*."

3. Add a new Section 3301.4 to read:

"3301.4 Reuse and reprocessing of construction and demolition site material.

Surplus materials and debris resulting from construction and demolition activities shall be reused or reprocessed in accordance with Article XIV of Chapter 11-4 of the *Municipal Code*."

14B-33-3302 Construction safeguards.

The provisions of Section 3302 of IBC are adopted by reference with the following modifications:

1. Revise Section 3302.1, excluding the exceptions, to read:

"3302.1 Alterations, repairs and additions.

Required *means of egress* components, existing structural elements, fire protection devices and sanitary safeguards shall be maintained at all times during *alterations, repairs* or *additions* to any *building* or *structure*."

2. Delete Section 3302.2.
3. Delete Section 3302.3.
4. Add a new Section 3302.4 to read:

"3302.4 Construction loads.

Provisions shall be made to ensure that stresses due to wind loads, dead loads and loads due to material storage and construction equipment imposed during the construction, *alteration, repair, addition* to or demolition of any *structure* shall not exceed the allowable stresses for materials as limited by the provisions of this code."

5. Add a new Section 3302.5 to read:

"3302.5 Protection of openings.

All stairways, elevator shafts, flues, and other vertical openings in the floors and roof shall be covered or properly protected."

14B-33-3303 Demolition.

The provisions of Section 3303 of IBC are adopted by reference with the following modifications:

1. Revise Section 3303.1 to read:

"3303.1 Procedure.

Demolition of any *building* or *structure* shall begin at the top thereof, and each *story* shall be completely razed or demolished and the material therefrom completely removed before demolition work begins on the next lower *story*, unless an alternative procedure is *approved* by the *building official*."

2. Revise Section 3303.3 to read:

"3303.3 Means of egress.

A *horizontal exit* shall not be destroyed unless and until a substitute *means of egress* has been provided."

3. Revise Section 3303.4 to read:

"3303.4 Filling and grading.

Where a *structure* has been demolished or removed, the vacant lot shall be filled, where necessary, with clean soil, cinders, or other inorganic material and graded to a level not lower than and not more than 12 inches (305 mm) above the level of sidewalks, alleys, or adjoining property, with proper allowance for settlement."

4. Revise Section 3303.6 to read:

"3303.6 Utility connections.

Service utility connections shall be discontinued and capped in accordance with the rules and requirements of the utility provider."

5. Revise Section 3303.7 to read:

"3303.7 Fire safety during demolition.

Fire safety during demolition shall comply with the applicable requirements of this chapter."

6. Add a new Section 3303.8 to read:

"3303.8 Debris removal.

Debris from the demolition of a *building* or *structure* shall be removed from the *site* as demolition progresses. Salvaged material, if left on the *site*, shall be neatly stored."

14B-33-3304 Site work.

The provisions of Section 3304 of IBC are adopted by reference with the following modifications:

1. Add a new Section 3304.2 to read:

"3304.2 Drainage across property lines.

Drainage across *property lines* shall not exceed that which existed prior to site work. Excess or concentrated drainage shall be contained on site or directed to a permitted drainage facility. Erosion of the ground in the area of discharge shall be prevented by installation of nonerosive down drains or similar devices."

2. Add a new Section 3304.3 to read:

"3304.3 Fencing of excavation work.

Where excavation work to any depth will occur within 5 feet (1524 mm) of a walkway on public or private property, temporary fencing, not less than 4 feet (1219 mm) in height shall be erected around the area of excavation. Such fencing shall be adequately supported to prevent it from collapsing due to wind. Where the depth of excavation work exceeds 30 inches (762 mm), the fencing shall meet the requirements for *guards* in Chapters 10 and 16. No person shall remove or open a required fence except to allow necessary delivery and removal of materials, tools and equipment or passage of workers, inspectors or other persons whose presence is required at the site."

14B-33-3305 Sanitary.

The provisions of Section 3305 of IBC are adopted by reference with the following modification:

1. Revise Section 3305.1 to read:

"3305.1 Facilities required.

Adequate sanitary facilities for workers shall be provided during construction, *rehabilitation* or demolition activities in accordance with the *Chicago Plumbing Code*."

14B-33-3306 Protection of pedestrians.

The provisions of Section 3306 of IBC are adopted by reference with the following modifications:

1. Revise Section 3306.1 to read:

"3306.1 Protection required.

Pedestrians shall be protected during construction, *rehabilitation* and demolition activities as required by this chapter and Article V-A of Chapter 10-28 of the *Municipal Code*."

2. Delete Table 3306.1.
3. Delete Sections 3306.3 through 3306.9.
4. Add a new Section 3306.10 to read:

"3306.10 Loads on pedestrian protection structures.

No additional *structures*, including field offices, construction equipment, materials, signs, displays, ornamentation or similar loads shall be erected or placed upon a construction canopy or similar pedestrian protection *structure* until plans identifying these loads are submitted to the *building official* and a *permit* is issued, authorizing the construction activity or loading."

14B-33-3307 Protection of adjoining property.

The provisions of Section 3307 of IBC are adopted by reference with the following modifications:

1. Delete the last two sentences of Section 3307.1.
2. Add a new Section 3307.2 to read:

"3307.2 Additional requirements.

Excavation work on private property shall comply with Section 406 of the *Chicago Construction Codes Administrative Provisions*."

14B-33-3308 Temporary use of streets, alleys and public property.

The provisions of Section 3308 of IBC are adopted by reference with the following modifications:

1. Revise Section 3308.1 to read:

"3308.1 General.

The *temporary* use of streets or public property for the storage or handling of materials or of equipment required for construction or

demolition, and the protection provided to the public shall comply with Chapter 10-28 of the *Municipal Code*."

2. Delete Sections 3308.1.1 and 3308.2.

3. Insert new Sections 3308.3 and 3308.4 to read:

"3308.3 Debris.

Debris shall not be allowed to accumulate in the *public way* in violation of Section 7-28-395 of the *Municipal Code*.

3308.4 Damage.

Costs and expenses incurred by the *City* to repair damage to the *public way* or city sewer system caused by construction, demolition or *rehabilitation* work on adjacent private property shall be reimbursed in accordance with Section 7-28-395 of the *Municipal Code*."

(Amend Coun. J. 10-7-20, p. 21791, Art. II, § 33)

14B-33-3309 Fire extinguishers.

The provisions of Section 3309 of IBC are adopted by reference with the following modification:

1. Revise the first sentence of Section 3309.1 to read:

"*Structures* under construction, *alteration* or demolition shall be provided with not fewer than one portable fire extinguisher in accordance with Section 906 and sized for not less than ordinary hazard as follows:"

14B-33-3310 Means of egress.

The provisions of Section 3310 of IBC are adopted by reference with the following modification:

1. Add a new Section 3310.3 to read:

"3310.3 Floor required.

In *buildings* where construction of the superstructure precedes construction of the permanent floor, a substantial temporary floor shall be constructed at a level not more than two floors below the highest level where work is being performed."

14B-33-3311 Standpipes.

The provisions of Section 3311 of IBC are adopted by reference without modification.

14B-33-3312 Automatic sprinkler system.

The provisions of Section 3312 of IBC are adopted by reference with the following modification:

1. Revise Section 3312.1 to read:

"3312.1 Completion before occupancy.

In *buildings* where an *automatic sprinkler system* is required by this code, it shall be unlawful to occupy any portion of a *building* or *structure* until the *automatic sprinkler system* installation has been tested and accepted by the *fire code official*, except as provided in Chapter 7 of the *Chicago Construction Codes Administrative Provisions*.

14B-33-3313 IBC Section 3313.

The provisions of Section 3313 of IBC are not adopted.

14B-33-3314 Owner's responsibility for fire protection.

The provisions of Section 3314 of IBC are not adopted. The following language is adopted as Section 3314:

"3314. OWNER'S RESPONSIBILITY FOR FIRE PROTECTION

3314.1 Program development and maintenance.

The *owner* or *owner's* authorized agent shall be responsible for the development, implementation and maintenance of a written plan establishing a fire prevention program at the project site applicable throughout all phases of the construction, repair, *alteration* or demolition work. The plan shall address the requirements of this chapter and other applicable portions of this code, the duties of staff, and staff training requirements. The plan shall be made available for review by the *building official* or *fire code official* upon request.

3314.2 Program superintendent.

The *owner* shall designate a person to be the fire prevention program superintendent who shall be responsible for the fire

prevention program and ensure that it is carried out through completion of the project. The fire prevention program superintendent shall have the authority to enforce the provisions of this chapter and other provisions as necessary to secure the intent of this chapter. Where guard service is provided in accordance with NFPA 241, the superintendent shall be responsible for the guard service.

3314.3 Prefire plans.

The fire prevention program superintendent shall develop and maintain a prefire plan in cooperation with the *fire code official*. The *fire code official* shall be notified of changes affecting the utilization of information contained in such prefire plans.

3314.4 Training.

Training of responsible personnel in the use of fire protection equipment shall be the responsibility of the fire prevention program superintendent. Records of training shall be kept and made a part of the written plan for the fire prevention program.

3314.5 Fire protection devices.

The fire prevention program superintendent shall determine that all fire protection equipment is maintained and serviced in accordance with this code. Fire protection equipment shall be inspected in accordance with the fire protection program.

3314.6 Hot work operations.

The fire prevention program superintendent shall be responsible for supervising the permit system for hot work operations in accordance with the *Chicago Fire Prevention Code*.

3314.7 Impairment of fire protection systems.

Impairments to any *fire protection system* in an *existing building* shall be in accordance with the *Chicago Fire Prevention Code*.

3314.7.1 Smoke detectors and smoke alarms.

Smoke detectors and smoke alarms located in an area where airborne construction dust is expected shall be covered to prevent exposure to dust or shall be temporarily removed. Smoke detectors and alarms that were removed shall be replaced upon conclusion of dust-producing work. Smoke detectors and smoke alarms that were covered shall be inspected and cleaned, as necessary, upon conclusion of dust-producing work.

3314.8 Temporary covering of fire protection devices.

Coverings placed on or over fire protection devices to protect them from damage during construction processes shall be immediately removed upon the completion of the construction processes in the room or area in which the devices are installed."

14B-33-3315 Temporary heating equipment.

The following language is adopted as a new Section 3315:

"3315. TEMPORARY HEATING EQUIPMENT

3315.1 Listed.

Temporary heating devices shall be *listed* and *labeled*. The installation, maintenance and use of *temporary* heating devices shall be in accordance with the listing and the manufacturer's instructions.

3315.2 Oil-fired heaters.

Oil-fired heaters shall comply with the *Chicago Fire Prevention Code*.

3315.3 LP-gas heaters.

Fuel supplies for liquefied-petroleum gas-fired heaters shall comply with the *Chicago Fire Prevention Code* and *Chicago Fuel Gas Code*.

3315.4 Refueling.

Refueling operations for liquid-fueled equipment or appliances shall be conducted in accordance with the *Chicago Fire Prevention Code*. The equipment or appliance shall be allowed to cool prior to refueling.

3315.5 Installation.

Clearance to combustibles from *temporary* heating devices shall be maintained in accordance with the *labeled* equipment. When in operation, *temporary* heating devices shall be fixed in place and protected from damage, dislodgement or overturning in accordance with the manufacturer's instructions.

3315.6 Supervision.

The use of *temporary* heating devices shall be supervised and maintained only by competent personnel."

14B-33-3316 Precautions against fire.

The following language is adopted as a new Section 3316:

"3316. PRECAUTIONS AGAINST FIRE

3316.1 Smoking.

Smoking shall be prohibited except in designated areas. "No smoking" signs shall be posted in conspicuous locations. In designated areas where smoking is permitted, noncombustible ashtrays shall be provided.

3316.2 Combustible debris, rubbish and waste.

Combustible debris, rubbish and waste material shall comply with the requirements of Sections 3316.2.1 through 3316.2.4.

3316.2.1 Combustible waste material accumulation.

Combustible debris, rubbish and waste material shall not be accumulated within *buildings*.

3316.2.2 Combustible waste material removal.

Combustible debris, rubbish and waste material shall be removed from *buildings* at the end of each shift of work.

3316.2.3 Rubbish containers.

Where rubbish containers with a capacity exceeding 5.33 cubic feet (40 gallons) (0.15 m³) are used for *temporary* storage of combustible debris, rubbish and waste material, they shall have tight-fitting or self-closing lids. Such rubbish containers shall be constructed entirely of materials that comply with either of the following:

1. Noncombustible materials.
2. Materials that meet a peak rate of heat release not exceeding 300 kW/ m² when tested in accordance with ASTM E1354 at an incident heat flux of 50 kW/m² in the horizontal orientation.

3316.2.4 Spontaneous ignition.

Materials susceptible to spontaneous ignition, such as oily rags, shall be stored in a *listed* disposal container.

3316.3 Burning of combustible debris, rubbish and waste.

Combustible debris, rubbish and waste material shall not be disposed of by burning on the *site*.

3316.4 Open burning.

Open burning shall comply with the *Chicago Fire Prevention Code*.

3316.5 Fire watch.

Where required by the *fire code official* or the prefire plan established in accordance with Section 3314.3, a fire watch shall be provided for demolition and construction work that is unusually hazardous in nature, such as *temporary* heating or hot work.

3316.5.1 Fire watch during construction.

Where required by the *fire code official*, a fire watch shall be provided during nonworking hours for new construction that exceeds 40 feet (12.2 m) in height above the lowest adjacent grade.

3316.5.2 Fire watch personnel.

Trained personnel shall be provided to serve as an on-site fire watch. Fire watch personnel shall be provided with not fewer than one means for notification of the fire department, and the sole duty of such personnel shall be to perform constant patrols and watch for the occurrence of fire. The combination of fire watch duties and site security duties is acceptable. Fire watch personnel shall be trained in the use of portable fire extinguishers.

3316.5.3 Fire watch location and records.

The fire watch shall include areas specified by the prefire plan established in accordance with Section 3314.3. The fire watch personnel shall keep a record of all time periods of duty, including a log entry each time the site was patrolled and each time a *structure* under construction was entered and inspected. The records and log entries shall be made available for review by the *fire code official* upon request.

3316.6 Cutting and welding.

Welding, cutting, open torches and other hot work operations and equipment shall comply with the *Chicago Fire Prevention Code*.

3316.7 Electrical.

Temporary wiring for electrical power and lighting installations used in connection with the construction, *alteration* or demolition of *buildings, structures*, equipment or similar activities shall comply with the *Chicago Electrical Code*.

3316.8 Cooking.

Cooking shall be prohibited except in designated cooking areas. Signs with a minimum letter height of 3 inches (76 mm) and a minimum brush stroke of 1/2 inch (13 mm) shall be posted in conspicuous locations in designated cooking areas and state:

DESIGNATED COOKING AREA.

COOKING OUTSIDE OF A DESIGNATED COOKING AREA IS PROHIBITED."

14B-33-3317 Flammable and combustible liquids.

The following language is adopted as a new Section 3317:

"3317. FLAMMABLE AND COMBUSTIBLE LIQUIDS

3317.1 Storage of flammable and combustible liquids.

Storage of flammable and *combustible liquids* shall be in accordance with the *Chicago Fire Prevention Code*.

3317.2 Class I and Class II liquids.

The storage, use and handling of flammable and *combustible liquids* at construction sites shall be in accordance with the *Chicago Fire Prevention Code*. *Ventilation* shall be provided for operations involving the application of materials containing flammable solvents.

3317.3 Housekeeping.

Flammable and combustible liquid storage areas shall be maintained clear of combustible vegetation and waste materials. Such storage areas shall not be used for the storage of combustible materials.

3317.4 Precautions against fire.

Sources of ignition and smoking shall be prohibited in flammable and *combustible liquid* storage areas. Signs shall be posted in accordance with the *Chicago Fire Prevention Code*.

3317.5 Handling at point of final use.

Class I and II liquids shall be kept in appropriate safety containers.

3317.6 Leakage and spills.

Leaking vessels shall be immediately repaired or taken out of service and spills shall be cleaned up and disposed of properly."

14B-33-3318 Flammable gases.

The following language is adopted as a new Section 3318:

"3318. FLAMMABLE GASES

3318.1 Storage and handling.

The storage, use and handling of flammable gases shall comply with the *Chicago Fire Prevention Code*.

3318.2 Cleaning with flammable gas.

Flammable gases shall not be used to clean or remove debris from piping open to the atmosphere.

3318.2.1 Pipe cleaning and purging.

The cleaning and purging of flammable gas piping systems, including cleaning new or existing piping systems, purging piping systems into service and purging piping systems out of service, shall comply with NFPA 56.

Exceptions:

1. Compressed gas piping systems other than fuel gas piping systems where in accordance with the *Chicago Fire Prevention Code*.
2. Piping systems regulated by the *Chicago Fuel Gas Code*.
3. Liquefied petroleum gas systems in accordance with the *Chicago Fire Prevention Code*."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 100)

14B-33-3319 Explosive materials.

The following language is adopted as a new Section 3319:

"3319. EXPLOSIVE MATERIALS

3319.1 Storage and handling.

Explosive materials shall be stored, used and handled in accordance with the *Chicago Fire Prevention Code*.

3319.2 Supervision.

Blasting operations shall be conducted in accordance with the *Chicago Fire Prevention Code*.

3319.3 Demolition using explosives.

Fire hoses for use by demolition personnel shall be maintained at the demolition site wherever *explosives* are used for demolition. Such fire hoses shall be connected to a water supply and shall be capable of being brought to bear on post-*detonation* fires anywhere on the site of the demolition operation."

14B-33-3320 Fire reporting.

The following language is adopted as a new Section 3320:

"3320. FIRE REPORTING

3320.1 Emergency telephone.

Emergency telephone facilities with *ready access* shall be provided at the construction site, or an equivalent means of communication shall be provided. The street address of the construction site and the emergency telephone number of the fire department shall be posted adjacent to the telephone. Alternatively, where an equivalent means of communication is used the site address and fire department emergency telephone number shall be posted at the main entrance to the site, in guard shacks and in the construction site office."

14B-33-3321 Motorized construction equipment.

The following language is adopted as a new Section 3321:

"3321. MOTORIZED CONSTRUCTION EQUIPMENT

3321.1 Conditions of use.

Internal-combustion-powered construction equipment shall be used in accordance with all of the following conditions:

1. Equipment shall be located so that exhausts do not discharge against combustible material.
2. Exhausts shall be piped to the outside of the *building*.
3. Equipment shall not be refueled while in operation.
4. Fuel for equipment shall be stored outside of the *building*."

14B-33-3322 Safeguarding roofing operations.

The following language is adopted as a new Section 3322:

"3322. SAFEGUARDING ROOFING OPERATIONS

3322.1 General.

Roofing operations utilizing heat-producing systems or other ignition sources shall be conducted in accordance with the *Chicago Fire Prevention Code*.

3322.2 Asphalt and tar kettles.

Asphalt and tar kettles shall be operated in accordance with the *Chicago Fire Prevention Code*.

3322.3 Fire extinguishers for roofing operations.

Fire extinguishers shall comply with the *Chicago Fire Prevention Code*. There shall be not less than one multiple-purpose portable fire extinguisher with a minimum 3-A 40-B:C rating on the roof being covered or repaired."

14B-33-3323 Scaffolding.

The following language is adopted as a new Section 3323:

"3323. SCAFFOLDING

3323.1 Construction.

Scaffolding shall be constructed and installed to protect the safety of persons working on, passing under or passing by the *scaffolding*.

3323.2 Material.

When the extent or location of *scaffolding* is such as to create an unusual fire hazard, the *fire code official* is authorized to require the use of noncombustible material or *fire-retardant treated wood*.

3323.3 Safeguards.

Scaffolding shall be constructed, operated, maintained and disassembled in a manner that follows accepted engineering practice and the manufacturer's specifications. ASSE A10.8 and ASME A120.1 shall be considered as accepted engineering practice, as shall

other national standards deemed applicable by the *building official*.

3323.3.1 Not in use.

When *scaffolding* is, or is expected to be, unoccupied for more than two hours, such *scaffolding* shall be safely and securely attached to the *building* in accordance with federal regulations, ASSE standards and the manufacturer's specifications.

3323.3.2 Wind.

When wind conditions actually do, or are predicted or expected to, or based on other measurements can reasonably be extrapolated to, exceed a velocity at the scaffold level identified in the manufacturer's specifications as posing a hazard, such *scaffolding* shall be lowered to the ground and secured or, if required by the manufacturers' specifications, raised to the roof and secured."

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 101)

14B-33-3324 Construction site cleanliness.

The following language is adopted as a new Section 3324:

"3324. CONSTRUCTION SITE CLEANLINESS

3324.1 General.

Work *sites* shall be maintained in accordance with Sections 3324.2 through 3324.5 during construction, demolition, excavation and *rehabilitation* activities.

Exception: The provisions of this section do not apply to activities conducted within a completely enclosed *structure*.

3324.2 Fencing.

Work *sites* shall be enclosed and secured by a continuous *fence* at least 6 feet (1829 mm) in height which shall be anchored to resist wind loads of 30 pounds per square foot (1436 Pa) without deflection of more than 3 inches (76.2 mm) between the top and bottom of the *fence*. The *fence*, including gates, shall be covered for its full height and length in fabric mesh that will allow the passage of air but contain dust and dirt. The *owner* and general contractor shall be responsible for keeping the fence and fabric mesh in good repair and free of graffiti and unauthorized advertising.

3324.2.1 Gate.

One gate, no larger than is reasonably necessary to allow truck access, may remain open when workers are actively working on the *site*.

3324.3 Storage.

All construction materials, dumpsters and debris collection devices shall be kept within the required fence unless a *permit* for use of the *public way* has been obtained from the Commissioner of Transportation. Dumpsters shall be regularly serviced to prevent contents from extending past the top of the dumpster. Construction materials and waste shall be gathered at the end of each work day and stored or disposed of in a neat and orderly manner. Where materials are stored for use on floors above grade, they shall be secured to prevent loosening or dispersion due to weather conditions or other phenomena.

3324.4 Litter and garbage.

Sealed containers for litter and garbage shall be provided throughout the work *site*. At least one container shall be provided for every floor and every 4,000 square feet (372 m²) of work area. All litter and garbage must be removed from the work *site* daily.

3324.5 Dust control.

The general contractor shall employ adequate wetting or other abatement measures to prevent the off-site dispersion of dust and debris from the work *site*. The general contractor shall take necessary steps to minimize the amount of dirt and debris that is transmitted by construction vehicles, including but not limited to stoning or paving of haul roads, wheel wash stations and street sweepers. All construction and demolition debris shall be removed from upper floors using dust-tight chutes or lowered in containers. Construction and demolition debris shall not be dropped or thrown from any floor. Where work occurs on a level that is four or more *stories above grade plane*, the work area shall be enclosed with fabric mesh as provided in Section 3324.2.

3324.5.1 Masonry work.

For masonry work, the immediate work area shall be enclosed to contain dust and debris from grinding or cutting operations.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, 102)

CHAPTER 14B-34

RESERVED

CHAPTER 14B-35

REFERENCED STANDARDS

14B-35-3500 Chapter 35.

14B-35-3500 Chapter 35.

The provisions of Chapter 35 of IBC are adopted by reference with the following modifications:

1. Delete the cross-reference to provisions of the IBC following each reference.
- 1.1. Insert a section to read:

" **AFSI**

Architectural Fabric Structures Institute c/o
Industrial Fabric Association International
1801 County Road B West
Roseville, MN 55113

FSAAS-16: Fabric Structures Associated Air Structures 2016"

- 1.2. Delete the entire section for ALI.
- 1.3. Delete the entire section for AMCA.
2. Insert the following standard under ASCE:
"41-17: Seismic Evaluation and Retrofit of Existing Buildings"
3. Insert a section to read:

" **ASHRAE**

ASHRAE
1791 Tullie Circle, NE
Atlanta, GA 30329

62.1-2016: Ventilation for Acceptable Indoor Air Quality"

4. Delete the following standards under ASME: ASME/A17.1-2016/CSA B44-16; A17.7-2007/CSA B44-07(R2012); A18.1-2014; and A90.1-2015.
5. Insert the following standard under ASME:
"A120.1-2014: Safety Requirements for Powered Platforms and Traveling Ladders and Gentries for Building Maintenance"
6. Insert the following standard under ASSE:
"ANSI/ASSE A10.8-2011: Scaffolding Safety Requirements"
7. Delete the following standards under ASTM: B101-12; D5019-07a; D7254-15; E90-09; E492-09; E1886-13A; E1996-14a; and E2599-15.
8. Insert the following standards under ASTM:
"A48/A48M-03(2016): Standard Specification for Gray Iron Castings"
"C1549-2016: Standard Test Method for Determination of Solar Reflectance Near Ambient Temperature Using a Portable Solar Reflectometer"
"D2974-14: Standard Test Methods for Moisture, Ash, and Organic Matter of Peat and Other Organic Soils"
"D4792/D4792M-13(2019): Standard Test Method for Potential Expansion of Aggregates from Hydration Reactions "

- "ASTM D8169/D8169M–18: Standard Test Methods for Deep Foundations Under Bi-Directional Static Axial Compressive Load"
- "E514/E514M–14a: Standard Test Method for Water Penetration and Leakage Through Masonry"
- "E903–20: Standard Test Method for Solar Absorptance, Reflectance, and Transmittance of Materials Using Integrating Spheres"
- "E1918–06(2016): Standard Test Method for Measuring Solar Reflectance of Horizontal or Low-sloped Surfaces in the Field"
9. Delete the following standard under AWC: ANSI/AWC PWF–2015.
- 9.1. Insert a new section to read:

Cool Roof Rating Council

" **CRRC**

2435 North Lombard Street

Portland, OR 97217

ANSI/CRRC S100–2020: Standard Test Methods for Determining Radiative Properties of Materials"

- 9.2. Delete the following standard under DASMA: ANSI/DASMA 115–2016.
10. Delete the following standards under ICC: ICC 500–14; ICC 600–14; IEBC–18; IECC–18; IFC–18; IFGC–18; IMC–18; IPC–18; IPMC–18; IPSDC–18; IRC–18; ISPSC–18; and IWUIC–18.
- 10.1. Delete the entire section for MHI.
11. Delete the following standards under NFPA: 4–18; 70–17; and 221–18.
12. Insert the following standards under NFPA:
- "25–17: Standard for the Inspection, Testing and Maintenance of Water-based Fire Protection Systems"
- "914–15: Code for Fire Protection of Historic Structures"
13. Delete the entire section for PTI.
14. Delete the following standard under UL: 1975–06.
- 14.1. Insert the following standards under UL:
- "521–99: Heat Detectors for Fire Protective Signaling Systems – with Revisions through March 2021
- 539–17: Single and Multiple Station Heat Alarms – with Revisions through May 2018."
15. Delete the entire section for WRI.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 103; Amend Coun. J. 10-7-20, p. 21791, Art. II, § 34; Amend Coun. J. 6-25-21, p. 32180, Art. II, § 28; Amend Coun. J. 9-21-22, p. 52080, Art. II, § 7)

CHAPTER 14B-36

APPENDICES

14B-36-3600 Appendices not adopted.

14B-36-3604 Appendix D – Fire limits.

14B-36-3605 Appendix E – Supplementary accessibility requirements.

14B-36-3619 Appendix S – Optional smoke control systems.

14B-36-3600 Appendices not adopted.

The provisions of Appendices A, B, C, F, G, H, I, J, K, L, M, and N of IBC are not adopted.

14B-36-3604 Appendix D – Fire limits.

The provisions of Appendix D of IBC are adopted by reference with the following modifications:

1. Throughout the provisions adopted by reference, replace the term "fire district" with "fire limits."

2. Revise Section D101.1 to read:

"D101.1 Scope.

The fire limits shall include that portion of the city bounded by: Division Street; Lake Michigan; the Stevenson Expressway; the CTA red line right-of-way; Cermak Road; the Metra SouthWest Service right-of-way; the South Branch of the Chicago River; 16th Street; the Dan Ryan Expressway; Roosevelt Road; Halsted Street, the Eisenhower Expressway; Ashland Avenue; Ogden Avenue; Hubbard Street; the Kennedy Expressway; Ogden Avenue; Chicago Avenue; North Halsted Street; and the North Branch Canal."

3. Delete Sections D101.1.1.
4. Delete Section D101.2 and its subsections.
5. Revise Section D102.1 to read:

"D102.1 Types of construction permitted.

Within the fire limits every building hereafter erected shall be either Type IA, IB, IIA, IIIA or IV, except as permitted in Section D105."

6. Delete Section D102.2.5.
7. Revise Section D102.2.6 to read:

"D102.2.6 Exterior walls.

Exterior load-bearing walls of Type IIA buildings and, where permitted, Type IIB buildings shall have a *fire-resistance rating* of 2 hours or more where such walls have a *fire separation distance* of 30 feet (9144 mm) or less. Where allowed by Section D105, exterior nonload-bearing walls of Type IIB buildings located within 30 feet (9144 mm) of an *abutting property line* or an assumed property line shall have *fire-resistance ratings* as required by Table 601, but not less than 1 hour. *Exterior walls* located more than 30 feet (9144 mm) from an *abutting property line* or an assumed property line shall comply with Table 601.

Exception: In the case of one-story buildings that are 2,000 square feet (186 m²) or less in area, *exterior walls* located more than 15 feet (4572 mm) from an *abutting property line* or an assumed *property line* need only comply with Table 601."

8. Add new Section D102.2.6.1 to read:

"D102.2.6.1 Exterior walls of Type IIB, VA and VB construction.

Where allowed by Section D105, *exterior walls* of buildings of Type IIB, VA and VB construction shall be located to provide a *fire separation distance* of not less than 3 feet (914 mm). Where the *fire separation distance* is 6 feet (1829 mm) or less, the exterior wall shall have a *fire-resistance rating* of 1 hour or greater."

9. Revise Section D102.2.7 to read:

"D102.2.7 Projections.

Projections extending beyond the *exterior wall* on buildings located in the fire limits shall be constructed of noncombustible materials."

10. Revise Section D102.2.8 to read:

"D102.2.8 Permanent canopies.

Permanent *canopies* are permitted to extend over adjacent open spaces provided that all of the following are met:

1. The *canopy* and its supports shall be of noncombustible material.

Exception: Any textile covering for the canopy shall be flame resistant as determined by tests conducted in accordance with NFPA 701 after both accelerated water leaching and accelerated weathering.

2. Any canopy covering, other than textiles, shall have a *flame spread index* not greater than 25 when tested in accordance with ASTM E84 or UL 723 in the form intended for use.
3. The *canopy* shall have at least one long side open.
4. The horizontal width of the canopy shall be not greater than 15 feet (4572 mm).
5. The *fire-resistance rating* of *exterior walls* shall not be reduced."

11. Revise Section D102.2.9 to read:

"D102.2.9 Rooftop structures.

Rooftop structures, except aerial supports 12 feet (3658 mm) high or less shall be of noncombustible material and shall be supported by construction of noncombustible material."

12. Revise the title of Section D103 to: "CHANGES TO BUILDINGS AND STRUCTURES."
13. Revise Section D103.1 to read:

"D103.1 Existing buildings and structures within the fire limits.

An *existing building* shall not be increased in *building height* or *building area* unless it is of a type of construction permitted for new *buildings* within the fire limits or is altered to comply with the requirements for such type of construction. An *existing structure*, other than a *building*, shall not be altered so as to increase its degree of nonconformity."

14. Delete Section D104.

15. Revise Section D105.1 to read:

"D105.1 General.

Nothing in this appendix shall prohibit within the fire limits, subject to the specified limitations, the erection of new *buildings* or *structures*, nor the enlargement of *existing buildings* or *structures*, nor the use of wood or other combustible veneers, as follows:

1. *Temporary* tents, canopies, platforms, stages reviewing stands and similar structures.
2. Trailers used in connection with duly authorized construction.
3. A detached *private garage*, not more than one *story* and 12 feet (3658 mm) in height, nor more than 500 square feet (60 m²) in area, located on the same *lot* with a *building* of Group R-5 occupancy.
4. *Fences* not over 10 feet (3048 mm) above adjoining grade.
5. Coal and material bins, water towers and trestles of Type IV construction.
6. Water tanks and cooling towers conforming to Sections 1510.3 and 1510.4.
7. Weather-protected entries not more than 12 feet (3658 mm) high and not more than 50 square feet (4.6 m²) in area.
8. *Greenhouses* less than 15 feet (4572 mm) high and less than 400 square feet (37.2 m²) in area.
9. *Porches* not over one *story* in height, and not over 10 feet (3048 mm) wide from the face of the *building*, provided that such *porch* does not come within 3 feet (914 mm) of any *abutting property line* and is not joined to more than one *building*.
10. Sheds open on a long side not over 15 feet (4572 mm) high and 400 square feet (37.2 m²) in area.
11. [Reserved.]
12. Wood or other combustible veneers on *exterior walls* conforming to Section 1404.5, not extending above the first *story above grade plane*.
13. *Fire-retardant treated wood* veneers conforming to Section 1404.5 not extending above the second *story above grade plane*.
14. Single-*story* buildings of Type IIB construction not exceeding 5,000 square feet (465 m²) in area. *Exterior walls* with a *fire separation distance* of less than 30 feet (9144 mm) shall have a *fire- resistance rating* of not less than 1 hour.
15. Roofs over parking lots and bus stations of Type IIB construction where the roof is at least 10 feet (3048 mm) above the floor and every 40 feet (12.2 m) there is an open roof *ventilation* area at least 6 feet (1829 mm) wide extending either the full length of the roof or the full width of the roof."
16. Delete Section D106.

(Amend Coun. J. 2-19-20, p. 14473, Art. II, § 104)

14B-36-3605 Appendix E – Supplementary accessibility requirements.

The provisions of Appendix E of IBC are adopted by reference with the following modifications:

1. Revise Section E101.1 to read:

"E101.1 Scope.

The provisions of this appendix shall control the supplementary requirements for the design and construction of facilities for *accessibility* for individuals with *disabilities*."

2. Revise the definition of "transient lodging" in Section E102.1 to read:

"**TRANSIENT LODGING.** A *building, facility* or portion thereof, excluding inpatient *medical care* facilities and *nursing homes*, that contains one or more *dwelling units* or *sleeping units* occupied on a *transient* basis. Examples of *transient lodging* include, but are not limited to, *group homes, hotels, homeless shelters, halfway houses* and social service lodging. *Transient lodging* also includes housing at a place of education required to comply with Section 1107.6.2.3."

3. Revise Section E103.1 to read:

"E103.1 Raised platforms.

See Section 1108.2.9.2."

4. Revise Section E104.1 to read:

"E104.1 General.

Transient lodging and assisted living facilities shall be provided with *accessible* features in accordance with Section E104.2. Group I-3 occupancies shall be provided with *accessible* features in accordance with Section E104.2."

5. Revise Section E104.2.1 to read:

"E104.2.1 Transient lodging and assisted living facilities.

In *transient lodging* and assisted living facilities, *sleeping units* with *accessible* communication features shall be provided in accordance with Table E104.2.1. Units required to comply with Table E104.2.1 shall be dispersed among the various classes of units."

6. Add new Section E104.2.1.1 to read:

"E104.2.1.1 Overlap.

At least one unit required to provide mobility features complying with Section 1107.5.1.1 or 1107.6.1.1 shall be used to satisfy the minimum number of units required to provide *accessible* communication features complying with Section E104.2.1. Not more than 10 percent of units required to provide mobility features complying with Section 1107.5.1.1 or 1107.6.1.1 shall be used to satisfy the minimum number of units required to provide *accessible* communication features complying with Section E104.2.1."

7. Revise Section E104.2.2 to read:

"E104.2.2 Group I-3.

In Group I-3 occupancies at least 3 percent, but no fewer than one of the total number of general holding cells and general housing cells equipped with audible *emergency alarm systems* and permanently installed telephones within the cell, shall comply with Section E104.2.4. Cells shall be provided in each classification level."

8. Add a new Section E111 to read:

"E111. PUBLIC WAY

E111.1 General.

Construction in the *public way* shall comply with regulations adopted by the Commissioner of Transportation."

9. Revise and renumber IBC Section E111 to read:

"E112. REFERENCED STANDARDS

DOJ 36 CFR Part 1192 Americans with Disabilities Act (ADA) Accessibility Guidelines for Transportation Vehicles (ADAAG). Washington, DC: Department of Justice, 1991"

14B-36-3619 Appendix S – Optional smoke control systems.

The following language is adopted as a new Appendix S:

"APPENDIX S. OPTIONAL SMOKE CONTROL SYSTEMS

S101. GENERAL

S101.1 Scope and purpose.

This appendix applies to mechanical or passive smoke control systems where they are installed. Where smoke control systems are proposed as an alternative to other requirements of this code, they shall be *approved* by the Committee on Standards and Tests in accordance with Section 1004 of the *Chicago Construction Codes Administrative Provisions*. The purpose of this appendix is to establish minimum requirements for the design, installation and acceptance testing of smoke control systems that are intended to provide a tenable environment for the evacuation or relocation of occupants. These provisions are not intended for the preservation of contents, the timely restoration of operations or for assistance in fire suppression or overhaul activities. Smoke control systems regulated by this appendix serve a different purpose than the smoke- and heat-removal provisions found in Section 910. Mechanical smoke control systems shall not be considered exhaust systems under Chapter 5 of the *Chicago Mechanical Code*.

S101.2 General design requirements.

Buildings, structures or parts thereof required by this code to have a smoke control system or systems shall have such systems designed in accordance with the applicable requirements of Section S101 and the generally accepted and well-established principles of engineering relevant to the design. The *construction documents* shall include sufficient information and detail to adequately describe the elements of the design necessary for the proper implementation of the smoke control systems. These documents shall be accompanied by sufficient information and analysis to demonstrate compliance with these provisions.

S101.3 Special inspection and test requirements.

In addition to the ordinary inspection and test requirements that buildings, structures and parts thereof are required to undergo, smoke control systems subject to the provisions of Section S101 shall undergo *special inspections* and tests sufficient to verify the proper commissioning of the smoke control design in its final installed condition. The design submission accompanying the *construction documents* shall clearly detail procedures and methods to be used and the items subject to such inspections and tests. Such commissioning shall be in accordance with generally accepted engineering practice and, where possible, based on published standards for the particular testing involved. The special inspections and tests required by this section shall be conducted under the same terms in Section 1704.

S101.4 Analysis.

A rational analysis supporting the types of smoke control systems to be employed, their methods of operation, the systems supporting them and the methods of construction to be utilized shall accompany the submitted *construction documents* and shall include, but not be limited to, the items indicated in Sections S101.4.1 through S101.4.7.

S101.4.1 Stack effect.

The system shall be designed such that the maximum probable normal or reverse stack effect will not adversely interfere with the system's capabilities. In determining the maximum probable stack effect, altitude, elevation, weather history and interior temperatures shall be used.

S101.4.2 Temperature effect of fire.

Buoyancy and expansion caused by the design fire in accordance with Section S101.9 shall be analyzed. The system shall be designed such that these effects do not adversely interfere with the system's capabilities.

S101.4.3 Wind effect.

The design shall consider the adverse effects of wind. Such consideration shall be consistent with the wind-loading provisions of Chapter 16.

S101.4.4 HVAC systems.

The design shall consider the effects of the heating, ventilating and air-conditioning (HVAC) systems on both smoke and fire transport. The analysis shall include all permutations of systems status. The design shall consider the effects of the fire on the HVAC systems.

S101.4.5 Climate.

The design shall consider the effects of low temperatures on systems, property and occupants. Air inlets and exhausts shall be located so as to prevent snow or ice blockage.

S101.4.6 Duration of operation.

All portions of active or engineered smoke control systems shall be capable of continued operation after detection of the fire event for a period of not less than either 20 minutes or 1.5 times the calculated egress time, whichever is greater.

S101.4.7 Smoke control system interaction.

The design shall consider the interaction effects of the operation of multiple smoke control systems for all design scenarios.

S101.5 Smoke barrier construction.

Smoke barriers required for passive smoke control and a smoke control system using the pressurization method shall comply with Section 709. The maximum allowable leakage area shall be the aggregate area calculated using the following leakage area ratios:

1. Walls $A/A_W = 0.00100$
2. Interior *exit stairways* and *ramps* and *exit passageways*: $A/A_W = 0.00035$
3. Enclosed *exit access stairways* and *ramps* and all other shafts: $A/A_W = 0.00150$
4. Floors and roofs: $A/A_F = 0.00050$

where:

A = Total leakage area, square feet (m^2).

A_F = Unit floor or roof area of barrier, square feet (m^2).

A_W = Unit wall area of barrier, square feet (m^2).

The leakage area ratios shown do not include openings due to gaps around doors and operable windows. The total leakage area of the *smoke barrier* shall be determined in accordance with Section S101.5.1 and tested in accordance with Section S101.5.2.

S101.5.1 Total leakage area.

Total leakage area of the barrier is the product of the *smoke barrier* gross area multiplied by the allowable leakage area ratio, plus the area of other openings such as gaps around doors and operable windows.

S101.5.2 Testing of leakage area.

Compliance with the maximum total leakage area shall be determined by achieving the minimum air pressure difference across the barrier with the system in the smoke control mode for mechanical smoke control systems utilizing the pressurization method. Compliance with the maximum total leakage area of passive smoke control systems shall be verified through methods such as door fan testing or other methods, as *approved* by the *fire code official*.

S101.5.3 Opening protection.

Openings in *smoke barriers* shall be protected by automatic-closing devices actuated by the required controls for the mechanical smoke control system. Door openings shall be protected by *fire door assemblies* complying with Section 716.

Exceptions:

1. Passive smoke control systems with automatic-closing devices actuated by spot-type smoke detectors *listed* for releasing service installed in accordance with Section 907.3.
2. Fixed openings between smoke zones that are protected utilizing the airflow method.
3. In Group I-1, Condition 2; Group I-2; and ambulatory care facilities, where a pair of opposite-swinging doors are installed across a corridor in accordance with Section S101.5.3.1, the doors shall not be required to be protected in accordance with Section 716. The doors shall be close-fitting within operational tolerances and shall not have a center mullion or undercuts in excess of 3/4 inch (19.1 mm), louvers or grilles. The doors shall have head and jamb stops and astragals or rabbets at meeting edges and, where permitted by the door manufacturer's listing, positive-latching devices are not required.
4. In Group I-2 and ambulatory care facilities, where such doors are special-purpose horizontal sliding, accordion or folding door assemblies installed in accordance with Section 1010.1.4.3 and are automatic closing by smoke detection in accordance with Section 716.2.6.5.
5. Group I-3.
6. Openings between smoke zones with clear ceiling heights of 14 feet (4267 mm) or greater and bank-down capacity of greater than 20 minutes as determined by the design fire size.

S101.5.3.1 Group I-1, Condition 2; Group I-2; and ambulatory care facilities.

In Group I-1, Condition 2; Group I-2; and *ambulatory care facilities*, where doors are installed across a *corridor*, the doors shall be automatic closing by smoke detection in accordance with Section 716.2.6.5 and shall have a vision panel with fire-protection-rated glazing materials in fire protection- rated frames, the area of which shall not exceed that tested.

S101.5.3.2 Ducts and air transfer openings.

Ducts and air transfer openings are required to be protected with a minimum Class II, 250°F (121°C) *smoke damper* complying with Section 717.

S101.6 Pressurization method.

The primary mechanical means of controlling smoke shall be by pressure differences across smoke barriers. Maintenance of a tenable environment is not required in the smoke control zone of fire origin.

S101.6.1 Minimum pressure difference.

The pressure difference across a *smoke barrier* used to separate smoke zones shall be not less than 0.05-inch water gage (0.0124 kPa) in fully sprinklered buildings.

In buildings permitted to be other than fully sprinklered, the smoke control system shall be designed to achieve pressure differences not less than two times the maximum calculated pressure difference produced by the design fire.

S101.6.2 Maximum pressure difference.

The maximum air pressure difference across a *smoke barrier* shall be determined by required door-opening or closing forces. The actual force required to open *exit* doors when the system is in the smoke control mode shall be in accordance with Section 1010.1.3. Opening and closing forces for other doors shall be determined by standard engineering methods for the resolution of forces and reactions. The calculated force to set a side-hinged, swinging door in motion shall be determined by:

$$F = F_{dc} + K(WA\Delta P)/2(W-d) \quad \text{(Equation S-1)}$$

where:

A = Door area, square feet (m²).

d = Distance from door handle to latch edge of door, feet (m).

F = Total door opening force, pounds (N).

F_{dc} = Force required to overcome closing device, pounds (N).

K = Coefficient 5.2 (1.0).

W = Door width, feet (m).

ΔP = Design pressure difference, inches of water (Pa).

S101.6.3 Pressurized stairways and elevator hoistways.

Where stairways or elevator hoistways are pressurized, such pressurization systems shall comply with Section S101 as smoke control systems, in addition to the requirements of Sections S101.20 of this code and the entire *Chicago Fire Prevention Code*.

S101.7 Airflow design method.

Where *approved* by the *fire code official*, smoke migration through openings fixed in a permanently open position, which are located between smoke control zones by the use of the airflow method, shall be permitted. The design airflow shall be in accordance with this section. Airflow shall be directed to limit smoke migration from the fire zone. The geometry of openings shall be considered to prevent flow reversal from turbulent effects. Smoke control systems using the airflow method shall be designed in accordance with NFPA 92.

S101.7.1 Prohibited conditions.

This method shall not be employed where either the quantity of air or the velocity of the airflow will adversely affect other portions of the smoke control system, unduly intensify the fire, disrupt plume dynamics or interfere with exiting. Airflow toward the fire shall not exceed 200 feet per minute (1.02 m/s). Where the calculated airflow exceeds this limit, the airflow method shall not be used.

S101.8 Exhaust method.

Where *approved* by the *fire code official*, mechanical smoke control for large enclosed volumes, such as in *atriums*, shall be permitted to utilize the exhaust method. Smoke control systems using the exhaust method shall be designed in accordance with NFPA 92.

S101.8.1 Smoke layer.

The height of the lowest horizontal surface of the smoke layer interface shall be maintained not less than 6 feet (1829 mm) above a walking surface that forms a portion of a required egress system within the smoke zone.

S101.9 Design fire.

The design fire shall be based on a rational analysis performed by the *registered design professional* and *approved* by the *fire code official*. The design fire shall be based on the analysis in accordance with Section S101.4 and this section.

S101.9.1 Factors considered.

The engineering analysis shall include the characteristics of the fuel, fuel load, effects included by the fire and whether the fire is likely to be steady or unsteady.

S101.9.2 Design fire fuel.

Determination of the design fire shall include consideration of the type of fuel, fuel spacing and configuration.

S101.9.3 Heat-release assumptions.

The analysis shall make use of best available data and shall not be based on excessively stringent limitations of combustible material.

S101.9.4 Sprinkler effectiveness assumptions.

A documented engineering analysis shall be provided for conditions that assume fire growth is halted at the time of sprinkler activation.

S101.10 Equipment.

Equipment including, but not limited to, fans, ducts, automatic *dampers* and balance *dampers*, shall be suitable for its intended use and suitable for the probable exposure temperatures that the rational analysis indicates.

S101.10.1 Exhaust fans.

Components of exhaust fans shall be rated and certified by the manufacturer for the probable temperature rise to which the components will be exposed. This temperature rise shall be computed by:

$$T_s = (Q_c / mc) + (T_a) \quad \text{(Equation S-2)}$$

where:

c = Specific heat of smoke at smoke layer temperature, Btu/lb°F (kJ/kg • K).

m = Exhaust rate, pounds per second (kg/s).

Q_c = Convective heat output of fire, Btu/s (kW).

T_a = Ambient temperature, °F (K).

T_s = Smoke temperature, °F (K).

Exception: Reduced T_s as calculated based on the assurance of adequate dilution air.

S101.10.2 Ducts.

Duct materials and joints shall be capable of withstanding the probable temperatures and pressures to which they are exposed as determined in accordance with Section S101.10.1. Ducts shall be constructed and supported in accordance with the *Chicago Mechanical Code*. Ducts shall be leak tested to 1.5 times the maximum design pressure in accordance with nationally accepted practices. Measured leakage shall not exceed 5 percent of design flow. Results of such testing shall be a part of the documentation procedure. Ducts shall be supported directly from fire-resistance-rated structural elements of the building by substantial, noncombustible supports.

Exception: Flexible connections, for the purpose of vibration isolation, complying with the *Chicago Mechanical Code* and that are constructed of fire-resistance-rated materials.

S101.10.3 Equipment, inlets and outlets.

Equipment shall be located so as to not expose uninvolved portions of the building to an additional fire hazard. Outside air inlets shall be located so as to minimize the potential for introducing smoke or flame into the building. Exhaust outlets shall be so located as to minimize reintroduction of smoke into the building and to limit exposure of the building or adjacent buildings to an additional fire hazard.

S101.10.4 Automatic dampers.

Automatic *dampers*, regardless of the purpose for which they are installed within the smoke control system, shall be *listed* and conform to the requirements of recognized standards.

S101.10.5 Fans.

In addition to other requirements, belt-driven fans shall have 1.5 times the number of belts required for the design duty, with the minimum number of belts being two. Fans shall be selected for stable performance based on normal temperature and, where applicable, elevated temperature. Calculations and manufacturer's fan curves shall be part of the documentation procedures. Fans shall be supported and restrained by noncombustible devices in accordance with the requirements of Chapter 16.

Motors driving fans shall not be operated beyond their nameplate horsepower (kilowatts), as determined from measurement of actual current draw, and shall have a minimum service factor of 1.15.

S101.11 Standby power.

Smoke control systems shall be provided with standby power in accordance with Section 2702.

S101.11.1 Equipment room.

The standby power source and its transfer switches shall be in a room separate from the normal power transformers and switch gears and ventilated directly to and from the exterior. The room shall be enclosed with not less than 1-hour *fire barriers* constructed in accordance with Section 707 or *horizontal assemblies* constructed in accordance with Section 711, or both.

S101.11.2 Power sources and power surges.

Elements of the smoke control system relying on volatile memories or the like shall be supplied with uninterruptable power sources of sufficient duration to span 15-minute primary power interruption. Elements of the smoke control system susceptible to power surges shall be suitably protected by conditioners, suppressors or other *approved* means.

S101.12 Detection and control systems.

Fire detection systems providing control input or output signals to mechanical smoke control systems or elements thereof shall comply with the requirements of Section 907. Such systems shall be equipped with a control unit complying with UL 864 and *listed* as smoke control equipment.

S101.12.1 Verification.

Control systems for mechanical smoke control systems shall include provisions for verification. Verification shall include positive confirmation of actuation, testing, manual override and the presence of power downstream of all disconnects. A preprogrammed weekly test sequence shall report abnormal conditions audibly, visually and by printed report. The preprogrammed weekly test shall operate all devices, equipment and components used for smoke control.

Exception: Where verification of individual components tested through the preprogrammed weekly testing sequence will interfere with, and produce unwanted effects to, normal building operation, such individual components are permitted to be bypassed from the preprogrammed weekly testing, where *approved* by the *fire code*.

official and in accordance with both of the following:

1. Where the operation of components is bypassed from the preprogrammed weekly test, presence of power downstream of all disconnects shall be verified weekly by a listed control unit.
2. Testing of all components bypassed from the preprogrammed weekly test shall be in accordance with the *Chicago Fire Prevention Code*.

S101.12.2 Wiring.

In addition to meeting requirements of the *Chicago Electrical Code*, all wiring, regardless of voltage, shall be fully enclosed within continuous raceways.

S101.12.3 Activation.

Smoke control systems shall be activated in accordance with this section.

S101.12.3.1 Pressurization, airflow or exhaust method.

Mechanical smoke control systems using the pressurization, airflow or exhaust method shall have completely automatic control.

S101.12.3.2 Passive method.

Passive smoke control systems actuated by spot-type detectors *listed* for releasing service shall be permitted.

S101.12.4 Automatic control.

Where completely automatic control is required or used, the automatic-control sequences shall be initiated from an appropriately zoned *automatic sprinkler system* complying with Section 903.3.1.1, manual controls provided with *ready access* for the fire department and any smoke detectors required by engineering analysis.

S101.13 Control air tubing.

Control air tubing shall be of sufficient size to meet the required response times. Tubing shall be flushed clean and dry prior to final connections and shall be adequately supported and protected from damage. Tubing passing through concrete or masonry shall be sleeved and protected from abrasion and electrolytic action.

S101.13.1 Materials.

Control-air tubing shall be hard-drawn copper, Type L, ACR in accordance with ASTM B42, ASTM B43, ASTM B68, ASTM B88, ASTM B251 and ASTM B280. Fittings shall be wrought copper or brass, solder type in accordance with ASME B16.18 or ASME B16.22. Changes in direction shall be made with appropriate tool bends. Brass compression-type fittings shall be used at final connection to devices; other joints shall be brazed using a BCuP-5 brazing alloy with solidus above 1,100°F (593°C) and liquids below 1,500°F (816°C). Brazing flux shall be used on copper-to-brass joints only.

Exception: Nonmetallic tubing used within control panels and at the final connection to devices provided that all of the following conditions are met:

1. Tubing shall comply with the requirements of the *Chicago Mechanical Code*.
2. Tubing and connected devices shall be completely enclosed within a galvanized or paint- grade steel enclosure having a minimum thickness of 0.0296 inch (0.7534 mm) (No. 22 gage). Entry to the enclosure shall be by copper tubing with a protective grommet of neoprene or Teflon or by suitable brass compression to male barbed adapter.
3. Tubing shall be identified by appropriately documented coding.
4. Tubing shall be neatly tied and supported within the enclosure. Tubing bridging cabinets and doors or moveable devices shall be of sufficient length to avoid tension and excessive stress. Tubing shall be protected against abrasion. Tubing connected to devices on doors shall be fastened along hinges.

S101.13.2 Isolation from other functions.

Control tubing serving other than smoke control functions shall be isolated by automatic isolation valves or shall be an independent system.

S101.13.3 Testing.

Control air tubing shall be tested at three times the operating pressure for not less than 30 minutes without any noticeable loss in gauge pressure prior to final connection to devices.

S101.14 Marking and identification.

The detection and control systems shall be clearly marked at all junctions, accesses and terminations.

S101.15 Control diagrams.

Identical control diagrams showing all devices in the system and identifying their location and function shall be maintained current

and kept on file with the *fire code official*.

S101.16 Fire fighter's smoke control panel.

A fire fighter's smoke control panel for fire department emergency response purposes only shall be provided and shall include manual control or override of automatic control for mechanical smoke control systems. The panel shall be part of the *fire command center* in *high-rise buildings* or buildings with smoke-protected assembly seating. In all other buildings, the fire fighter's smoke control panel shall be installed in a location adjacent to the fire alarm control panel. The fire fighter's smoke control panel shall comply with Sections S101.16.1 through S101.16.3.

S101.16.1 Smoke control systems.

Fans within the building shall be shown on the fire fighter's control panel. A clear indication of the direction of airflow and the relationship of components shall be displayed. Status indicators shall be provided for all smoke control equipment, annunciated by fan and zone, and by pilot-lamp-type indicators as follows:

1. Fans, *dampers* and other operating equipment in their normal status – WHITE.
2. Fans, *dampers* and other operating equipment in their off or closed status – RED.
3. Fans, *dampers* and other operating equipment in their on or open status – GREEN.
4. Fans, *dampers* and other operating equipment in a fault status – YELLOW/AMBER.

S101.16.2 Smoke control panel.

The fire fighter's control panel shall provide control capability over the complete smoke control system equipment within the building as follows:

1. ON-AUTO-OFF control over each individual piece of operating smoke control equipment that can be controlled from other sources within the building. This includes *stairway* pressurization fans; smoke exhaust fans; supply, return and exhaust fans; elevator shaft fans and other operating equipment used or intended for smoke control purposes.
2. OPEN-AUTO-CLOSE control over individual *dampers* relating to smoke control and that are controlled from other sources within the building.
3. ON-OFF or OPEN-CLOSE control over smoke control and other critical equipment associated with a fire or smoke emergency and that can only be controlled from the fire fighter's control panel.

Exceptions:

1. Complex systems, where *approved*, where the controls and indicators are combined to control and indicate all elements of a single smoke zone as a unit.
2. Complex systems, where *approved*, where the control is accomplished by computer interface using *approved*, plain English commands.

S101.16.3 Control action and priorities.

The fire-fighter's control panel actions shall be as follows:

1. ON-OFF and OPEN-CLOSE control actions shall have the highest priority of any control point within the building. Once issued from the fire fighter's control panel, automatic or manual control from any other control point within the building shall not contradict the control action. Where automatic means are provided to interrupt normal, nonemergency equipment operation or produce a specific result to safeguard the building or equipment including, but not limited to, duct freezestats, duct smoke detectors, high-temperature cutouts, temperature-actuated linkage and similar devices, such means shall be capable of being overridden by the fire fighter's control panel. The last control action as indicated by each fire fighter's control panel switch position shall prevail. Control actions shall not require the smoke control system to assume more than one configuration at any one time.

Exception: Power disconnects required by the *Chicago Electrical Code*.

2. Only the AUTO position of each three-position fire-fighter's control panel switch shall allow automatic or manual control action from other control points within the building. The AUTO position shall be the NORMAL, nonemergency, building control position. Where a fire fighter's control panel is in the AUTO position, the actual status of the device (on, off, open, closed) shall continue to be indicated by the status indicator described in Section S101.16.1. Where directed by an automatic signal to assume an emergency condition, the NORMAL position shall become the emergency condition for that device or group of devices within the zone. Control actions shall not require the smoke control system to assume more than one configuration at any one time.

S101.17 System response time.

Smoke-control system activation shall be initiated immediately after receipt of an appropriate automatic or manual activation command. Smoke control systems shall activate individual components (such as *dampers* and fans) in the sequence necessary to prevent physical damage to the fans, *dampers*, ducts and other equipment. For purposes of smoke control, the fire fighter's control

panel response time shall be the same for automatic or manual smoke control action initiated from any other building control point. The total response time, including that necessary for detection, shutdown of operating equipment and smoke control system startup, shall allow for full operational mode to be achieved before the conditions in the space exceed the design smoke condition. The system response time for each component and their sequential relationships shall be detailed in the required rational analysis and verification of their installed condition reported in the required final report.

S101.18 Acceptance testing.

Devices, equipment, components and sequences shall be individually tested. These tests, in addition to those required by other provisions of this code, shall consist of determination of function, sequence and, where applicable, capacity of their installed condition.

S101.18.1 Detection devices.

Smoke or fire detectors that are a part of a smoke control system shall be tested in accordance with Chapter 9 in their installed condition. Where applicable, this testing shall include verification of airflow in both minimum and maximum conditions.

S101.18.2 Ducts.

Ducts that are part of a smoke control system shall be traversed using generally accepted practices to determine actual air quantities.

S101.18.3 Dampers.

Dampers shall be tested for function in their installed condition.

S101.18.4 Inlets and outlets.

Inlets and outlets shall be read using generally accepted practices to determine air quantities.

S101.18.5 Fans.

Fans shall be examined for correct rotation. Measurements of voltage, amperage, revolutions per minute (rpm) and belt tension shall be made.

S101.18.6 Smoke barriers.

Measurements using inclined manometers or other *approved* calibrated measuring devices shall be made of the pressure differences across *smoke barriers*. Such measurements shall be conducted for each possible smoke control condition.

S101.18.7 Controls.

Each smoke zone equipped with an automatic-initiation device shall be put into operation by the actuation of one such device. Each additional device within the zone shall be verified to cause the same sequence without requiring the operation of fan motors in order to prevent damage. Control sequences shall be verified throughout the system, including verification of override from the fire fighter's control panel and simulation of standby power conditions.

S101.18.8 Testing for smoke control.

Smoke control systems shall be tested by a special inspector in accordance with Section 1705.18.

S101.18.8.1 Scope of testing.

Testing shall be conducted in accordance with the following:

1. During erection of ductwork and prior to concealment for the purposes of leakage testing and recording of device location.
2. Prior to occupancy and after sufficient completion for the purposes of pressure-difference testing, flow measurements, and detection and control verification.

S101.18.8.2 Qualifications.

Approved agencies for smoke control testing shall have expertise in fire protection engineering, mechanical engineering and certification as air balancers.

S101.18.8.3 Reports.

A complete report of testing shall be prepared by the *approved agency*. The report shall include identification of all devices by manufacturer, nameplate data, design values, measured values and identification tag or *mark*. The report shall be reviewed by the responsible *registered design professional* and, when satisfied that the design intent has been achieved, the responsible *registered design professional* shall sign, seal and date the report.

S101.18.8.3.1 Report filing.

A copy of the final report shall be filed with the *fire code official* and an identical copy shall be maintained in the building.

S101.18.9 Identification and documentation.

Charts, drawings and other documents identifying and locating each component of the smoke control system, and describing its proper function and maintenance requirements, shall be maintained on file at the building as an attachment to the report required by Section S101.18.8.3. Devices shall have an identifying tag or *mark* on them consistent with the other required documentation and shall be dated indicating the last time they were successfully tested and by whom.

S101.19 [Reserved.]

S101.20 Smokeproof enclosures.

Where provided, a smokeproof enclosure shall be constructed in accordance with this section. A smokeproof enclosure shall consist of an *interior exit stairway* or *ramp* that is enclosed in accordance with the applicable provisions of Section 1023 and an open exterior balcony or ventilated vestibule meeting the requirements of this section. Where access to the roof is required by the *Chicago Fire Prevention Code*, such access shall be from the smokeproof enclosure where a smokeproof enclosure is required.

S101.20.1 Access.

Access to the *stairway* or *ramp* shall be by way of a vestibule or an open exterior balcony. The minimum dimension of the vestibule shall be not less than the required width of the *corridor* leading to the vestibule but shall not have a width of less than 44 inches (1118 mm) and shall not have a length of less than 72 inches (1829 mm) in the direction of egress travel.

S101.20.2 Construction.

The smokeproof enclosure shall be separated from the remainder of the building by not less than 2-hour *fire barriers* constructed in accordance with Section 707 or *horizontal assemblies* constructed in accordance with Section 711, or both. Openings are not permitted other than the required *means of egress* doors. The vestibule shall be separated from the *stairway* or *ramp* by not less than 2-hour *fire barriers* constructed in accordance with Section 707 or *horizontal assemblies* constructed in accordance with Section 711, or both. The open exterior balcony shall be constructed in accordance with the *fire-resistance rating* requirements for floor assemblies.

S101.20.2.1 Door closers.

Doors in a smokeproof enclosure shall be self- or automatic closing by actuation of a smoke detector in accordance with Section 716.2.6.6 and shall be installed at the floor-side entrance to the smokeproof enclosure. The actuation of the smoke detector on any door shall activate the closing devices on all doors in the smokeproof enclosure at all levels. Smoke detectors shall be installed in accordance with Section 907.3.

S101.20.3 Natural ventilation alternative.

The provisions of Sections S101.20.3.1 through S101.20.3.3 shall apply to ventilation of smokeproof enclosures by natural means.

S101.20.3.1 Balcony doors.

Where access to the *stairway* or *ramp* is by way of an open exterior balcony, the door assembly into the enclosure shall be a *fire door assembly* in accordance with Section 716.

S101.20.3.2 Vestibule doors.

Where access to the *stairway* or *ramp* is by way of a vestibule, the door assembly into the vestibule shall be a *fire door assembly* complying with Section 716. The door assembly from the vestibule to the *stairway* shall have not less than a 20-minute *fire protection rating* complying with Section 716.

S101.20.3.3 Vestibule ventilation.

Each vestibule shall have a minimum net area of 16 square feet (1.5 m²) of opening in a wall facing an outer *court*, *yard* or *public way* that is not less than 20 feet (6096 mm) in width.

S101.20.4 Mechanical ventilation alternative.

The provisions of Sections S101.20.4.1 through S101.20.4.4 shall apply to ventilation of smokeproof enclosures by mechanical means.

S101.20.4.1 Vestibule doors.

The door assembly from the building into the vestibule shall be a *fire door assembly* complying with Section 716.2.2.1. The door assembly from the vestibule to the *stairway* or *ramp* shall not have less than a 20-minute *fire protection rating* and shall meet the requirements for a smoke door assembly in accordance with Section 716.2.2.1. The door shall be installed in accordance with NFPA 105.

S101.20.4.2 Vestibule ventilation.

The vestibule shall be supplied with not less than one air change per minute and the exhaust shall be not less than 150 percent of supply. Supply air shall enter and exhaust air shall discharge from the vestibule through separate, tightly constructed ducts used only for that purpose. Supply air shall enter the vestibule within 6 inches (152 mm) of the floor level. The top of the exhaust register shall be located at the top of the smoke trap but not more than 6 inches (152 mm) down from the top of the trap, and shall be entirely within the smoke trap area. Doors in the open position shall not obstruct duct openings. Duct openings with controlling *dampers* are permitted where necessary to meet the design requirements, but *dampers* are not otherwise required.

S101.20.4.2.1 Engineered ventilation system.

Where a specially engineered system is used, the system shall exhaust a quantity of air equal to not less than 90 air changes per hour from any vestibule in the emergency operation mode and shall be sized to handle three vestibules simultaneously. Smoke detectors shall be located at the floor-side entrance to each vestibule and shall activate the system for the affected vestibule. Smoke detectors shall be installed in accordance with Section 907.3.

S101.20.4.3 Smoke trap.

The vestibule ceiling shall be not less than 20 inches (508 mm) higher than the door opening into the vestibule to serve as a smoke and heat trap and to provide an upward-moving air column. The height shall not be decreased unless *approved* and justified by design and test.

S101.20.4.4 Stairway or ramp shaft air movement system.

The *stairway* or *ramp* shaft shall be provided with a dampered relief opening and supplied with sufficient air to maintain a minimum positive pressure of 0.10 inch of water (25 Pa) in the shaft relative to the vestibule with all doors closed.

S101.20.5 Stairway and ramp pressurization alternative.

Where the building is equipped throughout with an *automatic sprinkler system* in accordance with Section 903.3.1.1, the vestibule is not required, provided that each interior *exit stairway* or *ramp* is pressurized to not less than 0.10 inch of water (25 Pa) and not more than 0.35 inches of water (87 Pa) in the shaft relative to the building measured with all *interior exit stairway* and *ramp* doors closed under maximum anticipated conditions of stack effect and wind effect.

S101.20.6 Ventilating equipment.

The activation of ventilating equipment required by the alternatives in Sections S101.20.4 and S101.20.5 shall be by smoke detectors installed at each floor level and at each entrance to the smokeproof enclosure. When the closing device for the *stairway* and *ramp* shaft and vestibule doors is activated by smoke detection or power failure, the mechanical equipment shall activate and operate at the required performance levels. Smoke detectors shall be installed in accordance with Section 907.3.

S101.20.6.1 Ventilation systems.

Smokeproof enclosure ventilation systems shall be independent of other building ventilation systems. The equipment, control wiring, power wiring and ductwork shall comply with one of the following:

1. Equipment, control wiring, power wiring and ductwork shall be located exterior to the building and directly connected to the smokeproof enclosure or connected to the smokeproof enclosure by ductwork enclosed by not less than 2-hour *fire barriers* constructed in accordance with Section 707 or *horizontal assemblies* constructed in accordance with Section 711, or both.
2. Equipment, control wiring, power wiring and ductwork shall be located within the smokeproof enclosure with intake or exhaust directly from and to the outside or through ductwork enclosed by not less than 2-hour *fire barriers* constructed in accordance with Section 707 or *horizontal assemblies* constructed in accordance with Section 711, or both.
3. Equipment, control wiring, power wiring and ductwork shall be located within the building if separated from the remainder of the building, including other mechanical equipment, by not less than 2-hour *fire barriers* constructed in accordance with Section 707 or *horizontal assemblies* constructed in accordance with Section 711, or both.

Exceptions:

1. Control wiring and power wiring located outside of a 2-hour fire barrier construction shall be protected using any one of the following methods:
 - 1.1. Cables used for survivability of required *critical circuits* shall be listed in accordance with UL 2196 and shall have a fire-resistance rating of not less than 2 hours.
 - 1.2. Where encased with not less than 2 inches (51 mm) of concrete.
 - 1.3. Electrical circuit protective systems shall have a fire-resistance rating of not less than 2 hours. Electrical circuit protective systems shall be installed in accordance with their listing requirements.

S101.20.6.2 Standby power.

Mechanical vestibule and *stairway* and *ramp* shaft ventilation systems and automatic fire detection systems shall be provided with standby power in accordance with Section 2702.

S101.20.6.3 Acceptance and testing.

Before the mechanical equipment is put into operation, the system shall be tested in the presence of the *fire code*

official to confirm that the system is operating in compliance with these requirements.

S101.21 Elevator hoistway pressurization alternative.

Where elevator hoistway pressurization is provided in lieu of required enclosed elevator lobbies, the pressurization system shall comply with Sections S101.21.1 through S101.21.11.

S101.21.1 Pressurization requirements.

Elevator hoistways shall be pressurized to maintain a minimum positive pressure of 0.10 inch of water (25 Pa) and a maximum positive pressure of 0.25 inch of water (67 Pa) with respect to adjacent occupied space on all floors. This pressure shall be measured at the midpoint of each hoistway door, with all elevator cars at the floor of recall and all hoistway doors on the floor of recall open and all other hoistway doors closed. The pressure differentials shall be measured between the hoistway and the adjacent elevator landing. The opening and closing of hoistway doors at each level must be demonstrated during this test. The supply air intake shall be from an outside, uncontaminated source located a minimum distance of 20 feet (6096 mm) from any air exhaust system or outlet.

Exceptions:

1. On floors containing only Group R occupancies, the pressure differential is permitted to be measured between the hoistway and a *dwelling unit* or *sleeping unit*.
2. Where an elevator opens into a lobby enclosed in accordance with Section 3007.6 or 3008.6, the pressure differential is permitted to be measured between the hoistway and the space immediately outside the door(s) from the floor to the enclosed lobby.
3. The pressure differential is permitted to be measured relative to the outdoor atmosphere on floors other than the following:
 - 3.1. The fire floor.
 - 3.2. The two floors immediately below the fire floor.
 - 3.3. The floor immediately above the fire floor.
4. The minimum positive pressure of 0.10 inch of water (25 Pa) and a maximum positive pressure of 0.25 inch of water (67 Pa) with respect to occupied floors are not required at the floor of recall with the doors open.

S101.21.1.1 Use of ventilation systems.

Ventilation systems, other than hoistway supply air systems, are permitted to be used to exhaust air from adjacent spaces on the fire floor, two floors immediately below and one floor immediately above the fire floor to the building's exterior where necessary to maintain positive pressure relationships as required in Section S101.21.1 during operation of the elevator shaft pressurization system.

S101.21.2 Rational analysis.

A rational analysis complying with Section S101.4 shall be submitted with the *construction documents*.

S101.21.3 Ducts for system.

Any duct system that is part of the pressurization system shall be protected with the same *fire-resistance rating* as required for the elevator shaft enclosure.

S101.21.4 Fan system.

The fan system provided for the pressurization system shall be as required by Sections S101.21.4.1 through S101.21.4.4.

S101.21.4.1 Fire resistance.

Where located within the building, the fan system that provides the pressurization shall be protected with the same *fire-resistance rating* required for the elevator shaft enclosure.

S101.21.4.2 Smoke detection.

The fan system shall be equipped with a smoke detector that will automatically shut down the fan system when smoke is detected within the system.

S101.21.4.3 Separate systems.

A separate fan system shall be used for each elevator hoistway.

S101.21.4.4 Fan capacity.

The supply fan shall be either adjustable with a capacity of not less than 1,000 cfm (0.4719 m³/s) per door, or that specified by a *registered design professional* to meet the requirements of a designed pressurization system.

S101.21.5 Standby power.

The pressurization system shall be provided with standby power in accordance with Section 2702.

S101.21.6 Activation of pressurization system.

The elevator pressurization system shall be activated upon activation of either the building fire alarm system or the elevator lobby smoke detectors. Where both a building fire alarm system and elevator lobby smoke detectors are present, each shall be independently capable of activating the pressurization system.

S101.21.7 Testing.

Testing for performance shall be required in accordance with Section S101.18.8. System acceptance shall be in accordance with Section S101.19.

S101.21.8 Marking and identification.

Detection and control systems shall be marked in accordance with Section S101.14.

S101.21.9 Control diagrams.

Control diagrams shall be provided in accordance with Section S101.15.

S101.21.10 Control panel.

A control panel complying with Section S101.16 shall be provided.

S101.21.11 System response time.

Hoistway pressurization systems shall comply with the requirements for smoke control system response time in Section S101.17.

S101.22 Maintenance.

Smoke control systems shall be maintained to ensure to a reasonable degree that the system is capable of controlling smoke for the duration required. The system shall be maintained in accordance with the manufacturer's instructions and Sections S101.22.1 through S101.22.6.

S101.22.1 Schedule.

A routine maintenance and operational testing program shall be initiated immediately after the smoke control system has passed the acceptance tests. A written schedule for routine maintenance and operational testing shall be established.

S101.22.2 Records.

Records of smoke control system testing and maintenance shall be maintained. The record shall include the date of the maintenance, identification of the servicing personnel and notification of any unsatisfactory condition and the corrective action taken, including parts replaced.

S101.22.3 Testing.

Operational testing of the smoke control system shall include all equipment such as initiating devices, fans, dampers, controls, doors and windows.

S101.22.4 Dedicated smoke control systems.

Dedicated smoke control systems shall be operated for each control sequence semiannually. The system shall be tested under standby power conditions.

S101.22.5 Nondedicated smoke control systems.

Nondedicated smoke control systems shall be operated for each control sequence annually. The system shall be tested under standby power conditions.

S101.22.6 Components bypassing weekly test.

Where components of the smoke control system are bypassed by the preprogrammed weekly test required by Section S101.12.1, such components shall be tested semiannually. The system shall be tested under standby power conditions."

(Amend Coun. J. 10-7-20, p. 21791, Art. II, § 35)